

TestMAX ATPG and TestMAX Diagnosis Messages

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About This Book

The *TestMAX ATPG and TestMAX Diagnosis Messages* book describes TestMAX ATPG and TestMAX Diagnosis messages.

This preface includes the following sections:

- [Conventions](#)
- [Customer Support](#)

Conventions

The following conventions are used in Synopsys documentation.

Convention	Description
Courier	Indicates syntax, such as <code>write_file</code> .
<i>Courier italic</i>	Indicates a user-defined value in syntax, such as <code>write_file design_list</code>
Courier bold	Indicates user input—text you type verbatim—in examples, such as prompt> write_file top
Purple	• Within an example, indicates information of special interest. • Within a command-syntax section, indicates a default, such as <code>include_enclosing = true false</code>
[]	Denotes optional arguments in syntax, such as <code>write_file [-format fmt]</code>
...	Indicates that arguments can be repeated as many times as needed, such as <code>pin1 pin2 ... pinN</code> .
	Indicates a choice among alternatives, such as <code>low medium high</code>
\	Indicates a continuation of a command line.
/	Indicates levels of directory structure.
Bold	Indicates a graphical user interface (GUI) element that has an action associated with it.
Edit > Copy	Indicates a path to a menu command, such as opening the Edit menu and choosing Copy .
Ctrl+C	Indicates a keyboard combination, such as holding down the Ctrl key and pressing C.

Customer Support

Customer support is available through SolvNetPlus.

Accessing SolvNetPlus

The SolvNetPlus site includes a knowledge base of technical articles and answers to frequently asked questions about Synopsys tools. The SolvNetPlus site also gives you access to a wide range of Synopsys online services including software downloads, documentation, and technical support.

To access the SolvNetPlus site, go to the following address:

<https://solvnetplus.synopsys.com>

If prompted, enter your user name and password. If you do not have a Synopsys user name and password, follow the instructions to sign up for an account.

If you need help using the SolvNetPlus site, click REGISTRATION HELP in the top-right menu bar.

Contacting Customer Support

To contact Customer Support, go to <https://solvnetplus.synopsys.com>.

1

TestMAX ATPG and TestMAX Diagnosis Messages

The following sections describe the Messages in TestMax ATPG and TestMAX Diagnosis:

2

Messages M1 - M99

This section contains the following error messages and warnings:

- [Error and Warning Messages M1 - M9](#)
- [Error and Warning Messages M10 - M19](#)
- [Error and Warning Messages M20 - M29](#)
- [Error and Warning Messages M30 - M39](#)
- [Error and Warning Messages M40 - M49](#)
- [Error and Warning Messages M50 - M59](#)
- [Error and Warning Messages M60 - M69](#)
- [Error and Warning Messages M70 - M79](#)
- [Error and Warning Messages M80 - M89](#)
- [Error and Warning Messages M90 - M99](#)

Error and Warning Messages M1 - M9

This section contains the following error messages and warnings:

- [M1](#)
- [M2](#)
- [M3](#)
- [M4](#)
- [M5](#)
- [M6](#)
- [M7](#)

- [M8](#)
- [M9](#)

M1

Message Text

Invalid argument "*keyword_or_argument*". (M1)

Severity

Error

Description

One of the keywords or arguments specified with the command was not recognized or an item from a list was used that is no longer defined for that list.

What Next

Use `helpcommand_name` to obtain a list of supported arguments for the command. Check your command line to check that you spelled all arguments correctly.

Use `help` without additional arguments to get a list of accepted commands.

If your command contains a list element, such as the name of a clock, or an alias, or an ATPG function or primitive label, make sure this list item is still defined by using an appropriate report command.

Remember the following when determining the validity of your argument:

Signal names (wires) are not leaf cell pin names.

During the build process, database flattening might remove the level of hierarchy you are trying to access with your argument.

In Tcl mode, you might need to compare the instance name in the command with the same instance name in the reported error message, then look for missing back slashes(\) and append the additional escape character(s), as needed. For example, you might see the following message:

```
Error: Invalid argument "chip_top/blk1_top_reciver/reciver_top/blk3".  
(M1)
```

In this case, you would need to change the following command line:

```
add_capture_mask \chip_top/blk1_top_reciver/reciver_top/blk3
```

To the following:

```
add_capture_mask \\chip_top/blk1_top_reciver/reciver_top/blk3
```

As an alternative, you can remove backslashes entirely. In this case, TestMAX ATPG commands automatically match specified identifiers that have no backslashes to identifiers in the database that have backslashes.

M2

Message Text

Missing argument. (M2)

Severity

Error

Description

A required argument is missing.

What Next

Use `helpcommand_name` to obtain a list of supported command arguments and switches. Review this information for required command arguments.

M3

Message Text

Unused argument <argument>. (M3)

Severity

Warning

Description

Additional command arguments were specified which were not used.

What Next

Type `help<command_name>` to obtain a summary of supported command arguments and switches for the command in question. Review the supported syntax and modify your command to avoid the extra item.

M4

Message Text

File *file_name* already exists. (M4)

Severity

Error

Description

An output file that was specified already existed.

What Next

If the file already exists, you can delete it, rename it, or use the `-replace` command option to overwrite it. Some commands also support the `-append` option. Type `helpcommand_name` to see this information.

M5

Message Text

Unable to open read file *file_name*. (M5)

Severity

Error

Description

The specified file cannot be opened for reading.

What Next

- Check spelling and case sensitivity of the file specified.
- Check that the file exists.
- Check that the path name specified for the file is correct.
- Check the file ownership.
- Check the file access controls (ACL's).

M6

Message Text

Unable to open write file *file_name*. (M6)

Severity

Error

Description

The file specified cannot be opened or cannot be written to.

What Next

- Check for sufficient disk space.
- Verify that the file does not already exist and is write protected.
- Verify that the directory where the file is to be created exists.
- Check that the directory access controls (ACL's) allow writing.
- Verify that the directory ownership allows writing.

M7

Message Text

Unable to open append file *file_name*. (M7)

Severity

Error

Description

The specified file could not be written to in append mode.

What Next

- Check for sufficient disk space.
- Verify that the file is not write protected.
- Check that the path name to the file is correct.
- Verify that the file's access controls (ACL's) allow writing.
- Verify that the file's ownership allows writing.

- Verify that the directory access controls (ACL's) allow writing.
 - Verify that the directory ownership allows writing.
-

M8

Message Text

Memory allocation error allocating N entries of N bytes each. (M8)

Severity

Error

Description

Memory was requested from the operating system and could not be obtained.

What Next

If you have multiple applications running, try quitting all of them but TestMAX ATPG and trying again.

You might need to add more RAM memory to your workstation or computer, or try a different machine that has more RAM.

M9

Message Text

Memory allocation error allocating N bytes. (M9)

Severity

Session termination error

Description

Memory was requested from the operating system and could not be obtained.

What Next

If you have multiple applications running, try quitting all of them but TestMAX ATPG and trying again.

You might need to add more RAM memory to your workstation or computer or try a different machine that has more RAM.

Error and Warning Messages M10 - M19

This section contains the following error messages and warnings:

- [M10](#)
- [M11](#)
- [M12](#)
- [M13](#)
- [M14](#)
- [M15](#)
- [M16](#)
- [M17](#)
- [M18](#)
- [M19](#)

M10

Message Text

User interrupt issued, *process_name* process is terminated. (M10)

Severity

Error

Description

You halted operations by using the `STOP` button or issuing an interrupt signal to the process.

What Next

No corrective action is needed.

M11

Message Text

Invalid command `<command>`. (M11)

Severity

Error

Description

The command you entered was not recognized.

What Next

Check the spelling of the command as well as the allowed switches and arguments. Use `help` to get a brief summary of commands then `help<command_name>` for a summary of supported switches and arguments.

M12

Message Text

`<license> license could not be acquired. (M12)`

Severity

Error

Description

An operation was attempted that required a license feature not currently checked out and the attempt to check out this feature failed. Your site might have all licenses in use, your site might not be licensed for the required feature, or there might be a license server problem.

What Next

- Use the `report_licenses` command to observe the licenses that you have successfully checked out.
- Use the `get_licenses` command to test manually checking out the license needed. The `get licenses` command also describes the licenses required for various ATPG operations.
- Check with your System Administrator for assistance in determining if all licenses are in use, no licenses exist, or your license server is having problems.

M13

Message Text

`Requested report contained no entries. (M13)`

Severity

Warning

Description

There were no entries found for the requested report.

What Next

Report was successfully completed with no entries. No corrective action is needed.

M14

Message Text

Unterminated quotation marks for string *string*. (M14)

Severity

Error

Description

A quoted string was found, but the matching end quote was not found.

What Next

Review your command or input file for strings with unmatched quotation marks. Make the necessary changes and try again.

M15

Message Text

Maximum arguments (N) in command line exceeded. (M15)

Severity

Error

Description

The maximum limit of arguments was exceeded for the command. The default limit of arguments for a command is 10,000. You can increase this limit as described in the What Next section.

What Next

Increase the command argument limit using the `-command_words` option of the `set_workspace_sizes` command, and specify the command again.

M16

Message Text

Maximum size (N) of command line exceeded. (M16)

Severity

Error

Description

Arguments involving long pin pathnames, instance pathnames, or other items exceeded the command line buffer size of 50000 bytes.

What Next

Increase the command line length limit using the `-command_line` option of the `set_workspace_sizes` command and try the command again.

M17

Message Text

`license_name` license cannot be returned due to minimum license restrictions. (M17)

Severity

Error

Description

An attempt was made to return a license that would have resulted in a failure to satisfy the minimum license restrictions. The minimum licensing requires an ability to perform DRC.

What Next

If you want to exchange one license with another, sometimes you can get the new license before removing the old one. You cannot return a Test-Analysis license from the TEST mode; however, you can return to DRC mode using the `drc -force` command, and then return this license.

M18

Message Text

Line of length N1 exceeds current limit of N2. (M18)

Severity

Error

Description

A line whose length exceeded the internal working buffer size was encountered while reading or writing a file.

What Next

Increase the internal buffer size using the `-line` option of the `set_workspace_sizes` command and try the command again.

M19

Message Text

Command cannot be used during `<mode_name>` command mode. (M19)

Severity

Error

Description

This message indicates that the specified command is not allowed in the current command mode. Another possibility is that you are attempting to change the environment or restrictions for pattern generation after patterns have already been created. For example, you cannot use the `-nomulti_clocks_per_load` option of the `set_atpg` command if patterns have already been created that contain multiple clocks per load.

What Next

Check the command modes allowed for a command by specifying `command_name -help`. Usually, changing command modes corrects this problem. However, be aware that switching to DRC command mode will cause a loss of previously generated patterns and the patterns will need to be regenerated.

Error and Warning Messages M20 - M29

This section contains the following error messages and warnings:

- [M20](#)
- [M21](#)
- [M22](#)

- [M23](#)
- [M24](#)
- [M25](#)
- [M26](#)
- [M27](#)
- [M28](#)
- [M29](#)

M20

Message Text

Command can only be used after model is built. (M20)

Severity

Error

Description

Many commands can only be issued after the simulation model has been built to produce an in-memory working image.

What Next

Use the `run_build_model` command before attempting the command that caused a problem. For information on whether a simulation model is required for a command, use `help <command_name> -usage`.

M21

Message Text

Invalid logic value <command_argument>. (M21)

Severity

Error

Description

An invalid logic value was used as an argument for a command.

What Next

Use a valid logic value (0, 1, X, or Z) for the argument and rerun the command.

M22

Message Text

`Schematic cannot be modified while viewer is in current state. (M22)`

Severity

Error

Description

The session is in a state where the schematic viewer is not allowed to change.

What Next

Allow all current operations to finish then perform schematic functions.

M23

Message Text

`Fault codes cannot be retained when deleting faults. (M23)`

Severity

Error

Description

When reading a fault file with the `-delete` option, the `-retain_code` option was also selected.

What Next

Remove the `-retain_code` option and rerun the command.

M24

Message Text

`The -force option must be selected to go from TEST to BUILD command mode. (M24)`

Severity

Error

Description

This message is generated when you are in TEST mode and attempt to change to BUILD mode. It is a protection feature to ensure that you do not accidentally delete unsaved patterns.

What Next

Reissue the command, but use the `-force` option.

M25

Message Text

Unsuccessful write operation on file "*file_name*". (M25)

Severity

Error

Description

A write operation to the indicated file was not successfully completed.

What Next

- Check for sufficient disk space.
- Check write permissions for the directory and file used.
- Use the UNIX `limits` command to see if there is a limit on the size of a file that can be written.

M26

Message Text

Number of sources of feedback path (*loop_id*) exceeds limit (*number*).
(M26)

Depth of unranked gates in feedback path (*loop_id*) exceeds limit
(*number*). (M26)

Severity

Error

Description

This message appears when the number of source gates for a combinational feedback loop (`loop_id`), or the depth of unranked gates, exceeds the preset program limit (`number`).

What Next

This error usually indicates that your vendor library is modeled at the switch and transistor level. TestMAX ATPG does not support this type of library.

This error sometimes indicates a sequential device modeling error in which a combinational model is inserted when a sequential model is expected. If the number of source gates in the feedback loop meets your design expectations, you should increase the program limit using the `-max_feedback_sources` option of the `set_learning` command, and specify the `run_build_model` command again.

When you resolve M26 violations by increasing the number of feedback source gates, you should inspect the feedback paths using the `report_feedback_paths` command. Although a simulation model is successfully built and ATPG runs, the model still might be inefficient. As a result, DRC and ATPG results might be compromised. For best results, you should simplify the model. For example, library cells modeled at the switch or transistor level should be modeled at the gate level.

You should use the `set_learning-max_feedback_sources` command to increase the preset program limit only when a M26 message appears. The loop ID reported in this message is an important part of the analysis since it enables you to validate that the identified path is combinatorial. You should apply the `report_feedback_paths -verbose` command to the loop ID identified in this message to validate the contents of that loop. Moderate increases to this limit value (for example, 2x or 3x, and 200 or 300) are likely to identify large combinatorial loops. Magnitude increases (10x or more) might indicate modeling problems that need to be reviewed.

M27

Message Text

Severity of fatal rules cannot be changed from error. (M27)

Severity

Error

Description

An attempt was made to change the severity of a rule that creates a fatal error.

What Next

If a fatal error rule had violations, it is necessary to correct the problem that caused the rule to fail.

M28

Message Text

N faults on bidi connections were nofaulted. (M28)

Severity

Warning

Description

In the simulation model, there were potential fault locations on bidirectional ports which can't hold fault data because of ATPG algorithm restrictions and which were nofaulted.

There is a very slight drop in test coverage which on a large design is only about 2 faults per bidirectional port.

Why? The ATPG algorithm requires a placeholder for holding present state/next state data. Internally, there is usually a gate present to satisfy this need and if not, a BUF primitive with no additional fault sites might be inserted as needed. On I/O pins a BUF is also often required. For a bidirectional I/O pin a BUF presents a problem because it is a unidirectional gate and can't be placed on a bidirectional net/port without breaking the bidirectional nature. The ATPG algorithm can deal with this bidirectional port problem in one of two ways:

Method #1 - Split the internal representation of the bidirectional port into separate input and output ports. This is somewhat awkward because the user does not think of their design this way if and is not expecting to see two ports in graphics views and reports, one for each direction.

Method #2 - Show the bidirectional port as a single object to the user in both graphic views and data reports. To avoid the problem of a fault on the bidirectional section of the net exclude all faults there by inserting a non-removable nofault. Issue an appropriate warning.

TestMAX ATPG uses method #2, which has the side affect of reducing the potential fault population. So an M28 warning is issued.

In the grand scheme of things, the number of faults dropped from the fault population is trivial compared to the core logic of most designs. In addition, these faults are probably detected by ATPG patterns, its just that ATPG tool doesn't take credit for them. Any reasonable set of functional patterns should make up for the dropped faults.

What Next

No corrective action is required. However, if you are a test purist, you should review the plan for using functional patterns to make sure the bidirectional ports are fully tested.

M29

Message Text

Unsaved test patterns (M29) - confirm session exit? (Yes, No) "

Severity

Warning

Description

You receive this warning message if you attempt to exit the tool and ATPG patterns have been generated without saving them, or if you have written a subrange of patterns.

TestMAX ATPG does not keep track of which patterns have been saved by writing subranges using the `-first` and `-last` options, so as a safeguard this warning is given for all cases except when a write without any `-first` or `-last` option has been used.

What Next

- If you do not wish to save patterns, select "Yes" to confirm that you want to exit.
 - Respond "No" if you wish to cancel the exit and remain in TestMAX ATPG.
 - To avoid this exit query in the future, you can use the `-force` option of the `exit` or `quit` commands.
 - You should write binary patterns for all patterns with an M29 message.
-

Error and Warning Messages M30 - M39

This section contains the following error messages and warnings:

- [M30](#)
- [M31](#)
- [M32](#)
- [M33](#)
- [M34](#)
- [M35](#)

- [M36](#)
- [M37](#)
- [M38](#)
- [M39](#)

M30

Message Text

Contention status of bus gates has been reset to not_analyzed. (M30)

Severity

Warning

Description

When the "multiple_on" setting for contention checking is changed, the contention status of all BUS gates is reset to not analyzed.

What Next

It is recommended that BUS gate analysis be performed before running simulation or test generation. This analysis is automatically performed during DRC. You can also perform this manually using the `analyze_buses` command.

M31

Message Text

Invalid primary input name `port_name`. (M31)

Severity

Error

Description

The port name you specified does not match an input port of the top-level module.

What Next

- Check that the top module is set to the module name you expected.
- Obtain a list of the ports of the module in question by using the `report_modulesmodule_name -verbose` command or the `report_primitives-ports` command.
- Verify that the port you attempted to use is considered an input or bidirectional port.

M32

Message Text

Invalid primary output name `port_name`. (M32)

Severity

Error

Description

The port name you specified does not match an output port of the top-level module.

What Next

- Check that the top module is set to the module name you expected.
- Obtain a list of the ports of the module in question by using the `report_modulesmodule_name -verbose` command.
- Verify that the port you attempted to use is considered an output or bidirectional port.

M33

Message Text

Clock port `port_name` cannot be constrained to its on state. (M33)

Severity

Error

Description

A PI constraint was attempted, which would have resulted in a port that was previously defined as a clock being held in its "on" state.

What Next

Review the defined clock port to verify that it was defined with the correct on state using the `report_clocks` command.

If the clock was defined correctly, avoid attempting to constrain the clock to it's on state in the future.

M34

Message Text

`port_name` is already in clock list. (M34)

Severity

Error

Description

The port referenced in the `add_clocks` command was already defined as a clock.

What Next

Use the `report_clocks` command to see a list of defined clocks.

If you want to redefine the clock, first remove the clock definition using the `remove_clocks` command before trying the `add_clocks` command.

M35

Message Text

Scan chain input pin `name` cannot be a clock. (M35)

Severity

Error

Description

A pin that was defined as a "clock" cannot also be used as a scan chain input or output.

What Next

Verify that the pin needs to be defined as a clock. It is possible that you can skip the declaration of the pin as a clock and continue.

If the pin is truly acting as a clock to scan cells, you might need to make design modifications to move the scan chain input to another pin.

M36

Message Text

`Equivalent pin name also added to clock list. (M36)`

Severity

Warning

Description

If an equivalent pin is added to the clock list, all of its equivalent pin are also added to the clock list.

What Next

No action is necessary unless the added clock was not intended to be a clock.

M37

Message Text

`Capture clock <clock_name> is now removed. (M37)`

Severity

Warning

Description

The clock that was selected to be the ATPG capture clock was removed.

What Next

The ATPG capture clock is reset to "none". No corrective action is needed if this is required. Otherwise, a new capture clock must be selected using the `-clock` option of the `set_drc` command.

M38

Message Text

`clock_name is not in clock list. (M38)`

Severity

Error

Description

A command requiring that a clock be specified was given the name of a port that was not in the defined clock list.

What Next

Review the list of declared clock port using the `report_clocks` command. Retry the command using a declared clock.

If the port originally specified is correct, consider adding it to the declared clock list.

M39

Message Text

```
Current simulation model is now deleted. (M39)
```

Severity

Warning

Description

This warning is displayed when a simulation model is deleted. Possible causes are build errors or reentering BUILD command mode.

What Next

No action is required. This warning is just a reminder that the simulation model has been deleted. Until a new simulation model is constructed, the BUILD mode cannot be exited which is required to perform many commands.

Error and Warning Messages M40 - M49

This section contains the following error messages and warnings:

- [M40](#)
- [M41](#)
- [M42](#)
- [M43](#)
- [M44](#)
- [M45](#)
- [M46](#)

- [M47](#)
- [M48](#)
- [M49](#)

M40

Message Text

`Build process aborted: module=M, imodule=I, nprimitives=N. (M40)`

Severity

Error

Description

This message is given if any violation of a rule with severity error occurred during the building of a simulation model from the named module.

What Next

Correct the netlist causing the problem and read it again, followed by a rebuild. Or, lower the severity of the rules violated and rebuild.

M41

Message Text

`All netlist and library module data are now deleted. (M41)`

Severity

Warning

Description

Using the `-delete` option of the `read_netlist` command causes this warning.

What Next

No action is required. This warning is just a reminder that all previous netlists read in have been deleted.

M42

Message Text

`name is already in chain list. (M42)`

Severity

Error

Description

A name was assigned to a scan chain in the test protocol file that was a previously used scan chain name.

What Next

Change the assigned name for the chain that had the duplicate name and rerun DRC.

M43

Message Text

name is already an alias name. (M43)

Severity

Error

Description

An alias was attempted to be created using the same name of a previously created alias.

What Next

Either select a different name for the alias or delete the alias with that name.

M44

Message Text

Text of alias *alias_name* cannot be recognized as a command. (M44)

Severity

Error

Description

An alias was attempted to be created with text that could not be recognized as a valid command or which violated the restrictions for defining aliases.

What Next

Review the restrictions listed for defining aliases in the TestMAX™ ATPG and TestMAX Diagnosis Commands guide. Place a valid command at the beginning of alias text.

M45

Message Text

`Interrupt command can only be used in a top level command file. (M45)`

Severity

Error

Description

This is an error message that is given when an interrupt command is not issued from a top-level command file. Command files that are run from another command file cannot contain an interrupt command.

What Next

You must remove the interrupt from the lower-level command file.

M46

Message Text

`Resume command can only be used after a command file interrupt. (M46)`

Severity

Error

Description

The resume command can only be used when an interrupt command is pending.

What Next

The resume command is ignored, you need do nothing.

M47

Message Text

`Multiple (N) memory gates are derived from instance name name. (M47)`

Severity

Error

Description

A `read_memory_file` command was attempted for an instance name that was associated with more than one memory gate.

What Next

Identify the required memory gate using its gate ID number and try the command again using this gate ID reference.

M48

Message Text

No memory cell values defined for memory gate I (N). (M48)

Severity

Error

Description

A report was requested for memory cell values of a memory gate that did not have a memory table.

What Next

Remove the request for memory values and repeat the command.

M49

Message Text

No memory cell values defined for address A of memory gate I (G). (M49)

Severity

Error

Description

A report was requested for memory cell values of a memory gate that did not have a memory table.

What Next

Remove the request for memory values and repeat the command.

Error and Warning Messages M50 - M59

This section contains the following error messages and warnings:

- [M50](#)
- [M51](#)
- [M52](#)
- [M53](#)
- [M54](#)
- [M55](#)
- [M56](#)
- [M57](#)
- [M58](#)
- [M59](#)

M50

Message Text

`port is already constrained. (M50)`

Severity

Error

Description

An `add_pi_constraints` was attempted on a port that already had been constrained.

What Next

Review the existing PI constraints using the `report_pi_constraints` command. If you intend to change the constraint of a port, remove the existing constraint using a `remove_pi_constraintsport_name` command before the new add pi constraints command.

M51

Message Text

`Non-tristateable pin name cannot be constrained to a Z. (M51)`

Severity

Error

Description

An attempt was made to add a PI constraint to a top-level port that was not connected to circuitry that can accept a "Z" state. These ports are limited to be constrained to 0/1/X states.

What Next

Review the list of bidirectional ports using the `report_primitives-pio` command. If the port should have been three-stateable, review the simulation model using a `report_primitivesport_name` command as a starting point for further exploration.

M52

Message Text

Clock port P can only be constrained to its off state. (M52)

Severity

Error

Description

An attempt was made to define a PI constraint on a port that was previously declared as a clock and the constrained value attempted was not the clock's defined off state.

What Next

Use the `report_clocks` command to review the defined off state of the clock port. If the off state is not correct, change the definition by deleting the clock definition then redefining it.

If the PI constraint is correct but still conflicts with the defined state of the clock, then remove the port as a defined clock. The PI constraint would have dominated over the clock in any event, so defining this port as a clock is not necessary.

M53

Message Text

Capture clock P cannot be constrained. (M53)

Severity

Error

Description

An attempt was made to define a PI constraint on a port that was previously declared as a clock and that has been identified with behavior of a capture clock as opposed to an asynchronous set or reset. Having both a clock definition and a PI constraint on the same port is only allowed if the port is an asynchronous set or reset.

What Next

If the port must be constrained, remove the clock declaration using the `remove_clocks` command. Check to see if there are `set_drc-clock pin_name` settings in your command file.

M54

Message Text

Equivalence on pin P was removed due to constraint. (M54)

Severity

Warning

Description

A port that was previously declared to be equivalent is now being constrained. The equivalence is removed from this and all equivalent pins. The constraint specified for this pin will override the equivalence behavior.

Be aware that you should add a constraint on all pins previously defined to be equivalent in this situation, as the other equivalent pins will not be constrained and will not be equivalent after the first equivalent pin is constrained -- unless they are explicitly constrained.

What Next

If ports are equivalent as well as constrained, remove the PI equivalence declaration and add an appropriate PI constraint to both ports.

M55

Message Text

P is not constrained. (M55)

Severity

Error

Description

A `remove_pi_constraints` command was attempted on a port that did not have a PI constraint defined for it.

What Next

No action is necessary, however, if the port should have been previously constrained, investigate why it no longer has a constraint defined.

M56

Message Text

New command files cannot be executed during an interrupted command file.
(M56)

Severity

Error

Description

This is an error which occurs when an interrupt command is pending and you attempt to run another command file. While the first command file is paused/interrupted no other command file can be started.

What Next

Avoid using the command `source` while a command file is paused.

M57

Message Text

P1 is already equivalent to P2. (M57)

Severity

Error

Description

An `add_pi_equivalences` command was attempted between two ports, P1 and P2, that were already declared as equivalent.

What Next

No corrective action is required. Review your command sequence and avoid attempting to doubly declare port equivalences in the future.

M58

Message Text

P1 has conflicting equivalence with P2. (M58)

Severity

Error

Description

An `add_pi_equivalences` command was attempted between two ports that were already declared as equivalent and whose current equivalence conflicts with the newly requested equivalence.

What Next

The current equivalence between the two ports must be removed using the `remove_pi_equivalences` command. After removal, retry adding the PI equivalence.

M59

Message Text

Pin P cannot be equivalent to itself. (M59)

Severity

Error

Description

A PI equivalence declaration included the same port twice and a port cannot be declared equivalent to itself.

What Next

Correct the command and try again.

Error and Warning Messages M60 - M69

This section contains the following error messages and warnings:

- [M60](#)
- [M61](#)
- [M62](#)

- [M63](#)
- [M64](#)
- [M66](#)
- [M67](#)
- [M68](#)
- [M69](#)

M60

Message Text

Constrained port P cannot be equivalent. (M60)

Severity

Error

Description

Defining ports that are both equivalent and constrained is not supported.

What Next

Remove the PI equivalence declaration and define PI constraints on both ports using appropriate values.

M61

Message Text

Clock port P1 cannot be equivalent to nonclock port P2. (M61)

Severity

Error

Description

An attempt was made to declare two ports to be equivalent and one of those ports had been previously declared as a clock but the other was not.

Note: Although you can define two ports to be equivalent and then define one to be a clock you can't do this in the reverse order of: define one to be a clock and then two ports to be equivalent. Why? When you declare a clock and a non-clock to be equivalent do you

wish to make the clock into a non-clock or the non-clock into a clock? Rather than guess, enforce a certain order restriction on defining clocks first and PI equivs second.

What Next

If you intend for both ports be equivalent, declare both ports to be clocks before attempting to define them as equivalent ports. You can also declare both ports to be equivalent before defining one of them as a clock and the other will also become a clock, although a warning message is issued.

M62

Message Text

Clock ports P1 and P2 have inconsistent off-states. (M62)

Severity

Error

Description

Two ports that were previously declared as clocks have been attempted to be defined as equivalent and there is an inconsistency between the clock's off state and the declared equivalence.

What Next

If the clocks have opposite off-states, use the `-invert` option when declaring them as equivalent ports.

M63

Message Text

Maximum number of ATPG primitives (N) has been exceeded. (M63)

Severity

Error

Description

Defining ATPG primitives requires some memory space to keep track of these "virtual" gates. This message indicates the RAM space used for this purpose is full.

What Next

Use the `set_workspacesizes -atpg_gates` option to increase the RAM space for tracking virtual gates. Then repeat the `run_build_model` command.

M64

Message Text

P is not an equivalence port. (M64)

Severity

Error

Description

A `remove_pi_equivalences` command was attempted on a port, which was not in the list of defined PI equivalences.

What Next

To see which ports are defined as equivalent, use the `report_pi_equivalences` command.

If the port attempted should have been in the equivalence list, review the transcript and prior commands to attempt to discover why it is no longer in the list.

Avoid attempting this command in the future for ports that are not previously defined as equivalent.

M66

Message Text

I is already in cell constraint list. (M66)

Severity

Error

Description

An `add_cell_constraints` command was attempted on a scan cell that was already constrained.

What Next

No corrective action is required if the scan cell is already constrained to the wanted value. Otherwise, remove the current cell constraint using the `remove_cell_constraints` command, and retry adding the cell constraint.

M67

Message Text

Position N in chain S is already in cell constraint list. (M67)

Severity

Error

Description

An `add_cell_constraints` command was attempted on a scan cell that was already constrained.

What Next

No corrective action is required if the scan cell is already constrained to the wanted value. Otherwise, remove the current cell constraint using the `remove_cell_constraints` command and retry adding the cell constraint.

M68

Message Text

I is not in cell constraint list. (M68)

Severity

Error

Description

A `remove_cell_constraints` command was attempted on a scan cell that was not already constrained.

What Next

No corrective action is required if the scan cell was correctly identified. Otherwise, correct the scan cell identification and rerun the command.

M69

Message Text

Position N in chain S is not in cell constraint list. (M69)

Severity

Error

Description

A `remove_cell_constraints` command was attempted on a scan cell that was not already constrained.

What Next

No corrective action is required if the scan cell was correctly identified. Otherwise, correct the scan cell identification and rerun the command.

Error and Warning Messages M70 - M79

This section contains the following error messages and warnings:

- [M70](#)
- [M71](#)
- [M72](#)
- [M73](#)
- [M74](#)
- [M75](#)
- [M76](#)
- [M77](#)
- [M78](#)
- [M79](#)

M70

Message Text

Simulation with pattern storage cannot be performed for sequential simulation. (M70)

Severity

Error

Description

A `run_simulation` or `run_fault_simulation` command was attempted using both the `-sequential` and `-store` options.

What Next

Storage of patterns into the internal pattern buffer is not supported when the sequential option is used. Reissue the `run_simulation` command without the `-store` option.

M71

Message Text

`N gate (G) can only be constrained to 1. (M71)`

Severity

Error

Description

An attempt was made to constrain a SEL1, SEL01, or EQUIV gate to a 0 or Z state. These are special gates that are only allowed to be constrained to a 1 state.

What Next

Find an equivalent way to logically constrain the circuit without these constraints.

M72

Message Text

`site is already in ATPG constraint list. (M72)`

Severity

Error

Description

An `add_atpg_constraints` command was attempted on a site that was already constrained.

What Next

No corrective action is required if the site is already constrained to the wanted value. Otherwise, remove the current ATPG constraint using the `remove_atpg_constraints` command and retry adding the atpg constraint.

M73

Message Text

DRC ATPG constraints can only be added or removed in DRC command mode.
(M73)

Severity

Error

Description

A DRC ATPG constraint was attempted to be added or removed outside of DRC command mode.

What Next

Return to DRC command mode and rerun the command.

M74

Message Text

Invalid pattern source (*source*). (M74)

Severity

Error

Description

An operation was attempted that could not be performed given the current selection of the pattern source.

What Next

Use the `set_patterns` command to change the pattern source to be compatible with the selected operation and rerun the command.

M75

Message Text

ATPG primitive (N) cannot have an input connected to G gate. (M75)

Severity

Error

Description

An attempt was made to create an ATPG primitive with an input that was connected to a SEL1, SEL01, or EQUIV gate. These gates are special gates that are not allowed to connect to ATPG primitives.

What Next

Determine an equivalent way to create the logical behavior without using these gates.

M76

Message Text

`name` is already in ATPG primitive list. (M76)

Severity

Error

Description

An `add_atpg_primitives` command was attempted with a name that was previously used.

What Next

Select a different name for the ATPG primitive and rerun the command.

M77

Message Text

Gate connection name `name` is not unique. (M77)

Severity

Error

Description

An attempt was made to create an ATPG primitive that had an input which was not uniquely specified.

What Next

Determine a unique way to specify the ATPG primitive inputs (such as gate ID number) and rerun the command.

M78

Message Text

ATPG primitive can't be removed due to ATPG constraint (*constraint*) usage. (M78)

Severity

Error

Description

An attempt was made to remove an ATPG primitive that was currently being used to define an ATPG constraint.

What Next

Remove all ATPG constraints that are affected by the ATPG primitive using the `remove_atpg_constraints` command and then rerun the command.

M79

Message Text

ATPG primitive can't be removed due to a connection to an existing ATPG primitive (%s-N). (M79)

Severity

Error

Description

An attempt was made to remove an ATPG primitive that was currently being used to define another ATPG primitive.

What Next

Remove all ATPG primitives that are affected by the ATPG primitive using the `remove_atpg_primitives` command and then rerun the command.

Error and Warning Messages M80 - M89

This section contains the following error messages and warnings:

- [M80](#)
- [M81](#)

- [M82](#)
- [M83](#)
- [M84](#)
- [M85](#)
- [M86](#)
- [M87](#)
- [M88](#)
- [M89](#)

M80

Message Text

Maximum schematic size exceeded. (M80)

Severity

Error

Description

This message is reported when the schematic is too large to be stored by the application. The schematic will not be fully displayed.

What Next

Zoom out to reduce the size of the schematic. Reduce the number of gates displayed in the schematic.

M81

Message Text

Maximum zoom factor reached. (M81)

Severity

Error

Description

This message is reported when a Zoom In operation causes the zoom factor to exceed the maximum allowable amount.

What Next

Zoom Out the schematic to reduce the current zoom factor.

M82

Message Text

Minimum zoom factor reached. (M82)

Severity

Error

Description

This message is reported when a zoom out operation causes the zoom factor to fall below the minimum allowable amount.

What Next

Zoom in the schematic to increase the current zoom factor.

M83

Message Text

Unable to resolve wire layout. All wires are drawn but some wires can have been overlapped by other wires (M83)

Severity

Error

Description

This message is reported when the gates and wire connections required to be drawn in the schematic were too complicated to be determined by the schematic viewer. It completed the draw, but overlapped some graphics objects or nets.

What Next

Not much you can do but to try to reduce the number of gates displayed in the schematic.

M84

Message Text

Unable to complete schematic within the allocated time. The display list has been deleted. (M84)

Severity

Error

Description

The list of instances, nets, and graphics objects to be displayed was large enough that the dynamic drawing algorithm could not find a solution within the default time limit. When the time limit is reached the algorithm gives up trying and displays this message.

What Next

The timeout for this drawing effort can be adjusted using the "Time Out" field found in the Edit->Preferences->Viewer dialog box tab. The default is 20 seconds. Note that allowing more time does not necessarily result in better results. The command-line entry equivalent to the pull-down menu is:

```
set_environment_viewer -time_out 100
```

You can interactively draw parts of the schematic by picking the SHOW button and entering instance pathnames or gate ID numbers involved with the items which you wished to be drawn. Refer to the transcript window for some instance paths and gate ID for starting points. After you have a starting point you can add other gates by SHOW->NAMED, or expand from the current gate by clicking on the net diamonds.

You can have better luck by changing to PRIMITIVE view with buffers and inverters hidden and then re-attempting whatever you were working on which tried to draw and failed.

Perhaps there is a textual method for reviewing the data that cannot be displayed by using the `report` commands.

M85

Message Text

\$dumpportsoff occurs at T=<time>, all events after this point is ignored. (M85)

Severity

Warning

Description

When reading an Extended VCD pattern file, a `$dumpportsoff` block was encountered. TestMAX ATPG ends its reading effort and will not process any future events that might or might not exist in the file.

What Next

It is possible, but rare, to have pairs of `$dumpportsoff` and `$dumpportson` directives in the pattern file. If your pattern file contains only the off directive, then you may ignore this warning. If however, the pattern file contains a series of off/on directives, then the complete pattern file has not been read. A workaround for this issue is to edit the pattern file (or a copy of it) and to delete all lines between and including the `$dumpportsoff` keyword and the `$end` directive for the corresponding `$dumpportson` keyword.

M86

Message Text

Unsupported YDF version 1.1 for class-based diagnosis candidates. YDF version must be 1.2 or greater. (M86)

Severity

Error

Description

Diagnosis candidates using the class-based diagnosis reporting are supported only by YDF version 1.2 or later for further analysis in Yield Explorer.

What Next

To enable YDF version 1.2, use the following command:

```
set_ydf -version 1.2
```

For more information, refer to the following topics:

- Class-Based Diagnosis Reporting in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.
- `set_ydf` in the TestMAX™ ATPG and TestMAX Diagnosis Commands Guide.

M87

Message Text

Basic-scan enabled and -auto mode is required when an observe file is used. (M87)

Severity

Error

Description

This message is issued when you pass an observe point file to the `run_atpg` command with `-observe_file` option but did not select the `-auto_compression` option as well.

What Next

For doing ATPG analysis with observe points, you need to use the `-auto_compression` option of the `run_atpg` command.

For more information, refer to the topic `run_observe_analysis` in the TestMAX™ ATPG and TestMAX Diagnosis Commands Guide.

M88

Message Text

Variable TMAX_UG is undefined. Unable to find User's Guide. (M88)

Severity

Error

Description

An attempt to open the TestMAX ATPG User Guide was made but the environment variable `TMAX_UG` which should point to the file location was not defined. This variable is set when TestMAX ATPG is first started and if the file is not found at it's expected location the variable will not be set.

What Next

Recheck the file installation. The file "tmax_ug.pdf" should be located within the `$SYNOPSIS/doc/test/tmax` directory. Alternatively you can define the shell environment variable `TMAX_UG` to point to the PDF file before invoking TestMAX ATPG.

M89

Message Text

Unable to open Acrobat Reader. (M89)

Severity

Error

Description

The acrobat reader was not found in your current search path when you attempted to open the user's guide.

What Next

Check your login environment to make sure the 'acroread' utility can be found in your default PATH setting before re-invoking TestMAX ATPG.

If the PDF reader is no longer named 'acroread', try defining an alias relink with the name 'acroread' so that TestMAX ATPG can still invoke using this executable name.

Error and Warning Messages M90 - M99

This section contains the following error messages and warnings:

- [M90](#)
- [M91](#)
- [M92](#)
- [M93](#)
- [M94](#)
- [M95](#)
- [M96](#)
- [M97](#)
- [M98](#)
- [M99](#)

M90

Message Text

`String string not found. (M90)`

Severity

Warning

Description

This message is issued after a text search operation couldn't find the string symbolized by *string* in the given text window.

What Next

Enter the proper search parameters in the Find dialog box.

M91

Message Text

`Command(s) not submitted. (M91)`

Severity

Error

Description

This message is issued when a command or action was attempted when the application was busy performing an operation.

What Next

Wait until the application is idle before submitting a command or performing an operation.

M92

Message Text

`Window limit exceeded. (M92)`

Severity

Error

Description

This message is issued when the number of open report windows exceeds the maximum allowable amount.

What Next

Close unnecessary report windows and resubmit command.

M93

Message Text

Unable to exit application. (M93)

Severity

Error

Description

This message is issued when an application exit request is not able to be performed.

What Next

Wait until the application is idle before requesting an application exit operation.

M94

Message Text

The last N lines were not reported. (M94)

Severity

Error

Description

This message is issued when the number of lines displayed in a report window exceeds the maximum number.

What Next

Limit the number of lines reported in `report` commands.

M95

Message Text

`Invalid Parameters (M95)`

Severity

Error

Description

This message is issued when a parameter is entered or ignored in a dialog box.

What Next

Enter the correct parameters into the dialog box.

M96

Message Text

`Block ID# N not found. (M96)`

Severity

Error

Description

This message is issued when a Block ID parameter is not found.

What Next

Enter the correct Block ID.

M97

Message Text

`Invalid or unrecognized file format. (M97)`

Severity

Error

Description

Reading a file failed because the file was either in an unsupported format or the format was not recognized.

What Next

Check the file for corruption.

Check the documentation on the command you were attempting to use to see if there is any information on supported file formats or listed restrictions to file formats, such as not accepting gzip and so on.

M98

Message Text

Multiple cell names are not supported in `add_cell_constraints`. (M98)

Severity

Error

Description

The `add_cell_constraints` command was called with multiple cell name references. This command accepts only one cell name reference per call.

What Next

Separate the cell names into distinct calls to the `add_cell_constraints` command. For example, if you assemble a Tcl collection called "all_cells," the following Tcl operation calls the `add_cell_constraints` command for each cell in the collection:

```
foreach_in_collection obj ${all_cells} {  
    add_cell_constraints ox [get_attribute ${obj} cell_name]  
}
```

M99

Message Text

Creation of directory <path> failed. (M99)

Severity

Error

Description

Use of the `mkdir` command returned a failure status.

What Next

- Perhaps the directory already exists.
- If a complex directory pathname is used, such as 'aaa/bbb/ccc', check that 'aaa/bbb' already exists.
- Verify that the parent directory where the new directory is to be created allows writing by your user or group ID.
- Check for sufficient disk space.

3

Messages M100 - M199

This section contains the following error messages and warnings:

- [Error and Warning Messages M100 - M109](#)
- [Error and Warning Messages M110 - M119](#)
- [Error and Warning Messages M120 - M129](#)
- [Error and Warning Messages M130 - M139](#)
- [Error and Warning Messages M140 - M149](#)
- [Error and Warning Messages M150 - M159](#)
- [Error and Warning Messages M160 - M169](#)
- [Error and Warning Messages M170 - M179](#)
- [Error and Warning Messages M180 - M189](#)
- [Error and Warning Messages M190 - M199](#)

Error and Warning Messages M100 - M109

This section contains the following error messages and warnings:

- [M100](#)
- [M101](#)
- [M102](#)
- [M103](#)
- [M104](#)
- [M105](#)
- [M106](#)
- [M107](#)

- [M108](#)
- [M109](#)

M100

Message Text

Design rule checking failed: cannot exit DRC command mode. (M100)

Severity

Error

Description

An error condition occurred during the design rule checking process. This prevents transitioning out of DRC mode into TEST mode.

What Next

The cause of the error condition must be corrected or the severity of the error must be decreased. You can then repeat the DRC process using the `run_drc` command.

M101

Message Text

Design rule checking aborted due to user interrupt: cannot exit DRC command mode. (M101)

Severity

Error

Description

A user interrupt caused the design rule checking process to be terminated.

What Next

No action is required. It is still necessary to perform a successful design rules check before entering TEST.

M102

Message Text

Virtual scan chain name "_sc1" conflicts with a current chain name.
(M102)

Severity

Error

Description

The virtual scan chain name _sc1 was used as the name of an existing scan chain.

What Next

To successfully use the "what if" capability of the `set_scan_ability` command, you need to rename your top level pin to something other than _sc1.

M103

Message Text

Feedback path N failed to stabilize after 100 passes. (M103)

Severity

Warning

Description

During the simulation of a pattern, a feedback path failed to reach a stable value after 100 passes. Gates that fail to stabilize are set to X. This indicates a potential oscillation is occurring and could result in a significant impact on simulation performance.

What Next

No action is required. However, to improve performance both within ATPG and potentially in your digital logic simulator (Verilog, VHDL, and so forth) analyze the feedback path and determine why it does not stabilize. It would appear the feedback path is oscillating and this could have a large performance impact in simulation. Block the oscillation either by applying a PI constraint or by adding gates to block the feedback loop while in ATPG test mode. To successfully use the "what if" capability of the `set_scan_ability` command, you need to rename your top level pin to something other than _sc1.

M104

Message Text

Faulting cannot be changed when internal patterns are active. (M104)

Severity

Error

Description

An attempt was made to change the fault list while internal patterns were active. This is not allowed because it can change any potential fault grade relationship between the current fault list and the current patterns.

What Next

Use the `-delete` option of the `set_patterns` command to remove the internal patterns and rerun the command. The faults continue to retain their previous classification. To reset faults back to undetected, use the `reset_state` command.

M105

Message Text

Nofaulting cannot be changed when faults are in current fault list.
(M105)

Severity

Error

Description

An attempt was made to change nofaulting behavior when faults have already been placed in the current fault list.

What Next

Remove all faults from the current fault list using the `remove_faults` command and rerun the nofault change.

M106

Message Text

Fault model cannot be changed when faults are in current fault list.
(M106)

Severity

Error

Description

The fault model type attempted to change while faults were in the current fault list.

What Next

Remove all faults from the fault list and rerun the command.

M107

Message Text

N fault sites were ignored (first occ. at line N). (M107)

Severity

Warning

Description

During reading of a fault file with the `-retain_code` option, at least one invalid fault site or fault code was encountered. Invalid fault sites are ignored. Invalid fault codes on legitimate fault sites are ignored and the associated fault site is considered UD = undetected.

What Next

No action is required. However, if the number of fault sites ignored or considered UD is high you should investigate the cause and correct it if possible.

M108

Message Text

IDDQ selection can only be performed for combinational fault simulation.
(M108)

Simulation with pattern storage cannot be performed for internal
patterns. (M108)

Severity

Error

Description

The `-store` option of the `run_simulation` or `run_fault_sim` command cannot be used when the pattern source is set to internal patterns. The `-store` option saves patterns in

the internal pattern set and trying to simultaneously simulate from and store to the same pattern set is not allowed.

What Next

Reissue the command without the `-store` option.

Save the current internal patterns to a file and then set the pattern source to this external file to use the `-store` option.

M109

Message Text

```
Simulation aborted: #patterns=N (M109)
```

```
Simulation aborted: #patterns=N, #fail_pats=N(N), #failing_meas=N(N)  
(M109)
```

Severity

Error

Description

A user interrupt was issued to terminate the current simulation process.

What Next

No corrective action is required. Simulation results reflect only what occurred before the interrupt.

Error and Warning Messages M110 - M119

This section contains the following error messages and warnings:

- [M110](#)
- [M111](#)
- [M112](#)
- [M113](#)
- [M114](#)
- [M115](#)
- [M116](#)

- [M117](#)
- [M118](#)
- [M119](#)

M110

Message Text

Simulation aborted: #patterns=N, CPU time=T (M110)

Severity

Error

Description

A user interrupt was issued to terminate the current simulation process.

What Next

No corrective action is required. Simulation results only reflect what occurred before the interrupt.

M111

Message Text

There are no patterns in the <internal | external> pattern set. (M111)

Severity

Error

Description

Use of the internal patterns was requested but there were no patterns in the current internal pattern set.

What Next

If the pattern source is incorrect, set the pattern source to the intended patterns using the `set_patterns` command and rerun the simulation.

You can still write the chain test and optional test_setup related patterns by using the `-exclude patterns` option.

M112

Message Text

There are no patterns in the external pattern set. (M112)

Severity

Error

Description

Use of the external patterns was requested but there were no patterns in the current external pattern set.

What Next

If the pattern source is incorrect, set the pattern source to the intended patterns using the `set_patterns` command and rerun the simulation.

You can still write the chain test and optional `test_setup` related patterns by using the `-exclude_patterns` option.

M113

Message Text

Invalid *type* connection between gates G1 and G2. (M113)

Severity

Error

Description

Sequential simulation was attempted when the circuit contained a direct connection between two different latch or flip-flop gates. Sequential simulation cannot be performed when this condition exists due to algorithm restrictions. Where *<type>* is either DFF/DFF, DLAT/DLAT, DFF/DLAT, or DLAT/DFF.

What Next

Enable the automatic insertion of buffers between such devices by specifying the `-add_buffer` option of the `set_build` command. Then repeat the `run_build_model` command and `run_drc` commands before attempting sequential fault simulation. The added buffer will not affect the fault population as it is added with `nofault` attributes.

M114

Message Text

There are no active faults in the fault list. (M114)

Severity

Error

Description

An attempt was made to perform fault simulation or test generation when there are no active faults in the fault list. Detected faults and undetectable faults are not considered active.

What Next

If the fault list is not complete, add faults to the fault list using the `add_faults` or `read_faults` commands, and rerun the command.

M115

Message Text

Fault simulation terminated: #patterns=N, CPU time=N (M115)

Severity

Error

Description

A user interrupt was issued to terminate the current fault simulation process.

What Next

No corrective action is required. Fault simulation results only reflect what occurred before the interrupt.

M116

Message Text

Pattern compression has been aborted, results are restored to original values. (M116)

Severity

Error

Description

A user interrupt was issued to terminate the current pattern compression process.

What Next

No corrective action is required. Fault simulation results and patterns are restored to their original values before the pattern compression.

M117

Message Text

Class of fault <fault path> stuck <sa0|sa1> changed from <fault_name1> to <fault_name2>. (M117)

Severity

Warning

Description

During pattern compression, the classification of at least one fault was changed. There are a number of situations where this change in classification is normal, but there might also be a few situations where this change indicates a mismatch between the fault credit given by the original ATPG test generator and the retesting performed during compression by the fault simulator.

No danger of producing bad patterns exists. In most cases, the change in test coverage is negligible.

What Next

Contact your Synopsys Test Specialist. If you see large numbers of faults (>30) changing class, send your circuit as a testcase. After analysis, it is possible to determine if the change in fault classification was proper or improper. Over time this will help in tuning the ATPG test generation and fault simulation algorithms.

M118

Message Text

Pattern source is not set to "<internal | external>". (M118)

Severity

Error

Description

An operation was attempted for which the pattern source needed to be external and it was internal, or vice-versa. For example, diagnostics requires an external pattern source, `run_atpg-auto` requires an internal pattern source.

What Next

Use the `set_patterns` command to change the pattern source to what is needed and retry your failing command.

M119

Message Text

No scan chains were extracted. (M119)

Severity

Warning

Description

This message is printed during DRC (scan chain operation checking) when no scan chains are extracted and under the following conditions: the message level is set to expert mode (using the `set_messages -level expert` command), you did not use pre DFT DRC, and you specified the SPF.

What Next

To avoid this message, make sure the M119 message level is not set to expert mode.

Error and Warning Messages M120 - M129

This section contains the following error messages and warnings:

- [M120](#)
- [M121](#)
- [M122](#)
- [M123](#)
- [M124](#)
- [M125](#)
- [M126](#)

- [M127](#)
- [M128](#)
- [M129](#)

M120

Message Text

Line N, invalid pattern number P. (M120)

Severity

Error

Description

When reading a failure file during a failure diagnosis, an invalid pattern number, P, was identified on the indicated line, L. This can be caused by not, or wrongly specifying, failure log file directives for split pattern cases. This can also be caused by using a STIL or WGL pattern file written by a version before 2005.09 for cycle-based failure log files.

What Next

There is an inconsistency between the pattern source and the failure file. Determine the cause of the inconsistency and correct it.

If the pattern or cycle count is not reset from one failures log file to the next one on the tester, use the `.cycle_offset` or `.first_pattern` directives.

Cycle-based diagnostics requires certain annotations in STIL or WGL pattern files. This requires the pattern file to be generated by TestMAX ATPG 2005.09 or later. A possible workaround would be to regenerate patterns with the correct version, or try to read binary patterns from the earlier version of TestMAX ATPG into a correct version, and write out a new STIL or WGL pattern file..

M121

Message Text

Line d, invalid pin name *string*. (M121)

Line d, chain name cannot be used in compression mode. (M121)

d invalid nets were ignored (M121)

Line d, invalid net driver|fanout name *string*. (M121)

Line d, invalid data format *string*. (M121)

No direct path between *pinpath1* (*id1*) and *pinpath2* (*id2*), net deleted.
(M121)

Skipping a net defined in lines *d1-d2*. (M121)

Fanout *name* was defined in more than one net. (M121)

Severity

Error or Warning

Description

The various versions of this message are described as follows:

Line *d*, invalid pin name *string*. (M121)

This message is printed when reading a failure file during a failure diagnosis. It indicates that an invalid port name was identified on the indicated line.

What Next: An inconsistency exists between the external port names of the current netlist and the failure file. You need to determine the cause of the inconsistency and correct it.

Line *d*, chain name cannot be used in compression mode. (M121)

This message indicates that the failure file or mask file contains one failure specified with its chain name. This format is not supported for a failure file for diagnostics or a mask file for per-cycle pattern masking feature. The first occurrence of the bad format is indicated by line *d*.

What Next: Correct the failure or mask file, and rerun the flow.

d invalid nets were ignored (M121)

This is a warning message related to reading a subnet file. It indicates that the subnet defined for *d* nets was ignored.

There are multiple reasons this can occur. The subnet information contains the description of the net driver and its receivers, which are called fanout nets (these definitions are in the .net section). It contains also the description of the subnet definitions (these definitions are in the .subnet section). When this message appears, the either the net driver name or receiver names are incorrect, or the subnet definition is incorrect.

The net driver or receivers are incorrect if you cannot find one of them in the design database. This usually occurs because some type of optimization was performed during the build step.

Similarly, it's possible that a subnets definition is incorrect in the `.subnet` section because a fanout was removed during optimization. For example, the `.subnet` section might appear as follows:

```
.subnet
1 2
2 3
```

If the fanout with the `id=2` is not present in the design database, the `.subnet` section is recognized as follows:

```
.subnet
1
3
```

Because a subnet is defined for at least two fanout IDs, the entire `.subnet` section is invalid. Thus, the net is ignored. A net is incorrect if at least one of its subnets is incorrect. A subnet is incorrect if it defines at least two correctly mapped fanouts.

What Next: This is an informational warning message. If you want to know which nets are ignored and why, you should use the `read_layout -max_reported d` command. This causes additional messages to be printed for debugging purposes.

```
Line d, invalid net driver|fanout name string. (M121)
```

This is a warning message that indicates that the net name, specified at line `d` of the subnet file, is invalid. This might be because the name is extracted from the physical layout database. However, TestMAX ATPG expects the logical name as it is defined in the logical netlist. Or, it might be because the net was removed during optimization during the build process and the net name cannot be found. Or, TestMAX ATPG might be unable to find the pin name or map the net driver pin to an input pin instead of an output.

What Next: You can use the GSV to debug this message. To do this, add the net driver to the GSV using the `add_display_gates` command. If the net driver cannot be found, the `add_display_gates` command will fail. If the net fanout name is invalid, expand the fanout of the net driver to display all the drivers, and search for the missing net name.

```
Line d, invalid net driver|fanout name string. (M121)
```

This is an error message related to reading the subnet file. It indicates that the format of line `d` is invalid. In this case, the keywords `.net` and `.subnet` might be misspelled, or the order of the sections are inverted, or a section is missing. For example, a format error might occur if the `.net` keyword is written as `.nett`. The `string` parameter identifies which data format is causing the problem.

What Next: Go to line `d` and change the data format to comply with the subnet file format. The ASCII format for the net topology is defined in the FAQ topic Subnet Format For Diagnosis.

`Line d, invalid data format string. (M121)`

This is a warning message related to reading the subnet file. It indicates that a path is missing between the net driver (with pin path `pinpath1` and primitive ID `id1`) and the fanout (with pin path `pinpath2` and primitive ID `id2`), as indicated in the subnet file. This issue can occur if a path includes BUF or INV gates. This problem can also occur if a fanout net name is present in the fanout list of a net driver, but the fanout actually belongs to another net.

What Next: Change either the net driver or fanout name, or both. To verify that the net does not belong to the net driver fanout, use the `add_display_gate` command in the GSV to add the net driver indicated by `pinpath1`. You can then expand the display of the net by clicking on the cell output diamond. When all fanouts are displayed, search for `pinpath2`.

`Line d, invalid data format string. (M121)`

This is a warning message related to reading a subnet file. It indicates that the subnet defined between the lines `d1` and `d2` is rejected. There are multiple reasons this can occur. One possible reason is that the relationship of the subnets defined in the `.subnet` section are invalid. This can occur if the subnets contain fanouts that are not in the design database.

What Next: You should examine any errors or warnings printed before this message since they might explain the rejection of this subnet.

This is a warning message related to reading a subnet file. It indicates that fanout `name` is defined in more than one net. As a result, the first net with the fanout `name` is recognized, but the other redundant fanouts are ignored.

What Next: You should review the subnet generation flow. This error should not occur if you use the supported flow.

M122

Message Text

`Line number, invalid scan cell position number. (M122)`

`Line number, invalid format: bit position is missing. (M122)`

Severity

Error

Description

The first variation of this message appears when an invalid scan cell position is identified when reading a failure file during failure diagnostics.

The second variation of this message appears when the bit position information is missing from a pattern-based failure file.

What Next

For the first variation of this message, there is an inconsistency between the scan chain lengths of the current netlist and the failure file. Use the `report_scan_chains` and `report_scan_cells-all -reverse` command to determine the cause of the inconsistency and correct it.

For the second variation of this message, you must edit the format of the failure file so it includes the position of the failing bit of the serializer. For more information, see the topic Failure Data Files in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M123

Message Text

Failure file contains no failing patterns. (M123)

All failing patterns were skipped due to unsuccessful mapping. Not running diagnosis. (M123)

Severity

Error

Description

For the first message variation, when reading a failure file during a failure diagnosis, no failing basic scan or fast-sequential patterns were found. This could be because of a pattern truncation option or because the failing patterns are full-sequential (which is not supported by diagnosis).

For the second message variation, none of the failures were mapped. This is likely due to a complex or systematic defect that cannot be diagnosed with compressed patterns.

What Next

For the first message variation, perform diagnosis only on failure files that contain at least one failing basic scan or fast-sequential pattern. If you used the `-incomplete_failures` option of the `set_diagnosis` command, the failure log file might have contained only a single failing pattern. If you are confident that this failure data is complete, remove the `-incomplete_failures` option and try again.

If the failing patterns are full-sequential, remove the corresponding failures from the failure file. Full-sequential pattern failures are not supported by diagnosis.

For the second message variation, you should try running diagnosis with uncompressed patterns.

For more information, see the topic `set_diagnosis` in the TestMAX™ ATPG and TestMAX Diagnosis Commands Guide.

M124

Message Text

`Chain diagnosis cannot be performed when there are no scan chains. (M124)`

Severity

Error

Description

A scan chain failure diagnosis was attempted when there were no defined scan chains.

What Next

Only run a standard diagnosis if there are no scan chains.

M125

Message Text

`Requested command cannot be performed for fault model set to <model type>. (M125)`

Severity

Error

Description

A command was attempted that could not be performed with the current fault model type setting.

What Next

Set the fault model type to a value compatible with the command and rerun the command.

M126

Message Text

There were N faultable pins lost due to <reason> optimizations. (M126)

There were N faultable pins lost due to PIO pullup/downs removal. (M126)

Severity

Warning

Description

Design optimizations were performed that resulted in fewer gates but also had a side effect of dropping some fault sites. This is a warning that test coverage is slightly affected due to a small reduction in total faults. The <reason> text corresponds to the possible optimization algorithms. For more details, see the topic Understanding Flattening Optimizations in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

What Next

Review any B22 violations for more information on the fault sites that were dropped. If the loss of the faults sites is not acceptable, you might be able to disable the gate level optimization using an appropriate option of the `set_build -merge` command; otherwise, no user action is necessary.

M127

Message Text

Fault simulation terminated: #faults_simulated=N, test_coverage=T% (M127)

Severity

Error

Description

A user interrupt was issued to terminate the current fault simulation process.

What Next

No corrective action is required. Fault simulation results reflect only the processing that occurred before the interrupt.

M128

Message Text

Simulation terminated: #patterns=N/N, #fail_pats=N(N), #failing_meas=N(N)
(M128)

Severity

Error

Description

A user interrupt was issued to terminate the current simulation process.

What Next

No corrective action is required. Simulation results reflect only what the simulation that was completed before the interrupt.

M129

Message Text

[Good|Fault] Simulation for patn P-I not stabilized after N passes.
(M129)

Severity

Warning

Description

During the Fast-Sequential or Full-Sequential simulation of a pattern, gates failed to reach a stable value after the allowed number of simulation passes. Gates that failed to stabilize are set to X. This indicates a potential oscillation is occurring and might result in a significant impact on simulation performance, both for TestMAX ATPG and for a pattern simulation.

P is the pattern number. *N* is the simulation pass limit. *I* is the pattern event sequence (0-based) for the pattern given. You can gain a better understanding of what is happening during the pattern by using the `report_patterns` command. A typical Basic-Scan pattern event sequence may consist of: 0 = force PI's, 1 = measure PO's, 2 = pulse clock. Other types of patterns are more complex. Note that oscillation might occur during the setup for sequential simulation. When this occurs, the pattern number *P* is displayed as -1.

What Next

The `-oscillation` option of the `set_simulation` command might be used to increase the iteration limit. There is no guarantee that increasing the limit will eliminate this warning message. If a true oscillatory circuit exists (B23, X1 violations), then simulation will not stabilize no matter how high the oscillation limit is set.

No other action is required. However, if ATPG performance is slow you can remove the source of the oscillation in test mode by using a constrained pin to block the oscillation path.

For more information, see the following topics in the TestMAX™ ATPG and TestMAX Diagnosis Commands Guide:

`report_feedback_paths`

`analyze_feedback_path`

Understanding the Report Feedback Paths Output

Error and Warning Messages M130 - M139

This section contains the following error messages and warnings:

- [M130](#)
- [M131](#)
- [M132](#)
- [M133](#)
- [M134](#)
- [M135](#)
- [M136](#)
- [M137](#)
- [M138](#)
- [M139](#)

M130

Message Text

Unable to find pattern P. (M130)

Severity

Error

Description

The pattern number selected to be reported was not a valid pattern number for the selected pattern type.

What Next

Make sure the pattern type is selected properly and choose a pattern number that is contained in that pattern set. Rerun the command with those selections.

M131

Message Text

Write patterns terminated after writing N patterns. (M131)

N not supported type patterns were ignored. (M131)

Severity

Error/Warning

Description

In the first version of the message, a user interrupt was issued that terminated the writing of patterns to an external file.

In the second version, you wrote a netlist-independent pattern file with pattern types other than basic-scan and fast-sequential. (The Adaptive Scan pattern translation flow supports only basic-scan and fast-sequential patterns.)

What Next

For the first version of the message, no corrective action is required. Only those patterns written before the interrupt are stored in the file.

For the second version, no corrective action is required. The netlist-independent pattern file will contain only the supported pattern types.

M132

Message Text

Read patterns terminated after reading N patterns. (M132)

Severity

Error

Description

A user interrupt terminated the reading of patterns from an external file.

What Next

No corrective action is required. Only those patterns read before the interrupt are stored internally.

M133

Message Text

Internal pattern set is now deleted. (M133)

Severity

Warning

Description

Deleting faults when internal patterns have been generated caused this warning.

Switching from TEST command mode to DRC command mode after generating patterns cause this warning.

What Next

No action is required.

M134

Message Text

Current external pattern set is now deleted. (M134)

Severity

Warning

Description

An action occurred that caused the current external pattern set to be deleted. This occurs when a new external pattern set is read in, DRC command mode is re-entered, or the `-delete` option is used for the `set_patterns` command.

What Next

No action is required.

M135

Message Text

`Clockon_measure pattern failed to stabilize in 10 passes. (M135)`

Severity

Warning

Description

This is a warning message that indicates that the clockon measure pattern did not stabilize in the allowed number of passes. Change activity at this point is set to X and the simulation continues until the circuit is stable.

What Next

The inability to stabilize values during simulation indicates an oscillation or other feedback. A large number of these messages could affect test coverage and should be investigated. Use the `report_violations` command to list any C17 violations. This is a starting point for identifying sections of circuitry that are involved with clockon measures.

You can disable the generation of patterns with clockon measures using the `set_atpg-noallow_clockon_measures` command.

M136

Message Text

`N1(N2) faults were unsuccessfully detected. (M136)`

Severity

Warning

Description

`N1(N2) faults were unsuccessfully detected. (M136)\n`

Where,

- *N1* is the number of unsuccessful faults at the end of ATPG
- *N2* is the number of recorded unsuccessful faults during ATPG

During the ATPG process, the test generation was completed successfully for a fault, but the fault simulation failed to identify the fault as detected. This could be attributed to a user interrupt, pattern rejection due to failure to meet constraints, pattern rejection due to contention, sequential effects, an SDC exception, or a program error. You can view these faults by using the `-unsuccessful` option of the `report_faults` command.

N2 may vary from *N1* due to:

- Several faults in *N2* were discovered at a later time, resulting in *N2* surpassing *N1*.
- Certain faults were not considered failures in the ATPG process, resulting in *N2* being less than *N1*.

What Next

No action is required unless the loss of test coverage is excessive. If there is no apparent reason for the unsuccessful faults, submit a problem report with a test case.

M137

Message Text

`N faults were terminated due to excess number of decisions. (M137)`

Severity

Warning

Description

When the number of decision nodes that occur during test generation exceeds the space allowed to store the data, test generation is immediately terminated for that fault. A summary message is given at the end of the ATPG process indicating the total number of faults that were terminated due to this condition.

What Next

No corrective action is required if the test generation termination is acceptable. Otherwise, increase the number of allowed decision nodes using the `-decision` option of the `set_workspace_sizes` command. Doubling the current value is recommended. Use the `report_settings` command to see the current value.

M138

Message Text

`N faults were terminated due to excess change activity. (M138)`

Severity

Warning

Description

When the amount of change activity that occurs during test generation exceeds the space allowed to store the data, test generation is immediately terminated for that fault. A summary message is given at the end of the ATPG process indicating the total number of faults that were terminated due to this condition.

What Next

No corrective action is required if the test generation termination is acceptable. Otherwise, submit a problem report with a test case.

M139

Message Text

Number faults aborted during [post-capture] contention prevention. (M139)

Number faults aborted during PLL clock conditioning. (M139)

Number faults aborted due to failure to map care bits to load compressor. (M139)

Number1 faults aborted, *Number2* unsatisfiable, *Number3* successful during clock gating ATPG. (M139)

Severity

Warning

Description

When creating test patterns, fault detection might be possible, but contention is sometimes unavoidable. The first message reports the number of faults that are abandoned because contention occurred in generated patterns during ATPG. The test generator might eventually find a pattern that detects the target fault without causing contention, but this process is usually slow.

After completing fault detection during fast-sequential test generation, the test generator attempts to set the parameters for the clocking control cells of a detected fault. If this is not possible, the fault is considered aborted. The second M139 message reports the total number of faults that aborted during clock conditioning during the current simulation interval. These aborted faults will reduce the test coverage of the ATPG process. If the loss of test coverage becomes excessive, you should file a STAR report.

After completing fault detection during fast-sequential test generation of DFTMAX circuits, the test generator attempts to map the care bits to load-compressor values. If this mapping fails, the fault is considered aborted. The third M139 message reports the total number of faults from the tool's failure to map care bits to the load compressor during the current simulation interval. These aborted faults will likely reduce the test coverage of the ATPG process. If the loss of test coverage becomes excessive, you should file a STAR report.

The fourth M139 message reports on clock gating conditioning during the current simulation interval. During certain phases of test generation, clock-gating decisions can result in aborted (*Number1*) or unsatisfiable (*Number2*) conditions. Typically, these numbers are relatively small compared to the number of successful (*Number3*) attempts. This is an informational message, and no action is required.

What Next

Z1 rule violations identify the potential for contention. The more Z1 rule violations reported, the more the need for contention checks. Reducing or removing the Z1 violations typically involves a design change. In rare cases, you might be able to constrain a top-level input pin to avoid Z1 violations, but this often reduces total test coverage. Note that the Z1 rule violations only apply to the first variation of the M139 message. Neither Z1 violations or BUS primitives affect the second, third, or fourth variations of the M139 message.

You can determine if any BUS gates caused an aborted BUS analysis, which can contribute to a slower ATPG effort. BUS analysis is performed at the start of DRC, and you should look for "#abort=" or "#not_analyzed" statements on the "Contention status" line of the BUS analysis report. If any of these two fields contain non-zero values, you should increase the ATPG abort limit and rerun BUS analysis and DRC.

If you identify BUS gates with an aborted status, increase the ATPG abort limit and use the `analyze_buses` command to recheck the BUS gates. If you try this several times and the aborted BUS gates remain, use the `report_buses-contention abort` command to identify some or all of the failing primitives. Investigate the design for contributing drivers for these BUS gates to determine why contention cannot be avoided.

The M139 message is often accompanied by [M181](#) messages that identify specific BUS primitives by their gate ID for which contention is detected. You can review the description of the [M181](#) message for how to set the contention checking severity to error and halt ATPG. This allows you to investigate the BUS gate in which contention is frequently occurring.

Error and Warning Messages M140 - M149

This section contains the following error messages and warnings:

- [M140](#)
- [M141](#)

- [M142](#)
- [M143](#)
- [M144](#)
- [M146](#)
- [M147](#)
- [M148](#)
- [M149](#)

M140

Message Text

Invalid <data> for instruction "<instruction>" in pattern P. (M140)

Severity

Error

Description

During the reading of external patterns in binary format, invalid data was discovered in the indicated instruction of the indicated pattern.

What Next

There is a problem with the binary pattern data. Check that it is consistent with the current netlist, clock definitions, PI/PO definitions, and scan cells. If possible, redefine data to become consistent with the patterns.

It is possible that the design has been changed and is no longer a match for when the pattern data file was created. Another possibility is that the data file has become corrupted.

If you succeed in getting the patterns into memory, you might be able to obtain some information using the `report_patterns` command that helps you decide if the patterns should match the design.

M141

Message Text

There were no events for pattern P. (M141)

Severity

Error

Description

During the processing of external patterns in binary format, there was no event data during pattern P.

What Next

There is a problem with the binary pattern data. It is likely that the file is corrupted and is no longer usable.

If you succeed in getting the patterns into memory, you might be able to obtain some information using the `report_patterns` command which helps you decide if the patterns still match the design.

M142

Message Text

Test generation terminated during contention prevention. (M142)

Severity

Warning

Description

A justification process was performed where test generation was terminated during contention prevention.

What Next

Make sure no bus gates were terminated during the contention ability checking (rule Z1). If failures occurred, increase the atpg abort limit and run bus analysis to perform contention ability checking.

Increase the failure limit and rerun ATPG. If the contention prevention failure is still present, submit a problem report with a test case.

M143

Message Text

Test generation terminated during <phase> for fault <id> pin N stuck <0|1>. (M143)

Severity

Warning

Description

A fault analysis was performed where test generation was terminated during one of: contention prevention, post-capture contention prevention, or capture new justification.

What Next

Make sure no bus gates were terminated during the contention ability checking (rule Z1). If failures occurred, increase the ATPG abort limit and run bus analysis to perform contention ability checking. Increase the failure limit and rerun ATPG. If the contention prevention failure is still present, submit a problem report with a test case.

M144

Message Text

Incomplete netlist processing: *N* gates cannot be processed. (M144)

Severity

Error

Description

During the build model process, some information in the original netlist could not be flattened. This information comes from unsupported modeling constructs in the TestMAX ATPG gate-level netlist environment, such as behavioral modeling statements. This condition is reported as an error because it identifies an incomplete representation of the design. However, it is possible that the supported constructs are still accessible and the build model process continues as normal. In this case, be aware that the design behavior of the design does not reflect the full model behavior.

Scan enable gate %d (%s) is set to `scan_enable_value=%c` to prevent the pipelines from changing state (M144).

The M144 message is issued when DFTMAX input pipelines do not hold state during capture (failed R18) and `scan_enable` is set to system mode value might prevent pipelines from changing state. However, setting `scan_enable` to system mode value may prevent detection of certain faults that require `scan_enable` at shift value; these faults are classified as AU.

What Next

If possible, the missing data is ignored when the model is defined, and TestMAX ATPG generates an incomplete model. You must manually review the TestMAX ATPG-derived

model and validate that it sufficiently represents the desired behavior or review the netlist constructs and define modeling constructs compatible with ATPG. If this error is generated, each leaf-node module in the design should be individually reviewed using the library validation process to validate that the modules are appropriate and complete for ATPG. To allow generation of patterns with `scan_enable` at shift value, use `set_atpg -nouse_scan_enable`; see R18 documentation for more information.

M146

Message Text

Buffer/Inverter hiding is now set to off. (M146)

Severity

Warning

Description

The schematic viewing level was selected to be design when the buffer/inverter hiding was set on. Buffer/inverter hiding is supported only at the primitive level and is automatically turned off when the design level is selected.

What Next

No action is required. When the schematic viewing level is returned to primitive, the buffer/inverter hiding can be turned on.

M147

Message Text

Buffer/Inverter hiding can only be used during primitive level viewing.
(M147)

Severity

Error

Description

An attempt was made to select buffer/inverter hiding with the schematic viewing level set to design.

What Next

No action is required. To activate buffer/inverter hiding, set the schematic viewing level to Primitive, as follows:

1. Click the SETUP button on the GSV toolbar.

The GSV Setup dialog box appears. The Hierarchy selection lets you specify whether to display primitives or design components.

2. Select the Primitive check box in the Hierarchy section.
3. Optionally, select the BUF/INVs check box in the Hide section.
4. Click OK.

M148

Message Text

Traversing hierarchy can only be used during design level viewing. (M148)

Severity

Error

Description

A hierarchical traversing operation was attempted on the schematic viewer with the display level set to primitive.

What Next

Set the schematic viewing level to design and repeat the operation.

M149

Message Text

Traversing hierarchy up can only be used for a gate in a design level instance. (M149)

Severity

Error

Description

An upward hierarchal traversing operation was attempted on the schematic viewer with a gate that was not in a design level instance.

What Next

There is no way to perform this operation on the selected gate. You can select other gates and rerun the operation on them.

Error and Warning Messages M150 - M159

This section contains the following error messages and warnings:

- [M150](#)
- [M151](#)
- [M152](#)
- [M153](#)
- [M154](#)
- [M155](#)
- [M156](#)
- [M157](#)
- [M158](#)
- [M159](#)

M150

Message Text

Invalid instance name *instance_name* was used for show_block instruction.
(M150)

Severity

Error

Description

An invalid instance name was used to identify an instance to add to the schematic view. The name is either not an instance name or the name of an instance that is not at the design level. An instance at the design level can contain only primitives underneath it.

What Next

Correct the name and repeat the operation.

M151

Message Text

Instance name *instance_name* for show_block instruction was not at design level. (M151)

Severity

Error

Description

An invalid instance name was used to identify an instance to add to the schematic view. The name is a valid instance name, but the associated instance is not at the design level. An instance at the design level can contain only primitives underneath it.

What Next

Correct the name and repeat the operation.

M152

Message Text

Invalid primitive name *name* was used for show_block instruction. (M152)

Severity

Error

Description

An invalid name was used to identify a primitive to add to the schematic view. A valid name can be a primitive ID number or a pin pathname.

What Next

Correct the name and repeat the operation.

M153

Message Text

Traversing hierarchy down can only be used for a design level instance. (M153)

Severity

Error

Description

A downward hierarchical traversing operation was attempted on the schematic viewer with a cell that was not a design level instance.

What Next

There is no way to perform this operation on the selected cell. You can select other cells and retry the operation on them.

M154

Message Text

Invalid input number N was used for show_block_input instruction. (M154)

Severity

Error

Description

An invalid input pin number was used to identify an input gate to add to the schematic view. The input pin number is 0 based.

What Next

Correct the input pin number and repeat the operation.

M155

Message Text

Test-setup simulation data is not currently selected to be stored. (M155)

Severity

Warning

Description

A graphical display analysis was performed for a violation that used test-setup simulation values. However, the storage of the test-setup data is not currently enabled, resulting in the inability to display the relevant pin data.

What Next

Return to DRC mode and select the `-store_setup` option of the `set_drc` command and rerun DRC. After running DRC, an analysis that uses test-setup data can now display that data. You can either re-analyze or use the `Refresh` button.

M156

Message Text

Unconnected fanout list was truncated due to insufficient space. (M156)

Severity

Error

Description

The unconnected fanout list is an internal table that is created as gates are added to the schematic viewer draw list. This table allows you to click on a net diamond from a gate's output pin and add the next connected net and gate to the draw list. This internal table is limited to 256 additional connections. When a net has a high fanout, this message might be generated.

What Next

You might not be able to see a wanted gate for adding a fanout connection by using a popup menu when the list is truncated. However, if you know the instance path name or gate ID of the next connect point you would like to add to the schematic view, you can do so by other means such as the `add_display_gates` command.

M157

Message Text

Invalid output number N was used for `show_block_output` instruction.
(M157)

Severity

Error

Description

An invalid output pin number was used to identify an output gate to add to the schematic view. The output pin number is 0 based.

What Next

Correct the output pin number and repeat the operation.

M158

Message Text

Invalid fanout number N was used for show_block_output instruction.
(M158)

Severity

Error

Description

An invalid fanout number was used to identify an output gate to add to the schematic view. The fanout number is 0 based.

What Next

Correct the fanout number and repeat the operation.

For more information, see the following topics in the TestMAX™ ATPG and TestMAX Diagnosis Commands Guide:

```
set atpg
```

```
run_atpg
```

M159

Message Text

Invalid primitive name *name* was used for show_path_blocks instruction.
(M159)

Severity

Error

Description

An invalid name was used to identify a primitive to add to the schematic view. A valid name can be a gate ID number or a pin path name.

What Next

Correct the name and repeat the operation.

Error and Warning Messages M160 - M169

This section contains the following error messages and warnings:

- [M160](#)
- [M161](#)
- [M162](#)
- [M163](#)
- [M164](#)
- [M165](#)
- [M166](#)
- [M167](#)
- [M168](#)
- [M169](#)

M160

Message Text

Number of gates to display N1 > max allowed N2 - draw list is truncated.
(M160)

Severity

Warning

Description

An attempt was made to add gates to the schematic that exceeded the allowed number of gates. N2 is the current limit, N1 is how many gates were trying to be drawn.

What Next

If the schematic is adequate, no corrective action is required. If you want more gates on the schematic, increase the maximum number of gates allowed using the `Edit->Preferences` menu command (or the `set_environment_viewer-max_gates` command), and rerun the operation.

M161

Message Text

No blocks found in path. (M161)

Severity

Error

Description

An attempt was made to add all blocks between two selected points to the schematic, but no connection was found between the points.

What Next

No action is required if the gates at the tail of the path were identified correctly. The path is order dependent and contains all gates between the output of the first gate and the input of the second gate.

M162

Message Text

No graphical support for selected rule <rule here>. (M162)

Severity

Warning

Description

You attempted to analyze a rule that did not have automatic schematic viewing support.

What Next

No corrective action is required. You must analyze the rule manually.

M163

Message Text

Rule violation analysis cannot be performed without a simulation model.
(M163)

Severity

Error

Description

A rule analysis was attempted without a simulation model.

What Next

Create a simulation model using the `run_build_model` command and repeat the operation.

M164

Message Text

Invalid rule violation ID <violation ID here> (M164)

Severity

Error

Description

A rule analysis was attempted on an invalid violation ID number.

What Next

Correct the violation ID number and repeat the operation.

M165

Message Text

Invalid selection of feedback path. (M165)

Severity

Error

Description

An invalid name was used to identify a feedback path that was to be selected for schematic viewing.

What Next

Correct the name and repeat the operation. You can use the `report_feedback_paths` command to determine a valid name.

M166

Message Text

Invalid chain name "<name here>". (M166)

Severity

Error

Description

An invalid name was used to identify a scan chain that was to be selected for schematic viewing.

What Next

Correct the name and repeat the operation. You can use the `report_scan_chains` command to determine a valid name.

M167

Message Text

Invalid position N in chain <chain name>. (M167)

Severity

Error

Description

An invalid scan cell position was used to identify a scan cell that was to be selected for schematic viewing.

What Next

Correct the position and repeat the operation. The `report_scan_chains` command can be used to determine a valid position.

M168

Message Text

Line N: Missing fault arguments. (M168)

Severity

Error

Description

When reading a fault file, the indicated line number did not contain the required number of arguments. There must be at least three arguments for a non-comment line. The first argument is the fault type, the second argument is the fault code, and the third argument is the pin path name that identifies the fault site.

What Next

There is a format problem with the fault file that must be corrected before repeating the operation.

M169

Message Text

Line N: Invalid fault type <fault_type>. (M169)

Severity

Error

Description

When reading a fault file, the indicated line number did not contain a valid entry for the fault type field. The valid fault types are "sa0" and "sa1".

What Next

There is a format problem with the fault file that you must correct before repeating the operation.

Error and Warning Messages M170 - M179

This section contains the following error messages and warnings:

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- [M171](#)
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- [M178](#)
- [M179](#)

M170

Message Text

Line N: Invalid fault pin_pathname *pin_pathname*. (M170)

Severity

Error

Description

When reading a fault file, the indicated line number did not contain a valid entry for the fault site field. The fault site must be a valid pin path name.

What Next

There is a format problem with the fault file that must be corrected before repeating the operation.

M171

Message Text

Simulation of pattern N not allowed due to invalid pattern type. (M171)

Severity

Error

Description

An attempt was made to perform a good machine simulation or fault machine simulation in combinational mode while using patterns that require the sequential mode.

Or, an attempt was made to perform the `run_diagnosis` command on patterns that contain Fast-Sequential or Full-Sequential patterns.

What Next

Diagnosis for patterns containing Fast-Sequential or Full-Sequential ATPG patterns are not supported.

If you were attempting to perform a good machine simulation using the `run_simulation` command, then repeat the command but add the `-sequential` option.

M172

Message Text

There were no retention cells identified in the netlist; retention test will not test any scancell (M172)

Severity

Error

Description

When the `set_drc -clock -retention_test` command is specified, no retention cells are identified in the scan chain. The retention test generated in this context unloads only unknown values, and no scan cell states are tested.

What Next

Make sure the Verilog netlist uses ``define retention` statements for retention cells. Remove the `set_drc -clock -retention_test` command or consider using the `set_drc -clock -chain_capture` command and defining non-retention cells with explicit `add_cell_constraints` commands.

M173

Message Text

The `source` command cannot be used in a multi command line. (M173)

Severity

Error

Description

Multi-line commands and aliases defined to execute multi line commands are restricted and cannot the `source` command because of the ways commands are processed. When you attempt to execute a multi line command containing `source` or an alias defined as a multi line command containing `source`, you receive this error message.

What Next

Separate the `source` command, and perform it individually.

M174

Message Text

Previous capture procedures are not written because clocks have been removed (M174)

Severity

Warning

Description

This message might occur after you perform a `run_drc` command that defines one or more clocks and then you return to DRC mode and use a `remove_clocks` command before a `write_drc_file` command.

The definition of the clock comes from the original DRC file and the `remove_clocks` command is not sufficient to completely eliminate the clock from the DRC file that has just been written. Manual edits might be needed to fully adjust the DRC file for the removal of the clock.

What Next

Use an editor to review the DRC file just created for the following items:

- The SignalGroups block must be modified to remove the former clock from the defined list of clock0 and clock1 signals.
- The timing block must be modified to remove the definition of the 'P' character for the port that is no longer a clock.
- The Shift procedure should be reviewed to see if the former clock was pulsed. If so, it should be removed from the Shift procedure.
- The `capture_*` procedures need to be redefined. This is most easily accomplished by copy/paste methods from the original DRC file. However, leave out the capture procedure for the former clock.

M175

Message Text

PI constraint P (G) set to C not satisfied for N patterns (pat1=P).
(M175)

Severity

Selectable

Description

During scan-based parallel pattern simulation, the indicated PI constraint was not satisfied. For the current 32 pattern simulation pass, the number of patterns that experienced this failure and the first occurrence in the pass are given. The message also indicates if the failure occurred at preclock or postclock time.

What Next

This message is displayed only if you selected the `-verbose` option for the `set_contention` command. The default is `-noverbose`.

The severity of this failure is set using the `-severity` option of the `set_contention` command.

Patterns that fail to satisfy PI constraints are not given credit for fault detection and are not used for ATPG.

If the severity is error, the current operation immediately terminates. The simulation values of all gates can then be displayed for any of the 32 patterns in the simulation pass. You can display the first occurring pattern by selecting that pattern using the `pattern` option of the `set_pindata` command.

M176

Message Text

PI equivalence for P (G) not satisfied for N patterns (pat1=P). (M176)

Severity

Selectable

Description

During scan-based parallel pattern simulation, the indicated PI equivalence was not satisfied. For the current 32 pattern simulation pass, the number of patterns that experienced this failure and the first occurrence in the pass are given. The message also indicates if the failure occurred at preclock or postclock time.

What Next

This message is displayed only if the `-verbose` option is selected for the `set_contention` command. The default is `-noverbose`.

The severity of this failure is set using the `-severity` option of the `set_contention` command.

Patterns that fail to satisfy PI equivalences are not given credit for fault detection and are not used for ATPG.

If the severity is error, the current operation immediately terminates. The simulation values of all gates can then be displayed for any of the 32 patterns in the simulation pass. You can display the first occurring pattern by selecting that pattern using the `pattern` option of the `set_pindata` command.

M177

Message Text

```
Cell constraint %s-N set to %c not satisfied for N %spatterns (pat1=P).  
(M177)
```

Severity

Selectable

Description

During scan-based parallel pattern simulation, the indicated cell constraint was not satisfied. For the current 32 pattern simulation pass, the number of patterns that experienced this failure, and the first occurrence in the pass are given. The message also indicates if the failure occurred at preclock or postclock time.

What Next

This message is displayed only if the `-verbose` option is selected for the `set_contention` command. The default is `-noverbose`.

The severity of this failure is set using the `-severity` option of the `set_contention` command.

Patterns that fail to satisfy cell constraints are not given credit for fault detection and are not used for ATPG.

If the severity is error, the current operation immediately terminates. The simulation values of all gates can then be displayed for any of the 32 patterns in the simulation pass. You can display the first occurring pattern by selecting that pattern using the `pattern` option of the `set_pindata` command.

M178

Message Text

```
ATPG constraint N=<constraint> not satisfied for N %spatterns (pat1=P).  
(M178)
```

Severity

Selectable

Description

During scan-based parallel pattern simulation, the indicated atpg constraint was not satisfied. For the current 32 pattern simulation pass, the number of patterns that experienced this failure and the first occurrence in the pass are given. The message also indicates if the failure occurred at preclock or postclock time.

What Next

This message is displayed only if the `-verbose` option is selected for the `set_contention` command. The default is `-noverbose`.

The severity of this failure is set using the `-severity` option of the `set_contention` command.

Patterns that fail to satisfy atpg constraints are not given credit for fault detection and are not used for ATPG.

If the severity is error, the current operation immediately terminates. The simulation values of all gates can then be displayed for any of the 32 patterns in the simulation pass. You can display the first occurring pattern by selecting that pattern using the `pattern` option of the `set_pindata` command.

M179

Message Text

```
N patterns rejected due to M constraint violations (ID=J, pat1=K). (M179)
```

Severity

Selectable

Description

During the pattern simulation that accompanies ATPG pattern generation, one or more constraints were not satisfied. These constraints include: PI constraints, PI equivalences, cell constraints, and ATPG constraints.

This message might also occur during use of the `run_justification` command, which also uses pattern simulation to check any patterns it produces.

For the current group of 32 patterns, the number of patterns that experienced this failure is reported (N) as well as the first pattern of occurrence (K) and a gate ID (J) where the contention was detected. The message might also indicate if the violation occurred at preclock or postclock time.

Failure to satisfy this rule results in the patterns with violations being discarded. When the severity is warning, ATPG pattern generation continues and other patterns are tried. When

the severity is error the operation is halted. The search for other patterns to detect faults that also satisfy the constraints defined can or cannot be successful so the presence of constraint violations might lead to a degraded test coverage.

What Next

If this message occurs in conjunction with the use of the `run_justification` command, then review the list of PI equivalences and make sure that at least one pin in each equivalence relationship is represented in the list of set points used with the `run_justification` command.

If this message occurs during ATPG pattern generation, then you can investigate the cause of the constraint violation by using the `set_contention` command to adjust severity to error. Then restart the ATPG pattern generation. When the violation comes up again the ATPG pattern generation will stop. Use the SETUP button to select a pin data type of "error_data" and a design level of primitive. Then SHOW the gate ID from the message. Review the circuitry around the gate displayed for insight into the cause of the problem.

Error and Warning Messages M180 - M189

This section contains the following error messages and warnings:

- [M180](#)
- [M181](#)
- [M182](#)
- [M183](#)
- [M184](#)
- [M185](#)
- [M186](#)
- [M187](#)
- [M188](#)
- [M189](#)

M180

Message Text

```
Bus gate G failed <type> contention check for chain test pattern
(pat1=P). (M180)
```


Bus gate G failed <type> contention check for G patterns (pat1=P). (M180)

Bus gate G failed contention check for fast-sequential pattern (time=n).
(M180)

Bus gate G failed <type> float check for chain test pattern (pat1=P).
(M180)

Bus gate G failed <type> float check for G patterns (pat1=P). (M180)

Bus gate G failed float check for fast-sequential pattern (time=n).
(M180)

Bidi gate G failed <type> contention check for G patterns (pat1=P).
(M180)

Bidi gate G failed contention check for fast-sequential pattern (time=n).
(M180)

Bus gate G failed IDDQ float check for G patterns (pat1=P). (M180)

Bus gate G failed IDDQ float check for fast-sequential pattern (time=n).
(M180)

Bus gate G failed IDDQ weak contention check for G patterns (pat1=P).
(M180)

Bus gate G failed IDDQ weak contention check for fast-sequential pattern
(time=n). (M180)

RAM gate G failed IDDQ write check for G patterns (pat1=P). (M180)

RAM gate G failed IDDQ write check for fast-sequential pattern (pat1=P).
(M180)

Severity

Selectable

Description

During scan-based parallel pattern simulation, the indicated bus gate was at a potential contention condition. For the current 32 pattern simulation pass, the number of patterns that experienced this failure and the first occurrence in the pass are given. The message also indicates by the <type> field if the failure occurred at preclock or postclock time.

With contention checking enabled, patterns that fail to avoid bus contention are not given credit for fault detection and are not saved.

This check is performed if either:

- The `bus` option of the `set_contention` command is in effect.
- The `float` option of the `set_iddq` command is in effect.

This message appears only if the `-verbose` option of the `set_contention` command has been selected. By default, verbose messages are not displayed.

What Next

The severity of this failure can be adjusted by using the `severity` option of the `set_contention` command.

To investigate the source of the reported contention, then select a severity of error before using `run_atpg`. When the severity is set to error, ATPG effort will stop when the contention situation occurs. You can then use the pattern number indicated by the `pat1=` portion of the M180 message to display a pattern showing the net states leading to the contention situation. This is done by using the `pattern` option of the `set_pindata` command and then adding the gate ID identified by `G` to the graphical schematic viewer (GSV) window using the show by name menu option.

M181

Message Text

N patterns rejected due to M <type> contentions (ID=G, pat1=P). (M181)

Fast-sequential pattern rejected due to M <type> contentions (ID=G, time=P). (M181)

N patterns rejected due to PLL int-ext clock usage (clk1=D, clk2=D, pat1=P). (M181)

Pattern rejected due to incorrect clocking group. (M181)

Pattern <pattern_id> can't be simulated due to incorrect clocking group. (M181)

N patterns rejected (patn1=P) due to invalid clock grouping (clk1=G, clk2=G). (M181)

Severity

Warning

Description

This message is used for various types of pattern rejection, including contention and PLLs.

During scan-based parallel pattern simulation, one of the first two messages above are issued if a contention condition occurred for at least one BUS gate. For the current simulation pass, the number of patterns that experienced this failure and the first occurrence in the pass are given. For a Basic-Scan pattern, the message also indicates if the failure occurred at preclock or postclock time.

N is the number of patterns rejected.

M is the number of contentions occurring on that pattern.

<type> is one of IDDQ, BIDI, BUS, or FLOAT, and could also be qualified with a preclock or postclock attribute in front of "contentions". *G* is the representative gate ID where contentions occur. *P* is the pattern number (out of 32).

For contention issues, this check is performed only if the `bus` option is enabled for the `set_contention` command. The default is `bus` (checking is enabled).

For contention issues, the severity of this failure is controlled using the `-severity` option of the `set_contention` command. By default, a severity of warning allows the ATPG effort to continue.

Patterns that fail to avoid bus contention are not given credit for fault detection and are not saved. These patterns are discarded and a search for other patterns continues.

When PLLs are used, the third message above is issued (note that *D* is the clock ID). This message means that conditions necessary for fault detection might conflict with the clocking intended by the test generator. In this case, patterns might be rejected, possibly resulting in coverage loss.

The fourth and fifth versions of the messages above are issued by full-sequential ATPG or simulation when internal clocking is used but the pattern contains pulses of different clocks that are asynchronous to each other.

The sixth version of the message is issued by threaded ATPG when the clocking violates the clock disturb matrix identified by DRC.

What Next

In the case of contention issues, if too many patterns are being discarded, you might want to explore the cause of the contention. To do this, return to DRC mode and use the `set_contention` command to adjust the severity of the contention to error. Then issue the `run_atpg` command. The pattern generation will stop upon the first occurrence of the contention and then circuit values that show a contention condition might be observed as follows.

1. If the pattern is a Basic-Scan pattern, then use `set_primitive_report-time all` to select the display of the preclock, clock, and postclock simulation values on gates displayed in the schematic viewer. You could also choose to display just the preclock or the postclock values depending on the *<type>* of contention reported in the

message. For a Fast-Sequential pattern, use the default for the `-time` option of the `set_primitive_report` of clock.

2. Use the `SETUP` button to select primitive view, hide inverters and buffers, and select a Pin Data type of "pattern". Specify a pattern number between 0 and 31 based on the value P reported in the M181 message text. For a Fast-Sequential pattern, P is a time event corresponding to a pattern number in the internal scratchpad area used to generate patterns. So think of "event" as a pattern number as well.
3. Use the `SHOW->Named...` option to draw the gate ID displayed in the M181 message. This gate is the ATPG BUS primitive where the contention is occurring. You will then look for BUS input pins at conflicting states and click net expansion diamonds to add gates to the display and trace back to the source of the contention values.

In the case of PLLs, the design must ensure independence between the clock chain bits controlling internal clocks and fault detection conditions in the main design logic.

M182

Message Text

```
Bidi driver gate G failed direction hold check for N patterns (pat1=P).  
(M182)
```

Severity

Selectable

Description

During scan-based parallel pattern simulation, the indicated bidirectional driver failed to hold direction after the capture clock. For the current 32 pattern simulation pass, the number of patterns that experienced this failure and the first occurrence in the pass are given. The message also indicates if the failure occurred at preclock or postclock time.

What Next

This check is performed only if the `-retain_bidi` option is selected for the `set_contention` command. The default is `-noretain_bidi`.

The severity of this failure is set using the `-severity` option of the `set_contention` command.

Patterns that fail to retain bidi direction are not given credit for fault detection and are discarded.

If the severity is error, the current operation immediately terminates. The simulation values of all gates can then be displayed for any of the 32 patterns in the simulation pass.

You can display the first occurring pattern by selecting the `error_data` option of the `set_pindata` command.

M183

Message Text

```
N patterns rejected due to N bidi drivers failing to hold direction
(ID=G, pat1=P). (M183)
```

Severity

Selectable

Description

During scan-based parallel pattern simulation, at least one bidi driver failed to hold direction after the capture clock. For the current 32 pattern simulation pass, the number of patterns that experienced this failure and the first occurrence in the pass are given. The message also indicates if the failure occurred at preclock or postclock time.

What Next

This check is performed only if the `bus` option is selected for the `set_contention` command. The default is `bus`.

The severity of this failure is set using the `-severity` option of the `set_contention` command.

Patterns that fail to retain bidi direction are not given credit for fault detection and are not used for ATPG.

If the severity is error, the current operation immediately terminates. The simulation values of all gates can then be displayed for any of the 32 patterns in the simulation pass. You can display the first occurring pattern by selecting the `error_data` option of the `set_pindata` command.

M184

Message Text

```
Wire gate G failed <type> contention check for N patterns (pat1=P).
(M184)
```

Severity: selectable

Severity

Selectable

Description

During scan-based parallel pattern simulation, a contention condition occurred for at least one wire gate. For the current 32 pattern simulation pass, the number of patterns that experienced this failure and the first occurrence in the pass are given. The message also indicates if the failure occurred at preclock or postclock time.

What Next

This check is performed only if the `wire` option is selected for the `set_contention` command. The default is `nowire`.

This message is displayed only if the `-verbose` option is selected for the `set_contention` command. The default is `-noverbose`.

The severity of this failure is set using the `-severity` option of the `set_contention` command.

Patterns that fail to avoid wire contention are not given credit for fault detection and are not used for ATPG.

If the severity is error, the current operation immediately terminates. The simulation values of all gates can then be displayed for any of the 32 patterns in the simulation pass. You can display the first occurring pattern by selecting that pattern using the `pattern` option of the `set_pindata` command.

M185

Message Text

N patterns rejected due to N wire %s contentions (ID=G, pat1=P). (M185)

Severity

Selectable

Description

During scan-based parallel pattern simulation, a contention condition occurred for at least one wire gate. For the current 32 pattern simulation pass, the number of patterns that experienced this failure and the first occurrence in the pass are given. The message also indicates if the failure occurred at preclock or postclock time.

What Next

This check is performed only if the `wire` option is selected for the `set_contention` command. The default is `nowire`.

The severity of this failure is set using the `-severity` option of the `set_contention` command.

Patterns that fail to avoid wire contention are not given credit for fault detection and are not used for ATPG.

If the severity is error, the current operation immediately terminates. The simulation values of all gates can then be displayed for any of the 32 patterns in the simulation pass. You can display the first occurring pattern by selecting the `error_data` option of the `set_pindata` command.

M186

Message Text

```
Gate G failed port_name port %scontention check for N patterns (pat1=P).  
(M186)
```

```
Severity: selectable
```

Severity

Selectable

Description

During scan-based parallel pattern simulation, a contention condition occurred for at least one DLAT/DFP gate. For the current 32 pattern simulation pass, the number of patterns that experienced this failure and the first occurrence in the pass are given. The message also indicates if the failure occurred at preclock or postclock time.

What Next

This check is performed only if the `bus` option is selected for the `set_contention` command. The default is `nobus`.

This message is displayed only if the `-verbose` option is selected for the `set_contention` command. The default is `-noverbose`.

The severity of this failure is set using the `-severity` option of the `set_contention` command.

Patterns that fail to avoid DLAT/DFP contention are not given credit for fault detection and are not used for ATPG.

If the severity is error, the current operation immediately terminates. The simulation values of all gates can then be displayed for any of the 32 patterns in the simulation pass. You can display the first occurring pattern by selecting that pattern using the `pattern` option of the `set_pindata` command.

M187

Message Text

N patterns rejected due to N port %s contentions (ID=G, pat1=P). (M187)

Severity: selectable

Severity

Selectable

Description

During scan-based parallel pattern simulation, a contention condition occurred for at least one DLAT/DFF gate. For the current 32 pattern simulation pass, the number of patterns that experienced this failure and the first occurrence in the pass are given. The message also indicates if the failure occurred at preclock or postclock time.

What Next

This check is performed if `set_contention` bus is in effect.

The severity of this failure is set using the `-severity` option of the `set_contention` command.

Patterns that fail to avoid DLAT/DFF contention are not given credit for fault detection and are not used for ATPG.

If the severity is error, the current operation immediately terminates. The simulation values of all gates can then be displayed for any of the 32 patterns in the simulation pass. You can display the first occurring pattern by selecting the `error_data` option of the `set_pindata` command.

M188

Message Text

Atpg constraint N=%c failed contention check at pattern P. (M188)

Severity

Selectable

Description

During Fast-Sequential pattern simulation, the indicated ATPG constraint was not satisfied at the given pattern.

What Next

This message is displayed only if the `-verbose` option is selected for the `set_contention` command. The default is `-noverbose`.

The severity of this failure is set using the `-severity` option of the `set_contention` command.

If the severity is error, the current operation immediately terminates. You can display the simulation values of all gates for preselected patterns using the `seq_sim_data` option of the `set_pindata` command. The pattern data must have been stored during simulation, which requires setting values for the `-data` option of the `set_simulation` command.

M189

Message Text

Pattern P contained N ATPG constraint violations (ID=G). (M189)

Severity

Selectable

Description

During Fast-Sequential pattern simulation, at least one ATPG constraint was not satisfied at the given pattern.

What Next

The severity of this failure is set using the `-severity` option of the `set_contention` command.

If the severity is error, the current operation immediately terminates. You can display the simulation values of all gates for preselected patterns using the `seq_sim_data` option of the `set_pindata` command. The pattern data must have been stored during simulation, which requires setting values for the `-data` option of the `set_simulation` command.

Error and Warning Messages M190 - M199

This section contains the following error messages and warnings:

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- [M192](#)
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- [M194](#)
- [M195](#)
- [M196](#)
- [M197](#)
- [M198](#)
- [M199](#)

M190

Message Text

Bus gate G failed contention check at pattern P. (M190)

Severity

Selectable

Description

During Fast-Sequential pattern simulation, the indicated bus gate was at a potential contention condition.

What Next

This check is performed only if the `bus` option is selected for the `set_contention` command. The default is `bus`.

This message is displayed only if the `-verbose` option is selected for the `set_contention` command. The default is `-noverbose`.

The severity of this failure is set using the `-severity` option of the `set_contention` command.

If the severity is error, the current operation immediately terminates. You can display the simulation values of all gates for pre-selected patterns using the `seq_sim_data` option of the `set_pindata` command. The pattern data must have been stored during simulation, which requires setting values for the `-data` option of the `set_simulation` command.

M191

Message Text

Pattern P contained N bus contentions (ID=G). (M191)

Severity

Selectable

Description

During Fast-Sequential pattern simulation, a contention condition occurred for at least one bus gate at the given pattern.

What Next

This check is performed only if the `bus` option is selected for the `set_contention` command. The default is `bus`.

The severity of this failure is set using the `-severity` option of the `set_contention` command. The default is `-noverbose`.

If the severity is error, the current operation immediately terminates. You can display the simulation values of all gates for preselected patterns using the `seq_sim_data` option of the `set_pindata` command. The pattern data must have been stored during simulation, which requires setting values for the `-data` option of the `set_simulation` command.

M192

Message Text

Wire gate G failed contention check at pattern P. (M192)

Severity

Selectable

Description

During Fast-Sequential pattern simulation, the indicated wire gate was at a potential contention condition.

What Next

This check is performed only if the `wire` option is selected for the `set_contention` command. The default is `nowire`.

This message is displayed only if the `-verbose` option is selected for the `set_contention` command. The default is `-noverbose`.

The severity of this failure is set using the `-severity` option of the `set_contention` command.

If the severity is error, the current operation immediately terminates. You can display the simulation values of all gates for preselected patterns using the `seq_sim_data` option of

the `set_pindata` command. The pattern data must have been stored during simulation, which requires setting values for the `-data` option of the `set_simulation` command.

M193

Message Text

Pattern P contained N wire contentions (ID=G). (M193)

Severity

Selectable

Description

During Fast-Sequential pattern simulation, a contention condition occurred for at least one wire gate at the given pattern.

What Next

This check is performed only if the `wire` option is selected for the `set_contention` command. The default is `nowire`.

The severity of this failure is set using the `-severity` option of the `set_contention` command.

If the severity is error, the current operation immediately terminates. You can display the simulation values of all gates for preselected patterns using the `seq_sim_data` option of the `set_pindata` command. The pattern data must have been stored during simulation, which requires setting values for the `-data` option of the `set_simulation` command.

M194

Message Text

Gate G failed `port_name` port contention check at pattern P. (M194)

Severity

Selectable

Description

During Fast-Sequential pattern simulation, the indicated DLAT/DFF gate was at a potential contention condition.

What Next

This check is performed only if the `port` option is selected for the `set_contention` command. The default is `noport`.

This message is displayed only if the `-verbose` option is selected for the `set_contention` command. The default is `-noverbose`.

The severity of this failure is set using the `-severity` option of the `set_contention` command.

If the severity is error, the current operation immediately terminates. You can display the simulation values of all gates for preselected patterns using the `seq_sim_data` option of the `set_pindata` command. The pattern data must have been stored during simulation, which requires setting values for the `-data` option of the `set_simulation` command.

M195

Message Text

Pattern P contained N port contentions (ID=G). (M195)

Severity

Selectable

Description

During Fast-Sequential pattern simulation, a contention condition occurred for at least one DLAT/DFF gate at the given pattern.

What Next

This check is performed if the `port` option is selected for the `set_contention` command. The default is `noport`. It is also performed for functional patterns if your RAM model has write/write contention selected as FORBIDDEN. Since the forbidden setting might be applied to ATPG pattern generation but not to functional patterns, when the write/write contention is set to forbidden and functional patterns are detected that cause contention this violation message is issued.

The severity of this failure is set using the `-severity` option of the `set_contention` command.

If the severity is error, the current operation immediately terminates. You can display the simulation values of all gates for preselected patterns using the `seq_sim_data` option of the `set_pindata` command. The pattern data must have been stored during simulation, which requires setting values for the `-data` option of the `set_simulation` command.

M196

Message Text

Event collapsing blocked due to PO measure with clock '<port>' active at T=<time>, pat=N. (M196)

Severity

Warning

Description

When reading an Extended VCDE pattern file with clock recovery enabled, a measure of a Primary Output (PO) was called for while a clock was active. The clock port name is identified in the message as well as the time reference within the VCDE pattern file when the measure is to be made.

The pattern number *P* refers to the TestMAX ATPG pattern to which this event is being assigned.

Clock recovery rules do not support making measures while any clocks are active. The sequence of VCDE events that would be collapsed into a single clock pulse are left in expanded form. This might cause problems if some cycles succeed in having clock recovery while others do not. This is because it might be difficult to map the resulting patterns to a tester time set where the port is a pulsed pin in some cycles but non-pulsed in others. For the highest probability of success, all cycles where defined clock pins are asserted should be recoverable into clock pulses, or they should all be left exploded into separate events.

What Next

If your purpose for reading in patterns is to perform fault grading, then you can ignore this warning. The fault simulator is an event-based simulator and will handle the expanded event sequence as easily as a pulsed clock. You might however, see the internal pattern count higher than it would otherwise be if the clock were recoverable.

If the choice of the measure points came from a -strobe period/offset selection, you might wish to try moving the measure point by changing the offset. If you know, for example, that the cycle is 100ns and the clocks are from 50-80ns, then you should select an offset so that the measures are either before or after the clock's on time in each cycle.

If attempts to adjust the time at which measures are made is not successful, you might want to remove the problem port from the defined vcd_clock list. If you have used the -vcd_clock auto mode, then switch to an explicit list and do not define the troublesome port as a vcd_clock. In this way, clock collapsing will not be attempted, because the port is not considered a clock. A measure PO while the port is asserted will not result in this

warning. More patterns or cycles might result, as event collapsing cannot be as effective if one or more clocks are removed from the vcd_clock list.

M197

Message Text

Event collapsing blocked due to PI changes with clock '<port>' active at T=<time>, pat=N. (M197)

Severity

Warning

Description

When reading an Extended VCDE pattern file with clock recovery enabled, a change to a Primary Input (PI) was called for while a clock was asserted (on). The clock port name is identified in the message as well as the time reference in the VCDE pattern file at which the input port changes.

The pattern number, N, refers to the TestMAX ATPG pattern to which this event is being assigned.

Clock recovery rules do not support changing inputs while any clocks are active. The sequence of VCDE events that would be collapsed into a single clock pulse is left in expanded form. This might cause problems if some cycles succeed in having clock recovery while others do not. This is because it might be difficult to map the resulting patterns to a tester time set where the port is a pulsed pin in some cycles but non-pulsed in others. For the highest probability of success, all cycles where defined clock pins are asserted should be recoverable into clock pulses, or they should all be left exploded into separate events.

What Next

If your purpose for reading in patterns is to perform fault grading, then you can ignore this warning. The fault simulator is an event-based simulator and will handle the expanded event sequence as easily as a pulsed clock. You might, however, see the internal pattern count higher than it would otherwise be if the clock were recoverable.

If you wish to eliminate the warning you have two choices:

- change the patterns so the input change does not occur while a clock is asserted, or
- remove the port from the list of vcd_clock pins and reread the patterns.

M198

Message Text

Event collapsing blocked due to staggered clocks 'C1' and 'C2' at
T=<time>, pat=N. (M198)

Severity

Warning

Description

When reading an Extended VCDE pattern file with clock recovery enabled, two or more clocks were active that did not have the same starting time for their leading edges.

C1 and C2 are the port names of the clocks involved. The time reference is the event time within the VCDE pattern file of the second clock going active. The pattern number P refers to the TestMAX ATPG pattern being constructed to hold this event sequence.

Clock recovery rules do not support staggered clocks. When multiple clocks are active, they must begin at the same time and have the same duration. The sequence of VCDE events that would be collapsed into a clock pulse is left in expanded form for this pattern. This might cause problems if some cycles succeed in having clock recovery while others do not. This is because it might be difficult to map the resulting patterns to a tester timeset where the port is a pulsed pin in some cycles but non-pulsed in others. For the highest probability of success, all cycles where defined clock pins are asserted should be recoverable into clock pulses, or they should all be left exploded into separate events.

What Next

If your purpose for reading in patterns is to perform fault grading, then you can ignore this warning. The fault simulator is an event-based simulator and will handle the expanded event sequence as easily as a collapsed sequence involving clock pulses. You might, however, see the internal pattern count higher than it would otherwise be if the clock were recoverable, and the fault simulation might take a little longer.

If you wish to eliminate the warning, you have two choices:

1. change the patterns so that clocks are not staggered and they are either identically overlapped or separate and non-overlapped, or
2. remove one or more of the clocks from the list of defined vcd_clock ports and reread the patterns.

M199

Message Text

Event collapsing blocked due to different width overlapping 'C1' and 'C2' at T=<time>, pat=N. (M199)

Severity

Warning

Description

When reading an Extended VCDE pattern file with clock recovery enabled, two or more clocks were active that had identical starting times, but different durations (one ended before another).

C1 and C2 are the port names of the clocks involved. The time reference is the event time within the VCDE pattern file near the end of one of the clocks. The pattern number P refers to the TestMAX ATPG pattern being constructed to hold this event sequence.

Clock recovery rules do not support overlapped clocks that do not have identical timing. When multiple clocks are active, they must begin at the same time and have the same duration. The sequence of VCDE events that would be collapsed into a clock pulse is left in expanded form for this pattern. This might cause problems if some cycles succeed in having clock recovery while others do not. This is because it might be difficult to map the resulting patterns to a tester timeset where the port is a pulsed pin in some cycles but non-pulsed in others. For the highest probability of success, all cycles where defined clock pins are asserted should be recoverable into clock pulses, or they should all be left exploded into separate events.

What Next

If your purpose for reading in patterns is to perform fault grading, then you can ignore this warning. The fault simulator is an event-based simulator and will handle the expanded event sequence as easily as a collapsed sequence involving clock pulses. You might, however, see the internal pattern count higher than it would otherwise be if the clock were recoverable, and the fault simulation might take a little longer.

If you wish to eliminate the warning you have two choices:

1. change the patterns so that clocks are not staggered and they are either identically overlapped or separate and non-overlapped, or
2. remove one or more of the clocks from the list of defined vcd_clock ports and reread the patterns.

4

Messages M200 - M299

This section contains the following error messages and warnings:

- [Error and Warning Messages M200 - M209](#)
- [Error and Warning Messages M210 - M219](#)
- [Error and Warning Messages M220 - M229](#)
- [Error and Warning Messages M230 - M239](#)
- [Error and Warning Messages M240 - M249](#)
- [Error and Warning Messages M250 - M259](#)
- [Error and Warning Messages M260 - M269](#)
- [Error and Warning Messages M270 - M279](#)
- [Error and Warning Messages M280 - M289](#)
- [Error and Warning Messages M290 - M299](#)

Error and Warning Messages M200 - M209

This section contains the following error messages and warnings:

- [M200](#)
- [M201](#)
- [M202](#)
- [M203](#)
- [M204](#)
- [M205](#)
- [M206](#)
- [M207](#)

- [M208](#)
- [M209](#)

M200

Message Text

Full-sequential <ATPG | fault simulation> cannot be performed on the <IDDQ | bridging | IDDQ bridging> fault model. (M200)

Severity

Error

Description

This message is issued when you try to run Full-Sequential ATPG (`run_atpgfull_seq`) or fault simulation (`run_fault_sim_seq`) is attempted on an unsupported fault model. Full-Sequential ATPG and fault simulation does not currently support the IDDQ, bridging, or IDDQ bridging fault model.

What Next

If running ATPG, select a different ATPG engine such as Basic-Scan or Fast-Sequential. If running fault simulation, select a different fault simulator engine if the patterns are compatible with that engine.

M201

Message Text

Random pattern source cannot be used for sequential simulation. (M201)

Severity

Error

Description

Sequential simulation is not permitted when the pattern source is random.

What Next

Change your pattern source using the `set_patterns` command.

M202

Message Text

Pattern source contains N sequential patterns which will not be simulated. (M202)

Severity

Warning

Description

This is a warning message that is given when Basic-Scan or Fast-Sequential simulation is attempted on a pattern set that contains Full-Sequential patterns.

What Next

Try the simulation again with the `-sequential` option.

M203

Message Text

T pin P of instance I not in model. (M203)

Severity

Error

Description

During pattern mapping, the pattern data referenced a pin which could not be found on the instance for which the data was to be mapped.

T is the pin type of input, output, or bidirectional. P is the name of the pin. I is the instance pathname to the module for which the data mapping was being applied when the mismatch occurred.

What Next

This mismatch must be corrected before data mapping can proceed. Compare the data file to the module port list to identify where the mismatch is coming from.

This error might also result from improper connection of instance I in the design, causing it to be optimized away. Use the `report_instances` and `report_primitives` commands to ensure that instance I and pin P of the instance are present in the design after run build.

For more information, refer to the topic `run_build_model` in the TestMAX™ ATPG and TestMAX Diagnosis Commands Guide.

M204

Message Text

T pin P of instance I (G) has no potential correspondence points. (M204)

Severity

Error

Description

Pin P of type T (input, output, bidi) from instance I that mapped to gate G had no potential correspondence points. A valid correspondence point must be an external pin or a scan cell. For a pulsed pin, the correspondence point must be a defined clock pin.

What Next

This mismatch must be corrected before data mapping can proceed. Compare the data file to the module port list to identify the cause of the mismatch.

You might also try switching to pattern dependant mapping using the `run_mapping -dependent_pattern_mapping` option or modifying the netlist by adding logic to create a correspondence point for pin P.

M205

Message Text

Correspondence point for T pin I/P (gate ID - num) could not be found.
(M205)

Severity

Error

Description

During pattern mapping, a discrepancy was found between the source file and the netlist built by TestMAX ATPG. No correspondence point could be found for pin P of type T (input, output, bidi) from instance I.

What Next

This mismatch must be corrected before data mapping can proceed. Compare the data file to the module port list to identify the cause of the mismatch.

You might also try switching to pattern dependant mapping using the `-dependent_pattern_mapping` option of the `run_mapping` command, or modifying the netlist by adding logic to create a correspondence point for pin P.

M206

Message Text

Input pin *P* of instance *I* (*G*) is constrained to always be *V*. (M206)

Severity

Warning

Description

During pattern mapping, a problem was found. Pin *P* of type *T* (input, output, bidi) from instance *I* was constrained to always be at a binary state *V*. It is not possible to find a valid correspondence point from which it can be independently exercised.

What Next

This is a warning message, no action is required.

M207

Message Text

Clock *C* was on for *T* conditioning pattern of instance *I*. (M207)

Severity

Error

Description

A clock pin *C* was "on" (asserted) for a type *T* (input or clock) conditioning pattern of instance *I*. The clock pin should not be in the "on" state at this time.

What Next

This situation must be corrected before pattern mapping can proceed by adjusting the patterns or the netlist.

You might also want to try switching to dependent pattern mapping using the `-dependent_pattern_mapping` option of the `run_mapping` command.

M208

Message Text

Clock pin (*P*) not at off-state during `FORCE_ALL_PIS` event of pattern *N*. (M208)

Severity

Error

Description

A clock pin *P* was not at its off state for a force-PI event that occurred in pattern *N* of the module test patterns.

What Next

Review the module test patterns for the pattern number indicated. This data must be corrected before pattern mapping can proceed or the netlist must be adjusted to be consistent with the pattern.

You might also wish to try switching to dependent pattern mapping using the `-dependent_pattern_mapping` option of the `run_mapping` command.

M209

Message Text

Inconsistent clock off-state for pin I/P corresponding to clock C. (M209)

Severity

Error

Description

The off state of pulsed pin *P* of instance *I* was not consistent with the off state of its top level correspondence clock *C*.

What Next

This situation must be corrected before pattern mapping can proceed. You should check the polarity of the module's clock input as well as the top level correspondence clock input. The source of the problem might be in the netlist, in the clock definition in the protocol file, or in the clock definition in the pattern file.

Error and Warning Messages M210 - M219

This section contains the following error messages and warnings:

- [M210](#)
- [M211](#)
- [M212](#)

- [M213](#)
- [M214](#)
- [M215](#)
- [M216](#)
- [M217](#)
- [M218](#)
- [M219](#)

M210

Message Text

Pattern compression cannot be performed for Full-Sequential patterns.
(M210)

Severity

Error

Description

This is a message that is issued at the end of a Basic-Scan simulation pass. It indicates that a total of N Basic-Scan patterns failed Fast-Sequential verification and were not given credit for detection of faults. A failure occurs when the Basic-Scan and Fast-Sequential simulations have a different answer, except where the Basic-Scan value was X.

This message will only appear if you have selected to use the transition fault model.

What Next

This is an informational message and requires no action by the user unless an excessive number of patterns are being rejected.

M211

Message Text

Selected checkpoint time (T1) is less than minimum allowed time (T2).
(M211)

Severity

Error

Description

You specified a checkpoint interval time T_1 which was less than the minimum supported time T_2 .

What Next

Reissue the command and specify a checkpoint time of T_2 or greater CPU seconds.

M212

Message Text

`N1 (N2) Basic-scan patterns failed current pass simulation check and is treated as ignored measures. (M212)`

Severity

Warning

Description

This is message that is issued during ATPG pattern generation, but at the end of each Basic-Scan simulation pass. It indicates that a total of N_1 Basic-Scan patterns failed Fast-Sequential verification during the current pass and will now have the mismatched expected values masked as a corrective action. A failure occurs when the Basic-Scan and Fast-Sequential produce a different answer, including answers in which one of the expected values is X. N_2 indicates the total number of measures that had differences and had to be ignored (masked).

This check is performed whenever it is determined that the design can produce basic scan results which are at risk of being wrong. This will occur under many situations a few of which include the presence of transparent latches in the design, or a Leading Edge (LE) to Trailing Edge (TE) clock situation, or the presence of a ClockPO (C17), or most all of the clock rule violations such as: C5, C6, C8, C9, and so forth.

This fast-sequential simulation requires more CPU effort and only a single pattern at a time is processed instead of the 32 at once of the Basic-Scan algorithm, but the fast-sequential is a more accurate algorithm.

This is an expert level message, which is not displayed unless the message level has been selected as expert. See the `set_messages` command.

What Next

This is an informational message and requires no action by the user.

M213

Message Text

`N1(N2) Basic-scan patterns failed simulation check and were treated as ignored measures. (M213)`

Severity

Warning

Description

This is a summary message which occurs at the end of ATPG pattern generation which indicates that a number of Basic Scan patterns failed a simulation recheck using the Fast-Sequential simulator. `N1` is the number of patterns involved, and `N2` is the total number of failing measures.

This recheck was most likely activated due to a clock rule violation such as C5, or C12, and so forth., which existed when ATPG pattern generation was started. The failing pattern is adjusted to mask the measure value where the mismatch was detected. Test coverage is reduced slightly, but there is no danger of generating a pattern which later fails in simulation.

There can also be detailed occurrence messages in the form of M212 messages which accompany the actual ATPG pattern generation. This more detailed messages are only shown when expert level messaging is enabled, see the `set_messages` command.

What Next

This is an information only message which usually requires no action. It is an indicator that an automatic masking operation, which is enabled by default in TestMAX ATPG, was used.

If there are a severe number of basic scan patterns which are being reclassified (>30% of total patterns) you might wish to investigate the Cxx violation messages which are the fundamental cause of this corrective action.

M214

Message Text

`Simulation cannot be performed with basic scan selected off. (M214)`

Severity

Error

Description

This message is issued if a simulation requiring Basic-Scan was attempted but the `-basic_scan` simulation option was disabled by a prior use of the `-nobasic_scan` option of the `set_simulation` command.

This message can occur when attempting either good machine simulation of a random pattern source set patterns random; run simulation or selecting pindata to be `fault_sim_results` (`set_pindata`) while the `-nobasic_scan` option of the `set_simulation` command was selected.

What Next

You might be able to use the `-basic_scan` option of the `set_simulation` command to continue. However, this can invalidate the patterns you might be investigating or the data you might wish to view. For example, if you were trying to review the `fault_sim_results` from pattern 47, which was generated under the environment of `-nobasic_scan`, then switching this setting back to `-basic_scan` can present data that does not match pattern 47 and therefore mislead you in your investigation.

M215

Message Text

Tied pin (P) not at tied_value (V) during FORCE_ALL_PIS event of pattern N. (M215)

Severity

Error

Description

For pattern mapping, this is an error message that is given when a module input connected to tied circuitry is forced to a different value in the module patterns.

What Next

The netlist and the patterns must be consistent before you can successfully continue. Review the netlist and the patterns for the pin specified and adjust one to be consistent with the other.

M216

Message Text

Input pin P1 of instance I (G1) has the same source as pin P2 (G2). (M216)

Severity

Error

Description

For pattern mapping, this is a warning message that is given when two module input pins share a common source. The module patterns must then satisfy this relationship by always providing the same (or possibly complimentary) value on both module inputs.

What Next

This is warning unless the patterns do not satisfy this criteria in which case it is reported along with a M217 error. If an error occurs you must either adjust the patterns to satisfy the requirement or modify the design so that the common source relationship is removed.

M217

Message Text

Common pins (P1=0 and P2=1) not at same_value during FORCE_ALL_PIS event of pattern N. (M217)

Severity

Error

Description

For pattern mapping, this is an error message that is given when two module inputs which have a common source are forced to inconsistent values in the module patterns. This error appears in conjunction with warning M216.

What Next

Two pins that have a common source cannot be forced to inconsistent values. You should adjust the patterns to remove the inconsistency or change the design so that the common source is removed.

M218

Message Text

Clock pin *pin_name* has been selected to be not mapped. (M218)

Severity

Error

Description

For pattern mapping, this is an error message that is given when a `-nomap` option is attempted for a clock pin.

What Next

Clock pins cannot be specified with the `-nomap` option. Adjust your command and remove the reference to the clock pin.

M219

Message Text

Conditional input gate G1 (inv=%d) is used instead of G2 for pin P.
(M219)

Severity

Error

Description

For pattern mapping, this is a warning message that is given when a conditional input has been used. There is no checking that the module patterns are consistent with these conditions.

What Next

Be aware that no checking is performed.

Error and Warning Messages M220 - M229

This section contains the following error messages and warnings:

- [M220](#)
- [M221](#)
- [M222](#)
- [M223](#)
- [M225](#)
- [M226](#)
- [M227](#)

- [M228](#)
- [M229](#)

M220

Message Text

Force_pi/measure_po events of pattern N not satisfied due to
<description>. (M220)

Severity

Error

Description

For pattern mapping, this is an error message that is given to indicate the patterns supplied could not be mapped. Indicates the gate that was unsuccessfully mapped to an observe point. TestMAX ATPG wasn't able to satisfy force_pi or measure_po events because of one of the following conditions:

abort: the logic is too complicated to justify all the force_pi or measure_po

redundant: when functional dependencies between the module inputs or outputs never allow certain combination of values on these ports.

ATPG_untestable: when force_pi or measure_po can't be satisfied because of blocked paths caused by, for example, nonscan flip-flops or black boxes

What Next

Either the patterns must be adjusted to fix the problem or the design must be adjusted.

M221

Message Text

Force_pi/measure_po events of pattern N resulted in clock C on. (M221)

Severity

Error

Description

For pattern mapping, this is an error message that is given to indicate pattern N could not be mapped because it resulted in clock C being left at an "on" (asserted) state.

What Next

The patterns must be adjusted to fix the problem.

M222

Message Text

No new changes for event at T=<time>. (M222)

Severity

Warning

Description

While reading Extended VCD patterns, a new event time was encountered that did not define any changes to inputs or outputs from previous events.

What Next

You may ignore this warning. It is issued because VCDE and VCD pattern formats are records of "changes" and it is curious that the pattern contains no changes.

M223

Message Text

Strobe comment and other events occur at same time of T, other events are ignored. (M223)

Severity

Warning

Description

While reading Extended VCD patterns, the `set_patterns-strobe` comments command was specified that uses the location of `$comment ... $end` blocks in the VCDE file as indicators of where a measure of the patterns is desired. At the time T where the comment block was found, there were also other events involving input or output changes. These other events are ignored.

What Next

You should investigate the pattern source. Insertion of a `$comment` block at the same time as other events and using it to mark a measure point can have undesired effects. It is best if the `$comment` blocks have their own unique time of occurrence.

M225

Message Text

Number instance connectors exceeded <N>. (M225)

Severity

Warning

Description

An internal working limit for the maximum number of connections for a gate in the schematic viewer was reached. You probably clicked on a net diamond with a very large fanout or fanin.

What Next

The `set_workspace_sizes` command can be used to increase this limit.

M226

Message Text

Instance `I1` of undefined module `M1` called by module `M2` has unconnected vector ports. (M226)

Severity

Warning

Description

A `write_netlist` command was performed with one or more modules undefined or defined as black boxes or empty boxes. When this situation exists TestMAX ATPG does not know the actual width or direction of the vectored ports (bussed nets). The netlist created will have the bussed nets dropped from the instantiation and so this netlist cannot be used without some edits to correct the problem. This is a warning that modifications to the netlist is required before the netlist is usable.

`M1` is the name of the undefined module. `I1` is the instance name of this undefined module, which can be found within module `M2`.

What Next

Before using the netlist you should correct the problem by locating within module `M2` the instantiation of module `M1` with the instance name reported. Then edit the port connection list to restore the correct net connectivity.

An alternative solution is to define the undefined module by reading in a file where this module is defined, or at least has its port list defined. After this has been done you can issue another `write_netlist` command. With the module defined and the width of ports and port directions fully known, you should no longer get this warning message.

M227

Message Text

Full-sequential ATPG is required when selecting `seq_capture` procedure.
(M227)

Severity

Error

Description

This message is issued if you are trying to issue the `run_atpg` command and you have selected the sequential capture procedure without enabling the Full-Sequential ATPG.

What Next

Enable Full-Sequential ATPG before issuing another `run_atpg` command or remove the use of the sequential capture procedure.

M228

Message Text

Cannot map output A to B. (M228)

Severity

Error

Description

An attempt was made to define a bidirectional port WGL character mapping that cannot be supported.

What Next

Not all mappings are allowed. For example, you cannot map the dash for scan input or scan output to any other character. Also, you can map "zz" to "-z", but you cannot map "zz" to "z-". Why? It has to do with a loss of a measure and with being able to unambiguously read back in the WGL which is written out. The "zz" to "-z" mapping still indicates a measure is to be performed but a "zz" to "z-" mapping could be confused with

a "zx" to "z-" mapping which generally is interpreted to mean there is no force and no measure.

The ability to use some bidi mappings is affected by whether the tester can measure Z values or not. If the tester can measure Z values then the default setting of `set_buses-external_zZ` should be used and the WGL patterns can contain both ZZ and ZX data (no force, measure Z and no force, no measure). If the tester cannot measure Z values or you wish to generate patterns for which no Z-measure is needed you would set the option `set_buses-external_z X` before generating patterns. This would result in WGL patterns with "ZX" data for bidirectional pins but no "ZZ". If "ZZ" does not appear in the WGL you can define a bidi map of "ZX"->Z-" or "ZX"->-Z" which you could not do if the Z measure were enabled and "ZZ" were possibly present.

M229

Message Text

User-specified `vcd_clock` pin 'P' not found on target module. (M229)

Severity

Error

Description

A port defined using a `set_patterns-vcd_clock` option was not found on the target module. This is an error and further process of patterns stops and patterns are discarded.

What Next

Double-check the list of ports of the target module as well as the spelling used in the original command before trying again.

Error and Warning Messages M230 - M239

This section contains the following error messages and warnings:

- [M230](#)
- [M231](#)
- [M232](#)
- [M233](#)
- [M234](#)
- [M235](#)

- [M236](#)
- [M237](#)
- [M238](#)
- [M239](#)

M230

Message Text

Command not supported for non-GUI version. (M230)

Severity

Error

Description

You attempted a command that in the shell mode version of TestMAX ATPG that is only supported in the GUI version.

What Next

If you are using a command file, you might want to adjust your commands so they do not attempt the unsupported commands. Or, you might want to restart using TestMAX ATPG GUI mode if use of the command is not an option.

M231

Message Text

Sequential patterns (N1) exist before first saved pattern (N2). (M231)

Severity

Warning

Description

You wrote a subset of the total patterns available and in doing so skipped over some Full-Sequential patterns (=N1) that occurred before the block of patterns saved. Since Full-Sequential patterns expect the design state to be retained from pattern to pattern, your subset of patterns cannot properly initialize the design and so can produce simulation mismatches or fail on the tester.

What Next

Use the `-types` option of the `report_patterns` command to determine where the first Full-Sequential patterns exist and save patterns starting with this pattern or prior.

M232

Message Text

Force values were changed for N1 bidis in N2 patterns. (M232)

Severity

Warning

Description

This is an informational message that indicates that the patterns being created were changed to force Z on bidi pins to avoid postclock contention rather than discarding the patterns and searching for others. N1 is the number of bidirectional pins involved, N2 is the number of patterns involved.

What Next

No action is required. This information is an indication that good things are happening during pattern generation.

This situation occurs frequently due to postclock contention on a bidirectional pin. TestMAX ATPG is trying to create a pattern in which the bidi pin is driven at the beginning of the capture procedure, but the clock event causes the bidi driver to turn on and this would cause contention. The TestMAX ATPG default behavior is to step back and change the pattern so that the bidi is not driven, and so the contention is avoided.

M233

Message Text

No instances of selected module (I) were found. (M233)

Severity

Error

Description

The instance (=I) specified with the `run_mapping` command was not found in the design.

What Next

Check your spelling. Sometimes the use of escaped names in the design might be a little tricky. Try adding faults to the parent module in which the instance resides and then reporting the fault list to get an idea of the valid instance pathnames known to TestMAX ATPG.

If your original module was an empty box or a module definition with no internal gates, it can have been stripped away by the default TestMAX ATPG unused gate removal algorithm. You might want to explicitly list the module as a black box using the `set_build` command.

M234

Message Text

ATPG terminated due to meeting <type> limit. (M234)

ATPG terminated after successful chain_test generation. (M234)

ATPG terminated due to flat progress (0.08/333) (M234)

Severity

Warning

Description

The first version of this message is generated when a user-specified limit such as pattern count, test coverage, or CPU time has been reached.

The second version of the message indicates that ATPG halted because only the chain test patterns were specified (by the user) to be generated.

The third version of the message indicates that ATPG halted when detecting flat tail, that is, the trend of coverage increase per pattern falls below some threshold.

What Next

No action is necessary for first and second versions of the messages. These messages are intended to be informative and to indicate what caused the ATPG pattern creation effort to halt.

For third version, see the `set_atpg -allow_flat_tail_atpg` command for further information.

M235

Message Text

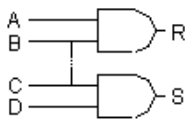
Diagnosis cannot be performed due to <more info here>. (M235)

Severity

Warning

Description

Consider the following circuit where a defect bridges the B and C nets:



Assume this bridge manifests itself logically as an AND bridge. The behavior as on tester would be as follows:

Pattern	A	B	C	D	Good		Faulty		Status
					R	S	R	S	
1	1	0	0	1	0	0	0	0	Pass
2	1	0	1	1	0	1	0	0	Fail
3	1	1	0	1	1	0	0	0	Fail
4	1	1	1	1	1	1	1	1	Pass

However, in TestMAX ATPG the main fault model used for diagnosis is the single stuck-at, with enhancements to detect complex defects. This presents some limitations. The algorithm attempts to combine behavior of multiple single stuck-at faults to explain the failure.

In this example, TestMAX ATPG may use B stuck-at 0 and C stuck-at 0 as candidate faults to explain the measurements. The behavior of these faults during fault simulation would be the following (where "A/B" denotes "A" being the good circuit value, and "B" denoting the faulty circuit value).

Pattern	A	B	C	D	B sa0		C sa0	
					R	S	R	S
1	1	0	0	1	0	0	0	0
2	1	0	1	1	0	1	0	1/0
3	1	1	0	1	1/0	0	1	0
4	1	1	1	1	1/0	1	1	1/0

TestMAX ATPG may decide that these two faults are the best candidates for explaining the failures seen by the tester. The problem arises with pattern 4, where the fault simulation expects failures for both these faults. Notice that the tester did not show a failure for this pattern. Therefore, TestMAX ATPG issues the warning indicating that the fault candidates cause the patterns that pass on the tester to fail in fault simulation.

What Next

This is an informational message only.

M236

Message Text

Transition fault ATPG requiring clock launch cannot be performed without fast-seq ATPG enabled. (M236)

Severity

Warning

Description

This message is generated when transition fault model has been chosen and a clock launch is allowed, but Fast-Sequential patterns have not been enabled. The creation of clock launch patterns will require the use of the multiple capture cycles of the Fast-Sequential patterns.

What Next

If you want to generate clock launch transition fault patterns, use the `-capture_cycles` option of the `set_atpg` command to select a capture_cycle depth of 2 or more, and then re-issue the `run_atpg` command.

If you do not want to generate Fast-Sequential patterns you can either ignore this warning or use the `-launch_cycle` option of the `set_delay` command to restrict the transition faults to the `last_shift_launch` type.

M237

Message Text

Effectiveness of using last shift to launch transitions is affected
by load postamble. (M237)

Severity

Warning

Description

This message is generated when transition fault model has been chosen and the `load_unload` procedure contains vectors after the scan shift (postamble) cycles. It is possible that the postamble cycles will interfere with the timing of the last-shift-launch strategy.

What Next

Review the `load_unload` postamble to see whether it can be eliminated.

M238

Message Text

Multicycle capture procedure will reduce effectiveness of transition
fault patterns. (M238)

Severity

Warning

Description

This violation message is generated when the `run_atpg` command is used, the fault model is transition, and the defined clock capture procedures in the DRC/STIL procedure file use multiple test cycles.

What Next

For the most effective transition fault patterns the time interval between the launch event and the capture event should be as small as possible. When a multi cycle capture procedure is used, this cannot be accomplished. If possible, modify the various

capture_XXX procedures to be a single test cycle. This cannot be possible if you are using an end-of-cycle protocol, in which case a 2 cycle procedure is the minimum you can create. Consider switching to a preclock measure protocol that will allow you to define a one cycle capture procedure.

If you are unsure which clocks have multi cycle capture procedure after running DRC successfully, write out a STIL procedure file with the `write_drc_file` command. Search for "capture_" in your STIL procedure file. You will then see if any clock capture procedure contains multiple-cycles.

For more information, refer to the topic Capture Procedures in STIL in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M239

Message Text

Line N: Invalid pattern number (P). (M239)

Severity

Error

Description

Pattern number P found in line N of the pattern reorder file specified with the `-reorder` option of the `write_patterns` did not exist in the pattern buffer (internal or external) being written.

What Next

Correct the reorder file to remove the reference to the non-existent pattern and try again.

Error and Warning Messages M240 - M249

This section contains the following error messages and warnings:

- [M240](#)
- [M241](#)
- [M242](#)
- [M243](#)
- [M244](#)
- [M245](#)

- [M246](#)
- [M247](#)
- [M248](#)
- [M249](#)

M240

Message Text

Line N: Duplicate pattern number (P) was ignored. (M240)

Severity

Warning

Description

Pattern number P found in line N of the pattern reorder file specified with the `-reorder` option of the `write_patterns` occurred more than after in the reorder file and the duplicate entries were ignored when writing patterns.

What Next

Correct the reorder file to remove the duplicate pattern reference and try again.

M241

Message Text

Fast-sequential ATPG cannot be performed with capture cycles
set to 0. (M241)

Severity

Error

Description

An attempt was made to generate patterns using either the `-fast_sequential_only` option of the `run_atpg` command or a transition fault model of `clock_launch` with the capture cycle depth at it's default of zero. A value of 2 or more must be in effect to create Fast-Sequential patterns.

What Next

Use the `-capture_cycles` option of the `set_atpg` command to select a value of 2 or greater and try the original command again.

M242

Message Text

No defined bidi control pin. (M242)

Severity

Error

Description

This violation is issued when the `capture_XXX` procedures defined in the DRC protocol file describe a ReflectIO protocol but the bidi control pin referenced cannot be found in the design or does not appear to control any tristate drivers.

What Next

Review the `capture_XXX` procedures and verify that the bidi control pin which is set to a state to turn bidi's off in the second cycle of the capture procedures is really a pin which controls the bidi's. Check that the bidi control pin is not blocked from its control function by some other pin held constant.

M243

Message Text

Multiple bidi control pins P1(G1) and P2(G2). (M243)

Severity

Warning

Description

One or more `capture_XXX` procedures in the DRC file use the ReflectIO protocol with more than one port used for the bidi control pin. TestMAX ATPG supports only a single bidi control pin.

What Next

You can have to change from a ReflectIO protocol to a Bidi dead-cycle protocol or change the design to have only a single control pin.

M244

Message Text

Different bidi control pins P1(G1) and P2(G2). (M244)

Severity

Warning

Description

One or more capture_XXX procedures in the DRC file use the ReflectIO protocol with a different bidi control pin named from the others. TestMAX ATPG supports only a single bidi control pin.

What Next

You can have to change from a ReflectIO protocol to a Bidi dead-cycle protocol or change the design to have only a single control pin.

M245

Message Text

Unused gate deletion affects fault coverage calculation. (M245)

Severity

Warning

Description

You specified that a fault coverage should be reported but unused gate deletion is in effect. Please be aware that the fault coverage calculation is affected by deleting unused gates and for the most accurate fault coverage unused gates should not be deleted. Test Coverage is the recommended figure of merit and it is not affected by unused gate deletion.

What Next

No user action is required. This message is for information only.

M246

Message Text

Force_all_PIs was preceded by an ignored force_all_PIs in pattern P.
(M246)

Severity

Warning

Description

When processing patterns for the `run_mapping` command the pattern sequence encountered forced the PI's (primary inputs) of the target instance twice. The second set of forced values is used and the first set is ignored.

P identifies the pattern number on which this double forcePI event occurred.

What Next

You should run the `report_patterns` command, then review the input patterns to see if there was a problem with the patterns and correct it if possible. Having two succeeding patterns change the inputs with no intervening measure PO or clock might be as intended, in which case you can ignore this warning.

M247

Message Text

Pulse_clock was preceded by measure_all_POs which reloaded scancells in pattern P. (M247)

Severity

Warning

Description

While processing patterns for a `run_mapping` command, a `measure_all_POs` event was encountered. This could potentially affect the scan chain and invalidate any setup conditions satisfied by the prior scan chain load. This potential invalidation is because measuring all PO's can involve unloading a scan chain which would destroy the conditions assumed setup by the scan load.

What Next

Review the pattern in question. If necessary, add an additional scan load to the patterns after the measure to restore the conditions established by the first scan load.

M248

Message Text

There were N modules changed by net connections. (M248)

Severity

Warning

Description

Some net connections were previously defined using the `-all` or `-module` options. These type of net connections change the netlist in memory and cannot be reversed. There were *N* modules affected. When you perform a `write_netlist` command the resulting file will contain some modules affected by this change.

What Next

If you do not want to use the modified modules you should identify and delete the changed modules from the output file and use the original version of those modules.

M249

Message Text

There were *N* net connections not removed because they changed netlist data. (M249)

Severity

Warning

Description

Some net connections were previously defined using the `-all` and `-module` options of the `add_net_connections` command. These type of net connections change the netlist in memory and cannot be reversed. There were *N* net connections that were not removed and they can be reviewed with the `report_net_connections` command.

What Next

To completely remove the net connections remaining it is necessary to return to build mode and reread the affected modules to restore the original unmodified netlist.

Error and Warning Messages M250 - M259

This section contains the following error messages and warnings:

- [M250](#)
- [M251](#)
- [M252](#)

- [M253](#)
- [M254](#)
- [M256](#)
- [M257](#)
- [M258](#)
- [M259](#)

M250

Message Text

Chain test not written because <more info>. (M250)

Severity

Warning

Description

When clock restrictions are in place such that a single clock must always be used, or a one-hot clocking is requested, or the sequential_capture procedure is called for then no chain tests is written. This is because the chain test requires a non-clocking capture between the chain load and the chain unload. The restrictions on clocking do not allow a pattern with a non-clocking capture procedure.

What Next

If a chain test is required, you must remove the clock restrictions before proceeding by adjusting the `set_drc -clock` option. If this is not possible because it causes other failures, then you can have to create your chain test patterns in a separate TestMAX ATPG session and write them to a unique pattern file.

M251

Message Text

Chain path multiple sensitization was truncated due to insufficient workspace size (1000). (M251)

Severity

Error

Description

During the scan chain trace, a multiply sensitized path for the shift data was encountered. When this happens, the TestMAX ATPG algorithm attempts to find the proper source of the data by analyzing all possible state element candidates that might be the source of this data. There is an internal limit 1000 candidates that can be analyzed, and this limit was reached.

What Next

You can increase the limit by specifying `set_workspace_size-drc_buffer_size 10000`.

M252

Message Text

Patterns are written using artificial scan ability. (M252)

Severity

Warning

Description

Patterns were created and a prior `set_scan_ability` command was still in effect which creates virtual scan elements not present in the actual design.

What Next

No action is required. This is a warning that the patterns will not be usable on the design until the design is altered to match the virtual scan chain created and used in the patterns.

M253

Message Text

The <license_name> license cannot be checked in when the <fault_model> is selected. (M253)

Severity

Error

Description

An attempt to free a license was made for either the Test-IDDQ or Test-Delay feature without first restoring the fault model to stuck with the `set_faults-model stuck` command.

What Next

Change the fault model, then retry the `remove licenses` command.

M254

Message Text

Path-delay ATPG can only be performed with the full-seq ATPG. (M254)

Severity

Error

Description

This is an error message that is issued when `run_atpg` is attempted with a fault model of path-delay without enabling the use of Full-Sequential ATPG. Path-delay is only supported for Full-Sequential ATPG.

What Next

Use the `set_atpg` command to enable the Full-Sequential ATPG algorithm and consider using the `-full_sequential_only` option of the `run_atpg` command.

M256

Message Text

`port_name` is already in scan enable list. (M256).

Severity

Error

Description

This is an error message that is issued when the port name associated with an `add_scan_enables` command is already in a scan-enable list.

What Next

Use the `report_scan_enables` command to see the list already defined. If you need to change the initialization value first remove the scan enable definition using `remove_scan_enables` and then redefine it with the new value.

M257

Message Text

`port_name` is not in scan enable list. (M257)

Severity

Error

Description

This is an error message that is issued when a `remove_scan_enables` command is attempted for a port name that is not contained in the current list of defined scan enables.

What Next

Use the `report_scan_enables` command to review the list of defined scan enables.

M258

Message Text

Clock period data is invalid. (M258)

Severity

Error

Description

This error occurs when a clock is defined with a period or time units which is different from previously defined clocks. After a clock is defined, all other clocks must use the same period, unit of time, and measure point.

Defining timing with different periods requires manually editing the test procedure file and creating multiple timing blocks.

What Next

Use the `report_clocks` command to review the previously specified period and time units and then repeat the failing command with consistent times and units to what was previously specified.

M259

Message Text

Clock `measure_time` data is invalid. (M259)

Severity

Error

Description

This is an error message that is issued when a clock is attempted to be defined with a `measure_time` that is different from a previously defined measure time. After being specified, all measure times must be the same.

Defining timing with different measure times on various ports requires manually editing the test procedure file timing block.

What Next

Use the `report_clocks` command to review the previously specified measure times and repeat the failing command with a measure time that matches what was previously used.

Error and Warning Messages M260 - M269

This section contains the following error messages and warnings:

- [M260](#)
- [M261](#)
- [M262](#)
- [M263](#)
- [M264](#)
- [M265](#)
- [M266](#)
- [M267](#)
- [M268](#)
- [M269](#)

M260

Message Text

Delay paths cannot be removed when path-delay faults are active. (M260)

Severity

Error

Description

This is an error message that is issued when delay paths are attempted to be removed while there are still active path-delay faults.

What Next

The fault list must be deleted before delay paths can be altered. The `remove_faults-all` command will delete all path delay faults from the fault list.

M261

Message Text

`Fault simulation identified fault as possible detected. (M261)`

Severity

Warning

Description

This is a warning message that is issued if the fault simulation detection verification of a fault analysis determines that a pattern intended to detect the selected fault was only possible detected. The simulation data for the pattern is placed in parallel pattern 1 (or Fast-Sequential pattern if fast-seq ATPG was performed).

What Next

Review the pattern for information as to why the fault is possibly detected.

M262

Message Text

`Fault simulation failed to detect fault. (M262)`

Severity

Warning

Description

This is a warning message that is issued if the fault simulation detection verification of a fault analysis determines that a pattern intended to detect the selected fault was not

detected. The simulation data for the pattern is placed in parallel pattern 1 (or Fast-Sequential pattern if fast-seq ATPG was performed).

What Next

You can draw the various gates involved with the fault and display the pattern saved in an attempt to try to understand why the fault was not detected by fault simulation.

M263

Message Text

```
<name|LFSR|PRPG shadow> chains are not in chain list. (M263)
```

Severity

Error

Description

This error message is issued if you try to remove a scan chain and the scan chain name specified has not been defined or is not in the scan chain list.

What Next

Use the `report_scan_chains` command to get a list of known scan chains. Also keep in mind that the command line form of the `remove_scan_chains` command will only work on scan chains defined by command and it will not remove scan chains defined in the DRC file. Also keep in mind that the `remove_scan_chains` command will only work on scan chains defined by the `add_scan_chains` command. It will not remove scan chains defined in the DRC file.

M264

Message Text

```
Duplicate scan chain input/output pin <P>. (M264)
```

Severity

Error

Description

This error message is generated when an `add_scan_chains` command attempts to use a pin for either scanin or scanout which has already been used in a previous scan chain definition.

What Next

Use the `report_scan_chains` command to get a list of known scan chains and their input/output pins.

M265

Message Text

Unused message. (M265)

Severity

Error

Description

Unused message.

What Next

Unused message.

M266

Message Text

Failure *failure_id* ignored due to X measure. (M266)

Number failures in the pipeline stages were ignored. (M266)

Number failure(s) were ignored due to X measures. (M266)

Severity

Warning

Description

The first variation of this message appears if a failure in the failure log file corresponds to an X measure in the external patterns. In this case, the diagnostic will ignore the failure and print an M266 warning message for each ignored failure.

The number of reported ignored failures is set by the command `set_diagnosis-max_report_failures <n>`. The default is 10.

The second variation of this message appears when tester failures in the pipeline stages are ignored during scan chain diagnosis in DFTMAX compression. In this case, the warning message reports the number of ignored failures in the pipeline stages.

The third variation of this message indicates the number of failures that are ignored because they occur during a masked cycle (X measure) in the external pattern file.

What Next

No action is required.

M267

Message Text

Number failure(s) were ignored in the last failing pattern. (M267)

Number failure(s) were read, *number* failure(s) were ignored. Failures during and after line *line_number* are ignored. (M267)

Number redundant failures were removed. (M267)

Number failure(s) exceed current pattern limit *pattern_number*. (M267)

Number failure(s) in cycle format exceed tester cycle limit *cycle_number*. (M267)

Failing gates exceeded *max_fail_gates=num_fail_gates*, *num_failures* failures were ignored. (M267)

Number in full-sequential patterns were ignored. (M267)

Severity

Warning

Description

The first variation of this message occurs when failure data is truncated. This occurs because either the `-incomplete` option of the `set_diagnosis` is specified or the `-failure_memory_limit` option is specified and the number of failures in the failure data file exceeds the specified limit.

The second variation of this message appears when one of the following occurs:

The failure log file reports more patterns than the limit specified by the `run_diagnosis-truncate` command.

Some of the failing patterns are not basic scan or fast-sequential patterns. TestMAX ATPG supports diagnosis only for these patterns.

The third variation of this message indicates that the same failure appears more than once in the failures log file. In the following example, the first two lines are redundant:

```
1 test_so4 2 (exp=1, got=0)
1 test_so4 2 (exp=1, got=0)
3 test_so4 2 (exp=1, got=0)
```

The fourth variation of this message relates to pattern-based failures and the fifth variation relates to cycle-based failures. These messages indicate that the failure log file reports failures for patterns that were not read in the external pattern buffer. The messages report the number of failures excluded from the diagnostics analysis and the maximum number of patterns or maximum number of cycles.

The sixth variation of this message appears if the number of failing gates (which are usually scan cells) exceeds the maximum value.

What Next

The first and second variations of this message are for informational purposes only and no additional action is required.

The third variation of this message indicates an erroneous data log conversion. You should check the flow used to convert your tester's failures to TestMAX ATPG failures.

The fourth and fifth variations of this message indicate that the input patterns and the failures log file used for diagnostics are not synchronized. You should verify and change either the patterns or the failures log file.

For the sixth variation of this message, the diagnostic accuracy might be impacted by this situation because failures were ignored.

M268

Message Text

Invalid image data (excessive gate size | format_code). (M268)

Severity

Error

Description

This message is issued during the reading of an image file containing invalid data.

What Next

This message normally indicates that the image file was either incorrect or corrupted. First, make sure that the image file was a successfully created image file. If the image file appears correct, then recreate the image file and try again.

M269

Message Text

`Patterns cannot currently be written when image was read. (M269)`

`Parallel Verilog cannot be written because image read did not include pin data. (M269)`

`Netlist cannot currently be written when image was read. (M269)`

Severity

Error

Description

These messages are issued when an unsupported operation is performed after TestMAX ATPG reads an image.

What Next

You must perform these operations in a new session that goes through the full build and DRC process without reading an image. To do this, you must use the `-netlist_data` or `-design_view` options of the `write_image` command to create image files that generate parallel patterns. Otherwise, the image file loses design information that affects the parallel patterns, and TestMAX ATPG issues an M269 message.

Error and Warning Messages M270 - M279

This section contains the following error messages and warnings:

- [M270](#)
- [M278](#)

M270

Message Text

`License Queuing is enabled (SNPSLMD_QUEUE string). (M270)`

Severity

Warning

Description

When this message is generated, TestMAX ATPG waits for a license if all copies of a license are currently checked out. The request for the license is placed in a queue and the license is provided in the queued order if multiple processes are waiting for the same license. TestMAX ATPG execution is suspended until the license is available and processing can continue.

The suspension ends if the license cannot be acquired within 72 hours, and the license check-out process fails (this occurs immediately if license queuing is not enabled). After a license is acquired, TestMAX ATPG holds that license until the TestMAX ATPG run finishes or the `remove_licenses` command is specified.

What Next

No further action is required. If you do not want license queuing, remove the definition of `SNPSLMD_QUEUE` from your environment (for example, in cshell, use “`unsetenv SNPSLMD_QUEUE`”) before invoking TestMAX ATPG. Review Flex-LM license operations for more information about reviewing license availability.

When a license is no longer needed for subsequent operation in long command sequences that pass through multiple modes of operation, you should use the `remove_licenses` command. TestMAX ATPG does not release licenses once they are acquired; you must do this manually. This will reduce license utilization in a single run, and, in turn, will reduce the potential for a deadlock between two parallel runs attempting to access licenses held (but not used) by the other run.

M278

Message Text

Input appears to be VCD. Only Extended VCD is supported. (M278)

Severity

Error

Description

The `set_patterns` command accepts only extended VCD (VCDE) patterns, not standard VCD. The two formats are not the same and only Extended VCD format is supported.

What Next

Select a pattern input file in VCDE format, regenerate the VCDE pattern input from a simulator. For more information on Extended VCDE refer to IEEE Verilog 1364-2001, chapter 18.3.

For more information, see the topic Pattern Formats Accepted by TestMAX ATPG in the TestMAX ATPG and TestMAX Diagnosis User Guide.

Error and Warning Messages M280 - M289

This section contains the following error messages and warnings:

- [M284](#)
- [M285](#)

M284

Message Text

```
<pattern_type> patterns and non<pattern_type> patterns cannot be used in  
the same pattern set. (M284)
```

Severity

Warning

Description

This message indicates that an invalid pattern set has been used that contains a mixture of different pattern types. TestMAX ATPG will never attempt to create pattern sets that combine incompatible pattern types.

What Next

The failing pattern set is not usable in TestMAX ATPG. The pattern set must be recreated, so that it contains a single pattern type.

M285

Message Text

```
Clocks (list) could not be reordered and is applied in parallel. (M285)
```

Severity

Warning

Description

This message indicates that the clocks that were applied in a pattern have a required serial relationship that cannot be satisfied. One situation that might lead to this message is when internal clocks are identified as being equivalent, but have a required order or

sequence when pulsing. Order relationships on internal clocks cannot be guaranteed by TestMAX ATPG; this message is generated to identify this situation. This situation also occurs when C41 violations are reported by DRC.

The list of clocks is a numeric ID, corresponding to gate IDs of the clock ports.

What Next

If this message results from a C41 violation in DRC, the patterns might be correct but the clocks must be examined carefully. If the clocks always pulse simultaneously and are skew-balanced in the design, patterns generated by ATPG will be correct. However, if the clocks are not both simultaneous and skew-balanced, the patterns generated with M285 messages will be incorrect and will fail at test or with full-timing simulation.

The pattern containing these clocks is used, but there is a risk that the measured values of the pattern might not be correct. Review the clock relationships between the identified clocks using the command `report_clocks -matrix`, and establish whether this relationship can be satisfied at test for these clocks. Downgrading C41 errors will likely generate patterns where these clocks cannot be pulsed together (because they cannot be grouped together), however the clocks will pulse together (because they are equivalent). This conflict generates patterns that cannot be used at test.

Consider removing conditions that cause these clocks to be equivalent. For instance, remove "always on" conditions for internal clocks, especially when these clocks cannot be grouped together.

If conditions that cause these clocks to be equivalent cannot be removed, consider disabling or inhibiting one of these clocks in a run to generate patterns that are not dependent on both clocks.

Error and Warning Messages M290 - M299

This section contains the following error messages and warnings:

- [M299](#)

M299

Message Text

All delay paths are now deleted. (M299)

Severity

Warning

Description

You have previously read in delay paths using the `add_delay_paths` command, but have now changed modes to DRC or BUILD, or performed a `run_drc` command, which causes the delay paths to be deleted from memory as they are only supported in TEST mode.

What Next

This is an informational message only and no action is necessary. If you return to TEST mode, you will have to reread the delay paths if you wish to generate path delay patterns.

You can avoid this message by using the `remove_delay_paths` command before changing modes.

5

Messages M300 - M399

This section contains the following error messages and warnings:

- [Error and Warning Messages M300 - M309](#)
- [Error and Warning Messages M310 - M319](#)
- [Error and Warning Messages M320 - M329](#)
- [Error and Warning Messages M330 - M339](#)
- [Error and Warning Messages M340 - M349](#)
- [Error and Warning Messages M350 - M359](#)
- [Error and Warning Messages M360 - M369](#)
- [Error and Warning Messages M370 - M379](#)
- [Error and Warning Messages M380 - M389](#)
- [Error and Warning Messages M390 - M399](#)

Error and Warning Messages M300 - M309

This section contains the following error messages and warnings:

- [M301](#)
- [M302](#)
- [M303](#)
- [M304](#)
- [M305](#)
- [M306](#)
- [M307](#)

- [M308](#)
- [M309](#)

M301

Message Text

Ignoring `--nonbuffered_io` flag. (M301)

Severity

Error

Description

This violation is reported when something went wrong while attempting to save image files under the work directory.

What Next

Check permissions of the work directory. It must be writable/readable and executable.

M302

Message Text

Cannot open the temporary command file under `<directory>`. (M302)

Severity

Error

Description

This violation is reported when a temporary command file (used during distributed fault simulation) could not be created in the work directory.

What Next

Check permissions of the work directory. It must be writable, readable, and executable.

M303

Message Text

Cannot dump temporary image under `D.` (M303)

Severity

Error

Description

An error occurred while TestMAX ATPG tried to write an image of the design to the work directory specified by D.

What Next

The work directory does not have sufficient space to store the image of the design. Change your work directory to a disk with available space.

M304

Message Text

Pattern source must be either external or internal. (M304)

Severity

Error

Description

This violation is reported when a pattern source is not one of external or internal.

What Next

Use only external or internal pattern sources.

M305

Message Text

List of processors to be used is empty. (M305)

Severity

Error

Description

This violation is reported when the list of processors is empty when running distributed ATPG.

What Next

Identify one or more slave processors, or revert to a single processor task.

M306

Message Text

`Cannot have more processors than faults. (M306)`

Severity

Error

Description

TestMAX ATPG issues this messages when the number of active faults is less than the number of processors.

What Next

Reduce the size of the processor list, or increase the size of the active fault list. You must have more active faults than processors.

M307

Message Text

`Ignoring previous definition of work dir <path> ... (M307)`

Severity

Warning

Description

This violation is reported when you are redefining the working directory. That is, the `set_distributed-work_dir` command is issued more than once in the same session.

What Next

This is an informational message only.

M308

Message Text

`name is not a directory. (M308)`

Severity

Error

Description

This violation is reported when the `set_distributed-work_dir` command contains a name that is not a directory.

What Next

Check to be sure that you have named a directory and not a file. The path to the work directory must be an absolute path.

M309

Message Text

Cannot create signature file under <directory_name> directory. (M309)

Severity

Error

Description

This violation is reported when the work directory does not have the correct permission to permit creation of a signature file. This file is used by all slaves to verify if the work directory as seen by the master is also visible by the slaves.

What Next

Check permissions of the work directory. It must be writable, readable, and executable.

Error and Warning Messages M310 - M319

This section contains the following error messages and warnings:

- [M310](#)
- [M311](#)
- [M312](#)
- [M313](#)
- [M314](#)
- [M315](#)
- [M316](#)
- [M317](#)

- [M318](#)
- [M319](#)

M310

Message Text

Setting current directory as the work directory. (M310)

Severity

Warning

Description

This is an informational message indicating that your current directory is used as the work directory by distributed processing. You can determine what your current work directory is by issuing a `pwd` command in TestMAX ATPG.

What Next

No action is required. But if you wish to change the work directory, you can do so with the `-work_dir` option of the `set_distributed` command.

M311

Message Text

<machine_name> is not a valid machine. (M311)

Severity

Error

Description

This violation is reported when you attempt to add a machine (<machine_name>) that is not supported by the tool to the pool of processors.

What Next

Check these possible causes and resolve:

machine not connected to the network

machine not responding to queries within predefined timeouts

machine is valid, but does not match the configuration of the master. For example,

if master is 64 bit, all slaves must be 64

if master is 32 bit, all slaves must be 32

The `rsh` UNIX command not working on your system. For example,

Should work as expected if `rsh` is setup properly.

M312

Message Text

`<directory> is not visible on machine <machine_name>.` (M312)

Severity

Error

Description

This violation is reported when the working directory is not visible on machine (`<machine_name>`).

What Next

Check to be sure that the working directory is defined so that it is visible across the master and all slave machines.

M313

Message Text

`<platform_name> is not a supported platform.` (M313)

Severity

Error

Description

This violation is reported when attempting to add a machine that belongs to an invalid (non-supported) platform.

What Next

Try another machine.

M314

Message Text

`<platform_name> is not a 64bit platform.` (M314)

Severity

Error

Description

This violation is reported when the master is a 64-bit machine while the slave is a 32-bit machine.

What Next

Use a 64-bit slave machine.

M315

Message Text

Cannot find *filename* script on machine *name*. (M315)

Severity

Error

Description

This violation is reported when the TestMAX ATPG environment is not set up properly on machine *name*, or when the script used to run TestMAX ATPG is not properly defined.

What Next

Log on the machine and try to start TestMAX ATPG. Check to be sure that you can start TestMAX ATPG using only the `tmax` command. The various environment variables such as `SYNOPSIS` and `PATH` must be properly set.

If the `set_distributed -script <scriptname>` command has been used, then make sure that this script exists and is executable from this machine. Log on the machine in question and try to execute this script; it should result in invoking TestMAX ATPG and there should not be any other messages or errors generated before TestMAX ATPG starts executing from this script. Any messages generated must be eliminated to invoke TestMAX ATPG in a distributed context.

M316

Message Text

Script *filename* is not an executable. (M316)

Severity

Error

Description

Either the script is not an executable or it does not exist.

What Next

Log on the machine and try to start TestMAX ATPG. Check that you can start TestMAX ATPG using only the `tmax` command.

Check that the environment variables such as `SYNOPSIS` and `PATH` are properly set.

If you have used the `set_distributed-script <scriptname>` command, then make sure that the script exists and is executable from this computer. Log on the computer and execute the script. It should invoke TestMAX ATPG and there should not be any other messages or errors generated before TestMAX ATPG starts executing from this script. Correct all errors and ensure that there are no error messages before you invoke TestMAX ATPG in a distributed context. TestMAX ATPG creates the exact output produced by the `rsh` UNIX command to help you determine the cause for the error. This problem is related to network access. If you have access to the computer, contact your network administrator.

M317

Message Text

```
Slave cannot open log file slave.log.*. (M317) Slave working directory is (<directory>).
```

Severity

Error

Description

This violation is reported when the working directory is not visible on machine (<machine_name>) or there are no write permissions in this directory (<directory>).

What Next

Check that the working directory and file permissions are defined so that it is visible across the master and all slave machines.

M318

Message Text

```
The current session is not in the correct mode. (M318)
```

Severity

Warning

Description

Description is TBD.

What Next

What next is TBD.

M319

Message Text

Please check your TestMAX ATPG environment on *name* machine. (M319)

Severity

Error

Description

TestMAX ATPG issues this message when the tmax script used to invoke the slaves does not match with the tmax script used to invoke the master session. This means that the two environments are different.

What Next

Execute `which tmax` in the xterm you launched the master and compare this path to the path after executing in the same xterm `rsh <slave machine> which tmax`. You must fix your environment.

Also, `which tmax` on the master will return a different path from `rsh <slave machine> -l <login> which tmax` (this runs `which tmax` on slave from the master's window). After editing your C-shell run control file, be sure to source your `.cshrc` file.

Error and Warning Messages M320 - M329

This section contains the following error messages and warnings:

- [M320](#)
- [M321](#)
- [M322](#)
- [M323](#)

- [M324](#)
- [M325](#)
- [M326](#)
- [M327](#)
- [M328](#)
- [M329](#)

M320

Message Text

Cannot save the current session. Am quitting as the session is corrupted.
(M320)

Severity

Error

Description

TestMAX ATPG encountered an error while saving the database to the work directory.

What Next

The session stopped. You have to start it again. If the same message constantly shows up for a given test case, please contact your Synopsys AE and provide the test case.

M321

Message Text

Following flags are not supported in distributed fault_sim mode. (M321)

Severity

Error

Description

Not all the options to the `run_fault_sim` command are supported. The one you tried to use is not supported.

What Next

This is an informational message only.

M322

Message Text

Patterns cannot be grouped because clocks are constrained. (M322)

Patterns cannot be grouped because clock constraints are specified.
(M322)

Patterns cannot be grouped because not all internal clocks can be turned off. (M322)

Severity

Warning

Description

When the `set_drc -clock -one_hot` command is used with the `set_atpg -group_clk_constraints { 100 0 0 }` command, ATPG must have the freedom to create clocking modes that implement the grouping. When one of these messages is printed, ATPG will not create clocking modes, but will generate ungrouped patterns.

The first M322 message reports that one or more clocks are constrained so that they cannot be separately controlled by ATPG.

The second M322 message reports that internal clocking procedures are enabled so that ATPG has no control over individual clocks.

The third M322 message reports that C37 violations exist, so that one or more clocks can't be disabled.

What Next

Remove the constraints of the reported type and re-run DRC so that the clocks can be separately and individually controlled.

M323

Message Text

Clock name is constrained to off state, ignoring for pattern grouping.
(M323)

Severity

Warning

Description

When the `set_drc -clock -one_hot` command is used with the `set_atpg -group_clk_constraints { 100 0 0 }` command, ATPG must have the freedom to create clocking modes that implement the grouping. When this message is printed, ATPG is unable to pulse the named clock, so it must be omitted from pattern grouping. All remaining clocks will be used for pattern grouping.

What Next

If this is intended, no action is required. If it is unexpected, remove the constraint and re-run DRC so that the identified clock can be included in the pattern grouping.

M324

Message Text

```
group_clk_constraints setting changed to {100 0 0} due to new pattern grouping support. (M324)
```

Severity

Warning

Description

The first implementation of the `set_atpg -group_clk_constraints { 100 0 0 }` command could result in very long ATPG runtimes. To limit the runtime overhead, incomplete pattern grouping could be specified. The new implementation reduces the runtime overhead, allowing complete pattern grouping to be applied in all cases. So whatever numbers are specified, they are converted to `{ 100 0 0 }`.

What Next

No action is required.

M325

Message Text

```
The -dmerge option is ignored. (M325)
```

Severity

Warning

Description

This message is only valid with the `run_atpg -distributed` command. This option is ignored during uniprocessor runs.

What Next

This is an informational message only.

M326

Message Text

Following flags are not supported in distributed atpg mode. (M326)

Severity

Error

Description

Only some flows of the `run_atpg` command are supported in the distributed environment. The rest is specifically not allowed.

What Next

Do not use unsupported options.

M327

Message Text

Work directory must have absolute path. (M327)

Severity

Error

Description

You specified a relative path as the working directory. This is not supported.

What Next

Change your work directory to an absolute path. Issue the `pwd` command to get the absolute path.

M328

Message Text

Please check your TestMAX ATPG environment on <machine_name> machine.
(M328)

Severity

Error

Description

TestMAX ATPG found something wrong with the environment of a distributed processor machine.

What Next

Log on that machine manually and verify that you are running the same version of TestMAX ATPG as on your master machine.

M329

Message Text

Cannot launch TestMAX ATPG shell executable (path) on *name* machine.
(M329)

Severity

Error

Description

TestMAX ATPG was unable to find a valid TestMAX ATPG executable on a distributed processor machine.

What Next

Log on that machine manually and try to start TestMAX ATPG with the `tmax` command. Verify that the TestMAX ATPG environment variables are set properly and that the `tmax` script is in your PATH.

To reproduce the problem, enter:

```
rsh -l login tmax -shell
```

If you still cannot invoke TestMAX ATPG using a `rsh` command, please contact your system administrator since you might have a network access problem.

Error and Warning Messages M330 - M339

This section contains the following error messages and warnings:

- [M330](#)
- [M331](#)
- [M332](#)
- [M333](#)
- [M334](#)
- [M335](#)
- [M336](#)
- [M337](#)
- [M338](#)
- [M339](#)

M330

Message Text

Net N in module M expanded to [I1:I2]. (M330)

Severity

Warning

Description

This violation is reported when using the `write_netlist` command to create a Verilog netlist and a fractional bus has been defined within the same module (such as BB[0:5] and BB[7:10]). The bus is written out in it's expanded range as BB[0:10], and the netlist written might be incorrect.

What Next

Carefully review the netlist created before trying to use it.

M331

Message Text

Cannot optimize net connections because some are below design level.
(M331)

Severity

Warning

Description

The message is issued if net connections are in effect, and one or more of the net connections targets a point which is below the "design-level". When this situation occurs, TestMAX ATPG cannot use its most optimal memory recovery algorithms during circuit flattening. The consequence of this is that RAM needs during `run_build_model` can peak to 2X-5X that of the post build amounts.

What Next

This is an informational message only. For a large circuit, this information can indicate that you can have a problem with memory capacity. If you are running out of memory on the current workstation and cannot access a machine with more memory, then consider removing the net connection.

Also note that the flattening algorithm is slower, perhaps 2X, when this message appears.

M332

Message Text

Parallel patterns can fail simulation because shadow cells have been identified. (M332)

Severity

Warning

Description

If TestMAX ATPG has identified during DRC that shadow cells exist in the design, and if a parallel testbench is written with a serial shift count of 0, then there is a danger of the simulation failing. This condition can also occur when specifying the `write_patterns -unified_stil_flow` option.

What Next

It is suggested that you select a serial shift count of at least 1. As an alternative, you could go back to DRC mode and disable the recognition of shadow registers using the `-noshadows` option of the `set_drc` command. Neither of these steps are necessary if your patterns pass in simulation.

M333

Message Text

`MUXClock options (D,E WFCs) will not be applied with scan compression designs (M333)`

Severity

Warning

Description

MUXClock events in the WaveformTable will not be applied in scan compression contexts. Scan compression environments are optimally structured for generic capture procedures, and the presence of generic capture procedures will override MUXClock formatting.

What Next

Consider whether these MUXClock options are required for your test environment. Remove the D and E pulse definitions to eliminate this warning.

M334

Message Text

`Incomplete set of path delay WaveformTables defined; expect all _launch_WFT_, _capture_WFT_, and _launch_capture_WFT_ (M334)`

Severity

Warning

Description

DRC checks for special WaveformTable names that are used in delay path testing. If the current DRC file contains WaveformTables named

```
"_launch_WFT_"
"_capture_WFT_"
"_launch_capture_WFT_"
```

Then this message is generated if only one or two (but not all three) of these WaveformTables are defined.

M335 messages is generated if you invoke delay path testing to define the missing WaveformTables when this message is generated.

What Next

If path delay testing is being used, consider defining all three WaveformTables to minimize risks of using the default WaveformTable to define the missing WaveformTables.

If path delay testing is not being used, consider renaming the WaveformTables to remove this warning and prevent future conflicts.

M335

Message Text

Adding "<table_name>" WaveformTable for path delay support, based on "<table_name>" (M335)

Severity

Warning

Description

When path delay testing is defined with the `add_delay_paths` command, the three path delay WaveformTables:

```
"_launch_WFT_"  
"_capture_WFT_"  
"_launch_capture_WFT_"
```

are checked. If they were not defined in the DRC file, then each missing WaveformTable is defined using the currently defined WaveformTable. The WaveformTable that is applied is determined by referencing the current WaveformTable (as specified in the `sequential_capture` procedure when present, or as specified in the capture procedure defined in the DRC File). If no capture procedures are present, then the first WaveformTable present in the DRC file is used to define the timing of the newly added WaveformTables.

WaveformTables will also be generated if the `write_drc-generic_captures` command is used AND `set_faults-model transition` was specified before the write command.

What Next

If the default WaveformTable definition is sufficient for the missing WaveformTable(s), then no action is necessary. If the missing WaveformTable needs to be defined to support unique timing necessary for path delay or transition testing, then add the definition to the DRC File by copying another WaveformTable and editing the copy to change the name and the necessary timing values. Repeat the `run_drc` command to process this additional information.

M336

Message Text

```
Incorrect event sequence in timing Waveform Table <wft-name>;  
  
PI time (<integer>) must be less than PO time (<integer>) (M336)  
  
PO time (<integer>) must be less than clock LE (<integer>) on  
<signal_name> (M336)
```

Severity

Warning

Description

The WaveformTables used for path delay testing ("_launch_WFT_", "_capture_WFT_", and "_launch_capture_WFT_") must follow certain timing restrictions. These restrictions are the same restrictions employed generally by capture operations, which are:

Primary Inputs need to be asserted first, followed by Primary Output strobes, followed by the Pulse timing for all clocks.

Since path delay test sequences are composed of Vectors applying different WaveformTables, these restrictions are applied to each of the path delay WaveformTables themselves. Events cannot be resequenced through multiple Vectors (as can be done with 2- or 3-cycle capture procedures). This error is generated when the timing defined in one of these WaveformTables violates these restrictions.

This error will not stop pattern generation; however, if the timing is not corrected before these Vectors are applied at a tester, failures can occur at this point due to improper timing.

What Next

The specified WaveformTable needs to be corrected in the STIL procedure file and the indicated timing relationship modified to correct the problem. If the specified WaveformTable does not exist in the STIL procedure file, then the timing was extracted from the default WaveformTable. In this situation, message M335 was generated to

identify what WaveformTable was used. Either the referenced WaveformTable can be corrected, or a new path delay specific WaveformTable can be created. After the STL procedure file has been modified, the `run_drc` command should be used to read this STL procedure file into the environment.

M337

Message Text

```
[Write patterns -input_spf_out | DRC -append] has unsupported STL  
procedure file for this operation; disabling option. (M337)
```

Severity

Error

Description

The `-input_spf_out` option of the `write_patterns-format <stil | stil99>` command, and the `-append` option on the `run_drc` command, pass the file used for the `run_drc` command directly into the STIL output. This feature is only supported when the STIL procedure file specified to the `run_drc` command is formatted as a STIL file and is not compressed. Binary, compressed, or other options on the format of the STIL procedure file cannot be used with these options. This message will also be generated if you fail to specify a file for the DRC run.

What Next

Either create an uncompressed STIL file for the `run_drc` command to use (and use these options), or do not use these options when writing STIL pattern output or when attempting to define a procedure with the `run_drc -append` option.

M338

Message Text

```
Bidirectional clock <clock_name> is not supported in WGL mux output.  
(M338)
```

Severity

Error

Description

Clocks defined on bidirectional signals cannot be generated properly in WGL patterns when the MUXClock option is active. This is caused by the need to define two WGL signals for mux operation and the potential need for two signals on each bidirectional

(when the `set_wgl-nogroup_bidis` option is set). This would generate four columns of data and the resolution process across all these columns is unclear.

What Next

Review whether the referenced signal needs to be declared a clock for ATPG, or whether it needs to be processed as a bidirectional in this context. Consider that ATPG clocks tend to be driven as inputs primarily, and that there are additional considerations in regards to mux operation that will further impact the behavior of this signal when muxed. Define this signal as a clock input if the MUXClock behavior is required.

M339

Message Text

Write patterns contains more than 1 pattern block, -patname option is not used. (M339)

Severity

Error

Description

The `-patname` option of the `write_patterns-format <stil | stil99>` command replaces the default STIL Pattern block name `"_pattern_"` with a name you specify. If the internal patterns contain multiple pattern blocks, then the name of each pattern block is used and this option is ignored. Note that all internally-generated data sets from TestMAX ATPG will only define one pattern block.

What Next

Review the situation that caused multiple pattern blocks to be defined, and do not use the `-patname` option in this situation.

Error and Warning Messages M340 - M349

This section contains the following error messages and warnings:

- [M340](#)
- [M341](#)
- [M342](#)
- [M343](#)
- [M344](#)

- [M346](#)
- [M347](#)
- [M348](#)
- [M349](#)

M340

Message Text

Pin *pin_name* used multiple times in clocking constraint (M340)

Severity

Error

Description

Issued when trying to use the same clock several times when defining a clocking constraint..

What Next

Correct your clocking constraint by having each clock mentioned only once in the constraint definition.

M341

Message Text

Clocking constrained pin *P* is not a clock. (M341)

Severity

Error

Description

Issued during DRC checking when a prior add clocking constraints command contains a reference to a pin that is not a recognized or defined clock. *P* indicates the name of the pin that is not a clock.

What Next

All the pins referenced in the add clocking constraint command must be defined as clocks within TestMAX ATPG. When you get this message, check to be sure you are using the intended pin name. If the pin *P* name is correct, verify that TestMAX ATPG recognize this pin as a clock.

M342

Message Text

Clocking constrained pin *pin_name* has equivalence relationships. (M342)

Severity

Error

Description

Issued when pin *pin_name* has an equivalence relationship already and you have tried to include it in a clocking constraint.

What Next

The pin *pin_name* is already part of an equivalence relationship previously defined with an `add_pi_equivalences` command in TestMAX ATPG. It cannot be included in a user-defined clocking constraint.

M343

Message Text

Clocking constrained pin *pin_name* is a constrained pin. (M343)

Severity

Error

Description

Issued when pin *pin_name* is constrained to its off state and you have tried to include it in a clocking constraint.

What Next

Since pin *pin_name* is constrained to its off state, it cannot be pulsed. Thus, it cannot be included in a user-defined clocking constraint unless the pin constraint is removed.

M344

Message Text

Selected clocking constraint is already in clocking constraint list.
(M344)

Severity

Error

Description

Issued when you try to define an already existing clocking constraint.

What Next

The clocking constraint already exists, check the validity of your pins list.

M346

Message Text

Pattern number *number* could not be generated; pattern is skipped. (M346)

Severity

Error

Description

This message indicates that the identified pattern was not generated during pattern write operations. As a result, the pattern set is missing at least one pattern and is an incomplete representation of the full test.

What Next

This situation is normally associated with a rejected pattern message, such as [M179](#), [M181](#), [M183](#), and so forth. You should review the applicable rejected pattern message to understand why this pattern was dropped.

You can replicate this condition by specifying the `run_simulation` command on the pattern set. This situation normally occurs during parallel pattern generation when the `run_simulation` command is applied to a set of patterns that are incompatible with the current state. .

To address the skipped pattern condition, you should consider changing the pattern generation process or remove the ungenerated pattern from the pattern set.

Note that the `set_patterns-external -append` command often creates situations that cause the `run_simulation` command to generate rejected pattern messages. After specifying the `set_patterns` command, you can use the `report_patterns-all -type -internal` command to observe the sequencing of the pattern types in the database.

M347

Message Text

Dynamic clocking cannot be changed in TEST command mode. (M347)

Severity

Warning

Description

Issued when you try to enable or disable the dynamic clock grouping feature while in TEST mode using a `set drc -clock` command.

What Next

Since TestMAX ATPG does extensive analysis regarding the clocks, you must select which clocking scheme you want before running DRC.

M348

Message Text

Differential PI Equivalencies can only be defined for groups of two PI's.
(M348)

Severity

Error

Description

Issued when you try to use the same pin in two differential equivalencies. For example, the following sequence of commands will result in a M348 message.

```
add pi equivalence Clock1 -differential Clock2
add pi equivalence Clock1 -differential Clock3
```

What Next

It is not possible to have a given pin involved in more than one differential equivalence. You can remove the PI equivalences.

M349

Message Text

[Autofix | Mask] behavior for rule violations cannot be changed in TEST mode. (M349)

Severity

Error

Description

You attempted to change the autofix or mask behavior of a DRC rule while in test mode. This is not supported once DRC has been completed because you would be changing the rules under which DRC was performed.

What Next

If you want to change the autofix or mask behavior, you must first return to DRC mode. After you change the settings, you must then specify the `run_drc` command to reanalyze DRCs with the new settings.

Error and Warning Messages M350 - M359

This section contains the following error messages and warnings:

- [M350](#)
- [M351](#)
- [M352](#)
- [M353](#)
- [M354](#)
- [M355](#)
- [M356](#)
- [M357](#)
- [M358](#)

M350

Message Text

Chain test cannot be created due to patterns constrained to apply a clock. (M350)

Severity

Warning

Description

The first pattern created by TestMAX ATPG (pattern #0) is intended to test the scan chains in the design. TestMAX ATPG will load values into the scan chains and then check the integrity of the chains by unloading those same values. No system clock should be pulsed during this vector to ensure that the values in the scan chains are not disturbed. If you force TestMAX ATPG to pulse a system clock in every vector (using a `set_drc -clock -one_hot` command), then this chain test cannot be created.

What Next

This is an information message.

M351

Message Text

Controller clock can only be selected in DRC command mode. (M351)

Severity

Warning

Description

Issued when you try to change the controller clock using the `set_drc-controller_clock` command in the BUILD or TEST mode. The effective mode for changing this option is DRC.

What Next

This is an information message.

M352

Message Text

Controller clock pin *pin_name* is not a clock. (M352)

Severity

Warning

Description

Pin *pin_name* must be defined as a clock to be used as the controller_clock.

What Next

Use the `report_clocks` command to verify the status of pin *pin_name*.

M353

Message Text

ATPG allowing clock-on measures cannot be performed when controller clock is selected. (M353)

Severity

Error

Description

This message is issued by `run_atpg` command checks if you have previously selected both `set_drc-controller_clock` and `set_atpg-allow_clockon_measures` options. These two settings cannot be achieved simultaneously as the `-allow_clockon_measures` option seeks to generate patterns with no active capture clocks and the `-controller_clock` option seeks to generate patterns where there is always at least one active capture clock.

What Next

You must remove one of these options before ATPG pattern generation is possible. If your design requires the `-controller_clock` option to pass DRC, then the `set_atpg-allow_clockon_measures` command must be removed. If your design can pass DRC checks without the `set_drc -controller_clock` restriction, then it is your choice as to which option to remove, and in fact you might want to consider a two-pass pattern generation approach where in the first pass you use `-controller_clock`, save faults and patterns, and then restart a second pass where only `-allow_clockon_measures` is used to target the remaining faults.

For more information, see the topic Supporting Clock ON Patterns in STIL in TestMAX ATPG and TestMAX Diagnosis User Guide.

M354

Message Text

ATPG allowing clock-on measures cannot be performed when ATPG clock is selected. (M354)

Severity

Error

Description

This message is issued by `run_atpg` command checks if you have previously selected both `set_drc -clock <pin>` and `set_atpg-allow_clockon_measures` options. These two settings cannot be achieved simultaneously as the `-allow_clockon_measures` option seeks to generate patterns with no active capture clocks and the `set drc -clock` command seeks to generate patterns where there is always at least one active capture clock.

What Next

You must remove one of these options before ATPG pattern generation is possible. If your design requires the `-clock` option to pass DRC, then the `set_atpg -allow_clockon_measures` command must be removed. If your design can pass DRC checks without the `set_drc -clock` restriction, then it is your choice as to which option to remove, and in fact you might want to consider a two-pass pattern generation approach where in the first pass you use `-clock`, save faults and patterns, and then restart a second pass where only `-allow_clockon_measures` is used to target the remaining faults.

For more information, see the topic Supporting Clock ON Patterns in STIL in the TestMAX ATPG and TestMAX Diagnosis User Guide.

M355

Message Text

Full-sequential ATPG is now enabled to support `path_delay` fault model. (M355)

Severity

Warning

Description

Issued if you select the Path delay fault model before turning on the Full-Sequential ATPG engine. Path delay test vectors require the Full-Sequential pattern generation algorithm.

What Next

This is an information message and TestMAX ATPG will turn on the relevant ATPG engine when giving this message. If you do not want to view this message, enter a `set_atpg-full_seq_atpg` command before a `set_faults-model path delay` command.

M356

Message Text

Changing capture clock in TEST mode can break scan chain operation.
(M356)

Severity

Warning

Description

During DRC, TestMAX ATPG does analysis regarding the clocks in the design. From TEST mode, if you change the clock parameters without running DRC again, then the scan chain operation (which relies on clocks) might be broken. This message is issued after a `set_drc-clock` command while in TEST mode.

What Next

This is an information message.

M357

Message Text

Clock equivalence relationships were removed to allow clock constraints.
(M357)

Severity

Warning

Description

Issued when pi equivalencies exists in the design and you have defined at least one clocking constraint. This warning message is issued during `run_drc` command.

What Next

All pi equivalencies, except the differential ones, are automatically removed when user-defined clocking is selected.

M358

Message Text

Clock equivalence relationships were removed to allow parallel clocking.
(M358)

Severity

Warning

Description

Issued when PI equivalencies exists in the design and you select the dynamic clock grouping scheme. This warning message is issued during `run_drc` command.

What Next

All PI equivalencies, except the differential ones, are automatically removed when dynamic clock grouping is enabled using the `set_drc-clock -dynamic` command. You can enable the use of pin equivalences for dynamic clock grouping by specifying the `set_drc -dynamic_clock_equivalencing` command.

In order to eliminate the ungrouping of equivalent clocks with the same polarity, when using the `set_drc -dynamic_clock_equivalencing` command, you also need to specify the `add_pi_equivalences -same_polarity` command. The `-same_polarity` option works the same as the `-differential` option, except that the two clocks must have the same off-state. Additionally, you might need to specify the `set_drc -dynamic_clock_equivalencingnodisturb` command.

For more information, see the topic `add_pi_equivalences` in the TestMAX ATPG and TestMAX Diagnosis Commands guide.

Error and Warning Messages M360 - M369

This section contains the following error messages and warnings:

- [M361](#)
- [M362](#)
- [M363](#)
- [M364](#)
- [M365](#)
- [M366](#)

- [M367](#)
- [M368](#)
- [M369](#)

M361

Message Text

Obsolete option `-exclude chain_test.` (M361)

Severity

Warning

Description

The `chain_test` selection of the `-exclude` option of the `write_patterns` command is obsolete and is being ignored.

Beginning with the 2001.08 release of TestMAX ATPG, chain test generation has changed from an operation performed while writing patterns, to an operation during ATPG pattern generation. By default, ATPG now creates a chain test as pattern 0 in the internal pattern set. The chain test is no longer a separate pattern added during the write pattern operation.

There are two important considerations of this change in chain test generation to consider:

With the previous chain test method it was possible to write patterns after entering TEST mode and before generation of any ATPG patterns. These patterns contained only the `test_setup` cycles and the chain test. Now that chain test creation is part of ATPG it is necessary to perform a `run_atpg` command and generate at least one pattern before a chain test exists. Creating a pattern file that contains only a chain test is now done with the command:

```
write_patterns filename -last 0
```

It was previously possible to use the `-first` and `-last` options to write a range of patterns, and that block would also include a chain test. A chain test will no longer appear when a range of patterns is written, unless that range includes pattern 0 and chain test creation has not been disabled.

What Next

Remove the `-exclude chain` option from your command sequence.

If you do not wish to include the chain test in your pattern output, use the `-first 1` option of the `write_patterns` command to begin writing patterns beginning at pattern 1. This excludes the chain test from the output written.

If you are already using the `-first` option, no addition changes are needed other than removal of the `-exclude chain` option.

If you wish to disable the creation of a chain test as pattern 0, or change the pattern used for the chain test, use the `-chain_test` option of the `set_atpg` command before any pattern generation.

If you are using both `-split` and `-exclude chain` options, then substitute `-first 1` for the `-exclude chain` option.

M362

Message Text

Sequential simulation for selected fault cannot be performed when internal patterns and faults exist. (M362)

Severity

Error

Description

The `run_simulation-sequential` command was used with the optional target fault path to simulate the effects of a stuck-at fault. This is supported only when the pattern source is external and there are no internal patterns or faults.

What Next

If you have a fault population, delete it and try again.

If you have only an internal pattern source, you can overcome this limitation by writing the patterns out, deleting internal patterns, then using the `set_patternsexternal` command to read the patterns back in.

If you have both internal and external pattern sources, you must delete the patterns, then reread the external patterns again before retrying this command.

For more information, see the topic `remove_faults` in the TestMAX™ ATPG and TestMAX Diagnosis Commands guide.

M363

Message Text

Controller clock can only be selected in DRC command mode. (M363)

Severity

Error

Description

This message is issued when you try to do a `set_drc -controller_clock` command and you are TEST mode.

What Next

The `set_drc -controller_clock` command affects the DRC analysis and needs to be set when you are in DRC mode. If you are TEST mode, go back to DRC mode (using `set drc -force`), set the `-controller_clock` option, and rerun DRC.

M364

Message Text

DRC clocking was selected to `seq_capture` without a `sequential_capture` procedure. (M364)

Severity

Error

Description

You get this message when you use the `set_drc-clock -seq_capture` and `run_drc` commands but do not define a Full-Sequential capture procedure in your STIL procedure file.

What Next

If you want to use a sequential capture procedure, you need to add one to your STIL procedure file. See the topic Sequential Capture Procedure in STIL in the TestMAX ATPG and TestMAX Diagnosis User Guide for understanding and creating a sequential capture procedure.

If you do not want to use a sequential capture procedure, use the `set drc -clock` any command and rerun DRC.

M365

Message Text

Unused message. (M365)

Severity

Error

Description

Unused message.

What Next

Unused message.

M366

Message Text

Unused message. (M366)

Severity

Error

Description

Unused message.

What Next

Unused message.

M367

Message Text

Viewing level can not set to design level when read image has been used.
(M367)

Severity

Warning

Description

When you read a design image into TestMAX ATPG (with the `read_image` command), you bring up the schematic in GSV in Primitive view only. This message is issued when you try to change the GSV to Design view.

What Next

This is an information message only.

M368

Message Text

`Observe file was not defined. (M368)`

Severity

Error

Description

The `-observe_file` option of the `run_observe_analysis` command was used without specifying an observe file.

What Next

Retry the command, specifying an output file for the `-observe_file` option.

M369

Message Text

`Insufficient buffer area for clock group tracing of clock C (G). (M369)`

Severity

Error

Description

This message is given when the buffer area that stores data for all the potential clock combinations is exceeded. This buffer area is very large and depends on the number of gates so that a change in circuit will have little affect on the situation.

What Next

The clock grouping analysis will continue with some truncation. If complete grouping is desired for this circuit, some clocks will have to be removed.

Error and Warning Messages M370 - M379

This section contains the following error messages and warnings:

- [M370](#)
- [M371](#)
- [M375](#)
- [M376](#)
- [M377](#)
- [M378](#)
- [M379](#)

M370

Message Text

Top-level module *module_name* is celldefine'd (M370)

Severity

Warning

Description

Specifying `read_netlist-library` on every netlist creates a problem, because it results in faults being placed only on the pins of the top-level design. This warning is issued when the top-level instantiates (directly or indirectly) at least 1000 modules (which avoids generating this warning when building the model of a library cell).

What Next

Reread at least the netlist file containing the top-level module without using the `-libraryoption`, then run `build` again.

M371

Message Text

Fault checkpointing can only be performed during sequential simulation.
(M371)

Severity

Warning

Description

The `-checkpoint` option of the `run_fault_sim` command was specified without also selecting the `-sequential` option. The `-checkpoint` option is supported only when the `-sequential` option is also used.

What Next

Repeat the command without the `-checkpoint` option or with the `-sequential` option.

M375

Message Text

Too many <input pins|clocks> required mapping - pattern mapping terminated. (M375)

Severity

Error

Description

Description is TBD.

What Next

What next is TBD.

M376

Message Text

Unused message. (M376)

Severity

Error

Description

Unused message.

What Next

Unused message.

M377

Message Text

Read image was not successful. (M377)

Severity

Error

Description

TestMAX ATPG displays this error message because:

The `read_image` command fails to read the image into memory, or

If the image is a secure image file, either you did not enter a password or entered an incorrect password (in this case, TestMAX ATPG generates this M377 message along with the M523 or M524 message).

What Next

The `read_image` command fails when you specify the wrong image file, a corrupted image file, or if your tool does not have proper licenses. If the image is a secure image file, check with the image file creator and enter the correct password.

TestMAX ATPG usually displays another error message before this M377 message that explains the why the error occurred. Correct the problem reported by the earlier message and try the `read_image` command again. If TestMAX ATPG does not display an error message preceding the M377 message, try writing out the image again using the `write_image` command. Ensure that the file size does not change when you retry the `read_image` command.

If the image has been zipped multiple times, unzip the image and try again.

Examples

```
BUILD> read_image testcase.img
Warning: All netlist and library module data are now deleted.
(M41)Error: Test-IDDQ license could not be acquired. (M12)Program error:
setting commands could not be read (12).Warning: Current simulation
model is now deleted. (M39)Warning: All netlist and library module data
are now deleted. (M41)
```

```
Error: Read image was not successful. (M377)
BUILD> read_image jpk.img
Warning: All netlist and library module data are now deleted. (M41)
Error: Unable to open read file "jpk.img". (M5)
Warning: All netlist and library module data are now deleted. (M41)
```

```
Error: Read image was not successful. (M377)
```

M378

Message Text

Build command mode cannot be entered when image was read. (M378)

Severity

Warning

Description

You cannot go back to the BUILD mode when you read an image saved by the `write_image` command.

What Next

If you want to go to the BUILD mode and make changes, you have to start from the original design files used to create the image in the first place.

M379

Message Text

conversion overflow, value > 64-bit integer range was truncated,
val='V' (M379).

Severity

Error

Description

While reading Verilog VCD or Extended VCD format functional patterns, an event time was encountered that either exceeded the range of a 64-bit unsigned integer, or when scaled due to the choices of `-strobe period` and `-strobe offset` exceeded this range.

`v` is the value from the input file which was being processed when the range was exceeded.

The maximum value supported is: 18,446,744,073,709,551,615

What Next

Review the selection of period and offset used with the `set_patternsexternal` command as well as the `$timescale` block found in the header of the input file.

Try to select numbers and units which reduce the need for scaling. For example, if the units in the VCDE input file are '10ps' then specify the strobe and offset as multiples of 10 and the units as 'ps'.

If the units in the input file are large, such as 10ns or 100ns, you might want to change this value down to a '1' in whatever time unit is used. This will provide a little scaling range room if number scaling is required. Note that in doing this, the internal time numbers for the patterns will not match the original. The event order is the same but the times is reduced by 100:1 or 10:1, depending on the change made. Since TestMAX ATPG simulates in a zero-delay mode, the times won't really matter. It is the order of events that is important!

Error and Warning Messages M380 - M389

This section contains the following error messages and warnings:

- [M380](#)
- [M381](#)
- [M383](#)
- [M384](#)
- [M385](#)
- [M387](#)
- [M388](#)
- [M389](#)

M380

Message Text

There was no debug data collected for selected pattern (N). (M380)

Severity

Error

Description

Description is TBD.

What Next

What next is TBD.

M381

Message Text

Fast sequential simulation failed to stabilize after 16 passes. (M381)

Severity

Warning

Description

When a design has certain `Snn` or `Cnn` rule violations that would otherwise result in bad patterns being produced by the Basic-Scan ATPG algorithm then the Fast-Sequential ATPG algorithm is used to double check those risky patterns.. When the two algorithms produce a different answer, the pattern is discarded and ATPG continues to find a pattern to detect the target fault.

During this resimulation of Basic-Scan patterns, the good machine logic simulator applies an event, and then propagates that logic change around the gates of the design using zero delay simulation. It allows 8 iterations for the logic values to stabilize. This should be plenty of iterations in a zero delay simulation.

If after 8 iterations the logic is not stable (oscillating), the algorithm changes the value it is propagating to an X and goes another 8 iterations to see if using an X brings about a stable value. Normally this works and then further analysis proceeds with the logic value's at X. Of course, an X means that ATPG pattern generation can't do much and fault simulation can't do much and so test coverage might be reduced, but the rest of the circuit is still up and running and a bad pattern is avoided.

If 16 iterations are completed and there is still no stability, then the algorithm failed to get all the affected logic nodes to an X value! Some of those 0/1's left are going to cause wrong answers somewhere, including bad patterns.

This message is also an indication that the use of fault simulation on functional patterns will give an incorrect answer, but this issue is harder to detect.

What Next

Use the `run_simulation` command after ATPG is completed to check for bad patterns.

One possible cause of this situation is a sensitized feedback loop in the design. Check for X1 violations and review feedback loops (B23 violations) using the `report_feedback_pathsreport` command. Review the causes and 'What Next' advice on the X1 and B23 help pages.

Consider altering the design to eliminate or reduce 'S' and 'C' violations. Concentrate first on any violations displayed in the banner when ATPG is first started.

M383

Message Text

Fast sequential simulation required N passes to stabilize. (M383)

Severity

Warning

Description

When a design has certain Snn or Cnn rule violations that would otherwise result in bad patterns being produced by the Basic-Scan ATPG algorithm, then the Fast-Sequential ATPG algorithm is used to double check those risky patterns. When the two algorithms produce a different answer, the pattern is discarded and ATPG continues to find a pattern to detect the target fault.

During this resimulation of Basic-Scan patterns, the good machine logic simulator applies an event, and then propagates that logic change around the gates of the design using 0 delay simulation. It allows nine iterations for the logic values to stabilize. This should be plenty of iterations in a 0 delay simulation.

If after nine iterations, the logic is not stable (oscillating), the algorithm changes the value it is propagating to an X and goes another nine iterations to see if using an X brings about a stable value. Normally this works and then further analysis proceeds with the logic value's at X. Of course an X means that ATPG pattern generation cannot do much and fault simulation cannot do much, so test coverage might be reduced, but the rest of the circuit is still up and running, and a bad pattern is avoided.

If 100 iterations are completed and there is still no stability, then the algorithm failed to get all the affected logic nodes to an X value. Some of those 0/1's left behind are going to cause wrong answers somewhere, including bad patterns.

This message is also an indication that the use of fault simulation on functional patterns will give an incorrect answer, but this issue is harder to detect.

What Next

Use the `run_simulation` command after ATPG is completed to check for bad patterns.

One possible cause of this situation is a sensitized feedback loop in the design. Check for X1 violations and review feedback loops (B23 violations) using the `report_feedback_paths` command. Review the causes and "What Next" advice on the X1 and B23 help pages.

Another cause of logic loops not settling is due to open latch gates in the design during that pattern. Likewise, Xs on flop (scan or nonscan) data and clock pins might cause a virtual oscillation as the X value will pass through the flop at that point.

The final simulated values might be different from the values anticipated by the test generator, so the pattern might not detect the intended faults and thus be dropped. Also look for an "unsuccessfully detected faults" message (expert-level only) at the end of ATPG.

Consider altering the design to eliminate or reduce 'S' and 'C' violations. Concentrate first on any violations displayed in the banner when ATPG is first started.

M384

Message Text

Clock analysis aborted due to input %d of %s (%d) set to %c when clocks are off. (M384)

Severity

Warning

Description

This message comes from the use of an undocumented command.

What Next

This message is informational only.

M385

Message Text

Pattern writing with common chain inputs not supported for selected format. (M385)

Severity

Error

Description

TestMAX ATPG generates this message when a design is using Shared ScanIn. The scan_in pin name is shared among scan groups.

```
ScanStructures { ScanChain c1 { ScanIn "si1"; ScanOut "so1"; } ScanChain
c2 { ScanIn "si1"; ScanOut "so2"; }

                ScanChain c3 { ScanIn "si1"; ScanOut "so3";
}
```

```
}
```

What Next

The STIL and WGL formats are the only formats supported for Shared ScanIn.

For more information, see the topic `write_patterns` in the TestMAX™ ATPG and TestMAX Diagnosis Commands guide.

M387

Message Text

```
Slave is running a different version of TestMAX ATPG. (M387)
```

Severity

Error

Description

TestMAX ATPG displays this message when the `tmax` version on the master machine and the slave machines are not the same. Check that the `tmax` executables (`tmax32` and/or `tmax64`) are from the same installation.

What Next

Check the way in which you installed TestMAX ATPG. Enter `which tmax` in the xterm where you launched the master and compare this path to the path after, in the same xterm, executing `rsh <slave machine> which tmax`. You must fix your environment.

Also, `which tmax` on the master returns a different path from `rsh <slave machine> -l <login> which tmax` (runs `which tmax` on the slave from the master's window).

After editing your c-shell run control file, be sure to source your `.cshrc` file.

M388

Message Text

```
Cannot launch rsh. (M388)
```

Severity

Error

Description

TestMAX ATPG issues this message when the capability to use the `rsh` command has been turned off on your machine.

What Next

TestMAX ATPG relies on the `rsh` to start processes on distributed processor machines. If `rsh` is not in your path, add it to your path. Use the `which rsh` command to find where it is located on your network. You need to contact your system administrator to solve this problem.

To reproduce the problem, enter:

```
rsh -l login tmax -shell
```

M389

Message Text

```
remsh timed out. (M389)
```

Severity

Error

Description

TestMAX ATPG issues this message when the `rsh` command timed out.

What Next

For distributed ATPG, the main process issues a `rsh` command to all processors specified by the `add_distributed_processors` command. The timeout for the `rsh` command is controlled by the `-rsh` option of the `set_distributed` command. Try passing an higher timeout value using the `-rsh` option. The default 6 seconds and you can specify any value between 6 and 500 (inclusive).

Error and Warning Messages M390 - M399

This section contains the following error messages and warnings:

- [M391](#)
- [M392](#)
- [M393](#)
- [M394](#)

- [M395](#)
- [M396](#)
- [M397](#)
- [M398](#)
- [M399](#)

M391

Message Text

A session with master on <processor_name> and slave on <processor_name> is not supported. (M391)

Severity

Error

Description

This combination of master and distributed processors is not supported.

What Next

Check the platform type. There must be a discrepancy between the master and the distributed processor architecture. The master is the golden environment.

M392

Message Text

Cannot generated when paths with severe P rule violations exist. (M392)

Severity

Error

Description

You get this message when you lower the severity of path delay (P) rules whose default severity is Error and run distributed Full-Sequential ATPG.

What Next

This currently a limitation of running distributed ATPG with the path delay fault model. You cannot have paths that violate P rules with default severity of Error.

Remove any paths from your path file that violate any of the following P rules: P1, P3, P5, P6, P7, P9, P10, P16, P18, P25

You can use the `report_violations` command to get the list of paths that violate the above rules. These rules are most likely the ones you are violating. After you delete these paths, use the `remove_delay_paths-all` command to remove the existing paths from memory and re-read the new paths file.

For more information, see the topic `add_delay_paths` in the TestMAX™ ATPG and TestMAX Diagnosis Commands guide.

M393

Message Text

Cannot have more than N slaves. (M393)

Severity

Error

Description

TestMAX ATPG displays the M393 error message when you use more than 30 slave processes with the `add_distributed_processors` command. There is a limitation that you cannot use more than 30 slave processes with this command.

What Next

This is a current program limitation.

M394

Message Text

Removing all entries from the processor list. (M394)

Severity

Warning

Description

This message appears when either the `-lsf`, `-grd`, or `-generic` option to the `add_distributed_processors` command is used and the processor list is not empty. You cannot mix processors from a load sharing application (like LSF and GRID) with normal processors.

You can also get this message if you redefine the work directory after you have run an `add_distributed_processors` command.

What Next

No action is required. The message is informational that the previous list of defined processors is ignored. To avoid this message in the future, use the `remove_distributed_processors` command. You can also get the current list of processors with by issuing a `report_distributed_processors` command.

M395

Message Text

User defined script must have an absolute path. (M395)

Severity

Error

Description

This message is issued when a relative path was specified with distributed ATPG.

What Next

Provide the filename with the complete path.

M396

Message Text

`file_name` is not a file. (M396)

Severity

Error

Description

This message is issued when the user-defined script specified does not exist.

What Next

Check your path and try again.

M397

Message Text

Ignoring previous definition of shell script %s ... (M397)

Severity

Warning

Description

This message is issued when you use the `set_distributed-work_dir` command more than once during the same session.

What Next

This is an information message only.

M398

Message Text

Unused message. (M398)

Severity

Error

Description

Unused message.

What Next

Unused message.

M399

Message Text

<command|option> is no longer a valid command/option. (M399)

Severity

Error

Description

For additional flexibility and to accommodate enhancements, starting with the V-2003.12 release, some of the commands/options for running Distributed ATPG have changed.

You get this message if you use the commands/options that were valid in previous version of the tool but are no longer valid.

What Next

The changed commands and options are:

Before V-2003.12	After V-2003.12

set_work_directory	
<work_dir>	set_distributed -work_dir
-script	set_distributed -script
set_distributed -rsh	set_distributed -shell_timeout
add_distributed_processors -lsf <str> \	add_distributed_processors
-lsf <str> \	
-queue <str> -nslaves d -options "-q <str>" -nslaves d	

For more information, see the following topics in the TestMAX™ ATPG and TestMAX Diagnosis Commands Guide:

set_distributed

add_distributed_processors

6

Messages M400 - M499

This section contains the following error messages and warnings:

- [Error and Warning Messages M400 - M409](#)
- [Error and Warning Messages M410 - M419](#)
- [Error and Warning Messages M420 - M429](#)
- [Error and Warning Messages M430 - M439](#)
- [Error and Warning Messages M440 - M449](#)
- [Error and Warning Messages M450 - M459](#)
- [Error and Warning Messages M460 - M469](#)
- [Error and Warning Messages M470 - M479](#)
- [Error and Warning Messages M480 - M489](#)
- [Error and Warning Messages M490 - M499](#)

Error and Warning Messages M400 - M409

This section contains the following error messages and warnings:

- [M400](#)
- [M401](#)
- [M402](#)
- [M404](#)
- [M405](#)
- [M406](#)
- [M407](#)

- [M408](#)
- [M409](#)

M400

Message Text

Cannot use -grd option without first defining user script
(set_distributed -script ...). (400)

Severity

Error

Description

This message is given when you tried to launch slaves with GRID without defining a launch script with the -script option of the set_distributed command.

What Next

Create a launch script and identify it with a set_distributed -script command.

For more information, see the topic `add_distributed_processors` in the TestMAX™ ATPG and TestMAX Diagnosis Commands Guide.

M401

Message Text

STIL parallel [compressor] patterns are only usable with MAX Testbench simulations (M401)

Severity

Warning

Description

This warning message appears when TestMAX ATPG writes parallel BIST patterns using the -format stil or -format stil99 options of the write_patterns command. Because parallel BIST patterns apply data to internal design cells only, you can use this pattern format only with MAX Testbench simulations.

What Next

Use these STIL patterns only with MAX Testbench simulations.

M402

Message Text

Integer values on `-parallel` are no longer supported. Use `+define +tmax_parallel=n` to override simulation serial-vector settings for parallel simulations. (M402)

Severity

Warning

Description

You have specified an integer value with the `-parallel` option of the `write_patterns` command. This syntax is not recognized.

What Next

To remove this warning, specify only the `-parallel` option.

M404

Message Text

N remodeled latches might fail MAX TestBench parallel simulation; use `-cellnames type` or `set_drc -nodslave_remodel -noreclassify_invalid_dslaves`. (M404)

Severity

Warning

Description

If remodeled latches exist in the design, and you write out STIL99 patterns for parallel simulation in MAX Testbench, then the Verilog simulation might report simulation failures around these remodeled elements due to a limitation of STIL information.

The parallel simulation is handled correctly for STIL (not `-format stil99`) pattern files when the `write_patterns -cellnames type` command is used. If the `-cellnames type` option is not used, or if you write STIL99 patterns, then the `set_drc-nodslave_remodel -noreclassify_invalid_dslaves` command is applied before the `run_drc` command to eliminate the parallel simulation issue in this context. Note that the second option requires that you rerun ATPG to generate patterns that satisfy this context.

This message is also generated after the `write_patterns-format stil99 -parallel` command (if remodeled latches are present), or when the `-usf` option is used. Although

this message is a warning, using these patterns in MAX TestBench in parallel mode will likely generate simulation mismatches.

What Next

Consider using the `-format stil` option to write out patterns for parallel simulation validation even if you use the `-format stil99` option at test. The information in both sets of patterns is the same, barring changes required to address syntax differences in the two formats. Note that patterns created using the `-format stil` option will simulate properly in parallel mode.

If you are going to write out STIL99 patterns and you plan to run MAX TestBench in parallel mode with these patterns, use the `set_drc-nodslave_remodel -noreclassify_invalid_dslaves` command to validate these patterns properly in parallel mode. Otherwise, you may ignore this warning message.

The patterns generated with the `set_drc-dslave_remodel -reclassify_invalid_dslaves` command are always correct for serial test operation and serial MAX Testbench simulation, and parallel simulation using STIL (not `-format stil99`) patterns.

If parallel simulation failures are reported, you can try specifying the `-cellnames type` option of the `write_patterns` command for STIL (but not STIL99) parallel files. Note that this format supports the `set_drc -dslave_remodel` command context for parallel simulation.

M405

Message Text

Update commands can be used only after successful Read_DRC. (M405)

Severity

Error

Description

Update commands (`update_clock` and `update_wft`) will operate only in DRC mode, after a `read_drc` command has successfully read an STIL procedure file to be updated.

What Next

Use the `read_drc` command before using an update command.

A specific sequence of commands is required to modify the timing data in an existing STIL procedure file.

M406

Message Text

Update timing on WaveformTable with ns timing but updates in ps. Lost resolution. (M406)

Severity

Error

Description

The change to the timing was specified in ps resolution, but the changed WaveformTable was defined in ns resolution. The requested value will attempt to be mapped onto this scale, but it will lose resolution; if the requested value is too small, the timing will not be applied.

Update commands (`update_clock` and `update_wft`) are intended to modify existing information, not generate a full rewrite of the timing information.

What Next

Re-write the original STIL procedure file using ps scaling, and then re-run this operation with the WaveformTable appropriately scaled.

M407

Message Text

P waveform on <clock> has *number* events; only the first two events changed. (M407)

Severity

Error

Description

The `update_clock` command supports the modification of single-pulsed clock definitions only. If the waveform has too many events present, then the `update_clock` command will affect only the first two events in this waveform. Be aware that the first two events might not be representative of the pulse itself.

What Next

Review the waveform that was requested to be changed, and consider directly modifying the pulse definitions when there are too many events present.

M408

Message Text

Update scale on WaveformTable with ps timing. Requested scale is ns. Lost resolution. (M408)

Severity

Warning

Description

The change to the timing scale was specified in ns resolution, but the changed WaveformTable was defined with ps resolution; as a result, some time values lost precision with this change (which means there were significant figures specified in the ps range).

What Next

Use the `update_wft` and `update_clock` commands to define appropriate ns values for the affected signals, if necessary.

M409

Message Text

Error: The Test-Power license cannot be checked in when the power budget is not set to zero. (M409)

Severity

Informational Message

Description

This message is related to the Power Aware ATPG feature. It is issued any time the `remove_licenses test-power` command is specified and the power budget specified by the `set_atpg -power_budget` command is not set to 0.

What Next

Before checking in the Test-Power license, specify the following command: `set_atpg -power_budget 0`.

Error and Warning Messages M410 - M419

This section contains the following error messages and warnings:

- [M410](#)
- [M412](#)
- [M413](#)
- [M414](#)
- [M415](#)

M410

Message Text

Serial shift value `shift_value` should be less than the maximum scan chain length value `max_length`; default load mode is set to serial. (M410)

Severity

Warning

Description

This message is displayed when the `write_patterns` command attempts to generate a parallel testbench with a serial shift count larger or equal than the maximum scan chain length. The tool will continue the generation of the testbench in serial load mode.

What Next

You should specify a serial shift count less than the maximum scan chain length value.

You can run parallel load mode simulation using MAX Testbench by specifying the `-tmax_parallel` option in the `vcs` command. For more information, see the *Test Pattern Validation User Guide*.

M412

Message Text

The Test-Accelerate-Max license cannot be checked in when distributed processors are selected. (M412)

Severity

Error

Description

This message displays when you have distributed processors added with the `add_distributed_processors` command and you try to remove the Test-Accelerate-Max license with the `remove_licenses` command.

What Next

If you would like to check in the Test-Accelerate-Max license, you can run the `remove_distributed_processors` command and then run the `remove_licenses` command. However, if some other process checks out the Test-Accelerate-Max license after you check it in, it will not be available for you to run distributed ATPG.

M413

Message Text

Full sequential ATPG flow is not supported for the `-parallel_strobe_data_file` option, disabling the generation of the parallel strobe data file.

Disabling option `-parallel_strobe_data` for multi-thread ATPG as it is not supported.

Use `run_sim PSD flow instead(M413)`.

Severity

Warning

Description

This message appears when you attempt to use the `-parallel_strobe_data_file` option of the `set_atpg` command in Full-Sequential mode.

What Next

If using full sequential ATPG switch to Basic Scan or Fast-Sequential mode, rerun the `run_atpg` command.

If using multi-threaded ATPG, refer to the TestMAX ATPG and TestMAX Diagnosis Documents > [TestMAX ATPG and TestMAX Diagnosis User Guide, version R-2020.09-SP4](#) > [Debugging Parallel Simulation Failures Using Combined Pattern Validation](#) > Using the `run_simulation` command to create a PSD file.

M414

Message Text

There were not enough ATPG licenses available. (M414)

Severity

Error

Description

This message appears when Multicore or Distributed ATPG cannot get enough additional licenses to run certain features.

What Next

See "Licensing" for information on licenses required for running Multicore or Distributed ATPG.

M415

Message Text

No pins could be found that were connected to input N of selected gate G.
(M415)

Severity

Error

Description

Description is TBD.

What Next

What next is TBD.

Error and Warning Messages M420 - M429

This section contains the following error messages and warnings:

- [M420](#)
- [M421](#)
- [M422](#)
- [M423](#)

- [M424](#)
- [M425](#)
- [M426](#)
- [M427](#)
- [M428](#)

M420

Message Text

There are no patterns in selected pattern set. (M420)

Severity

Error

Description

The selected pattern set is empty and has no patterns.

What Next

If the pattern source is incorrect, set the pattern source to the intended patterns with the `set_patterns` command and rerun the simulation.

M421

Message Text

Loop in pattern P ignored due to invalid <type> event (more info). (M421)

Severity

Error

Description

Description is TBD.

What Next

What next is TBD.

M422

Message Text

Loop in pattern P ignored due to missing END_LOOP statement. (M422)

Severity

Error

Description

Description is TBD.

What Next

What next is TBD.

M423

Message Text

Signal *signal_name* defined as `set_build -inout_force` was not an input for *module_name*. (M423)

Severity

Error

Description

The signal name identified is in the `run_build_model -inout_force` command, but is not an input in the port list of the top-level module. As a result, the signal was not changed to an inout signal. Signals that are already inout signals will also result in this message. If the reference name is one bit of a vector (for example, "`t[1]`") and the signal is declared to be multiple bits (for example, "`input [1:0] t;`"), you must modify all bits of the vector.

What Next

You should review all `run_build_model -inout_force` commands for misspellings or unnecessary use. Also, remove brackets on vectored signal references; all bits in a vector must be modified from one reference to the name (without brackets).

M424

Message Text

Excessive number of inputs for gate `id=%d`, `type=%s`, `nfi=%d`, `nfo=%d`, `nfio=%d`, `imodule=%s` (M424)

Severity

Error

Description

An attempt was made to create a model containing the identified gate with more than 65,535 inputs. This model cannot be created.

What Next

The source of the problematic gate (netlist, net connections, or build options that modify the model) must be changed. For more information, see the topic Building the ATPG Model in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M425

Message Text

```
Excessive number of outputs for gate id=%d, type=%s, nfi=%d, nfo=%d,  
nfio=%d, imodule=%s (M425)
```

Severity

Error

Description

An attempt was made to create a model containing the identified gate with more than 65,535 outputs. The model cannot be created.

What Next

The source of the problematic gate (netlist, net connections, or build options that modify the model) must be changed. For more information, see the topic Building the ATPG Model in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M426

Message Text

```
Unrecognized external pattern file contents (M426)
```

Severity

Error

Description

The contents of the external pattern file specified in the command `set_patterns-external` could not be identified as STIL, WGL, VCDE, TestMAX ATPG binary, Verilog or VHDL syntax.

What Next

Review the contents of the pattern file you used. Look for previous error messages, if present (such as "File not found"). If there is a large comment section before the STIL version statement (for STIL files), or the waveform statement (for WGL files), or the first module statement (for Verilog files), remove that comment. Remember that TestMAX ATPG can only read a subset of constructs in ASCII files which match the constructs that TestMAX ATPG will use when it generates these files.

M427

Message Text

Option `nodrop_faults` is supported only for the sequential fault simulator. (M427)

Severity

Error

Description

The `-nodrop_faults` option of the `run_fault_sim` command is only supported for standalone Full-Sequential. This option suspends removing detected faults from the fault list during fault simulation. In this way, the number of faults detected by each pattern can be reported using a `run_fault_sim -seq -nod` command.

When this option is activated, the `run_fault_sim -seq -nod` command additionally reports the total number of faults detected by each test pattern.

What Next

What next is TBD.

M428

Message Text

The last pattern must be greater than the first pattern. (M428)

Severity

Error

Description

The `-first` and `-last` option of the `write_patterns` command were used and the pattern number provided with the `-last` option was less than the pattern number used for the `-first` option.

What Next

Review your selected pattern range for `-first` and `-last` options and try again.

Error and Warning Messages M430 - M439

This section contains the following error messages and warnings:

- [M430](#)
- [M431](#)
- [M432](#)
- [M433](#)
- [M434](#)
- [M435](#)
- [M436](#)
- [M437](#)
- [M438](#)
- [M439](#)

M430

Message Text

Chain test pattern failed contention checking (ID=G). (M430)

Severity

Warning

Description

This message is issued when BUS contention cannot be avoided with the current pattern used for the chain test. For example, two adjacent cells might have to be "01" to avoid an internal bus contention issue, but the chain test calls for them to be "00", causing a conflict. `G` is the Gate ID where contention is occurring.

What Next

This is a warning message and the chain test will continue to be generated. However, you should review contention problem resulting from the chain test and determine how serious the problem is to you. TestMAX ATPG would normally avoid contention, but cannot do so and still achieve the fixed pattern currently selected for the chain test.

It might be possible to change the pattern used for the chain test or disable it completely (see the `-chain_test` option of the `set_atpg` command) and avoid the contention problem.

If you continue, you can exclude the chain test from your final patterns using the `-first 1` option of the `write_patterns` command.

You might find it helpful to view the gate ID mentioned (`G`) in the schematic viewer and select Pattern 0 to gain an idea of the contention source.

M431

Message Text

Chain test pattern failed constraint checking (ID=G). (M431)

Severity

Warning

Description

This message is issued when ATPG constraints cannot be met with the current pattern used for the chain test. For example, you might have constrained a scan cell to always load a 1, but this might be in conflict with a chain test pattern that calls for the same scan cell to be a 0. `G` is the Gate ID where constraints cannot be met.

What Next

This is a warning message and the chain test will continue to be generated. However, you should review the chain test pattern relative to your list of ATPG and cell constraints to see if it will cause unacceptable problems to have those constraints violated.

It might be possible to change the pattern used for the chain test or disable it completely (see the `-chain_test` option of the `set_atpg` command) and avoid the problem.

If you continue, you can exclude the chain test from your final patterns using the `-first 1` option of the `write_patterns` command.

You might find it helpful to view the gate ID mentioned (`G`) in the schematic viewer and select Pattern 0 or Constraint data to gain an idea of the conflict.

M432

Message Text

Chain test pattern failed float contention checking (ID=1260). (M432)

Severity

Warning

Description

This message is issued when chain test pattern results in floating condition on a BUS.

What Next

You can ignore this message if float condition is acceptable in your design, but the float conditions needs to be fixed if you plan to use IDDQ testing. To fix the problem, analyze the Z2 violations in the DRC report and fix them.

M433

Message Text

Inputs changed simultaneously with clock trailing edge at T=<time>.
(M433)

Severity

Warning

Description

While reading functional patterns in Extended VCD (VCDE) format with clock recovery enabled, inputs were found to be changing at the same time that a clock was going off (trailing edge). The time reported is the event time as found in the VCDE file where the simultaneous events occur.

The clock recovery algorithm prefers that clock events be completely separated from any other changes. This is a warning that the clock recovery will separate the events and you will see a clock pulse in the patterns followed by a force of primary inputs (PI's). Depending upon the design, this might be safe to do or it might result in bad patterns.

What Next

If your purpose for reading in patterns is to perform fault grading, then you might be able to ignore this warning. Perform a `run_simulation-sequential` command to see if any mismatches are reported. If there are no mismatches, then you are probably safe to accept the patterns as-is and continue with fault simulation.

If you wish to try to eliminate the warning, you can either change the pattern source so that the VCDE does not change inputs coincident with clock changes, or you might wish to disable clock recovery when reading in the patterns. This can be done by identifying which clock was involved and then dropping that port from the defined `vcd_clock` list when the patterns are read.

M434

Message Text

`N cells changed state during pipeline shift N. (M434)`

Severity

Warning

Description

This message is associated with a special analysis that is activated when the `-pipeline` option of the `set_drc` command is used.

When the `-pipeline` option is specified before DRC, the simulation of the load/unload procedure will include simulation of a sufficient number of applications of the shift procedure; scanin ports are set to X to stabilize any X propagation. If there are any cells which change to X for a single shift simulation, the `M434` message indicates the number of cells which were set to X for that shift application.

The purpose of this simulation is to ensure that all scan cells (including pipeline cells inside scan chains) are set to X before simulation of the load/unload procedure. Cells in a scan chain set to a binary value during the shift patterns will likely cause the tracing of the scan chain to fail and should be avoided.

The test setup procedure will sometimes place binary values on scan cells. Normally this is not a problem since the standard analysis will X-out cells which can change state during a capture cycle. Pipeline cells are normally constrained to hold state during the capture cycle.

What Next

If a constrained pipeline cell is in a scan chain and initialized to a binary value during test setup, the `-pipeline` option of the `set_drc` command should be used reset it to X.

M435

Message Text

`There were number module outputs coerced to inputs. (M435)`

Severity

Warning

Description

This message is generated at the end of a `set_build-coerce_port_directions` command. It indicates PO gates that connect to nets with one or more other sinks were transformed into PI gates and no B7 was issued. This transformation was done hierarchical, bottom-up. The model built reflects the new port direction. The netlist data was also changed: a subsequent `write_netlist` command will show the module having a PI instead of the PO; also, a subsequent `run_build_model` command will not show the transformation again, because the internal netlist data has already been changed.

What Next

This is an informational message only, no further action is necessary.

M436

Message Text

Observe procedure cannot be changed in TEST command mode. (M436)

Severity

Error

Description

This message is issued when you try to do a `set_drc-observe_procedure` command and you are in TEST mode.

What Next

The `set_drc-observe_procedure` command affects the DRC analysis and needs to be set when you are in DRC mode. If you are TEST mode, go back to DRC mode (using `set_drc-force`), and set the option to change the observer procedure.

M437

Message Text

Detected pattern storage can only be performed for external patterns.
(M437)

Severity

Error

Description

This message is generated when you issue a `run_fault_sim-detected_pattern_storage` command while the pattern source is set to internal. The `-detected_pattern_storage` argument can only be used on external patterns.

What Next

If you want to use the `-detected_pattern_storage` argument on an internal pattern set, then you can write out the pattern, read it back in, and fault simulate it. For example:

```
...run_atpg -auto  
  
...write_patterns pat.bin -format bin -rep  
  
...reset state  
  
...set_patterns -external pat.bin  
  
...run_fault_sim -detected_pattern_storage  
  
...report_faults -pattern_id 2
```

M438

Message Text

Detected pattern storage can only be used for full sequential simulation.
(M438)

Severity

Error

Description

This error occurs when the `run_fault_sim` command is issued with `-sequential` and `-detected_pattern_storage` options. The `-detected_pattern_storage` option is only supported for the Basic-Scan and Fast-Sequential fault simulators.

What Next

If the patterns are basic-scan and/or fast-sequential, removing the `-sequential` option can cause the detection status to be stored by the `-detected_pattern_storage` option.

If the patterns require the Full-Sequential fault simulator, consider using the `-nodrop_faults` option, which has slightly different functionality.

M439

Message Text

Pattern ID data is not currently available. (M439)

Severity

Error

Description

This message is generated when you issue a `report_faults-pattern_id` command while the fault simulation results have not been stored.

What Next

To get information about the faults detected by a particular pattern, you need to fault simulate the patterns with the `run_fault_sim -detected_pattern_storage` command to store the results in memory.

Note: You can use the `-detected_pattern_storage` argument only on an external pattern set. If you have an internal pattern set (which can be from an ATPG run), write out the patterns and read them back to fault simulate. For example,

```
run_atpg -auto

write_patterns pat.bin -format bin -rep

reset state

set_patterns -external pat.bin

run_fault_sim -detected_pattern_storage

report_faults -pattern_id 2
```

Error and Warning Messages M440 - M449

This section contains the following error messages and warnings:

- [M440](#)
- [M441](#)
- [M442](#)
- [M443](#)
- [M444](#)

- [M445](#)
- [M446](#)
- [M447](#)
- [M448](#)
- [M449](#)

M440

Message Text

EDCPV flow is disabled for this STIL due to incompatible `write_patterns` options. (M440)

Severity

Warning

Description

The `-first` and `-last` options of the `write_patterns` command are not supported when using a parallel strobe data file specified by the `set_atpg -parallel_strobe_data_file` command.

What Next

A pattern file containing a range of patterns created by the `write_patterns` command does not include links pointing to the parallel strobe data file. However, this pattern file can be still be used during simulation.

To use a parallel strobe data file and specify a pattern range for generation based on the first and last specified patterns:

Create a parallel strobe data file using the `set_atpg -parallel_strobe_data_file` command and the `run_atpg` command (for details, see the topic [Using the run_atpg Command to Create a PSD File in the TestMAX™ ATPG and TestMAX Diagnosis User Guide](#)).

Write the patterns using the `write_patterns` command without the `-first` and `-last` options.

Create a testbench that specifies range of patterns to be created using the `-first` and `-last` options of the `write_testbench` or `stil2verilog` command. The simulation also uses the parallel strobe data file.

For more information, see the topic [Debugging Parallel Simulation Failures for Combined Pattern Validation \(CPV\)](#) in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M441

Message Text

Increasing `set_workspace_sizes -string` to `<integer>`. (M441)

Severity

Warning

Description

The parameters of this design require an increase to the specified workspace size field (in this case to the `-string` parameter) to process this design. This message is generated to identify this change to the environment. This message is informational only. It indicates some internal adjustments that are necessary to complete processing, which might affect memory requirements as the processing continues.

What Next

No action is necessary. The parameter is properly adjusted for the required operation. To eliminate this message, you may use the command `set_workspace_sizes-string` before the operation that generated this message, and specify a value equal to or greater than the one specified in this message.

For more information, see the `set_workspace_sizes` command in the TestMAX™ ATPG and TestMAX Diagnosis Commands guide.

M442

Message Text

Cannot find unique chain `<input|output>` for chain `S` associated with `input|output` test. (M442)

Chain `S` has undefined `<input|output>` pin. (M442)

Severity

Error

Description

This message is generated when the automatic scan pin identification process fails.

M443

Severity

Error

Description

TestMAX ATPG generates the M443 error message when it cannot identify the PRPG/MISR chain during DRC and therefore it cannot find the PRPG ->scan_in or scan_out ->MISR connections.

What Next

Move the scan_in and scan_out for the PRPG/MISR chain up to the top of the chip and then run DRC.

M444

Message Text

```
Defined <input|output> gate G for chain S in CODEC N failed connection  
checking. (M444)
```

Severity

Error

Description

This message is issued when a mismatch has been detected between the STL procedure file description of internal chain inputs / outputs and the netlist.

What Next

Most commonly, the netlist is assumed to be correct, so either the STL procedure file must be changed to match the netlist, or the definitions of internal chain inputs/outputs must be removed from the STL procedure file so the tool automatically infers them.

M445

Message Text

```
Non-encrypted netlist cannot be written when encrypted gzip was read.  
(M445)
```

Severity

Error

Description

An attempt was made to perform a `write_netlist` command in a non-encrypted output format when one or more netlist modules in memory was read in with an encrypted format. As this would unlock the encryption, it is not allowed.

What Next

You might be able to write selected modules as output in a non-encrypted form, but not the entire netlist and library. You might also be able to write the netlist out in `-compress` binary form (an encrypted form).

M446

Message Text

Top module is not currently defined. (M446)

Severity

Error

Description

Description is TBD.

What Next

What next is TBD.

M447

Message Text

Unsuccessful chain trace due to repeated gate I (G). (M447)

Severity

Error

Description

Description is TBD.

What Next

What next is TBD.

M448

Message Text

Unsuccessful chain trace due to too many cells. (M448)

Severity

Error

Description

Description is TBD.

What Next

What next is TBD.

M449

Message Text

Unsuccessful chain trace due to sensitization on gate I (G). (M449)

Severity

Error

Description

TestMAX ATPG generates the M449 error message when it cannot trace the PRPG and MISR chain. This is usually because the scan_in and scan_out pins for the PRPG/MISR chain are not connected properly.

What Next

You must first debug the M449 error before proceeding. If TestMAX ATPG cannot identify the PRPG/MISR chain during DRC, then it cannot find the PRPG->scan_in or scan_out->MISR connections and it generates M443 error message. Move the scan_in and scan_out for the PRPG/MISR chain up to the top of the chip and then run DRC.

Error and Warning Messages M450 - M459

This section contains the following error messages and warnings:

- [M450](#)
- [M451](#)

- [M452](#)
- [M453](#)
- [M454](#)
- [M455](#)
- [M456](#)
- [M457](#)
- [M458](#)
- [M459](#)

M450

Message Text

Unused message. (M450)

Severity

Error

Description

Unused message.

What Next

Unused message.

M451

Message Text

LFSR chain *name* was previously used in CODE N. (M451)

Severity

Error

Description

Description is TBD.

What Next

What next is TBD.

M452

Message Text

Fault candidates will cause passing patterns to fail. (M452)

Severity

Warning

Description

The run diagnosis algorithm has two phases. In the first phase, fault candidates that explain some patterns that failed on the tester are found. In the second phase, these fault candidates are simulated using the patterns that passed on the tester to improve the precision.

In some cases, the best result obtained is a fault candidate that explains many patterns that failed on the tester, but does not match the observed behavior of the patterns that passed on the tester. That is, the fault candidate is detected by some of the patterns that passed on the tester. In this case, run diagnosis outputs the fault candidate, but also issues this warning.

What Next

No action required. This message lets you know that the ideal faults used in the run diagnosis algorithms do not exactly match the real defect's behavior on actual devices. The candidate fault incorrectly predicts some additional patterns to fail on the tester.

For more information, see the `run_diagnosis` command in the TestMAX™ ATPG and TestMAX Diagnosis Commands guide.

M453

Message Text

The added fault(s) affect scan chain load/unload operation. (M453)

Severity

Warning

Description

The fault or faults reported from the `run_simulation` command might affect the scan chain load and unload operations. A correct simulation of the scan shift process will result in scan cells downstream of the fault(s) to be uncontrollable, and the scan cells upstream of the fault(s) to be unobservable. TestMAX ATPG does not simulate the serial shifting

of scan chains, but can approximate the effect of shifting in the presence of the fault(s). However, this approximation is not guaranteed to be correct in cases of multiple faults.

This message typically results from faults added to scan cell inputs or outputs, clock or asynchronous set/reset lines, which are usually classified as DI (detected by implication).

What Next

This message is informational and requires no action. You are cautioned that the result of the simulation is accurate only for the capture sequence and might not correctly model the scan chain shifting operation. More accurate simulation results can be obtained by using an external simulator, such as VCS with a `-serial` option of the `write_patterns` command.

For more information, refer to the following commands in the TestMAX™ ATPG and TestMAX Diagnosis Commands guide:

```
run_simulation
write_patterns
Fault Classes
```

M454

Message Text

Added fault(s) affect scan chain load/unload operation and cannot be simulated. (M454)

Severity

Error

Description

The fault or faults reported by the `run_simulation` command affects the scan chain load and unload operations. The fault(s) cannot be correctly simulated. The faults causing this error is likely a DI (detected by implication) faults.

An example of a fault that can cause this error is a fault that causes a scan cell to select data only from the scan-in path and not from the functional logic.

What Next

The `run_simulation` command cannot be used to simulate this fault. An alternative is to use an external simulator, such as VCS, with a modified netlist (modeling the fault) with the patterns generated by the `write_patterns` command. This command's `-serial` option should be used if the correct simulation of the scan chain shifting is desired.

For more information, see the following commands in the TestMAX™ ATPG and TestMAX Diagnosis Commands guide:

`run_simulation`

`write_patterns`

Fault Classes

M455

Message Text

All faults from the fault list will be deleted during diagnosis. (M455)

Severity

Warning

Description

There were faults in the fault list when the `run_diagnosis` command was called. These faults are removed when the `run_diagnosis` command executes.

What Next

This is an informational message only. No action is required in a typical case. However, if you need the removed faults, you can save them by specifying the `write_faults` command before issuing the `run_diagnosis` command. If you would like the `run_diagnosis` command to consider only the faults in the active fault list, you should use the `-rank_faults` option.

M456

Message Text

Option <option> is incompatible with the previous options. (M456)

Severity

Error

Description

The command option you specified is incompatible with one of the other options of this command.

What Next

Consult the online help for the proper usage of the command. Specify only compatible options.

M457

Message Text

" *name*" option is not compatible with `-unique_clocks`. (M457)

Severity

Error

Description

A `write_patterns` command with the `-unique_clocks` option was combined with another option that is not supported.

What Next

Repeat the `write_patterns` command but leave off the unsupported option that is in conflict with `-unique_clocks`.

M458

Message Text

Testbench name exceeds 255 characters. (M458)

Severity

Error

Description

This message occurs when the `-verilog_testbench_name` option of the `write_patterns` command is used and the string provided exceeds 255 characters.

What Next

Retry the command providing a shorter testbench name string.

M459

Message Text

N clock pins have been previously defined and will continue to be used.
(M459)

Severity

Error

Description

The `-predict` option of the `add_clocks` command was used when some clock definitions have already been made.

What Next

This is an information message only. The pre-existing clock definitions are retained.

Error and Warning Messages M460 - M469

This section contains the following error messages and warnings:

- [M460](#)
- [M461](#)
- [M462](#)
- [M463](#)
- [M464](#)
- [M465](#)
- [M466](#)
- [M467](#)
- [M468](#)
- [M469](#)

M460

Message Text

Input N1 of <type> gate (G) could not be controlled when clocks are off.
(M460)

Severity

Error

Description

Description is TBD.

What Next

What next is TBD.

M461

Message Text

Input N1 of <type> gate (G) could be controlled by more than one pin.
(M461)

Severity

Error

Description

Description is TBD.

What Next

What next is TBD.

M462

Message Text

Regular expression exceeds internal limit of 24 [...] ranges. (M462)

Severity

Error

Description

A regular expression was supplied for a command match that contained more than 24 character ranges. These are square brackets enclosing one or more character, or a range of characters, such as [a-z0-9].

What Next

Try reforming your regular expression to use less character ranges. For example, two ranges in a row "[aA][bB]" can be combined as "[abAB]+". It is not exactly the same,

but might be close enough to still provide the desired matches and be within the limit of 24 character ranges. You may also convert a range into a wildcard match or a single character match, which do not have the same limits.

M463

Message Text

Verilog parallel compressor patterns do not support serial vectors, parallel <n> is handled as -parallel (M463)

Severity

Warning

Description

You have used the `-parallel <n>` (where <n> is an integer value) option in the write patterns command. For many pattern scenarios, this causes the last <n> bits of scan data to be treated as serial scan data, and generates <n> serial shifts after parallel-loading the rest of the data. However, in Verilog parallel patterns, because the scan chains are internal to the design and not accessible externally, this feature cannot serial-load the last <n> bits of data.

What Next

Change the specified option `-parallel <n>` to `-parallel` to remove this warning.

M464

Message Text

Compressor chain output checking could not identify the output of chain *chain_name* (M464)

Severity

Error

Description

This message is issued when chain outputs are not identified during the internal chain input and output inference process. This issue usually indicates a mismatch between the chain routing specified in the netlist and in the STIL procedure file. It usually occurs when using manually generated STIL procedure files or manually generated top-level netlists. This issue can also sometimes occur when the `read_drc` command flow generates STIL procedure files.

What Next

If the STIL procedure file is generated from the `read_drc` command flow, you should consider using the `-reorder_xtol_signals` option of the `read_drc` command to prevent these messages from occurring when running the `run_drc` command. In this case, you need to regenerate the STIL procedure file from the `read_drc` command.

If you are not using the `read_drc` command flow, you can edit the STL procedure file so it includes the beginning and ending point information for the scan chains, or make that information available in the scan chains report produced from TestMAX DFT.

The following example shows how to edit the STL procedure file to include the beginning scan input and ending scan output information for the scan chains:

STL procedure file Before Edits:

```
ScanStructures {  
  ScanChain "HASS1_1" {  
  
  }  
  ...  
  ...  
}
```

STL procedure file After Edits:

```
ScanStructures {  
  ScanChain "HASS1_1" {  
    ScanIn "Path/to/beginning/scanchain/FF/SI";  
    ScanOut "Path/to/ending/scanchain/FF/Q";  
  }  
  ...  
}
```

If M464 messages are seen, you can use the `analyze_chain` command to debug the issue. The `analyze_chain` command does not identify the root cause of the issue, but helps you debug the issue by providing additional information about single-observe simulation values for specific load and unload modes.

Note: The `analyze_chain` command is supported only for DFTMAX compression.

The following example describes how the `analyze_chain` command helps in debugging M464 messages.

1. To use the `analyze_chain` command, set the `set_drc -preserve_chain_analysis` command before running DRC.

The following M464 messages are displayed during DRC:

```
Error: Compressor chain output checking could not identify the output  
of chain des_unit_3_ScanCompression_mode_i_1 (M464)
```

```
Error: Compressor chain output checking could not identify the output  
of chain des_unit_3_ScanCompression_mode_o_1 (M464)
```

Error: Compressor chain output checking could not identify the output of chain des_unit_3_3 (M464)

Error: Compressor chain output checking could not identify the output of chain des_unit_3_4 (M464)

Error: Compressor chain output checking could not identify the output of chain des_unit_3_5 (M464)

Error: Compressor chain output checking could not identify the output of chain des_unit_3_6 (M464)

Error: Compressor chain output checking could not identify the output of chain des_unit_3_7 (M464)

Error: Compressor chain output checking could not identify the output of chain des_unit_3_8 (M464)

2. Choose one of the chain messages to analyze it further.

3. Use the `analyze_chain` command to report all the single- and multi-observe modes of the chosen chain.

```
DRC-T> analyze_chain des_unit_3_3
```

```
Analyzing chain des_unit_3_3 ...
```

```
Simulating single observe mode for unload_mode = 0, load_mode = 0, pin  
= test_solWarning: Multiple scan chain outputs in compressor single  
observe mode
```

```
Simulating single observe mode for unload_mode = 0, load_mode = 1, pin  
= test_sol
```

```
Simulating single observe mode for unload_mode = 6, load_mode = 0,  
pin = test_so8
```

```
Warning: Reconvergence path detected!Warning: Multiple scan chain  
outputs in compressor single observe mode
```

```
Simulating single observe mode for unload_mode = 6, load_mode = 1, pin  
= test_so8
```

```
Simulating multiple observe mode for unload_mode = 1, load_mode = 0,  
pin = test_so8
```

```
Simulating multiple observe mode for unload_mode = 3, load_mode = 0,  
pin = test_so8
```

```
Simulating multiple observe mode for unload_mode = 7, load_mode = 0,  
pin = test_so8
```

Chapter 6: Messages M400 - M499

Error and Warning Messages M460 - M469

```
Simulating multiple observe mode for unload_mode = 1, load_mode = 1,
pin = test_so8
```

```
Simulating multiple observe mode for unload_mode = 3, load_mode = 1,
pin = test_so8
```

```
Simulating multiple observe mode for unload_mode = 7, load_mode = 1,
pin = test_so8
```

4. Choose a single-observe mode to further analyze the behavior of that specific mode.

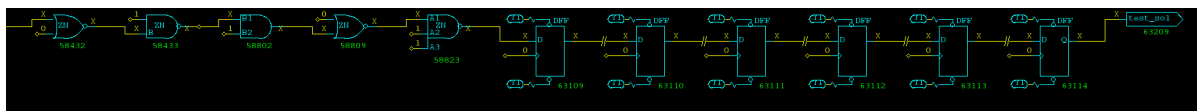
TestMAX ATPG simulates the single-observe mode and tries to infer the ScanOut for that chain by tracing back from the output pin.

The following is an example of a single-observe mode traced to multiple flops.

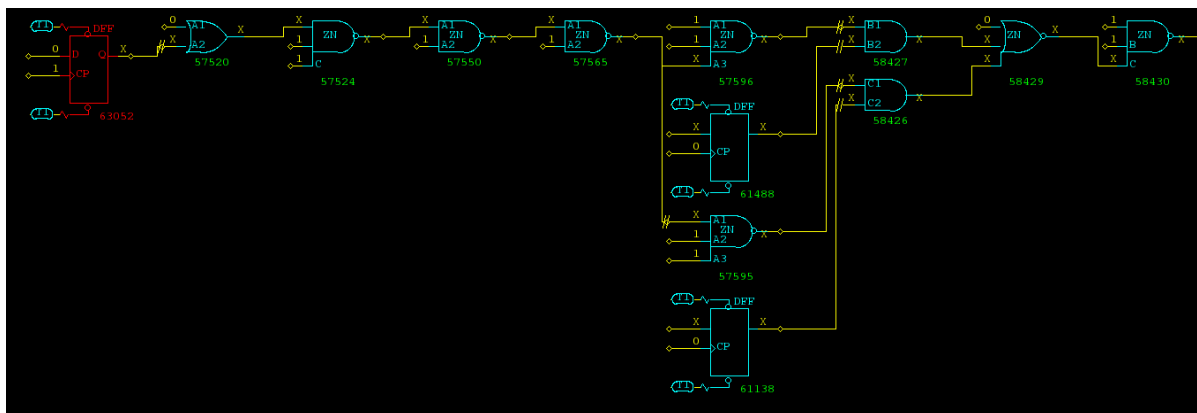
```
DRC-T> analyze_chain des_unit_3_3 -load_mode 0 -unload_mode 0
Simulating single observe mode for unload_mode = 0, load_mode = 0, pin = test_so1
Warning: Multiple scan chain outputs in compressor single observe mode

Traced to test_so1 (63209) (PO)
Traced to des_glue_1/SNPS_PipeTail_test_so1_6 (63114) (DFF) (unload_pipeline)
Traced to des_glue_1/SNPS_PipeTail_test_so1_5 (63113) (DFF) (unload_pipeline)
Traced to des_glue_1/SNPS_PipeTail_test_so1_4 (63112) (DFF) (unload_pipeline)
Traced to des_glue_1/SNPS_PipeTail_test_so1_3 (63111) (DFF) (unload_pipeline)
Traced to des_glue_1/SNPS_PipeTail_test_so1_2 (63110) (DFF) (unload_pipeline)
Traced to des_glue_1/SNPS_PipeTail_test_so1_1 (63109) (DFF) (unload_pipeline)
Traced to des_glue_1/SNPS_PipeHead_test_si2_9 (63052) (DFF)
Traced to des_unit_3/de_c/\o_data_reg[30] (61488) (DFF)
Traced to des_unit_3/ctl1/key_dd_reg[23] (61138) (DFF)
```

5. Open the GSV and use the -display option of the analyze_chain command to show the traced gates.



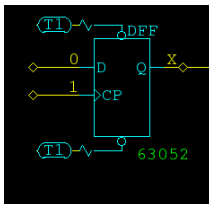
The following is the schematic of the last traced gates (continued from previous schematic):



The analyze_chain command reports scan-cells and the schematic shows that there are six unload pipelines. The last three cells are candidates for scan-outs. There should only be a single candidate for scan-out.

```
Traced to des_glue_1/SNPS_PipeHead_test_si2_9 (63052) (DFF)
Traced to des_unit_3/de_c/\o_data_reg[30] (61488) (DFF)
Traced to des_unit_3/ctl1/\key_dd_reg[23] (61138) (DFF)
```

One of the cell's (gate ID 63052) is a head pipeline. If you observe the pipeline cell simulation values, the scan-in value is 0, but the output is X for the scan cell as shown in the figure:



Tracing the head pipeline scan-in backward shows that the input scan-in is proper.

6. Check the SPF for differences in the number of pipeline stages in the actual design and SPF.

The TestMAX ATPG schematic shows nine head pipeline flops for the particular scan-in, but the following SPF pipeline information shows eight head pipeline flops:

```
UserKeywords CompressorStructures;

CompressorStructures {
  LoadPipelineStages 8;
  UnloadPipelineStages 6;
```

7. Correct the design by removing the additional head pipeline to resolve the M464 issue.

You can obtain the scan input and scan output points from either TestMAX DFT using the report_scan_path command or from an ASCII-readable CTL of the given submodels.

If this additional information does not resolve the problem, you should file a bug report.

M465

Message Text

```
Unsupported command <command_name> in procedure <proc_name>. (M465)
```

Severity

Warning

Description

This message is issued when Verilog or image files are being written, and indicates that there is an unsupported command in the named procedure. For example, a MEASURE command is not supported in the procedure load_unload.

What Next

The unsupported command is ignored and might lead to VCS simulation failures. You should review the STL procedure file and remove the unsupported command.

M466

Message Text

```
<type_name1> pattern added after <type_name2> pattern. (M466)
```

Severity

Warning

Description

This message is issued when using the `set_patterns-append` command, and indicates that the internal pattern set is no longer ordered for optimal simulation performance.

For example, DFTMAX patterns are always appended at the end. Thus, any new `basic_scan` patterns are placed after any `fast_sequential` patterns that already exist in the internal pattern set. When simulating (using the `run_simulation` command), the fast-sequential simulator is used for the basic-scan patterns placed after `fast_sequential` patterns. In this case, the fast-sequential simulator is less efficient than the basic-scan simulator for basic-scan patterns.

What Next

No action is required.

M467

Message Text

```
Instance <inst_name> of module <mod_name> written in TMAX-style Verilog  
(M467)
```

Memory instance <inst_name> of module <mod_name> written in TMAX-style Verilog (M467).

_BUSK0 in module <mod_name> written in TMAX-style Verilog (M467).

_BUSK1 in module <mod_name> written in TMAX-style Verilog (M467).

Severity

Warning

Description

The `write_netlist -format verilog_vcs` command writes a standard Verilog netlist, whenever possible. In some cases, the `verilog_vcs` option cannot write out an instance in standard Verilog, and will instead write it out as Verilog_Tmax style. The M467 messages notifies you when this case occurs.

What Next

No action is required.

M468

Message Text

Selected `-misr_data` option is not compatible with `-parallel`. (M468)

Severity

Error

Description

You you specify the `write_patterns` command, you cannot use the `-parallel` option when you select `-misr_dataper_pattern` or `at_end`.

What Next

Specify only `-misr_dataper_pattern` or `at_end`, or `-parallel` when using the `write_patterns` command.

M469

Message Text

X-chain <name1> has lower X-probability than non-X-chain <name2>. (M469)

Severity

Warning

Description

This check is performed similarly to the `analyze_compressors -xchain_analysis` command; it simulates random patterns and determines how frequently each scan cell captures an X value. The X-probability of a scan chain is computed as the average X-frequencies of its scan cells.

This message indicates that ATPG runs might result in excessive patterns because X-chains can only be observed in single-observability mode when very few chains are observable.

What Next

The design and models used when running TestMAX DFT to create X-chains appear to have been updated when running TestMAX ATPG. You should consider updating the TestMAX DFT flow.

Error and Warning Messages M470 - M479

This section contains the following error messages and warnings:

- [M470](#)
- [M471](#)
- [M472](#)
- [M473](#)
- [M474](#)
- [M475](#)
- [M476](#)
- [M477](#)
- [M478](#)
- [M479](#)

M470

Message Text

There are <num> non-X-cells in X-chains (first_id= *gate_id*). (M470)

Severity

Warning

Description

This is an expert-level message reported when checking for [M469](#) is performed. It indicates that <num> scan cells placed in X-chains have low X-probability, and might contribute to increased pattern count, especially if <num> is large. The *gate_id* is the ID of the first of these scan cells. For a complete list, and more analysis options, see the `analyze_compressors-xchain_analysis` command.

What Next

The design and models used when running TestMAX DFT to create X-chains appear to have been updated when running TestMAX ATPG. You should consider updating the TestMAX DFT flow.

M471

Message Text

There are <num> X-cells in non-X-chains (first_id= *gate_id*). (M471)

Severity

Warning

Description

This is an expert-level message reported when checking for [M469](#) is performed. It indicates that <num> scan cells placed in non-X-chains have high X-probability, and might contribute to increased pattern count, especially if <num> is large. The *gate_id* is the ID of the first of these scan cells. For a complete list, and more analysis options, see the `analyze_compressors-xchain_analysis` command.

What Next

The design and models used when running TestMAX DFT to create X-chains appear to have been updated when running TestMAX ATPG. You should consider updating the TestMAX DFT flow.

M472

Message Text

`Nonscancell ID used to control PLL internal clocking. (M472)`

Severity

Error

Description

This message indicates that the non-scan cell *ID* controls PLL internal clocking. Non-scan cells cannot be controlled through scan load, so the ability to control clocks may be severely degraded, which results in poor test coverage.

If pipelining is used and the clock chain does not have pipelines, this message might be erroneously issued for an incorrectly configured compressed (DFTMAX) design.

All chains must go through the compressors. Even if a chain (such as the clock chain) is not compressed, it is still recognized by TestMAX as going through the compressors. Further, all compressor inputs must have the same pipeline depth, and all compressor outputs must have the same pipeline depth.

For example, in a design with two stages of output pipeline on the unload compressor, the two cells on the output side of the clock chain are considered pipeline cells (see the topic `report_compressors -pipeline` command in the TestMAX ATPG and TestMAX Diagnosis Commands Guide.), and are not part of the chain. Pipeline cells are non-scan cells (thus, the reason for the M472 message). The expected usage would be to add two more DFFs on the output of the clock chain.

What Next

Modify the design so that all chains, even non-compressed chains, have the same input and the same output pipeline depth.

M473

Message Text

`STIL patterns defaulted to serial simulation mode. (M473)`

Severity

Warning

Description

This message is printed when the `write_patterns` command is not specified with either the `-parallel` or `-serial` option. It indicates that, unlike previous TestMAX ATPG releases before the 2008.09 release, the default STIL file is written in serial mode (regardless of the inserted DFT type).

What Next

If you want to write out STIL patterns in parallel mode, you must specify the `-parallel` option with the `write_patterns` command.

M474

Message Text

STIL patterns defaulted to parallel simulation mode. (M474)

Severity

Warning

Description

This message is printed when the `write_patterns` command is specified with the `-unified_stil_flow` option, but does not include either the `-parallel` or `-serial` option. It indicates that, by default, the unified STIL flow creates patterns in parallel mode.

What Next

If you want to write out STIL patterns in serial mode using the Unified STIL Flow (USF), you must specify the `-serial` option with the `write_patterns` command.

M475

Message Text

To run serial shift with STIL parallel patterns, for scan compression designs, please add `-unified_stil_flow` option (M475)

Severity

Warning

Description

This message is printed when the `write_patterns` command is not specified with the `-unified_stil_flow` option, but includes the `-parallel` option on a scan compression

design. It indicates that, in this case, the STIL file cannot be generated in parallel mode using serial shift.

What Next

Specify the `-unified_stil_flow` option with the `write_patterns` command when running serial shift with STIL parallel patterns.

M476

Message Text

Detected Scan Compression, generating serial-only STIL patterns. (M476).

Severity

Warning

Description

This message indicates that the generated STIL patterns are limited to serial simulation only, and are not USF-compliant.

M477

Message Text

The design contains X-TOL but the STL procedure file description of Compressor is incomplete (only compression mode). Generating serial only STIL patterns. (M477)

Severity

Warning

Description

This message indicates that the generated STIL patterns are limited to serial simulation only, and are not USF-compliant.

What Next

Check to see if the TestMAX DFT variable `test_sccomp_put_xtol_connections` was set properly.

M478

Message Text

`-unified_stil_flow` option is valid only for `stil` and `stil99` output formats, ignored. (M478)

Severity

Warning

Description

This message indicates that the `-unified_stil_flow` option of the `write_patterns` command is valid only when using the `-format stil` option. The `-unified_stil_flow` option is ignored if any other formats are specified.

M479

Message Text

`-nointernal_scancells` option is not compatible with the [default] `-parallel` option, serial patterns are generated. (M479)

`-scanstructures min` option is not compatible with the [default] `-parallel` option, serial patterns are generated. (M479)

Severity

Warning

Description

This message indicates that either the `-nointernal_scancells` or the `-scanstructures min` option of the `write_patterns` command conflicts with the `-parallel` option. The `-parallel` option specifies parallel simulation, and the `-nointernal_scancells` and `-scanstructures min` options work only for serial simulation.

What Next

The `-parallel` option is disabled, and TestMAX ATPG generates serial patterns. No further action is necessary if this behavior is sufficient. You can eliminate this warning by removing the `-parallel` option.

Error and Warning Messages M480 - M489

This section contains the following error messages and warnings:

- [M480](#)
- [M481](#)
- [M483](#)
- [M484](#)
- [M485](#)
- [M486](#)
- [M487](#)
- [M488](#)

M480

Message Text

There were no matches for wildcard string *string* (M480)

Severity

Warning

Description

This message indicates that regular expression string *string* did not return matching results.

What Next

See the list of regular expressions supported by TestMAX ATPG and makes sure your regular expression is correct. Double-check that you do have at least one match for the regular expression you are trying.

M481

Message Text

Warning: Test setup patterns are not stored. Please use "set_drc -store_setup" and rerun drc. (M481)

Severity

Warning

Description

TestMAX ATPG displays this warning message when you execute the `add_waveform_signals` command and you have not used the `store_setup` option (with the `set drc -store_setup` command) when running DRC.

What Next

Go back to the DRC mode, use the `set drc -store_setup` command and rerun DRC.

M483

Message Text

Unable to satisfy ATPG constraint @ <node>. (M483)

Severity

Error

Description

During Full-Sequential ATPG initialization, the specified ATPG constraint was found to be unsatisfiable. All faults are ATPG untestable in the presence of this constraint.

What Next

The constraint must be removed or changed to continue with ATPG.

M484

Message Text

Unable to satisfy <n> ATPG constraints. (M484)

Severity

Error

Description

During Full-Sequential ATPG initialization, the specified number of ATPG constraints were found to be unsatisfiable. All faults are ATPG untestable in the presence of these constraint(s).

What Next

The constraint(s) must be removed or changed to continue with ATPG.

M485

Message Text

Unable to prevent contention @ <node>. (M485)

Severity

Error

Description

During Full-Sequential ATPG initialization, ATPG was unable to prevent contention at the specified BUS. All faults are ATPG untestable with the specified contention options.

What Next

Correct the design to avoid contention at the BUS gate or relax the bus contention settings (via the `set_contention` command).

M486

Message Text

Unable to prevent contention at *number* BUS(es). (M486)

Severity

Error

Description

During Full-Sequential ATPG initialization, ATPG was unable to prevent contention at the specified number of BUS(es). All faults are ATPG-untestable with the specified contention options.

What Next

Correct the design to avoid contention at the BUS gate or relax the bus contention settings (via the `set_contention` command).

To identify the affected buses, specify `set_contention-severity error`. After a pattern fails, it can be displayed in the GUI by using the `set_pindata-error_data` command. You can also try the `analyze_buses` command; checking the Z violations, e.g., Z1, Z4, and so forth., would aid in this manner.

M487

Message Text

Unconstrained primary input <pin-name> used as
[SCAN ENABLE | SET | RESET] might change during at-speed cycles. (M487)

Severity

Warning

Description

Ensure that scan enable primary inputs are constrained to their inactive values during the system-clock launch mode of transition fault test and path delay test. This prevents TestMAX ATPG from generating the last-shift launch patterns and from switching these primary inputs during at-speed cycles. Similarly, ATPG should not pulse SET and RESET inputs during at-speed cycles. The SCAN ENABLE, SET, and RESET signals are typically slow signals that should not change during an at-speed test.

What Next

Use the `add_pi_constraints` command to keep the primary inputs at a static value for delay test.

For more information, see the following commands in the TestMAX™ ATPG and TestMAX Diagnosis Commands guide:

```
add_pi_constraints
analyze_buses
analyze_buses Report
report_buses
```

For more information, refer to the topic Contention Analysis in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M488

Message Text

Support for multi cycle Shift is limited, simulation might fail. (M488)

Severity

Warning

Description

This message will appear if you specify a `write_patterns` command using the STIL or STIL99 format and the Shift block in the STL procedure file contains more than one `v` statement.

MAX Testbench does not support all the configurations for multi cycle shift. For example, a situation in which statements such as “`si=###`” and “`so=###`” are on different `v` statements is not supported. The recommended flow for multi cycle Shifts, for both serial and parallel simulations, is to use MAX TestBench with the unified STIL flow (in this case, add the `-unified_stil_flow` option of the `write_patterns` command in the TestMAX ATPG script). For more information on the unified STIL flow, see the *TestMAX ATPG User Guide*.

What Next

If the cause of multiple vectors in the Shift block is related to additional (and necessary) refclock pulses, consider defining the refclock using the `add_clocks -ref_timing` command, instead of defining these pulses in additional vectors -- particularly during Shift. Alternately, you might be able to redefine the period of the Shift vector to encompass these additional pulses (you can also redefine the waveforms in the WaveformTable to contain multiple pulses). You can then create a single extended vector to cover these events.

Error and Warning Messages M490 - M499

This section contains the following error messages and warnings:

- [M494](#)
- [M495](#)
- [M496](#)

M494

Message Text

```
Merging enabled without dynamic clock grouping for multiple clocks.  
(M494)
```

Severity

Warning

Description

ATPG was started on a design with multiple clocks with pattern merging enabled, but without enabling dynamic clock grouping. By enabling merging, you apparently want a compact pattern set. However, on a design with multiple clocks, the most compact pattern

set is most likely obtained by also enabling dynamic clock merging. This message reminds you to consider enabling dynamic clock grouping.

What Next

Consider enabling dynamic clock grouping with the `set_drc-clock -dynamic` command to get the most compact pattern set. This message can be ignored if having a less compact pattern set is not a concern.

For more information, refer to the topic Understanding Clock Grouping in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M495

Message Text

Merging and pattern limits enabled without minimum detections per pattern. (M495)

Severity

Warning

Description

ATPG was started on a design after specifying limits on the number of patterns and merging enabled, but without minimum detections per pattern set. Having a pattern limit implies that you want the highest coverage possible under the pattern count constraint. This message reminds you to consider using the minimum detection per pattern setting to achieve the best results.

What Next

Consider using options `-basic_min_detects_per_pattern`, `-fast_min_detects_per_pattern`, or `-full_min_detects_per_pattern` of the `set_atpg` command to select patterns that detect a minimum set of faults. This should produce better results when pattern memory is limited. This message might be ignored if having a less compact pattern set is not a concern.

M496

Message Text

`Instance_modify instance_name gate_name` was not found and removed (M496)

`Instance_modify instance_name gate_name` cannot be applied to module `module_name` with `number` inputs and was removed (M496)

Severity

Error

Description

The specified instance was not found, or the specified instance has BIDI pins and cannot be remodeled as the Boolean gate chosen, or the specified instance has more than one input and cannot be remodeled as the Boolean gate chosen.

What Next

Check the path name to the specified instance, or change the instance, or model the instance as a TIE/multi input gate.

7

Messages M500 - M599

This section contains the following error messages and warnings:

- [Error and Warning Messages M500 - M509](#)
- [Error and Warning Messages M510 - M519](#)
- [Error and Warning Messages M520 - M529](#)
- [Error and Warning Messages M530 - M539](#)
- [Error and Warning Messages M540 - M549](#)
- [Error and Warning Messages M550 - M559](#)
- [Error and Warning Messages M560 - M599](#)

Error and Warning Messages M500 - M509

This section contains the following error messages and warnings:

- [M504](#)
- [M505](#)
- [M507](#)
- [M508](#)
- [M509](#)

M504

Message Text

(1) Could not find any single failure in *number* shift cycles for pattern *number*. (M504)

(2) Could not map failures for compressor <compressor_name> for pattern= *number*. (M504)

(3) Could not map failures for CODEC *number* during shift *number* of pattern *number* to a chain. (M504)

(4) Failure file reading aborted due to too many unmapped failures. (M504)

Severity

Warning

Description

Message (1) is printed during the run diagnosis of an adaptive scan design if the failures for a particular shift cycle could not be mapped to a scan cell.

Message (2) is printed during the run diagnosis of an adaptive scan design if all failures for a particular pattern could not be mapped to a scan cell.

Message (3) is printed during run diagnosis in a circuit, if the failures observed at the outputs could not be mapped back to a scan cell. This is usually due to the possibility of aliasing because of multiple fault effects.

Message (4) is printed for a circuit when TestMAX ATPG aborts reading the failure file if 10 failing patterns cannot be mapped. Diagnosis is performed after truncating the failure log file at the last successfully mapped failing pattern.

What Next

If the diagnostic resolution is low, the internal scan chains can be reconfigured as parallel scan chains (no output compression), scan patterns re-run on the device in the tester, and re-diagnosed.

M505

Message Text

Writing *design_style* patterns is not supported for the selected format. (M505)

Severity

Error

Description

The selected pattern format for the `write_patterns` command is not supported for designs with the identified constructs / design style enabled. For example, the parallel STIL format is not supported for sequential compressor patterns.

What Next

Review the restrictions for the identified design style, then select a supported format to write the patterns.

M507

Message Text

Writing parallel <design_style> patterns is not supported. (M507)

Severity

Error

Description

The `-parallel` option of the `write_patterns` command is not supported for designs with the identified constructs / design style enabled. For example, the parallel format is not supported for Sequential Compressor patterns.

What Next

Do not use the `-parallel` option with these patterns.

M508

Message Text

Launch-on-Shift patterns generated without `load_unload_pi/`
`reseed_load_unload_pi`. PI transitions might be lost. (M508)

Severity

Warning

Description

To preserve any primary input (PI) transition effects for launch-on-shift patterns, the specified procedures in the message must be present in the STL procedure file. Without these procedures, the tests will still operate, but might not effectively test PI transition effects in the patterns.

What Next

Review the V violations generated during DRC; there should be V7 warning messages generated at DRC for this situation. Add the specified procedures into the STL procedure file and rerun DRC to permit complete testing in this context.

M509

Message Text

Sequential Compression patterns with `[-misr_measure once_per_test | -seq_comp_accumulator_mode ]` do not support `-first`. (M509)

Sequential Compression patterns with `[-misr_measure once_per_test | -seq_comp_accumulator_mode ]` do not support `-last`. (M509)

Sequential Compression patterns with `[-misr_measure once_per_test | -seq_comp_accumulator_mode ]` cannot be split. (M509)

Sequential Compression requires 2 or more patterns per split file; please use a different value of `-split`. (M509)

Sequential Compression patterns with `-last` requires at least two patterns afterward. (M509)

Severity

Error

Description

Sequential compression patterns have restrictions on partitioning subsections of the pattern blocks. The various versions of this error message is generated if these restrictions are violated and the patterns will not be generated. These issues affect the use of the `-split`, `-first`, and `-last` options of the `write_patterns` command.

If the `set_atpg -misr_measure once_per_test` option is specified for the patterns, the patterns cannot be partitioned.

If the `set_drc -seq_comp_accumulator_mode` option ("MISR-less" sequential compression) is specified for the patterns, the patterns cannot be partitioned.

Sequential compression patterns must apply the next two patterns at the end of each block, as part of a split pattern set. In addition, the first two patterns at the start of a partitioned set will have all expect data set to X. If the last pattern block of the split set occurs with less than two patterns present or less than two patterns are present in a block, the blocks cannot be written properly. For instance, a pattern set that specifies `"-split 100"` will contain 102 patterns per block, with the last 2 patterns repeated at the start of the next block (without expect data). This requirement cannot be satisfied if the last pattern block has 1 pattern in it, for instance you have 501 patterns and request `"-split 50"`. However, `"-split 55"` will work because the last block will have 6 patterns in it.

Any partitioning scenario that results in less than 2 patterns per block, for instance using `"-split 1"`, generates an error.

What Next

Change the value of the partitioning option(s), or eliminate the option(s), to eliminate the error.

If `set_atpg -misr_measure once_per_test` is required, partitioning options cannot be used. Otherwise use `set_atpg -misr_measure once_per_pattern` to generate a pattern set that can be partitioned.

If `set_drc -seq_comp_accumulator_mode` is required, partitioning options cannot be used. Otherwise use `set_drc -noseq_comp_accumulator_mode` to generate a pattern set that can be partitioned.

If the `write_patterns -split` option will result in a modulo function over the pattern count that results with a remainder of 1, use a different value of `-split` to avoid this.

Error and Warning Messages M510 - M519

This section contains the following error messages and warnings:

- [M511](#)
- [M512](#)
- [M513](#)
- [M514](#)
- [M515](#)
- [M516](#)
- [M517](#)
- [M518](#)

M511

Message Text

Mapped patterns not merged at pattern N1 due to inconsistent events <E1 E2> in pattern N2. (M511)

Severity

Warning

Description

During the `run_mapping` command, the incoming device patterns were used to create pattern `N2` at the end of the current internal pattern set. Then to reduce the total patterns, an attempt was made to merge pattern `N2` into the existing set of internal patterns at position `N1`.

This message is an indication that merging into pattern `N1` was not possible, because the two patterns had inconsistent events, `E1` and `E2`. Events are actions such as: `scan_load`, `forcePI`, `pulse_clock`, `measurePO`, and `scan_unload`. Events must be identical or non-conflicting in order for the two patterns to merge.

The new pattern was successfully added to the internal pattern set as pattern `N2`, but was not successfully merged into prior pattern `N1`.

What Next

If this warning message occurs frequently, the end result is more patterns than you desire, because merging is not successful. There might be nothing that can be done about this condition, but it also might be possible by reviewing the event sequences between patterns `N1` and `N2` that you can adjust the incoming patterns to be more similar, and this will result in TestMAX ATPG having better success in merging them.

For more information, refer to the `report_patterns` command in the TestMAX™ ATPG and TestMAX Diagnosis Commands guide.

M512

Message Text

Mapped patterns not merged at pattern `N1` due to inconsistent `<type>` event size in pattern `N2`. (M512)

Severity

Warning

Description

During the `run_mapping` command, the incoming device patterns were used to create pattern `N2` at the end of the current internal pattern set. Then to reduce the total patterns, an attempt was made to merge pattern `N2` into the existing set of internal patterns at position `N1`.

This message is an indication that merging into pattern `N1` was not possible, because the two patterns had inconsistent events sequences. Events are actions such as: `scan_load`, `forcePI`, `pulse_clock`, `measurePO`, and `scan_unload`. Events must be identical or one pattern a subset of the other in order for the two patterns to be merged.

The new pattern was successfully added to the internal pattern set as pattern `N2`, but was not successfully merged into prior pattern `N1`.

What Next

If this warning message occurs frequently, the end result is more patterns than you desire, because merging is not successful. There might be nothing that can be done about this condition, but it also might be possible by reviewing the event sequences between patterns `N1` and `N2` that you can adjust the incoming patterns to be more similar, and this will result in TestMAX ATPG having better success in merging them.

For more information, refer to the topic `report_patterns` in the TestMAX™ ATPG and TestMAX Diagnosis Commands guide.

M513

Message Text

Mapped patterns not merged at pattern `N1` due to inconsistent load on cell `S-I` in pattern `N2`. (M513)

Severity

Warning

Description

During the `run_mapping` command, the incoming device patterns were used to create pattern `N2` at the end of the current internal pattern set. Then to reduce the total patterns, an attempt was made to merge pattern `N2` into the existing set of internal patterns at position `N1`.

This message is an indication that merging into pattern `N1` was not possible, because the two patterns had conflicting values that needed to be loaded into the scan cell identified by chain `S`, and position `I`. One pattern needed a 0, and the other a 1, or vice versa.

Although the new pattern was not successfully merged into pattern `N1`, it does remain in the internal pattern set as pattern `N2`.

What Next

If this warning message occurs frequently, the end result is more patterns than you desire, because merging is not successful. There might be nothing that can be done about this condition, but it also might be possible by reviewing the event sequences between patterns `N1` and `N2` that you can adjust the incoming patterns to be more similar, and this will result in TestMAX ATPG having better success in merging them.

For more information, refer to the topic `report_patterns` in the TestMAX™ ATPG and TestMAX Diagnosis Commands guide.

M514

Message Text

Mapped patterns not merged at pattern N1 due to inconsistent force on pin P (G) in pattern N2. (M514)

Severity

Warning

Description

During the `run_mapping` command, the incoming device patterns were used to create pattern N2 at the end of the current internal pattern set. Then to reduce the total patterns, an attempt was made to merge pattern N2 into the existing set of internal patterns at position N1.

This message is an indication that merging into pattern N1 was not possible, because the two patterns had conflicting values that needed to be forced on primary input P (gate ID = G). One pattern needed a 0, and the other a 1, or vice versa.

Although the new pattern was not successfully merged into pattern N1, it does remain in the internal pattern set as pattern N2.

What Next

If this warning message occurs frequently, the end result is more patterns than you desire, because merging is not successful. There might be nothing that can be done about this condition, but it also might be possible by reviewing the event sequences between patterns N1 and N2 that you can adjust the incoming patterns to be more similar, and this will result in TestMAX ATPG having better success in merging them.

For more information, refer to the topic `report_patterns` in the TestMAX™ ATPG and TestMAX Diagnosis Commands guide.

M515

Message Text

Mapped patterns not merged at pattern N1 due to inconsistent clock pulses <P1 P2> in pattern N2. (M515)

Severity

Warning

Description

During the `run_mapping` command, the incoming device patterns were used to create pattern `N2` at the end of the current internal pattern set. Then to reduce the total patterns, an attempt was made to merge pattern `N2` into the existing set of internal patterns at position `N1`.

This message is an indication that merging into pattern `N1` was not possible, because the two patterns had inconsistent clock events. `P1` and `P2` indicate the clocks involved. The patterns must use the same clocks to be merged and have the same number of clock pulses per clock.

Although the new pattern was not successfully merged into pattern `N1`, it does remain in the internal pattern set as pattern `N2`.

What Next

If this warning message occurs frequently, the end result is more patterns than you desire, because merging is not successful. There might be nothing that can be done about this condition, but it also might be possible by reviewing the event sequences between patterns `N1` and `N2` that you can adjust the incoming patterns to be more similar, and this will result in TestMAX ATPG having better success in merging them.

For more information, refer to the topic `report_patterns` in the TestMAX™ ATPG and TestMAX Diagnosis Commands guide.=

M516

Message Text

Mapped patterns not merged at pattern `N1` due to inconsistent pattern data size `<E1 E2>` in pattern `N2`. (M516)

Severity

Warning

Description

During the `run_mapping` command, the incoming device patterns were used to create pattern `N2` at the end of the current internal pattern set. Then to reduce the total patterns, an attempt was made to merge pattern `N2` into the existing set of internal patterns at position `N1`.

This message is an indication that merging into pattern `N1` was not possible, because the two patterns had a different sequence of events (size: `E1 E2`). Events are actions such as: `scan_load`, `forcePI`, `pulse_clock`, `measurePO`, and `scan_unload`. Event sequences should be identical to improve the possibility of pattern merging.

Although the new pattern was not successfully merged into pattern `N1`, it does remain in the internal pattern set as pattern `N2`.

What Next

If this warning message occurs frequently, the end result is more patterns than you desire, because merging is not successful. There might be nothing that can be done about this condition, but it also might be possible by reviewing the event sequences between patterns `N1` and `N2` that you can adjust the incoming patterns to be more similar, and this will result in TestMAX ATPG having better success in merging them.

For more information, refer to the topic `report_patterns` in the TestMAX™ ATPG and TestMAX Diagnosis Commands guide.

M517

Message Text

Mapped patterns not merged due to inconsistent clock usage for clock `C`.
(M517)

Severity

Warning

Description

During the `run_mapping` command, the incoming device patterns were used to create pattern `N2` at the end of the current internal pattern set. Then to reduce the total patterns, an attempt was made to merge pattern `N2` into the existing set of internal patterns at position `N1`.

This message is an indication that merging into pattern `N1` was not possible, because the two patterns had inconsistent clock events for clock `C`. The patterns must have identical clock pulses and presence or absence of scan unloads to be merged.

Although the new pattern was not successfully merged into pattern `N1`, it does remain in the internal pattern set as pattern `N2`.

What Next

If this warning message occurs frequently, the end result is more patterns than you desire, because merging is not successful. There might be nothing that can be done about this condition, but it also might be possible by reviewing the event sequences between patterns `N1` and `N2` that you can adjust the incoming patterns to be more similar, and this will result in TestMAX ATPG having better success in merging them.

For more information, refer to the topic `report_patterns` in the TestMAX™ ATPG and TestMAX Diagnosis Commands guide.

M518

Message Text

Mapped patterns not merged at pattern N1 due to unexpected events in pattern N2. (M518)

Severity

Warning

Description

During the `run_mapping` command, the incoming device patterns were used to create pattern N2 at the end of the current internal pattern set. Then to reduce the total patterns, an attempt was made to merge pattern N2 into the existing set of internal patterns at internal position N1.

This message is an indication that merging into pattern N1 was not possible, because the two patterns had different event sequences. Event sequences are a series of actions such as: `scan_load`, `forcePI`, `pulse_clock`, `measurePO`, and `scan_unload`. Event sequences must be identical for the two patterns to be potentially merged.

Although the new pattern was not successfully merged into pattern N1, it does remain in the internal pattern set as pattern N2.

What Next

If this warning message occurs frequently, the end result is more patterns than you desire, because merging is not successful. There might be nothing that can be done about this condition, but it also might be possible by reviewing the event sequences between patterns N1 and N2 that you can adjust the incoming patterns to be more similar, and this will result in TestMAX ATPG having better success in merging them.

For more information, refer to the topic `report_patterns` in the TestMAX™ ATPG and TestMAX Diagnosis Commands guide.

Error and Warning Messages M520 - M529

This section contains the following error messages and warnings:

- [M520](#)
- [M521](#)
- [M522](#)
- [M523](#)

- [M524](#)
- [M525](#)
- [M526](#)
- [M527](#)
- [M529](#)

M520

Message Text

The image file is at a newer level than the TestMAX ATPG build and cannot be used. (M520)

Severity

Error

Description

This error message is issued when a `write_image` command is issued in a version of TestMAX ATPG newer than the version that is currently trying to read the image.

What Next

Use the `report_version` command to determine what version you are trying to read with the image. Then use a newer version of TestMAX ATPG when reading the image.

Contact the image creator and ask what version was used to create the image, or ask for an image version with the same version as you are trying to read the image.

M521

Message Text

Selected command cannot be set to nosecur mode. (M521)

Severity

Warning

Description

The first version of this message is issued when a `set_commands-secure` command was used on a command that was already marked as allowed. In this case, the `set_commands` command had no effect.

The second version of this message indicates that the specified `set_commands-nosecure` command cannot be removed from the allowed command list as it is neutral command. In this case, the `set_commands` command had no effect.

What Next

This warning is for informational purposes only.

M522

Message Text

Selected command was not currently set to secure mode. (M522)

Severity

Warning

Description

This warning message is issued when a `set_commands -nosecure` command was attempted on a command that was not marked as allowed in a secure state. The `set_commands` command had no effect.

What Next

This warning is for informational purposes only.

M523

Message Text

Secure image requires a password. (M523)

Severity

Error

Description

This error message is issued when a `read_image` command was attempted on a secure image, but no password was supplied.

What Next

Use the `-password` option of the `read_image` command along with the correct password to read the image.

M524

Message Text

`Invalid password (<password>) for secure image. (M524)`

Severity

Error

Description

This error message is issued when an incorrect password was provided to a `read_image-password` command while reading a secure image file.

What Next

Be sure that the password is correct by checking with the image file creator. *Note:* Secure images created with TestMAX ATPG versions before V-2004.06 cannot be read by a previous version.

M525

Message Text

`Schematic viewing is not enabled for secure image. (M525)`

Severity

Error

Description

This error message is issued when a command that required the graphical schematic viewer (GSV) was issued while TestMAX ATPG was in a secure state where schematic viewing was not allowed. At the time the image file was created, the `-schematic_view` option of the `write_image` command was not used.

What Next

If schematic viewing is necessary, contact the image creator and ask for a new secure image file with schematic viewing enabled.

M526

Message Text

`Un-secure image does not require a password. (M526)`

Severity

Error

Description

This error message is issued when a password was supplied with a `read_image` command while attempted to read an unprotected image file.

What Next

Reissue the `read_image` command without the `-password` option.

M527

Message Text

Write image cannot be performed in secure command mode. (M527)

Severity

Error

Description

This error message is issued when a `write_image` command was attempted while TestMAX ATPG was in a secure state after reading a secure image. This is not permitted, because the `write_image` command enables creation of an unprotected image file. This implies that distributed ATPG is not allowed with a secure image file.

What Next

To issue a `write_image` command or to do distributed ATPG, an unprotected image must be used.

M529

Message Text

Chain test pattern required for accurate chain diagnosis. (M529)

Insufficient uniquely mapped chains failures, #chain_failures= d. (M529)

The first pattern chain test pattern is N. (M529)

Failed to identify a defective scan chain and defect type. (M529)

Severity

Warning

Description

The first version of the message indicates that a chain test pattern is required to diagnose scan chain failures accurately. The pattern set used for diagnosis does not have a chain test. A less accurate algorithm is used for diagnosis.

The second version of the message indicates that the given number of failures are insufficient to allow the chain diagnosis engine to accurately locate the fault candidate.

The third version of this message indicates that pattern 0 is not a chain test pattern and that the first chain test pattern was found at pattern *N*. The first chain test pattern is used by the `run_diagnosis` command to identify failing scan chain(s) and to identify scan chain defect behavior(s).

The fourth variation of this message occurs during chain diagnostics for DFTMAX Ultra design only when TestMAX ATPG diagnostics cannot find a defective scan and defect type. The set of required conditions for defect identification depends on the test architecture. This message may also occur if a truncation is incorrectly defined or if diagnostics encounters an unmodeled defect type.

What Next

If the first version of the message appears, no action is required. However, if the diagnosis results are not satisfactory, consider doing the following:

Generate a pattern set with the chain test

Re-run the new patterns on the device on the tester, and

Run diagnosis again with the new failure log.

If the second version of the message appears, you need to provide more failures in the data log file to allow the chain diagnosis to accurately locate the fault candidate.

If the third version of the message appears, no action is required. This is an informational message.

Error and Warning Messages M530 - M539

This section contains the following error messages and warnings:

- [M530](#)
- [M531](#)
- [M532](#)
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M530

Message Text

Expected value for failure (<pat_num> <output> <scan_cell_num> exp=<num>, got=<num>) did not match. (M530)

Severity

Warning

Description

This warning message is issued when the `-check_expected_data` option of the `set_diagnosis` command is used during diagnosis, and a mismatch exists between the expected value in the failure data file and the expected value in the pattern. The <pat_num> identifies the pattern on which the mismatch occurred. The <output> is the name of the output at which the failure was observed. The <scan_cell_num> is the scan cell number (if applicable). The `exp=<num>` is the stored and expected value in the failure data file, and the `got=<num>` is the value that TestMAX ATPG got from its simulation. Diagnosis was not performed.

What Next

This warning indicates that there might be a problem in the process of translating the failing tester cycles to the TestMAX ATPG failure data file format. You may attempt diagnosis anyway by removing the `-check_expected_data` option. However, if the failure data file is truly in error, the diagnosis results is untrustworthy. The recommended action is to investigate and correct the translation process, and then rerun diagnosis after correcting the failure data file. Review your `write_patterns` operation for warning messages.

For more information, refer to the topic `set_diagnosis` in the TestMAX™ ATPG and TestMAX Diagnosis Commands guide.

M531

Message Text

Specified subnet id is bigger than #subnets= *d* for this net. (M531)

Specified internal defect id is bigger than #defects= *d* for this pin.
(M531)

Severity

Error

Description

The first version of this message indicates that the subnet id specified in the command `run_simulation -subnet <id>` **does not exist**.

The second version of the message indicates that the internal defect id specified in the command `run_simulation-internal <fault_id>` **does not exist**.

What Next

For the first version of the message, change the subnet id to an existing id. You can use the command `report_layout -subnet` to print a list of existing subnet ids.

For second version of the message, change the internal defect id to an existing id. You can use the command `report_layout-internal_cell_defects` to print a list of existing internal defect ids.

For more information, refer to the following commands in the TestMAX™ ATPG and TestMAX Diagnosis Commands guide:

```
run_simulation
```

```
report_layout
```

M532

Message Text

Ranking is not supported for this fault model. (M532)

Ranking is not supported in this mode. (M532)

Option `-chain_failure` is not supported in this mode. (M532)

Severity

Error

Description

The `-rank_faults` option is only supported in scan mode for stuck-at, transition and bridging faults.

Chain diagnosis is not supported in compression mode.

What Next

If an unsupported fault model is being used, consider changing the fault model and fault list to a supported one before re-running diagnosis with the `-rank_faults` option.

Do not use the `-chain_failure` option when the design is in compression mode. Only functional logic diagnosis is supported.

M533

Message Text

The initial fault list is empty (M533)

Severity

Error

Description

You executed the `run_diagnosis` command with an option that expects faults in the active faults list, but the active faults list was empty.

What Next

You must generate the fault list before using the `-use_fault_list` or `-rank_fault` options. Use the `read_faults` or `add_faults` commands before you run this command again. A previous `run_diagnosis` command with the `-create_fault` option generates the fault list.

M534

Message Text

Unsupported platform (M534)

Severity

Warning

Description

This command was attempted on an unsupported hardware platform. So, the command was not successful.

What Next

Run TestMAX ATPG on a supported platform and specify the command again. Supported platforms include Solaris Sparc (32 and 64 bit) and amd64.

M535

Message Text

Incorrect cell name or version. (M535)

Incorrect library name: <library_name>. (M535)

Severity

Error

Description

The first version of this message is issued if you specified a non existing top-level Milkyway cell name using the `read_layout-cell_name` command or if you specified a non-existing version of the Milkyway cell to be loaded using the `read_layout-version` command.

The second version of this message is issued if you specified a non existing top-level Milkyway library name using the `read_layout` command.

What Next

Specify either a correct Milkyway top-level library, cell name, or version.

M536

Message Text

Diagnosis aborted due to CPU time exceeding the `set_diagnosis -time_limit`. (M536)

Diagnosis aborted due to inconsistent behavior during and after chain test. (M536)

Scan chain diagnosis aborted due to exceeding the maximum number of defective chains. (M536)

Severity

Warning

Description

The first variation of this message is issued if the user-specified CPU time limit is exceeded or if the diagnosis process is manually interrupted. This is an informational message only. No action is required if you intentionally aborted the diagnosis.

The second variation of this message is issued if the diagnosis engine stops because the failures encounter conditions that make the diagnosis inconclusive. This occurs when the chain test patterns are passing, but there are multiple failures in the capture patterns.

For DFTMAX compression, this behavior often results if only a few failing patterns are mapped, and several other patterns are skipped due to the excessive number of masked (unmapped) shift cycles.

The inconclusive diagnosis conditions indicate that it is unlikely these failures are produced by a logic defect. If these failure logs are used by the diagnosis, they consume considerable CPU time, and the results are often inconclusive. Thus, the diagnosis engine filters the log files of these bad failures.

The third variation of this message is issued if the chain diagnosis finds that the number of defective chains is greater than the maximum set by default.

What Next

If you want the diagnostics to continue, specify `set_diagnosis -notime_limit` to cancel any `set_diagnosis -time_limit` settings, and do not interrupt the process.

You can also try to increase the maximum number of defective scan chains to use for diagnostics using `set_diagnosis -max_defective_chains` command.

M537

Message Text

Pattern source contains patterns which cannot be simulated (M537)

Severity

Warning

Description

This message appears when you simulate patterns types that are not supported by the simulator.

For example the following are not supported:

"run simulation -seq" on patterns generated in compression mode

"run simulation -seq" on patterns generated with shared-scan in

What Next

No action needs to be performed if test-coverage is satisfactory. If you are getting lower test-coverage compared to regular scan ATPG, consider the following techniques:

Run an incremental ATPG. This will retarget any faults that were left as ND (not-detected) faults due to X's

Disable disturb clock grouping (`set_drc -nodisturb`) and rerun ATPG.

M538

Message Text

Pattern format is not supported for current operation. (M538)

Pattern type is not supported for `-split` option, `pat= name`. (M538)

Severity

Error

Description

A netlist-independent pattern file with pattern types other than basic-scan and fast-sequential was read. The Adaptive Scan pattern translation flow supports only basic-scan and fast-sequential patterns.

What Next

Create a new netlist-independent pattern file that includes only basic-scan and fast-sequential patterns. Use the `report_patterns` and `write_patterns` commands to examine pattern types and to limit the patterns written out.

Error and Warning Messages M540 - M549

This section contains the following error messages and warnings:

- [M540](#)
- [M546](#)
- [M547](#)
- [M548](#)

M540

Message Text

Parallel Verilog with serial shifts cannot be written after read_image.
(M540)

Severity

Error

Description

Parallel Verilog patterns with serial shift(s) [-parallel n_shifts] cannot be written after an image was read. Only serial and fully-parallel patterns are supported with images. The write_patterns operation is aborted after this message.

What Next

Consider writing the patterns in STIL. The serial and parallel mode can be overridden when invoking simulation by specifying simulation command-line options.

The command-line option is:

```
+define+txmax_parallel=N
```

This option forces parallel-load simulation of all scan data, with N bits extracted and serially simulated. This overrides the behavior of a testbench written with the -serial option, and will override the value of N of a Verilog DPV testbench written with the -parallel option. If N is not specified, then it is processed as zero (all bits parallel-loaded).

M546

Message Text

PLL internal clock cannot be defined on a <primitive_type> primitive.
(M546)

Severity

Error

Description

Internal clocks can only be defined on the output of a combinational primitive (buf, inv, and, nand, or, nor, xor, xnor, mux) or a TIEX.

What Next

Trace the design schematic and choose an accepted primitive to define the internal clock.

M547

Message Text

```
Invalid number of cycles (<number_of_cycles>) were specified for PLL  
internal clock (<clock_name>)
```

(M547)

Severity

Warning

Description

This message appears when there is a mismatch between the number of cycles defined with the `-cycle` option of the `add_clocks` command and the number of cycles specified with the `set_drc-num_pll_cycles` command.

What Next

Verify that you defined the number of PLL capture cycles (using the `set_drc-num_pll_cycles` command) before defining the internal clocks with the `add_clocks-cycle` command.

Verify that the number of PLL cycles defined with the `set drc -num_pll_cycles` command matches the number of cycles specified with the `-cycle` option of the `add_clocks` command.

M548

Message Text

```
net is already in the PLL internal clock list. (M548)
```

Severity

Warning

Description

The indicated `net` has already been defined as an Internal clock. This definition may appear in either or both of the following two places:

An `Internal` statement in the `ClockStructures` block of the STIL Procedure File.

The main argument of the `add_clocks-intclock` command.

What Next

This warning indicates that the two definitions are redundant. No immediate action is required. The last definition encountered takes precedence. If the STL procedure file contains these definitions, the last definition is defined from the STL procedure file constructs. This allows the STL procedure file data to be written out and read back as the primary definition. You should remove all redundant definitions for consistent operation. The definitions are easier to maintain when redundancies are deleted; this ensures that any future changes only have to be made in one place.

Error and Warning Messages M550 - M559

This section contains the following error messages and warnings:

- [M550](#)
- [M551](#)
- [M552](#)
- [M553](#)
- [M554](#)
- [M555](#)
- [M556](#)
- [M557](#)
- [M558](#)
- [M559](#)

M550

Message Text

PLL condition already exists for cycle *cycle_number* on gate *gate_number*.
(M550)

Severity

Warning

Description

The indicated block ID is already defined as a control for the indicated cycle number for the same internal clock. This definition might appear in either or both of two different places:

A `Cycle` statement in the `ClockStructures` block of the STIL Procedure File.

An argument of the `-cycle` option of the `add_clocks-intclock` command.

What Next

This warning indicates that the two definitions are redundant. No immediate action is required. The first definition takes precedence, and the `add_clocks` command always overrides the STL procedure file definition. You should remove all redundant definitions for consistent operation. The definitions are easier to maintain when redundancies are deleted; this ensures that any future changes only have to be made in one place.

M551

Message Text

```
PLL DRC clock cannot be defined on a <type> primitive (M551)
```

Severity

Error

Description

This message is issued when an attempt is made to add a PLL DRC clock on a node that is the output of an invalid primitive type (`<type>`). The following gate types are allowed: BUF, INV, AND, NAND, OR, NOR, XOR, XNOR, MUX, TIEX

A similar message ([M546](#)) is issued when an attempt is made to define a PLL internal clock on an invalid primitive type.

What Next

PLL DRC clocks are only used for DRC purposes and there is no requirement that they be defined. If they are defined, pulses on these clocks are simulated during the simulation of procedures in the test procedure file. Depending on the implementation and the modeling, it might be necessary to successfully pass other DRC rules, including tracing internal chains. These clocks are also checked to ensure that their signals successfully propagate to their associated PLL internal clock.

This problem can be corrected by defining a different node name for the PLL DRC clock that is logically equivalent, but on a valid primitive type. It is also possible to change the netlist to include a buffer that can carry this signal. Since PLL DRC clocks are not required

for ATPG, the problem can often be resolved by simply removing the PLL DRC clock definition.

M552

Message Text

`net` is already in the PLL DRC clock list. (M552)

Severity

Warning

Description

The indicated `net` is already defined as a PLL clock. This definition may appear in either or both of the following places:

A `PLL` statement in the `ClockStructures` block of the STIL Procedure File.

The argument of the `-pll_source` option of the `add_clocks-intclock` command.

What Next

The warning indicates that the two definitions are redundant. No immediate action is required. The last definition encountered takes precedence. If the STL procedure file contains these definitions, the last definition is defined from the STL procedure file constructs. This allows the STL procedure file data to be written out and read back as the primary definition. You should remove all redundant definitions for consistent operation. The definitions are easier to maintain when redundancies are deleted; this ensures that any future changes only have to be made in one place.

M553

Message Text

`-cellnames` option not supported for `-format stil99`, ignored. (M553)

Severity

Warning

Description

The `write_patterns-format stil99` option can only be used when `-cellnames < internal | verilog >` is set.

What Next

The default `write_patterns-cellnames verilog` setting is applied to the generated patterns. To eliminate the warning, remove the `-cellnames` setting or use a value supported for this format.

M554

Message Text

`Cannot write patterns from designs with no inputs defined. (M554)`

Severity

Error

Description

The current design has no primary inputs defined, and patterns cannot be written for this situation. Note that designs with no inputs have no controllability opportunity and any patterns possible could only represent a fixed behavior.

What Next

Either define a dummy input or present a meaningful design for TestMAX ATPG operation.

M555

Message Text

`PLL clock activity did not stabilize in number passes. (M555)`

Severity

Error

Description

During DRC, PLL clocks are simulated as free-running, internal clocks until there are no more events to simulate or a maximum number of passes (the default is 100) has been reached. In the latter case, an M555 warning is issued. DRC might still pass after this warning, although more severe checks might subsequently fail.

What Next

If there are counters in the clock controller which require more than 100 cycles to reach their final value, use `set_drc-max_pll_simulation_passes <N>` to increase the limit.

M556

Message Text

PLL clock simulation period was not defined. (M556)

Severity

Error

Description

This is an error that can result if DRC was run on an incomplete specification (STL procedure file or Quick-STIL).

What Next

Ensure that a period is defined for all events..

M557

Message Text

Invalid conditioning value (<value>) for PLL internal clock (*number*) in cycle *number*. (M557)

Severity

Error

Description

This error is issued if a value other than 0 or 1 was defined for the named internal clock.

What Next

Correct the definition using only values of 0 or 1.

M558

Message Text

Invalid PLL always-on conditioning: Number of cycles (*number*) is not set to 2. (M558)

Invalid PLL always-on conditioning: Always-on conditioning cannot be mixed with gate conditioning. (M558)

Invalid PLL always-on conditioning: Different conditioning for clock *name* in cycle pairs *name* and *name*. (M558)

Severity

Error

Description

In the first version of the messages listed above, the Always-on internal clocks are only supported for 2-clock ATPG; this error is issued for any other type of ATPG (basic-scan or fast-sequential with more than 2 cycles).

The second message listed above is issued if an internal clock definition mixes always-on conditioning with gate-conditioning.

The third message listed above is issued if an internal clock is defined with different conditions for the two cycles.

What Next

For the first message above, use only 2-clock ATPG or correct the definition of the internal clock.

For the second message above, a clock is either always on, or conditionally on; therefore, you should choose one only and correct the definition.

For the third message above, fix the clock definition so that so it doesn't have different conditions for the two cycles.

M559

Message Text

Unconstrained reference clock *clock_name* is now constrained to its off state. (M559)

Severity

Warning

Description

A reference clock was defined and is capable of pulsing during capture when internal clocks are defined in the design. Because the relative placement of this clock is unknown with respect to the internal clocks, this clock is constrained in the inactive state for test generation purposes. The clock pulses in the patterns, as required by its reference clock designation.

What Next

To remove the M559 message (in the presence of internal clocks), you need to constrain this signal to the inactive state using the `add_pi_constraintsinactive_valueclock`

command. Although the M559 message will not appear, the problems related to constraining this signal during the capture operations will still be present.

To use this reference clock as a capture clock for ATPG, you must convert it from a free-running clock to a clock that can be paused. This will prevent the internal clocks from working, so these clocks must be bypassed by external clocks. To do this, you must change the protocol to remove any FreeRunning blocks and extra pulses in the Procedures block, including the pre-shift vectors in the load_unload procedure and the F blocks in the capture procedures. If the protocol is an STIL procedure file written by TestMAX DFT, you can make these same changes using the `-patternexec` option of the `run_drc` command.

For more information, refer to the topic C39 in the TestMAX™ ATPG and TestMAX Diagnosis DRC Rules guide.

Error and Warning Messages M560 - M599

This section contains the following error messages and warnings:

- [M561](#)
- [M562](#)
- [M563](#)
- [M564](#)
- [M565](#)
- [M566](#)
- [M568](#)
- [M580](#)
- [M590](#)
- [M597](#)
- [M598](#)
- [M599](#)

M561

Message Text

Chain test can not be created with PLL clocking without the clock-off pattern. (M561)

Severity

Warning

Description

The chain test must be applied without capture clocks during the capture cycle. But internal clocks have been defined in a way that requires at least one of them to pulse in every capture sequence. The most common reason is an internal clock definition with ALWAYS ON conditioning.

This message warns that no chain test is generated in this pattern set. However, other patterns is generated as usual.

What Next

The message can be suppressed by setting the command `set_atpg -chain_test off`.

It is recommended that you generate a chain test for each configuration of the scan chains and shift clocks, so the chain test should only be turned off when it is impossible to generate it.

For more information, refer to the topic C37 in the TestMAX™ ATPG and TestMAX Diagnosis DRC Rules guide.

M562

Message Text

Number of defined PLL capture cycles (<number_of_cycles>) is less than currently allowed ATPG capture cycles (<number_of_cycles>). (M562)

Severity

Error

Description

This is error is issued when a `set_atpg` command attempts to set a number of ATPG capture cycles that exceeds the number of PLL capture cycles.

What Next

Correct the PLL capture cycles and/or the ATPG capture cycles.

M563

Message Text

No reference clocks have been defined. (M563)

Severity

Warning

Description

This warning is issued if PLLs are present but no reference clocks have been defined.

What Next

If there are reference clocks in the design, you must ensure they are pulsed in every tester cycle to keep the PLLs synchronized.

M564

Message Text

ATPG for PLL always-on clocking requires transition fault model with last shift launch. (M564)

ATPG for PLL always-on clocking requires transition fault model with system clock launch. (M564)

ATPG for PLL always-on clocking cannot be performed with custom clocking. (M564)

Severity

Error

Description

Internal clocks that are defined to be always-on (for example, from the command `add_clocks-intclock -cycle N -always_on`), will place restrictions on how the patterns are generated. These restrictions are critical with the transition fault model, which requires a particular number of capture cycles (and capability to define the capture clocks) due to the nature of the test. Also, custom clocking operations do not support the always-on option at this time.

What Next

This is a design restriction that affects the options for pattern generation. To remove the error, set the appropriate launch condition with the command `set_delay-launch_cycle (last_shift OR system_clock)`. If pattern coverage is not sufficient, consider modifying the design to remove the always-on condition.

M565

Message Text

Number of defined PLL capture cycles (1) is less than required for `system_clock launch (2); last_shift` used. (M565)

Severity

Error

Description

When `set_delay-launch_cycle (system_clock OR any)` is used, a minimum of 2 capture cycles is required when `system_clock` patterns are generated. If the number of PLL capture cycles is less than this requirement, `system_clock launch` cannot be used. In this situation, only `last_shift launch` can be used to create patterns. The user selection (of `launch_cyclesystem_clock`, or `any`) is overridden by the `launch_cyclelast_shift` option.

What Next

This is a design restriction that affects the options for pattern generation. To remove the error, change the `set_delay-launch_cycle` command to use `last_shift`. If `last_shift` pattern coverage is not sufficient, consider increasing the number of PLL capture cycles supported in the design, to allow `system_clock launch` to be supported.

M566

Message Text

Constraints *name* have no solution. No patterns are generated unless the constraints are removed or modified. (M566)

Severity: Warning

Clock constraints *name* have no solution. (M566)

Severity

Error

Description

The first version of this message is printed during the `run_drc` process.

The second version of this message is printed during the `run_atpg` process.

What Next

To resolve both versions of the M566 message, you should specify the `report_clocks-constraints` command, as shown in the following example:

```
TEST-T> report_clocks -constraints
```

```
-----  
Clock Constraints constraints1:  
  have no solution because of the following external constraints:  
    dutc1/FF_0_reg = 1  
    dutc1/FF_1_reg = 1  
    dutc2/FF_0_reg = 0  
    dutc2/FF_1_reg = 0  
    dutc2/FF_2_reg = 0
```

In the case of the example, the cells were constant-value nonscan cells, which were defined in the clock instruction register but whose constant state did not correspond to any of the defined clocking procedures. Note that clock constraints may have no solutions for reasons other than external constraints, that is, ill-defined, depth mismatches, and ambiguity.

M568

Message Text

Clock constraints *name* limits the maximum capture cycle to *number*. (M568)

Severity

Warning

Description

This message prints at the start of the `run_atpg` process when the "Maximum sequential depth" setting reported by the `report_clocks -constraints` command is smaller than the "capture_cycles" setting reported by the `report_settings atpg` command .

What Next

No action is required. Pattern generation will continue using the clocking procedures provided.

M580

Message Text

N faults were aborted due to using excessive care bits during simulation pass. (M580)

Severity

Warning

Description

This is an informational message only.

M590

Message Text

Pattern compression cannot be performed for circuits with compressors.
(M596)

Severity

Error

Description

This error is due to running the obsolete `run pattern_compression` command.

M597

Message Text

Compressors which use XOR circuitry are not currently supported. (M597)

Severity

Error

Description

This error appears in ScanCompression mode when there is a problem with the decompressor structure inside a codec. DFTMAX supports only decompressors constructed with MUXes.

What Next

Inspect the netlist and determine if a Decompressor section contains any XOR gates. This condition sometimes occurs during the post processing flow if you are using a scan-inserted ASCII netlist.

If this error appears with the other R violations (such as R7, R8, or R31), there might be a problem with the decompressor architecture in the STIL procedures file. In the following example decompressor, notice that chain 13 is driven from multiple decompressor inputs:

```
Mode 0 {  
  ModeSerializerControls {  
    ...  
  }  
  
  Connection 0 "9";  
  ...  
  ...  
  Connection 5 "13" "28" "38" "39" "45" "64" "67" "68" "92" "95" "114"  
    "396" "404" "412" "427" "439" "452" "472";  
  ...  
  ...  
  Connection 12 "13" "22" "32" "52" "66" "77" "94" "97" "118" "124"  
    "403" "426" "436" "443" "459" "463" "478";  
}
```

The architecture in this example is illegal. For any given mode in a decompressor (in this case, Mode 0), a single internal chain can be driven by only one input.

M598

Message Text

Warning: *number* scancells were not observable due to compressor effects during simulation interval. (M598)

Severity

Warning

Description

This message indicates that data captured in some scan cells was blocked from reaching external scan-output ports. This normally occurs if you have X-sources in the design. Normally, faults that are masked are available to be picked up by fault simulation as ATPG proceeds. M598 is reported after every simulation internal while running ATPG only when expert-level reporting is enabled (`set_messages -levelexpert`).

What Next

No action needs to be performed if test-coverage is satisfactory.

If you are getting lower test-coverage compared to regular scan ATPG, please consider the following techniques:

Run an incremental ATPG. This will retarget any faults which were left as ND (not-detected) faults due to X's.

Disable disturb clock grouping (`set_drc -nodisturb`) and rerun ATPG.

M599

Message Text

Pattern set does not contain pattern used for short chain unload values.
(M599)

Severity

Error

Description

At the end of the ATPG process for DFTMAX circuits, an analysis is performed to calculate the correct unload values that result from the unload compressor. For short chains, the unload value for a given pattern depends on the load value in the next pattern. If there is no succeeding pattern, then this message is issued. This analysis is also performed when using the `-store` option of the `run_simulation` command.

What Next

When this failure occurs, the compressor unload values are likely to be bad. The ATPG process should always place padding patterns to provide the additional pattern necessary to satisfy this check. If the message occurs, this normally indicates a bug and a STAR should be filed.

8

Error and Warning Messages M600 - M699

This section contains the following error messages and warnings:

- [Error and Warning Messages M600 - M629](#)
- [Error and Warning Messages M630 - M639](#)
- [Error and Warning Messages M640 - M649](#)
- [Error and Warning Messages M650 - M659](#)
- [Error and Warning Messages M660 - M669](#)
- [Error and Warning Messages M670 - M679](#)
- [Error and Warning Messages M680 - M689](#)
- [Error and Warning Messages M690 - M699](#)

Error and Warning Messages M600 - M629

This section contains the following error messages and warnings:

- [M601](#)
- [M604](#)
- [M609](#)
- [M610](#)
- [M620](#)
- [M622](#)
- [M623](#)
- [M624](#)
- [M625](#)
- [M626](#)

- [M627](#)
- [M628](#)

M601

Message Text

Pattern N does not contain correct compressor <load | unload> data.
(M601)

Severity

Error

Description

At the end of the ATPG process for DFTMAX circuits, an analysis is performed to calculate the correct unload values that result from the unload compressor. The patterns must contain compressor load and unload data. If this data is missing, then this message is issued. In the message text, N is the 0-based pattern ID number in which the failure occurred. This analysis is also performed when using the `-store` option of the `run_simulation` command.

What Next

When this failure occurs, the compressor unload values are likely to be bad. The ATPG process should always place compressor load and unload data into patterns to satisfy this check. If the message occurs, this normally indicates a bug and a STAR should be filed.

M604

Message Text

Compressor analysis cannot be performed for circuits which already
contain compressors. (M604)

Compressor analysis cannot be performed with simulation of last shift
cycles. (M604)

Severity

Error

Description

The first version of this message appears when you run the `analyze compressors` command after you run scan-compression mode DRC. This is not allowed. This message appears when you simulate patterns types that are not supported by the simulator.

The second version of this message appears when you run the `analyze compressors` command with a non-zero value specified by the `set_simulation -shift_cycles` command. This not allowed.

What Next

For the first version of this message, go back to DRC mode and rerun regular scan-mode DRC. You can then use the `analyze compressors` command to do scan compression what-if analysis.

For the second version, make sure the `set_simulation -shift_cycles` command is set to a value of 0 (the default), then rerun the `analyze compressors` command.

M609

Message Text

Internal pipeline gate *gate_id* stage *number* previously traced in stage *number*. (M609)

Severity

Error

Description

This message indicates that an incorrect design was identified during pipeline tracing.

What Next

To fix this problem, you might need to change the netlist or the STL procedure file. The STL procedure file might require changes to the PI settings in the `load_unload` procedure. If internal registers are used to hold the test mode state, you might need to change the `test_setup` procedure.

M610

Message Text

Specified power budget (*pb_percent*) is less than the minimum recommended low-power ATPG budget (*atpg_b_percent*) (M610).

Severity

Warning

Description

This message indicates that ATPG might not be able to achieve the specified power budget set by the `set_atpg -power_budget` command. If the `set_atpg -power_effort high` command is set, it likely that many or all patterns are rejected.

What Next

Either set the `set_atpg -power_budget` command to the recommended minimum percentage, or add more scan-controllable clock-gating cells to your design to lower the recommended minimum setting.

M620

Message Text

```
Warning: #candidate chains (d1) and #connecting chains (d2) for unload  
compressor U_compressor are different. (M620)
```

Severity

Warning

Description

This message indicates that a problem in tracing compressor mode scan chains. d2 is the number of chains specified in the protocol file and d1 is the number of scan-flops that TestMAX ATPG found when tracing back from scan-out ports.

What Next

This warning message appears when you do not set constraints to sensitize the compressor mode correctly. If you are using the initialization sequence to initialize a test-mode register, check the value of the register after `test_setup` procedure.

If d1 is 0, invoke any scan-out port in the GUI (using the `add_display_gates` command) and set up your pin data to shift (using the `set_pindatashift -shift_char S` command). Trace back from the scan-out port through the XORs and see if you reach any scan-flop.

For more information, see the following commands in the TestMAX™ ATPG and TestMAX Diagnosis Commands guide:

```
add display gates
```

```
set pindata
```

M622

Message Text

X candidate chain outputs for chain *chain_name* were found (assigned output is *gate_id*) (M622).

Severity

Warning

Description

This warning is issued when multiple (x) internal chains connect to the same POs (via the unload compressor) as chain *chain_name* as defined in the STL procedure file. The chains are indistinguishable based on their output connections. Therefore, the tool must pick one of the candidates (in this case it picked the internal chain whose output gate is gate ID *gate_id*) and this is considered chain *chain_name* there-on.

What Next

No action is required. However, you should be aware that the chain identified as chain *chain_name* by the tool might not be the chain you referred to as chain *chain_name*.

For more information, see the topic [M623](#).

M623

Message Text

Chain <chain_name1> has same output connections as chain <chain_name2> (assigned output is *gate_id*). (M623)

Severity

Warning

Description

This warning is issued when multiple internal chains (e.g., chains <chain_name1> and <chain_name2>) connect to the same POs (via the unload compressor) as defined in the STL procedure file. The chains are indistinguishable based on their output connections. Therefore, the tool must pick one of the candidates (in this case it picked the internal chain whose output gate is gate ID *gate_id*) and this is considered chain <chain_name1> there-on.

What Next

No action is required. However, you should be aware that the chain identified as chain `<chain_name1>` by the tool might not be the chain you referred to as chain `<chain_name1>`.

For more information, see the topic [M622](#).

M624

Message Text

Chain `chain_name` with output `output_gate_id` doesn't match load compressor connections. (M624)

Severity

Warning

Description

This message can be issued during internal chain IO identification in a DFTMAX design during ambiguity resolution. Ambiguity occurs when multiple chains have the same output (unload compressor) connections, so they cannot be differentiated based on output connections alone. In this case, the input (load decompressor) connections of all chains in an ambiguity class (chains with same output connections) are analyzed.

For example, take a case in which chains `c1` and `c2` both connect (based on STL procedure file) to outputs 1, 4 and 7; TestMAX ATPG starts tracing the netlist backwards from the outputs and finds two chains, with internal output gates `id1` and `id2`, respectively. Either one can be `c1` or `c2`. Next, say the STL procedure file further specifies that `c1` connects to inputs 0, 3, 5 in modes 0, 1, 2, respectively; while `c2` connects to inputs 1, 3, 6. This resolves the ambiguity, so the chain which connects, based on netlist tracing, to inputs 0, 3, 5 is termed `c1`, and the other `c2`. However, assume that netlist tracing reveals that one of the two chains connects to inputs 4, 5, 6 — this does not match either load compressor connections, as defined in the STL procedure file, and an M624 message is issued.

If R7 violations are generated, M624 messages might also occur. R7 violations need to be resolved, but tracing M624 messages is more direct than analyzing the R7 failures. The process for tracing M624 messages is explained in the "What Next" section below.

If M624 messages are generated without R7 violations, then you can ignore these messages, and the remaining flow, including pattern generation, is correct. However, M624 messages will probably identify conditions of redundant logic to the connections on internal scan-inputs. It might be valuable to review these reports for opportunities to improve testability of this logic. Note that DI credit is associated with the logic on scan inputs.

What Next

Analyze the netlist (in the schematic viewer) starting with the output and input gates of the chain `chain_name` and the STL procedure file (in an editor), and find the cause of the mismatch.

You can specify the `set_drc -compressor_debug_data` command and the `run_drc` command to generate more information about the initial M624 message you encounter. The `-compressor_debug_data` option will help isolate any disconnections between the STL procedure file and the design. The following sequence is representative of the operations used to trace an M624 message back to the root cause of the design elements and the design state:

After applying `set_drc -compressor_debug_data`, use `run_drc` to see detailed information on the first M624:

As shown above, `unload_mode=1`, `chain=44` connects to two inputs instead of one. Now you need two pieces of information:

Use `report_scan_chains` to identify the input gate-id of chain "44"; in this example the input of chain "44" is gate `id=23676`.

Perform the following sequence to store and see the state of `unload_mode 1`:

In the GSV, trace back from gate `id=23676` and observe the incorrect gating for `load mode=1`.

M625

Message Text

```
Chain <chain_name1> with output <gate_id1> has been reassigned output  
<gate_id2> from chain <chain_name2> without ambiguity. (M625)
```

Severity

Warning

Description

This warning is issued when multiple internal chains connect to the same POs (via the unload compressor). The chains are indistinguishable based on their output connections. However, the chains differ in their input connections. Therefore, the ambiguity is resolved based on distinct input connections and the previous association (chain `<chain_name1>` with output `<gate_id2>`) is re-done (chain `<chain_name1>` now with output `<gate_id2>`, previously assigned to chain `<chain_name2>`).

What Next

No action is required.

For more information, see topics [M622](#) and [M623](#).

M626

Message Text

Chain *chain_name* with output *<output_gate_id>* is still ambiguous with *number* other chains. (M626)

Severity

Warning

Description

This message is issued during the internal chain IO inferring phase of DRC. It indicates that the named chain could not be differentiated from a number of other chains since they all have the same compressor connections. This condition is common when the number of internal chains is larger than the number of unique combinations of output pins, which is $\text{power}(2, \# \text{outputs})$.

What Next

No action is required.

For more information, see the topic [M627](#).

M627

Message Text

Chain *chain_name1* with output *output_gate_id1* has been reassigned output *output_gate_id2* from chain *chain_name2* with a *number-chain* ambiguity. (M627)

Severity

Warning

Description

This message is issued during the internal chain IO inferring phase of DRC and indicates that chain *chain_name1* has been reassigned, as indicated.

This message is common when the number of internal chains is larger than the number of unique combinations of output pins, which is $\text{power}(2, \# \text{outputs}) - 1$.

In this case, a number of chains have the same unload compressor connections. DRC starts from the chain outputs, and, because no differentiation can be made at this point, arbitrarily assigns `chain_name1` to internal chain output `output_gate_id1` and `chain_name2` to internal chain output `output_gate_id2`. After tracing to the input of the chains, DRC finds that the load decompressor connections are different and now permits a resolution of the ambiguity, so `chain_name1` is reassigned output `output_gate_id2` from chain `chain_name2`.

What Next

No action is required.

For more information, see the topic [M626](#).

M628

Message Text

```
Number of ambiguous chains ( number) did not equal number of ambiguous  
chain outputs (number). (M628)
```

Severity

Error

Description

This error occurs when the name of a core chain in scan structure does not match a name in the compressor structure.

What Next

Make sure that your STIL procedure file is written out by DFT MAX, and that it has not been tampered.

For additional related information, see the article "[Debugging an M628 Message](#)" in *Synopsys SolvNet*.

The following example demonstrates a case in which a name of a core chain in the scan structure does not match a name in the compressor structure:

ScanStructures:

```
ScanChainGroups {  
    core_group {  
        "1" ;  
        "2" ;  
        "3" ;  
        "4_err" ;  
        "5" ;  
    }  
}
```

```
    "6" ;  
}
```

CompressorStructures:

```
Compressor U_decompressor {  
    ModeGroup mode_group;  
    LoadGroup load_group;  
    CoreGroup core_group;  
    Mode {  
        Connection 0 "1" ;  
        Connection 1 "2" ;  
        Connection 2 "3" ;  
        Connection 3 "4" ; <--- Note, should be 4_err  
        Connection 4 "5" ;  
        Connection 5 "6" ;  
    }  
}
```

Error and Warning Messages M630 - M639

This section contains the following error messages and warnings:

- [M630](#)
- [M632](#)
- [M633](#)
- [M634](#)
- [M636](#)
- [M637](#)
- [M638](#)
- [M639](#)

M630

Message Text

Pattern translation is not supported in this mode. (M630)

Severity

Error

Description

This error occurs when TestMAX ATPG tries to read a netlist/mode-independent pattern set when the design is not constrained to scan mode.

What Next

The Adaptive Scan pattern translation flow is only supported in scan mode. Use the appropriate STIL procedure file for scan mode and rerun DRC.

M632

Message Text

```
<N> patterns need more than 10 frames were skipped (pat=<pat_name>).  
(M632)
```

```
<N> patterns were skipped (pat=<pat_name>). (M632)
```

Severity

Error

Description

The first version of the message indicates that <N> patterns need more than 10 frames to be simulated. But the diagnosis simulator could simulate pattern with at maximum 10 clock frames. As a result, the failures for that pattern are skipped. (pat=<pat_name>) indicates the first skipped pattern.

The second version of the message indicates that the pattern format is unsupported for diagnostics. As a result, the failures for that pattern are skipped. (pat=<pat_name>) indicates the first skipped pattern.

What Next

For the first case, regenerate the patterns with less than 10 clock frames.

For the second case, the patterns need to be regenerated by taking into account the current diagnosis limitation for the scan technology used.

M633

Message Text

```
Diagnosis will use X PO masks and d capture masks. (M633)
```

```
N passing patterns were ignored during chain diagnosis (pat1=n). (M633)
```

Both values (0 and 1) were unloaded from the last scan cell in chain=C.
(M633)

Severity

Warning

Description

The first version is an informational message that indicates that the TestMAX ATPG session currently has PO or capture masks in place that will affect the diagnosis results. This is due to commands issued earlier in the session, such as `add_capture_masks` or `add_po_masks`. These masks might cause the diagnosis settings to be different from that on the tester. Some failures which were observable on the tester might be masked during diagnosis, resulting in inaccurate results.

The second version of the message indicates that `N` passing patterns were ignored during chain diagnosis. The first passing pattern number to be ignored is indicated by `pat1=n`.

If a chain defect exists, every pattern typically has at least a few failures. Due to the tester memory limit, the failure data is often incomplete. Therefore you should use the `set_diagnosis -incomplete` command by default. If you forget to set this option and the failures are incomplete, diagnostics will report several consecutive failing patterns followed by all “passing” patterns. This will increase the TPSF and will impact the matching score. For this reason, if the failures log file does not include failures for more than one pattern read in the external pattern buffer, in particular the last one, the chain diagnosis will automatically print this message.

The third version of this message indicates that the chain diagnosis engine failed to find the first position of the failing scan cell in chain C. For at least one case, the diagnosis found that both sides of the defect polarity are unloaded. When the chain diagnosis engine searches for a failing position for stuck-at faults, it tries to find the first cell in which the unload value is always 0 or 1. If both values from each cell are unloaded, there is some inconsistency in the failing data or some other issue. For example, some tester failures might be missing. Missing failures might imply that 0 and 1 were unloaded when actually not all failures were considered.

What Next

If you received the first version of this message, no action is required. This message is only informational, indicating that diagnosis results might be negatively affected due to masks in the TestMAX ATPG session.

You should consider using `remove_capture_masks` or `remove_po_masks` so that only the pattern set will determine detection of faults.

If you received the second version of this message, you should use the `set_diagnosis -incomplete` command.

If you received the third version of this message, you should use the maximum possible number of tester failures and select the valid truncation method. You can use the `set_diagnosis -incomplete` command or the `set_diagnosis -per_pin_limit` command.

M634

Message Text

Required debug data was not stored when layout was read. (M634)

Severity

Error

Description

This error indicates that the `-debug` option of `read_layout` was not used when the layout was read. You cannot specify `report_layout` before storing the necessary data during `read_layout`.

What Next

You must run `read_layout` with the `-debug` option to store data required by `report_layout`. Then, reissue your `report_layout` command.

M636

Message Text

Mixing different net pairs type is not allowed. (M636)

Severity

Error

Description

This error occurs when TestMAX ATPG tries to add new bridging net pairs to existing ones and the pair types are different. For example:

```
TEST-T> read_layout file1 -likely_pairs
```

```
TEST-T> read_layout file2 -all_pairs -add
```

What Next

Make sure the bridging net pairs have the same pairs type and reissue the command.

M637

Message Text

TestMAX ATPG and Milkyway top modules are different <tmax_top_module>
<milkyway_top_module>. (M637)

Severity

Warning

Description

This warning occurs when the TestMAX ATPG top module name is different from the Milkyway database top module name when you are reading a Milkyway database to extract layout information.

What Next

To avoid layout information extraction issues, make sure all databases are as compliant as possible.

M638

Message Text

number candidates having scores smaller than <min_report_score> were not reported. (M638)

Severity

Warning

Description

This message will appear when a *number* of fault candidates exist but were not reported. This is because the match scores for these candidates were less than the <min_report_score>. Note, however, that at least one fault candidate per defect is reported (the candidate with the maximum score).

What Next

To see the unreported fault candidates, change the `-min_report_score` option of the `set_diagnosis` command.

M639

Message Text

Multiple pattern files read requires directive `.pattern_file_name`. (M639)

Current split pattern data is now deleted. (M639)

Severity

Error

Description

The first version of this message indicates that one of the multiple failures log files does not use the mandatory directive `.pattern_file_name`. This directive is necessary to map a single failure file to a split pattern file.

The second version of this message occurs if split patterns were read in before the current operation.

What Next

In the case of the first version of this message, correct the faulty failures file by specifying the `.pattern_file_name` directive and its value.

For the second version of this message, no action is required.

Error and Warning Messages M640 - M649

This section contains the following error messages and warnings:

- [M644](#)
- [M646](#)
- [M647](#)
- [M648](#)
- [M649](#)

M644

Message Text

Shift simulation cannot be performed due to selected
`shift_cycles_simulated` exceeding maximum chain length. (M644)

Severity

Warning

Description

This message is related to the value specified by the `-shift_cycles` option of the `set_simulation` command. This option enables the simulation of nonscan state elements for a specified number of last shift cycles. If you specify a value of 0 (the default), nonscan state elements are not simulated. If you specify a value greater than or equal to the maximum chain length (checked when running ATPG), the M644 message is reported. For example, if your scan chain length is 5 and you set the `-shift_cycles` option to 5 or more, the M644 message is reported.

What Next

Use the `-shift_cycles` option to specify a value less than the maximum scan chain length.

M646

Message Text

```
[ -reorder | -sorted ] option is not compatible with compressor patterns.  
(M646)
```

Severity

Error

Description

The `-reorder` and `-sorted` options are not supported for Adaptive Scan Technology compression patterns.

This error message is always generated when Adaptive Scan technology compression patterns are written using the `-reorder` or `-sorted` options.

What Next

Do not use the `-reorder` or `-sorted` options for Adaptive Scan technology compression patterns.

M647

Message Text

```
Line L, Cycle C exceeds maximum pattern cycle P; skipped. (M647)
```

Severity

Warning

Description

TestMAX ATPG displays the M647 message when the cycle number in the failure log file exceeds the number of cycles it reads during the `set_patterns-external` operation. It identifies the offending line (L) in the failure log file and the highest-numbered cycle (P) found in the patterns. TestMAX ATPG ignores this failure.

What Next

Check that the pattern file and failure log files correspond (this warning indicates that they might not be synchronized). Ensure that the pattern file is complete, and the cycle numbers in the failure log file are correct. Use the `-cycle_offset` option of the `set_diagnosis` command to adjust the differences between the first cycle in the pattern and the first tester cycle.

For more information, refer to the `set_diagnosis` command in the TestMAX™ ATPG and TestMAX Diagnosis Commands guide.

M648

Message Text

```
Line L, Cycle C is before the unload or in test_setup (pat P, cell S).  
(M648)
```

Severity

Error

Description

TestMAX ATPG identifies the cycle number in the failure log file in a pattern region before the unload occurs. It also identifies the offending line in the failure log file.

What Next

Ensure that the patterns in the external pattern buffer and the failure log file correspond. Ensure the cycle numbers in the failure log file are correct. Use the `-cycle_offset` option of the `set_diagnosis` command to adjust the differences between the first cycle in the pattern and the first tester cycle.

For more information, refer to the `set_diagnosis` command in the TestMAX™ ATPG and TestMAX Diagnosis Commands guide.

M649

Message Text

Line L, Cycle-based failure entry not supported with current patterns.
(M649)

Line L, in file "F" incorrect directive "D". (M649)

Line L, Cycle-based format not supported for the selected mode. (M649)

Severity

Error

Description

The first version of this message indicates that the patterns in the external pattern buffer do not support cycle counting, but the failure log file used for run diagnosis is in terms of cycle numbers. The offending line in the failure log file is shown.

The second version of this message indicates that an incorrect directive has been specified in the failures log file.

The third version of this message indicates that the cycle-based format is not supported for the current diagnosis setting.

What Next

If cycle based failure log files must be used, either STIL or WGL patterns must be used with the `set_patterns-external` command. If STIL or WGL patterns are not available, the failure log file must be translated into a pattern-based format, as the cycle based failure log file feature cannot be used.

Correct the wrong directive syntax D in the faulty failures log file F line L and re-execute the diagnosis.

If you receive the third version of this message, the likely cause is that the diagnosis is waiting for a pattern-based format. This could be due to the usage of the `set_diagnosis -dftmax_chain_format` command. Unset this configuration using the switch `-nodftmax_chain_format`.

Error and Warning Messages M650 - M659

This section contains the following error messages and warnings:

- [M650](#)
- [M651](#)
- [M653](#)
- [M654](#)
- [M656](#)
- [M657](#)
- [M658](#)
- [M659](#)

M650

Message Text

Cannot write to simulation [invocation script | debug] file *filename*.
(M650)

Cannot locate or access the TestMAX ATPG simulation template *filename*.
(M650)

Cannot replace [NETLIST_FILES | LIB_FILES | TOP_TEST_MODULE] on line
number written into file *filename*. (M650)

No nodes to monitor in simulation debug file *filename*. (M650)

Severity

Error

Description

These errors identify problems during the process of creating simulation support files from the `write_simtrace` command. When these errors are generated, the resulting files will either be incomplete or not produced at all.

What Next

If the "Cannot write to simulation ... file" message appears, review the access rights to the specified file and location.

The "Cannot locate or access the TestMAX ATPG simulation template" message indicates an incomplete installation of software. Review your installation. Template files are found under the /admin/setup directory.

The "Cannot replace ... on line ... written into file" indicates an improper simulation template file. Review your template file (if you have made a local version) or the TestMAX ATPG installation.

The "No nodes to monitor" message identifies the lack of a `-gate` or `-scan` option on the `write_simtrace` command line, which would result in an empty file. Add at least one `-gate` or `-scan` option to this command.

M651

Message Text

Incomplete pattern *number* in `-core` mode, missing leading force PI in `luc load_unload call`. (M651)

Skipping pattern *number* in `-core` mode, does not start with a load or a PI force. (M651)

Skipping pattern *number* in `-core` mode, contains multiple different force-PI statements. (M651)

Skipping pattern *number* in `-core` mode, contains a non-X measure after a `[clock pulse | measure]`. (M651)

Skipping pattern *number* in `-core` mode, contains multiple loads. (M651)

Severity

Error

Description

M651 messages indicate a difference in functionality between the defined "luc" environment, and the patterns being written out. This difference will cause incomplete or incorrect pattern generation because of limitations of formatting flexibility in the "luc".

What Next

If TestMAX ATPG displays an M651 message, you must go back to the ATPG generation step or an earlier step in the ATPG process and set additional constraints to prevent the patterns from containing a construct that cannot be formatted for the "luc" operation.

If the M651 message states that the pattern "contains multiple different force-PI statements", ensure that you define the `set_atpg -nopi_changes` option before ATPG generation.

If the M651 message states that the pattern "contains a non-X measure", ensure that you define the `set_po_masks -all` option before ATPG generation.

If the M651 message states that the pattern "contains multiple loads", ensure that you define the `set_atpg -single_load_per_pattern` option before ATPG generation.

M653

Message Text

Line L, Cycle-based failure entry not supported with current patterns.
(M649)

Unable to satisfy cell constraints on position 336 using load compressors. (M653)

Severity

Error

Description

The load compressor creates dependencies between scan cells. Therefore, it is generally recommended that you avoid cell constraints, since cell constraints require care bits and decrease compression. In some cases, due to dependencies, not all cell constraints can be simultaneously satisfied using load compressors.

What Next

Remove some of the cell constraints.

M654

Message Text

Inconsistent PLL clocking corrected: clock= d, cell= d. (M654)

Inconsistent PLL clocking not corrected: clock= d, cell= d. (M654)

Severity

Warning

Description

This message appears when a bit (or bits) of a clock chain have constraints other than just the internal clock control. These constraints can be other logic that the bits fan out to that ATPG is trying to detect faults, or they can be decompressor constraints or bus conflict avoidance constraints.

In this case, ATPG attempts to generate a pattern, and the internal clock setting conflicts with the other constraints. The pattern is corrected to self-consistently meet the highest priority constraints. If the PLL clocking is corrected (as indicated in the first version of the message), the clocking works exactly as when it was created by ATPG. If the PLL clocking is not corrected (the second version of the message), the clocking changes from when it was created. In either case, fault simulation is run so that the pattern can be applied and the results are correct.

If the resulting pattern detects no faults, it might be discarded. The pattern is retained if it detects some faults (it is unlikely that the original target fault is detected).

What Next

This presence of this message does not indicate an increase in the risk of generating bad patterns. However, the fault coverage or pattern count might be affected. To prevent a lower fault coverage or pattern count, make sure the clock chain bits control only the internal clock without fanout to other logic.

M656

Message Text

```
Optimized compressor calculations cannot be used due to excessive modes.  
(M656)
```

Severity

Warning

Description

This message will appear when at least one load compressor has more than seven modes. It indicates that certain ATPG optimizations cannot be used, and that DFTMAX ATPG will be slower.

In most cases, using more than three load modes has no benefit since there is no meaningful reduction in input dependencies

M657

Message Text

```
Compressor pipeline data is not available. (M657)
```

Severity

Error

Description

This message is usually issued when the `report_compressors-pipeline` command is specified in DRC mode. The pipeline register information is not available until DRC has passed.

This error message is also reported if you specify the `report_compressors-pipeline` command when using serialized DFTMAX compression. This occurs even if a series of pipeline registers are connected to serializer registers, since these registers are not compressor pipelines.

What Next

Fix all errors that prevent DRC from passing. You can also try downgrading the errors to warnings and determine if the extracted information is included on the pipeline registers.

Since it is not possible to report the pipeline registers adjacent to serializer registers, you should not use the `report_compressors-pipeline` command in this case.

M658

Message Text

NT shifts of PT patterns used X tolerance modes during ATPG process.
(M658)

Severity

Warning

Description

Issued during X-tolerant test generation, this message provides a summary of all [M659](#) messages at the end of the test generation process -- that is, the total number of shift cycles (NT) and the total number of patterns (PT) that used X-tolerant modes.

During test generation, an X-tolerant mode is used for a shift cycle instead of the full-observability mode, if the latter would result in loss of observation because of X-masking. There are two situations in which an X-tolerant mode is selected:

- If targeted fault detection would be lost due to X-masking in full-observability mode.
- Or, if fortuitously using an X-tolerant mode yields higher overall observability than using the full-observability mode for that shift cycle.

What Next

No action is required. Note that if most shifts of most patterns use X tolerance mode, this is an indication that the design has many unknowns (Xs). This condition might increase pattern count and thus reduce compression.

M659

Message Text

N shifts of P patterns used X tolerance modes during simulation interval.
(M659)

D-F/T shifts were selected to use unshared compression modes during simulation interval.

N1-N2/N3 shifts were selected to use unshared compression modes during simulation interval.

Severity

Warning

Description

The first version of this message is issued during X-tolerant test generation. At the end of each simulation interval, it summarizes how many total shift cycles (N) in how many patterns (P) used X-tolerant modes.

During test generation, an X-tolerant mode is used for a shift cycle instead of the full-observability mode, if the latter would result in loss of observation because of X-masking. There are three situations in which an X-tolerant mode is selected:

If targeted fault detection would be lost due to X-masking in full-observability mode.

If fortuitously using an X-tolerant mode yields higher overall observability than using the full-observability mode for that shift cycle.

Improved diagnosis of failing scan chains is enabled with the `-xtol_chain_diagnosis` option of the `set_atpg` command.

The reported data (D-F/T) on the second line of the message refers the following:

The first number (D) is the number of times direct-observation (XTOL) modes were selected fortuitously during good-machine simulation to increase overall observability.

The second number (F) is the number of times single-fanout XOR modes were selected fortuitously during good-machine simulation to increase overall observability.

The third number (T) is the number of times single-fanout XOR modes were selected during test-generation to detect a targeted fault.

The third line is an expert-level message that is issued at the end of each simulation interval if any of the three reported counts are not zero. The reported counts reflect usage during the current simulation interval and indicate the following:

N1 indicates the number of times the direct observe mode was selected to improve total observability independent of targeted fault detection.

N2 indicates the number of times the multiple observe mode was selected to improve total observability independent of targeted fault detection.

N3 indicates the number of times the multiple observe mode was selected to detect targeted faults.

A similar expert level message is issued at the end of an ATPG run with the following text:

```
N1-N2/N3 shifts were selected to use unshared compression modes during
ATPG process
```

The [M658](#) message gives a summary of all M659 messages at the end of the test generation process -- that is, the total number of shift cycles (NT) and the total number of patterns (PT) that used X-tolerant modes.

What Next

No action is required. Note that if most shifts of most patterns use X tolerance mode, this is an indication that the design has many unknowns (Xs). This condition might increase pattern count and thus reduce compression.

Error and Warning Messages M660 - M669

This section contains the following error messages and warnings:

- [M660](#)
- [M661](#)
- [M663](#)
- [M664](#)
- [M666](#)
- [M667](#)
- [M668](#)
- [M669](#)

M660

Message Text

Reading [Verilog | VHDL][compressed] patterns is not supported (M660)

Severity

Error

Description

The `set_patterns-external` command cannot be used to read back patterns written for simulator contexts (Verilog Testbench or VHDL pattern sets) on designs containing logic to compress the pattern data. There is not sufficient information in these pattern sets to support the interpretation of the compressed data present for these designs.

What Next

You must read back a format supported, for instance a binary, STIL, or WGL pattern set written by TestMAX ATPG.

M661

Message Text

Simulation with pattern storage cannot be performed when internal patterns are present. (M661)

Severity

Error

Description

The `run_simulation -store` command cannot be used when internal patterns are present with a scan compression design or in any flow with pattern adjustment.

What Next

If you want to use the `run_simulation -store` command on a scan compression design, you also need to specify the `set_patterns -delete -external` command. Alternatively, do not use the `-store` option of the `run_simulation` command in this context.

M663

Message Text

There were *number* faultable pins without slack data. (M663)

Severity

Warning

Description

This message appears when some pins do not have slack data when reading in slack data files.

What Next

You may choose to ignore this message or read in slack data for all pins. This message does not prevent slack-based transition fault ATPG. If slack data is missing, TestMAX ATPG attempts to fill in those slacks with known fanin or fanout pin slacks. If it is an input pin, the minimum slack of the fanin pins is assigned as the pin slack. The computed slack number is used in guiding fault sensitization decision only. If it is an output pin, the minimum slack of its fanout pins is assigned as the pin slack. The computed slack number is used in guiding fault propagation decision only.

Although this message does not block slack-based transition fault ATPG, a complete slack data file might provide better results.

M664

Message Text

```
Reference Clock W ON_time=X <= measure_time=Y in waveformtable Z. All PO  
measures were masked. (M664)
```

Severity

Warning

Description

This message will appear when you are using post-clock strobe. End-of-cycle measure is NOT compatible with PLL reference clocks. With PLL reference clocks defined, ATPG can generate patterns with the following sequence of events:

forcePI - measurePO - pulse reference clocks

When writing such patterns out in STIL (or any other external format), the vector that contains the measurePO must also pulse reference clocks (by definition, reference clocks must be pulsed in every vector). But the end-of-cycle measure timing means the order of events is reversed in this vector:

pulse reference clocks - measurePO

This is incorrect and the pattern will likely fail on silicon and, possibly, VCS and PROOFS simulation. To avoid this problem, a special check is performed during PLL-DRC; if

reference clocks are defined, their timing is checked and, if needed, all PO measures are masked and a new message is issued:

```
Warning: Reference Clock <clock_name> ON_time=<time> <=  
measure_time=<measure_time>  
in waveformtable <table_name>. All PO measures were masked. (M664)
```

What Next

You must change your STL procedure file from end-of-cycle measure to pre-clock-measure timing. In the STL procedure file, correct both the output measure time in the WFT, and remove extra _so strobe in the pre-shift vector then run DRC again.

The default timing from TestMAX DFT or other ATPG tools use what is called "Post-Cycle Measure" or "End-of-Cycle Measure"; TestMAX ATPG uses "Pre-Clock Measure" or "Non-End-of-Cycle Measure."

For more information, see the topic Defining Scan Chain Load/Unload in STIL in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M666

Message Text

Parallel STIL format does not support postamble Vectors with scan compression designs. (M666)

Severity

Error

Description

This error message will appear if you attempt to generate parallel STIL patterns (using the `write_patterns -parallel` command) that require the `postamble v` statement in the `internal_load_unload` procedure. A `postamble v` statement implies an Nshift simulation which is not compatible with parallel STIL data. Some example cases that would enable this message include LOS and LOES with a `pipe > 1`.

What Next

Use a unified STIL file for parallel pattern (generated by the `write_patterns -unified_stil_flow` command).

M667

Message Text

Clock <clock_name> offstate <state> is incompatible with build optimizations. (M667)

Severity

Error

Description

After a build, when defining clocks, the offstate is checked for consistency with the performed optimization. This message (M667) is issued if the offstate is the opposite than the one assumed when deciding which DFFs to modify.

What Next

You need to change the clock offstate so that it is consistent with the performed optimization.

M668

Message Text

Pattern <pattern_name> rejected due to bad clock_off value in cycle <cycle> on PLL internal clock_gat= gate_id v=<value>, condition_gat=<gate_cond> v=<value> (M668)

Severity

Warning

Description

This warning is issued when the simulator rejects a pattern because the clock chain bit values are not consistent with the clocking intended by the test generator. This situation arises rarely, but can occur if the clock chain bits also feed logic other than the clock controller. In this case, fault detection might require values on the clock chain bits that are contrary to the values required for the intended clocks to pulse.

What Next

The patterns retained are correct. However, many occurrences of this warning indicate that the design has significant logic fed by clock chain bits, a design style that can result in untestable faults.

M669

Message Text

Some reference clock timing is not supported for selected format. The following free-running clocks must be manually transferred outside the patterns. (M669)

Some reference clock timing is not expanded due to `set_drc -nofreerunning_expansion`. The following free-running clocks must be manually transferred outside the patterns. (M669)

Severity

Warning

Description

This warning occurs when you are using the `write_patterns` command and you have a free-running reference clock. The first version of this message will appear if the selected format cannot represent all the clock information. The second version will occur if you have specified the command `set_drc-nofreerunning_expansion`. If the default is used, `set_drc-freerunning_expansion`, you will never get this warning for STIL or binary formats. For the WGL and STIL99 formats, you will get this warning if you have asynchronous reference clocks (synchronous reference clocks are okay). You also get this warning for all other formats.

What Next

When you get this message, you also get a printout of the reference clock data from the `report_clocks` command. You will need to manually carry this information along with the patterns. In the case of asynchronous reference clocks, the clock pin will not even appear in the patterns. For synchronous reference clocks, the pin appears in the patterns, but the clock definition might be incomplete, and the pattern data for the clock pin will not keep it running consistently. If these patterns are simulated, you will need to manually correct these warnings, including adding the clock events into the simulation.

Error and Warning Messages M670 - M679

This section contains the following error messages and warnings:

- [M670](#)
- [M674](#)
- [M675](#)
- [M676](#)

- [M677](#)
- [M679](#)

M670

Message Text

Could not create padding pattern data for X-tolerant fault detection when adjacent pattern cannot be used. (M670)

Severity

Warning

Description

Padding patterns are required during test generation when there is a conflict between scan-in values to satisfy load care bits for a pattern and scan-in values to satisfy the X-tolerant observability needs of the primary target fault of the previous pattern. The conflict is resolved by inserting a pre-computed padding pattern. This padding pattern is computed at the beginning of ATPG and must have no effect on the circuit under test; that is, all clocks must be off, all constraints must be satisfied, and no output values are measured.

This message is issued when test generation fails to create a padding pattern. Normally, the padding pattern requires no care bits and test generation immediately succeeds in creating it. However, care bits can be required to satisfy design constraints, such as placing all BUSES in a non-contention state and satisfying user-specified constraints. The care bits required for the padding pattern can be unsatisfiable due to dependencies introduced by the load compressor.

The presence of this message indicates that some loss of test coverage is possible.

What Next

Remove all unnecessary constraints.

M674

Message Text

number pipeline inputs changed values to prevent pipeline cells changing during capture (M674)

Severity

Warning

Description

This message is issued during ATPG when the value scanned into a DFT MAX scan data pipeline cell which does not hold state during shift is changed during simulation to ensure that predictable values are considering short chain effects when unloaded. It is a warning message that is only seen when expert level messaging is enabled. When this message appears, there is a small risk that some targeted faults might not be successfully detected, however there should be no other effects. During DRC, DFT MAX load compressor pipeline cells are checked to see if they hold state during the capture. Violations will result in R18 warnings.

What Next

It is recommended that pipeline cells are created so that they pass compressor rule R18 and avoid this risk of losing test coverage.

M675

Message Text

`number` invalid dependent slaves have been reclassified as MASTERS. (M675)

Severity

Warning

Description

During DRC, TestMAX ATPG has identified `number` devices that were originally configured as dependent slaves and has reclassified them to be master devices to avoid S29 violations. TestMAX ATPG identifies devices that are loaded first during the shift operation to be the master.

What Next

Check that the correct polarity is being used for the clock associated with the dslave device. If you believe that the dslave algorithm has misidentified all dslaves, you may use the command `set_drc -noreclassify_invalid_dslaves` to prevent TestMAX ATPG from reclassifying the specified devices. Note that the `set_drc -nodslave_remodel` command will automatically invoke the `-noreclassify_invalid_dslaves` option.

For more information, refer to the topic S29 in the TestMAX™ ATPG and TestMAX Diagnosis DRC Rules Guide.

M676

Message Text

Bridge strength effects are ignored for dynamic_bridge faults. (M676)

Severity

Warning

Description

This indicates that the `-optimize_bridge_strengths` option of `set_atpg` command was selected when running ATPG or Fault Simulation with the dynamic bridging faults model. This option, however, is not supported in this circumstance.

What Next

Turn off this option by specifying `set atpg-nooptimize_bridge_strengths`.

M677

Message Text

Pipeline stage values with scan enable ON causes parallel simulation failures; use `set_simulation -nopipeline_cells` to avoid problem. (M677)

Severity

Warning

Description

Using the `set_simulation-pipeline_cells` command in designs with input pipelines and scan enable not constrained to its system value can result in parallel testbench failing VCS because input pipe DFFs remain X in the parallel simulation.

What Next

Use the `set_simulation-nopipeline_cells` command when writing parallel patterns to match the number of serial shifts to the input pipeline register depth.

M679

Message Text

Load compressor pipeline for port `port_name` (<port_id>) had
<num_candidates> DFF candidates for stage <num_stage>. (M679)

Severity

Warning

Description

This message is issued during DRC when DFT MAX load compressor pipeline cells are identified. It indicates that tracing forward from *port_name* (<port_id>), on paths sensitized by simulating the shift procedure, to identify the DFF corresponding to the pipeline cell for the <num_stage> level of the pipeline has identified more than 1, that is <num_candidates>, DFFs. This situation occurs in designs in which the path from the input port to the scan chain has fanout to other, typically nonscan, DFFs.

What Next

If DRC fails to identify pipeline cells, or if R27 or S1 violations are reported, confirm that the pipeline cell identified and reported for cases with M679 is indeed the intended DFF. If not, check the design and identify possibly missing constraints in the shift procedure that would de-sensitize the path to the DFF that was mistaken for the pipeline cell.

Error and Warning Messages M680 - M689

This section contains the following error messages and warnings:

- [M680](#)
- [M681](#)
- [M682](#)
- [M683](#)
- [M684](#)
- [M685](#)
- [M686](#)
- [M687](#)
- [M688](#)
- [M689](#)

M680

Message Text

Multicycle paths with length less than 2 are not supported. (M680)

Severity

Warning

Description

TestMAX ATPG does not support definitions of paths using multi cycle length 1.

What Next

Multicycle paths of length 1 are sometimes used to remove subsets of exceptions that were previously defined with larger multicycle path multipliers. TestMAX ATPG does not use multicycle paths of length 1 at all, so the original exception remains. The only way to get the correct exception is to avoid over-specification of the original exception.

TestMAX ATPG ignores this exception but continues to read the SDC file and use the other exceptions.

For more information, see the topic [How TestMAX ATPG Interprets SDC File Commands](#) in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M681

Message Text

Object name is not a valid from point. (M681)

Severity

Warning

Description

An invalid pin was used as a from point. The valid from pins are Clock, PI, PIO, CLK of DFF and Data, CLK of DLAT.

What Next

Define the from point using one of the valid objects.

TestMAX ATPG ignores this exception but continues to read the SDC file and use the other exceptions.

For more information, see the topic [How TestMAX ATPG Interprets SDC File Commands](#) in the [TestMAX™ ATPG and TestMAX Diagnosis User Guide](#).

M682

Message Text

No valid from point found. Ignoring exception. (M682)

Severity

Error

Description

All from specified points are invalid. TestMAX ATPG has ignored the exception.

What Next

Define from points using the valid objects. The valid from points are Clock, PI, PIO, DFF and DLATCH.

TestMAX ATPG ignores this exception but continues to read the SDC file and use the other exceptions.

For more information, see the topic [How TestMAX ATPG Interprets SDC File Commands](#) in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M683

Message Text

Object <object_name> is not a valid to point. (M683)

Severity

Warning

Description

An invalid pin was used as a to point. The valid to pins are Clock, PO, PIO and all pins of DFF and DLAT except Q and CLK.

What Next

Define a to point using one of the valid objects.

TestMAX ATPG ignores this exception but continues to read the SDC file and use the other exceptions.

For more information, see the topic [How TestMAX ATPG Interprets SDC File Commands](#) in the [TestMAX™ ATPG and TestMAX Diagnosis User Guide](#).

M684

Message Text

No valid to point found. Ignoring exception. (M684)

Severity

Error

Description

All specified to points are invalid. TestMAX ATPG has ignored the exception.

What Next

Define to points using one of the valid objects. The valid objects are Clock, PO, PIO, DFF and DLATCH.

TestMAX ATPG ignores this exception but continues to read the SDC file and use the other exceptions.

For more information, see the topic [How TestMAX ATPG Interprets SDC File Commands in TestMAX™ ATPG and TestMAX Diagnosis User Guide](#).

M685

Message Text

Object `<object_name>` is not a valid through point (M685).

Severity

Warning

Description

An invalid object was used a through point. The valid through point gates are, any gate other than DFF, DLATCH or MEMORY. But Q and CLK are valid even in the case of DFF and DLAT.

What Next

Define a through point using one of the valid objects.

TestMAX ATPG ignores this exception but continues to read the SDC file and use the other exceptions.

For more information, see the topic [How TestMAX ATPG Interprets SDC File Commands in the TestMAX™ ATPG and TestMAX Diagnosis User Guide](#).

M686

Message Text

No valid through point found. Ignoring exception. (M686)

Severity

Error

Description

All specified through points are invalid. TestMAX ATPG has ignored the exception.

What Next

Define through points using one of the valid objects. The valid through point gates are: any gate other than DFF, DLATCH or MEMORY.

TestMAX ATPG ignores this exception but continues to read the SDC file and use the other exceptions.

For more information, see the topic How TestMAX ATPG Interprets SDC File Commands in TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M687

Message Text

More than one object matched for <pattern_name>. (M687)

Severity

Error

Description

The pattern(s) you specified matched more than one object. This command option accepts only a single object.

What Next

Check the values you entered.

TestMAX ATPG ignores this exception but continues to read the SDC file and use the other exceptions.

For more information, see the topic How TestMAX ATPG Interprets SDC File Commands in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M688

Message Text

Clock <clock_name> exists in more than one group. (M688)

Severity

Error

Description

The `set_clock_groups` SDC command has a restriction that each clock can be defined in only one clock group.

What Next

To define multiple groups related to the same clock, use multiple `set_clock_groups` commands.

TestMAX ATPG ignores this exception but continues to read the SDC file and use the other exceptions.

For more information, see the topic [How TestMAX ATPG Interprets SDC File Commands](#) in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M689

Message Text

Clock does not have waveform values monotonically increasing, so waveform has been adjusted. (M689)

Severity

Warning

Description

There must be an even number of edges that are interpreted as alternating rising and falling edges. The edges must be monotonically increasing, except for a special case when two edges are specified. When only two edge values are specified and the first value is greater than the second value, it is interpreted as a return-to-one waveform instead of the normal return- to-zero waveform, if the falling edge adding one period is still larger than rising edge.

What Next

This is an information message only.

TestMAX ATPG ignores this exception but continues to read the SDC file and use the other exceptions.

For more information, see the topic How TestMAX ATPG Interprets SDC File Commands in TestMAX™ ATPG and TestMAX Diagnosis User Guide.

Error and Warning Messages M690 - M699

This section contains the following error messages and warnings:

- [M690](#)
- [M691](#)
- [M692](#)
- [M693](#)
- [M694](#)
- [M695](#)
- [M696](#)
- [M697](#)
- [M698](#)
- [M699](#)

M690

Message Text

You can specify only a single object. (M690)

Severity

Error

Description

The `-source` option takes only a single object as an argument.

What Next

You can have a generated clock derived from a single clock. However, you cannot generate a clock from more than one master clock.

TestMAX ATPG ignores this exception but continues to read the SDC file and use the other exceptions.

For more information, see the topic How TestMAX ATPG Interprets SDC File Commands in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M691

Message Text

You cannot specify an output port ' *name*' to be a generated clock master source. (M691)

Severity

Error

Description

Generated clock master must be an input or inout port or a pin.

What Next

Specify an input or inout port or a pin as the generated master source.

TestMAX ATPG ignores this exception but continues to read the SDC file and use the other exceptions.

For more information, see the topic How TestMAX ATPG Interprets SDC File Commands in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M692

Message Text

Creating a generated clock on input or inout port ' *port_name*'. (M692)

Severity

Warning

Description

When creating a generated clock on an input or inout port, note that the generated clock will only derive its waveform characteristics from the master clock. Delay, on the other hand, is derived from delay information at the port itself. Additionally, creating a generated clock to drive an inout port can cause unpredictable circuit behavior. If the port is functioning as an output while the clock is driving, bus contention occurs. If the clock is three-stated, any registers in the fanout of the clock can lose state.

What Next

Verify that the circuit is safe according to your design rules. TestMAX ATPG assumes that the generated clock is valid.

TestMAX ATPG ignores this exception but continues to read the SDC file and use the other exceptions.

For more information, see the topic How TestMAX ATPG Interprets SDC File Commands in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M693

Message Text

The master clock source pin *pin_name* was not found in the design. (M693)

Severity

Error

Description

TestMAX ATPG could not find the Master Clock Source pin in the design.

What Next

Check the `-source` argument and change it to a valid clock source.

TestMAX ATPG ignores this exception but continues to read the SDC file and use the other exceptions.

For more information, see the topic How TestMAX ATPG Interprets SDC File Commands in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M694

Message Text

The number of edges specified ' *number*' is not an odd number larger than or equal to 3. (M694)

Severity

Error

Description

The number of edges to make one period of the generated clock waveform has to be an odd number larger than or equal to 3.

What Next

Carefully specify edges and ensure that you specify one full clock cycle using the edges.

TestMAX ATPG ignores this exception but continues to read the SDC file and use the other exceptions.

For more information, see the topic How TestMAX ATPG Interprets SDC File Commands in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M695

Message Text

The `-edges` spec of generated clock '`<clock_name>`' has edge number less than 1, the edge number should be from 1 up. (M695)

Severity

Error

Description

The `-edge` specification in `create_generated_clock` SDC command should use an edge number of the master clock from 1 up. Zero or negative numbers are not allowed.

What Next

Change your `-edge` spec in the `create_generated_clock` SDC command.

TestMAX ATPG ignores this exception but continues to read the SDC file and use the other exceptions.

For more information, see the topic How TestMAX ATPG Interprets SDC File Commands in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M696

Message Text

In the `-edge` specification of `create_generated_clock` '`<clock_name>`', the edge numbers must be in increasing order. (M696)

Severity

Error

Description

In the `-edge` specification of a `create_generated_clock` SDC command, the edge numbers specified must be in increasing order.

What Next

Check the `-edge spec` in `create_generated_clock` command and make sure the edge numbers are increasing.

TestMAX ATPG ignores this exception but continues to read the SDC file and use the other exceptions.

For more information, see the topic How TestMAX ATPG Interprets SDC File Commands in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M697

Message Text

The number of `edge_shifts` specified '*number*' using '`-EDGE_SHIFT`' option is not equal to the number of edges specified '*number*' using '`-EDGES`' option. (M697)

Severity

Error

Description

The number of `edge_shifts` specified using the `-edge_shifts` option must be equal to the number of edges specified using the `-edges` option.

What Next

Make the number of `edge_shifts` equal to the number of edges.

TestMAX ATPG ignores this exception but continues to read the SDC file and use the other exceptions.

For more information, see the topic How TestMAX ATPG Interprets SDC File Commands in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M698

Message Text

Clocks without source objects are not supported. (M698)

Severity

Warning

Description

TestMAX ATPG does not support clocks without source objects. TestMAX ATPG is not be able to interpret exceptions that reference this clock.

What Next

Define clocks using valid source objects.

TestMAX ATPG ignores this exception but continues to read the SDC file and use the other exceptions.

For more information, see the topic How TestMAX ATPG Interprets SDC File Commands in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M699

Message Text

```
Invalid slow path. Path doesn't exist between <instance_name1> and  
<instance_name2> gates. (M699)
```

Severity

Error

Description

There is no path between the <instance_name1> and <instance_name2> gates.

What Next

Redefine the path using valid points.

TestMAX ATPG ignores this exception but continues to read the SDC file and use the other exceptions.

For more information, see the topic How TestMAX ATPG Interprets SDC File Commands in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

9

Error and Warning Messages M700 - M799

This section contains the following error messages and warnings:

- [Error and Warning Messages M700 - M729](#)
- [Error and Warning Messages M730 - M739](#)
- [Error and Warning Messages M740 - M749](#)
- [Error and Warning Messages M750 - M759](#)
- [Error and Warning Messages M760 - M769](#)
- [Error and Warning Messages M770 - M779](#)
- [Error and Warning Messages M790 - M799](#)

Error and Warning Messages M700 - M729

This section contains the following error messages and warnings:

- [M701](#)
- [M706](#)
- [M710](#)
- [M717](#)
- [M718](#)
- [M719](#)
- [M720](#)
- [M721](#)
- [M722](#)
- [M724](#)
- [M725](#)

- [M726](#)
- [M727](#)
- [M729](#)

M701

Message Text

Invalid launch clock name specified. (M701)

Invalid capture clock name specified. (M701)

Severity

Error

Description

The first variation of this message appears when the `add_faults-launch` command is specified, but the launch clock name is not in the TestMAX ATPG database. The second variation of this message appears when the `-capture` option used, but the capture clock is not in the TestMAX ATPG database.

What Next

Specify the `add_faults` command again with a valid clock name.

Note that when using internal clocks (on-chip clocking), the name of the clock in the TestMAX ATPG database is often different from the name specified by the `add_clocks` command or in the STL procedure file. You can use the `report_clocks-verbose` command to print a list of accepted clock names.

M706

Message Text

`-profile` can only be used with stuck and transition fault models. (M706)

`-per_clock_domain` can only be used with stuck and transition fault models. (M706)

`-exclusive` can only be used with stuck and transition fault models. (M706)

`-shared` can only be used with stuck and transition fault models. (M706)

`-inter_clock_domain` can only be used with stuck and transition fault models. (M706)

`-intra_clock_domain` can only be used with stuck and transition fault models. (M706)
`-launch` can only be used with stuck and transition fault models. (M706)

`-capture` can only be used with stuck and transition fault models. (M706)

Add faults by clock domain can only be used with stuck and transition fault models. (M706)

Remove faults by clock domain can only be used with stuck and transition fault models. (M706)

Severity

Error

Description

The first error message appears when you try to use the `run_fault_sim` command with the `-profile` option. The `-profile` option only supports the stuck and transition fault models.

The second error appears when you use the `-per_clock_domain` option of the `report_summaries` command. This option reports all specified faults with extra information for their launch and capture clocks. All clocks are reported, even those classified as "shared" or "multiple."

The third error appears when you use the `-exclusive` option of the `remove_fault` command. It specifies that only the faults that are launched and captured exclusively (using a single launch and a single capture) are removed from the fault list.

The fourth error appears when you use the `-shared` option of the `remove_faults` command. This specifies that only the faults that are launched or captured by multiple clocks are removed from the fault list. This excludes all PI and PO faults.

The fifth error appears when you use the `-inter_clock_domain` option of the `add_faults` and `remove_faults` commands. This option adds and removes only exclusive faults that are launched and captured by different clock domains respectively in the two aforementioned commands.

The sixth error appears when you use the `-intra_clock_domain` option of the `add_faults` and `remove_faults` commands. This option adds and removes only exclusive faults from the fault list that are launched and captured by the same clock domains in the two commands respectively.

The seventh error appears when you use the `-capture` switch of the `add_faults` and `remove_faults` commands. The `-capture` switch adds and removes all faults to the fault list that are captured exclusively by the specified clock for the two commands, respectively. This switch also accepts the keywords PO, NO_CLOCK, or MULTIPLE, in either all upper

or all lowercase, to remove faults that are observed by PO/PIO, no clock domains, or multiple clock domains, respectively.

What Next

To eliminate this error, rerun using one of the supported fault models.

M710

Message Text

```
MISR cell <cell_id> in compressor <compressor_name> failed shift  
operation verification. (M710)
```

Severity

Error

Description

This message is issued for DFTMAX and MISR designs when the MISR fails the shift operation verification. The failing MISR cell is identified by both its position in the MISR (*cell_id*) and its gate ID. This message indicates that the value simulated during DRC did not match the STL procedure file description. The MISR must capture the unload compressor outputs during shift.

What Next

Use the `set_pindata shift` command to debug the problem in the GSV. Trace back the sensitized paths from the failing MISR cell to the internal nodes corresponding to compressor outputs.

M717

Message Text

```
There was no defined x-tolerant connection for chain chain_name using  
unload compressor <unload_compressor_name> (M717)
```

Severity

Error

Description

This message is issued during internal chain I/O identification. When the STL procedure file defines all modes, then single-observe modes are used to identify chain I/O. This is a more robust method than identifying chain I/O based on the full-observability mode, which is required when only the full-observability mode is defined in the STL procedure file.

The M717 message indicates that internal chain I/O identification could not find a single-observe mode for the named chain.

What Next

Check your STL procedure file and make sure all chains have a single-observe mode.

M718

Message Text

```
Seed not saved; reason (M718)
```

Severity

Error

Description

This message describes an error that occurred when adding seed values used by TestMAX ATPG to compute a set of expected seed and signature values. For example, seed data might be missing or incorrectly formatted when specifying the `add_lbist_seeds` command. In the following example, Q is not accepted as part of the seed value:

```
add_lbist_seeds -format binary_normal 00000000000000000000000000000000Q001
Error: Seed not saved; seed value 'Q' not supported for binary data.
(M718)
```

What Next

You can specify the `report_lbist_seeds` command to report the seed data accepted by TestMAX ATPG. If an error is reported, note the *reason* text of the error description, then specify the `remove_lbist_seeds` command to stop the seed addition process. Based on the error description, you can then make the required changes using the `add_lbist_seeds` command.

For more information on using TestMAX ATPG with the DFTMAX LogicBIST tool, see the *DFTMAX LogicBIST User Guide*.

M719

Message Text

```
Parallel simulation with DFTMAX Ultra Unified STIL Flow patterns is beta.
(M719)
```

Severity

Warning

Description

This message appears when you specify the `-unified_stil_flow` option of the `write_patterns` command when writing parallel patterns for a design generated using DFTMAX Ultra compression. For example:

```
write_patterns pats.stil -format stil -replace -unified_stil_flow
write_patterns pats.stil -format stil -replace -parallel
-unified_stil_flow
```

What Next

If your parallel patterns fail in VCS, you should use the `-nounified_stil_flow` and `-parallel` options of the `write_patterns` command for parallel simulation. For more information, see the topic "Pattern Types Required by DFTMAX Ultra." in the TestMAX ATPG and TestMAX Diagnosis User Guide

M720

Message Text

number clock pulses rejected. Clock <clock_name> has a *number* bit clock chain, but needs *number* bits. (M720)

Severity

Informational

Description

This message is issued at the end of ATPG when synchronized internal clocks are present (these clocks are declared in a SynchronizedClocks block of the STIL procedure file). The message indicates that TestMAX ATPG attempted to pulse an internal clock, but failed because the clock chain was too short.

This message reports the total number of times that this situation occurred for all clocks. It also reports the situation with the highest discrepancy between the desired clock chain length and the actual one.

What Next

For optimal performance, you should change your design to increase the length of the specified clock chain to the amount indicated. This might produce slightly better fault coverage and/or fewer patterns.

It is possible that you might get this message again, even after increasing the clock chain length. This can happen for two reasons:

- 1) Another clock chain is too short. The initial message only reports the worst case, but there might be another case with a smaller discrepancy.
- 2) When the initial rejected pulse is allowed, TestMAX ATPG might try to add another pulse that requires an even longer clock chain.

You can guarantee that you'll avoid this message if you ensure that your clock chains are long enough to handle the worst case scenario. This happens when all clocks of the sync group are active in the same pattern (which requires the maximum clock chain bits to account for latency variations). The worst-case scenario also pulses the slowest clock of the sync group in all frames but the last one, and pulses all clocks in the last frame.

Note that you can also safely ignore this message.

M721

Message Text

Full-sequential ATPG is ignoring synchronized internal clock timing.
(M721)

Severity

Warning

Description

This message appears when you run Full-Sequential ATPG and synchronized clock information is present in the ClockTiming block of the STL procedure file. Note that synchronized clocks are supported for Basic Scan, Fast-Sequential or Two Clock mode, but not for Full-Sequential mode.

What Next

If you want to use synchronized clock data, you must use Basic Scan, Fast-Sequential or Two Clock mode.

M722

Message Text

Internal clock <clock_name> latency is set to *number* cycles of period <time_period>. ClockController *name* has latency *number*. (M722)

Severity

Informational

Description

This message is displayed during DRC during the process of reading a test protocol file. It is issued when a clock controller controls multiple clocks of different periods.

Since the latency attribute of the clock controller is specified in terms of its fastest clock, the latencies for any slower clocks need to be adjusted. This message shows the absolute latency of the clock controller, which is calculated by multiplying the latency value in cycles times the period of the fastest clock.

The latency for a clock that is slower than the fastest clock on the clock controller is ceiling (absolute controller latency / clock period). TestMAX ATPG takes the ceiling to cover the case where the period of the slow clock is not aligned with the latency of the controller.

For example, this can happen when you have a clock controller with clocks of periods 10ns and 20ns, and you did not specify the latency. The default latency is 5, which produces a controller latency of 50ns (5 latency * 10ns fastest clock). The slow clock latency is 3 cycles of 20ns which makes an absolute latency of 60ns.

What Next

You can safely ignore this message. However, you should refer to the information in this message if you get C40 latency warning.

M724

Message Text

The following settings are ignored for threaded ATPG. (M724)

list_of_command_settings

Severity

Warning

Description

This message warns you that certain command settings are ignored by TestMAX ATPG II.

What Next

You can safely ignore this message. However, you should be aware of the impact of running ATPG without the listed commands.

Examples

```
TEST-T> run_atpg
The following settings are ignored for threaded ATPG. (M724)
set_atpg -decision random
set_atpg -lete_fastseq
set_atpg -basic_min_detects_per_pattern 17 40
Starting threaded ATPG with 8 threads. (M733)
```

M725

Message Text

Latch *latch_id* is considered as transparent during the pipeline tracing.
(M725)

Severity

Informational

Description

This message is reported when a latch has the same enable port signal as the clock port signal for the previous flip-flop on the scan data path. This latch is considered transparent during the pipeline tracing.

What Next

This is an information message only. For more information, see the description of the `-lockup_after_compressor` option of the `set_drc` command.

M726

Message Text

Lockup after compressor latch *latch_id* is driven by sequential elements having different level-triggered clock. (M726)

Severity

Informational

Description

This message is reported after the `set_drc -lockup_after_compressor` command is specified. This command might cause the simulation to fail.

What Next

This is an information message only.

M727

Message Text

Error: Feature requires use of tmax2. (M727)

Severity

Error

Description

You have asked for a feature only supported with threaded ATPG.

What Next

Invoke the tool using the tmax2 command instead of tmax to use this feature.

M729

Message Text

The following setting requires single-process/multi-core ATPG. (M729)

list_of_settings

Severity

Warning

Description

This message appears when you are running threaded TestMAX ATPG and have made certain settings that require the use of the non-threaded TestMAX ATPG engine. In this case, TestMAX ATPG does not run in threaded mode but reverts to non-threaded mode instead, as indicated by the [M733](#) message.

What Next

If you want to make sure you run threaded TestMAX ATPG , avoid making any settings that are not compatible with threaded ATPG. For a complete description of how incompatible settings, see "TestMAX ATPG Multithreading Command Option Support" in the *TestMAX ATPG and TestMAX Diagnosis User Guide*.

Examples

```
TEST-T> run_atpg
Warning: The following setting requires single-process/multi-core ATPG.
(M729)
set_delay -launch_cycle last_shift
Starting single-process ATPG. (M733)
```

Error and Warning Messages M730 - M739

This section contains the following error messages and warnings:

- [M730](#)
- [M731](#)
- [M732](#)
- [M733](#)
- [M734](#)
- [M735](#)
- [M738](#)
- [M739](#)

M730

Message Text

Integer values on parallel with nounified are not supported. Use the unified flow to allow for nshifts in parallel simulation. (M730)

Severity

Warning

Description

You specified an option with an integer value with the `-parallel` option of the `write_patterns` command. This syntax is not recognized.

What Next

Use the `-unified_stil_flow` option of the `write_patterns` command (the default) to enable a precomputed `n_shifts` value during parallel simulation. To remove this warning, specify only the `-parallel` option.

M731

Message Text

Ignoring the `-nounified_stil_flow` option (M731)

Severity

Warning

Description

You specified the `-serial` option with the `-nounified_stil_flow` option of the `write_patterns` command. The `-nounified_stil_flow` option works only for dual STIL parallel simulation, and is ignored.

What Next

When this message is generated, the `-nounified_stil_flow` option is automatically disabled. If the `-nounified_stil_flow` option is required, remove the `-serial` option. In this case, the `write_patterns` command recognizes the `-nounified_stil_flow` option and generates parallel patterns by default.

M732

Message Text

```
-nounified_stil_flow is not supported with [-scanstructures min |  
-nointernal_scancells] (M732)
```

Severity

Error

Description

This error message appears when you specify the `-nounified_stil_flow` option with either the `-scanstructures min` option or the `-nointernal_scancells` option of the `write_patterns` command. The `-nounified_stil_flow` option is used only for dual STIL parallel simulation and cannot be used with these options because they require serial simulation.

What Next

To eliminate this error, remove the `-nounified_stil_flow` option from the `write_patterns` command.

M733

Message Text

```
Starting single-process simulation. (M733)
```

```
Starting single-process fault simulation. (M733)
```

Starting single-process ATPG. (M733)

Starting parallel ATPG with *N* processes. (M733)

Starting parallel fault simulation with *N* processes. (M733)

Starting parallel good simulation with *N* processes. (M733)

Starting threaded simulation with *N* threads. (M733)

Starting threaded ATPG with *N* threads. (M733)

Starting threaded fault simulation with *N* threads. (M733)

All thread settings are disabled for the `-distributed` option. (M733)

All thread settings are disabled for the `-parallel` option. (M733)

Severity

Informational only

Description

The first four sets of messages describe the status of the various processes associated with a simulation, ATPG, or fault simulation run. The fifth set of messages is reported when thread settings are turned off because of overriding multicore or distributed settings.

What Next

No action is required.

M734

Message Text

An incomplete pattern was defined; subsequent use of these patterns will be incomplete. (M734)

Severity

Error

Description

At the end of processing the `set_patterns -external` command, the pattern data is checked for completeness. This message indicates there was an incomplete pattern defined; the `report_patterns -all -type` command identifies the incomplete pattern as “bad” in the type field. These patterns will not be applied in subsequent commands such as the `run_simulation` command.

What Next

Review the incoming pattern data sent to the `set_patterns -external` command. Be aware that TestMAX ATPG only fully supports TestMAX ATPG-generated pattern data; modifications or hand-generated patterns may create this situation.

M735

Message Text

```
run_simulation truncated due to incomplete pattern data. (M735)
```

Severity

Error

Description

During the processing of the `run_simulation` command, an irregular pattern was identified. The simulation stopped when it encountered this particular pattern. You can use the `report_patterns -all -type` command to identify the pattern as “bad” in the type field.

What Next

Review the source of the pattern data using the `report_patterns` command. If the pattern data came from an external source (using the `set_patterns -external` command) review that source. If the pattern data was generated by TestMAX ATPG with no external source, contact Synopsys.

M738

Message Text

```
Gate tracing truncated; if other errors occur, increase  
set_workspace_size -max_trace_gates above integer [location]. (M738)
```

Severity

Warning

Description

This message is generated when the maximum gate tracing depth is exceeded when tracing R17 DRC violations. You might be able to ignore this message because this behavior does not necessarily affect subsequent DRC behavior. The `[location]` reference is for Synopsys use to help identify issues related to this message.

What Next

You can set the `-max_trace_gates` option of the `set_workspace_sizes` command to determine if increasing the depth of the gate tracing affects subsequent DRC results and eliminates the R17 violations. Unless there are subsequent DRC issues, no change is required. The overall DRC runtime is affected if you make large increases of the gate-tracing value.

M739

Message Text

`set_atpg -capture_cycles number` exceeds the `min_ateclock_cycles` value; the `-capture_cycles` value will define the number of capture calls. (M739)

Severity

Warning

Description

This message appears if the value of the `-min_ateclock_cycles` option of the `set_atpg` command is less than the value set for the `-capture_cycles` option. In this case, the value specified for the `-capture_cycles` option is applied. This reflects the maximum number of possible capture cycles present in the ATPG patterns and provides consistent capture sequences for all generated patterns.

What Next

No change is required. You can increase the value of the `-min_ateclock_cycles` option to force a consistent capture sequence across the patterns. Alternatively, you can reduce the value of the `-capture_cycles` option to reflect the expected maximum number of external clock pulses for the capture sequences.

Error and Warning Messages M740 - M749

This section contains the following error messages and warnings:

- [M740](#)
- [M741](#)
- [M742](#)
- [M743](#)
- [M745](#)

- [M746](#)
- [M747](#)

M740

Message Text

Simulation terminated due to X capture from chain=*C* into MISR of seq_compressor *K* in pattern *P* during shift *S* using mode=*M*. (M740)

Severity

Error

Description

Note the following:

C is name of the failing chain

K is the name of the failing sequential compressor

P is the failing pattern number

S is the failing shift position

M is the observe mode type

This error occurs when an X value is used as an input to a MISR (multiple-input shift register) when you are simulating sequential-based compression patterns. Since an X value negates all MISR values, the current simulation or ATPG process terminates immediately. For debugging purposes, the error data automatically selects the simulated value for the failing pattern. The shift input value to a MISR depends on the values currently on the output of the chains and the current observe mode, as determined by the XTOLPRPG (X-tolerant pseudo-random pattern generator). These values select all chain outputs propagated to the MISR. The ATPG process uses the appropriate observe mode to ensure that X values are not propagated to the MISR.

What Next

During the ATPG process, an M740 message indicates a bug, except in LBIST mode. If this error occurs during ATPG in LBIST mode, you must fix the design because an LBIST design cannot contain Xs.

If this error is reported when you are using the `run_simulation` command, the problem might be caused by additional constraints applied to create the patterns (such as capture masks) but are not used during the ATPG process .

For debugging purposes, you can use the simulation values reported by the `-error_data` option of the `set_pindata` command. You can also specify the `set_simulation -analyze_x_sources` command so that each pattern is evaluated individually. In this case, TestMAX ATPG prints an analysis of the simulation of patterns that have at least one scan cell with a final simulated value of X.

M741

Message Text

Simulation terminated due to invalid single observe_mode of
seq_compressor S in pattern P during shift N. (M741)

Severity

Error

Description

Note the following:

`S` indicates name of seq_compressor associated with failure

`P` indicates pattern number during which the failure occurred

`N` indicates which shift of the unload during which the failure occurred

This message is issued during simulation of sequential compression patterns when an invalid single observe mode is selected by the XTOLPRPG. Since this will cause an X to propagate to the MISR which will destroy its values, the current simulation or ATPG process is immediately terminated. To provide debug assistance, the `error_data` is automatically set to select the simulated value for the failing pattern. The ATPG process should always ensure that no Xs are allowed to propagate to the MISR by using an appropriate observe mode.

What Next

When running ATPG or running simulation of patterns created by TestMAX ATPG, this should never happen. If it does occur during these operations, this should be considered a bug.

If this occurs during a `run_simulation` operation of external patterns not created by TestMAX ATPG, the patterns should not be used for any purpose. To aid in debug, the simulation values can be viewed by using the `error_data` option of the `set_pindata` command.

M742

Message Text

Simulation terminated due to invalid observe_mode of seq_compressor S in pattern P during shift N. (M742)

Severity

Error

Description

Note the following:

S indicates name of seq_compressor associated with failure

P indicates pattern number during which the failure occurred

N indicates which shift of the unload during which the failure occurred

This message is issued during simulation of seq_compressor patterns when an invalid observe mode is selected by the XTOLPRPG. Since this will cause an X to propagate to the MISR which will destroy its values, the current simulation or ATPG process is immediately terminated. To provide debug assistance, the error_data is automatically set to select the simulated value for the failing pattern. The ATPG process should always ensure that no Xs are allowed to propagate to the MISR by using an appropriate observe mode.

What Next

When running ATPG or running simulation of patterns created by TestMAX ATPG, this should never happen. If it does occur during these operations, this should be considered a bug.

If this occurs during a run_simulation operation of external patterns not created by TestMAX ATPG, the patterns should not be used for any purpose. To aid in debug, the simulation values can be viewed by using the error_data option of the set_pindata command.

M743

Message Text

Simulation terminated due to invalid XTOLPRPG hold of seq_compressor S in pattern P during shift N. (M743)

Severity

Error

Description

Note the following:

S indicates name of seq_compressor associated with failure

P indicates pattern number during which the failure occurred

N indicates which shift of the unload during which the failure occurred

This message is issued during simulation of sequential compression patterns when an invalid usage of the hold observe mode is selected by the XTOLPRPG. Since this will cause an X to propagate to the MISR which will destroy its values, the current simulation or ATPG process is immediately terminated. To provide debug assistance, the error_data is automatically set to select the simulated value for the failing pattern. The ATPG process should always ensure that no Xs are allowed to propagate to the MISR by using an appropriate observe mode. After a reseed operation, a valid non-hold observe mode must be used before the hold observe mode can be exercised.

What Next

When running ATPG or running simulation of patterns created by TestMAX ATPG, this should never happen. If it does occur during these operations, this should be considered a bug.

If this occurs during a `run_simulation` operation of external patterns not created by TestMAX ATPG, the patterns should not be used for any purpose. To aid in debug, the simulation values can be viewed by using the `error_data` option of the `set_pindata` command.

M745

Message Text

Sequential compression is not currently supported for a mixture of internal and external clocks. (M745)

Severity

Error

Description

This is an error message that is issued during DRC of DFTMAX Ultra circuits when there is a mixture of internal and external clocks. If there is a failure with this message, the DRC process will immediately terminate with an error.

What Next

This condition is not currently supported with DFTMAX Ultra. If possible, use only internal clocks or use only external clocks.

M746

Message Text

Sequential compression is not supported for alway_on/off internal clocking with cell constraints on clock control cells. (M746)

Severity

Error

Description

This is an error message that is issued during DRC of DFTMAX Ultra circuits when there are cell constraints placed on clock control cells and there is at least one occurrence of usage of alway_on/off internal clocking. If there is a failure with this message, the DRC process will immediately terminate with an error.

What Next

This condition is not currently supported with DFTMAX Ultra. If possible, remove the cell constraints on the clock control cells.

M747

Message Text

MISR lengths (N1) exceed #shift-cycles (N2): N3 initialization patterns are used. (M747)

Severity

Error

Description

Note the following:

N1 is the sum of the MISR lengths

N2 is the number shifts per load

N3 is the number of initialization patterns that is used during ATPG

This is a warning message that is issued during DRC of accumulator-mode DFTMAX Ultra circuits indicating that it is necessary to use additional patterns to initialize the accumulators and to also unload the accumulators at the end of the test patterns.

What Next

No action is required.

Error and Warning Messages M750 - M759

This section contains the following error messages and warnings:

- [M750](#)
- [M751](#)
- [M752](#)
- [M753](#)
- [M754](#)
- [M755](#)
- [M756](#)
- [M757](#)
- [M758](#)
- [M759](#)

M750

Message Text

Negative slack data for gate *number* (<pinpathname>). (M750)

There were *number* faultable pins with negative slack data. (M750)

Severity

Warning

Description

This message appears when the `set delay -negative_slack_checks` option is set and some pins have negative slack data when reading in slack data files.

What Next

You may choose to ignore this message or check your slack data file. This message does not prevent long-path ATPG. If slack data is negative, TestMAX ATPG will use it the same way as positive slacks. Although this message does not block long-path ATPG, negative slack data might indicate the PrimeTime run was not using the same timing exceptions that would be used in TestMAX ATPG. This inconsistency in timing exceptions might result in bad patterns.

M751

Message Text

There were *number* faultable pins with infinite slack data. (M751)

Infinite slack data for gate *number* (<pinpathname>). (M751)

Severity

Warning

Description

These messages appear when the `set_delay -infinite_slack_checks` option is set and some pins have infinite slack data when reading in slack data files.

What Next

You may choose to ignore these messages or check your slack data file. These messages do not prevent long-path ATPG. If slack data is infinite, TestMAX ATPG will use it the same way as normal slacks in guiding ATPG or calculating delta. Although these messages do not block long-path ATPG, infinite slack data might indicate the node is untestable.

M752

Message Text

Fault <location> <polarity> is detected with negative slack. (M752)

Severity

Warning

Description

This message appears when the `set_delay -negative_slack_checks` option is set and the fault is detected with a negative slack transition.

What Next

You may choose to ignore this message or check your pattern set. Negative slack detection might indicate the pattern detects the fault along a timing exception path.

M753

Message Text

Fault <location> <polarity> is detected with infinite slack. (M753)

Severity

Warning

Description

This message appears when the `set delay -infinite_slack_checks` option is set and the fault is detected with an infinite slack transition.

What Next

You may choose to ignore this message or check the fault and the detection pattern. Infinite slack detection might indicate the detection of a fault that was intended to be untestable.

M754

Message Text

Delta data is not currently available for reporting. (M754)

Severity

Error

Description

Delta data is generated after small delay defect ATPG or fault simulation. This data might be invalidated if you change the fault list. This message is issued when the requested delta data does not exist or if it has been invalidated by fault list changes.

What Next

Generate the delta data during small delay defect ATPG or fault simulation.

M755

Message Text

Simulating full sequential patterns with 1 process to preserve proper circuit states. (M755)

Severity

Warning

Description

This message appears when using the `run_simulation` command to perform a multicore simulation on full-sequential patterns. Because circuit states between patterns are preserved by the full-sequential simulator, the multicore process cannot partition the patterns to increase the simulation speed. In this case, a single slave process is used to simulate all full-sequential patterns.

What Next

This is an informational message only. You should not expect an increase in simulation speed.

M756

Message Text

Cannot fork child processes due to exceeding number of processes limit. (M756)

Cannot fork child processes due to insufficient swap space. (M756)

Cannot fork child processes. errno=<errno>. (M756)

Severity

Error

Description

This message appears when running multi core applications on a machine with insufficient resources to launch a user-specified number of processes.

What Next

In the first version of this message, you will need to check with your IT department to identify the limitations on the number of processes for the machine.

In the second version of this message, you will need to check with your IT department to identify the swap space limitations of the required swap space for forking a new process.

In the third version of this message, TestMAX ATPG cannot decide the meaning of the `errno`. You will need to look up the meaning of the `errno` in your operating system.

M757

Message Text

Host might not have enough CPUs for *number* slave processes. (M757)

Severity

Warning

Description

This message appears when the user-specified number of slave processes is larger than the currently available CPUs in the host. If you continue, you might encounter multicore ATPG or multicore simulation runs slower than expected.

What Next

You may choose to ignore this message or adjust the number of specified slave processes using the `set_atpg -num_processes` command (for multicore ATPG) or `set_simulation -num_processes` (for multicore simulation).

M758

Message Text

Host has single CPU. Starting single process ATPG. (M758)

Host has single CPU. Starting single process fault simulation. (M758)

Severity

Warning

Description

The first version of this message appears when the `set_atpg-num_processes max` command is specified and TestMAX ATPG can only detect one CPU on the machine.

The second version of this message appears when the `set_simulation -num_processes max` command is specified and TestMAX ATPG can only detect one CPU on the machine.

In both cases, TestMAX ATPG starts ATPG or fault simulation with one process rather than multicore ATPG or fault simulation with one or more slave processes.

What Next

This is an information message only.

M759

Message Text

Cannot detect the number of CPUs of the host. Please specify the number of slave processes. (M759)

Severity

Error

Description

This message appears when either the `set_atpg -num_processes max` command or `set_simulation -num_processes max` command is specified and TestMAX ATPG cannot detect the number of CPUs of the host.

What Next

Specify the number of slave processes using the `set_atpg -num_processes` command or the `set_simulation -num_processes` command.

Error and Warning Messages M760 - M769

This section contains the following error messages and warnings:

- [M763](#)
- [M764](#)
- [M767](#)
- [M768](#)

M763

Message Text

Path <path_identifier> has multiple (*number*) launch nodes. Only one is supported. (M763)

Severity

Warning

Description

This message appears when a single launch or capture node could not be found for hold time ATPG.

What Next

Specify the endpoints of the path. Generally, for hold time ATPG, the launch node must be the output of the launching flip-flop, and the capture node must be the input of the capturing flip-flop.

M764

Message Text

TIE object <instance> cannot be used with set_disable_timing command (M764)

Severity

Warning

Description

This message indicates that TestMAX ATPG is ignoring the exception because it is applied to a gate modeled with one of these primitives: TIEX, TIE0, TIE1 or TIEZ. A timing exception applied to a TIE primitive has no effect. The ATPG coverage is identical with or without the exception. The exception for this instance will not appear in the `report_sdc` output

What Next

No action is required.

M767

Message Text

No logical path from through point <gate_name> (gate_id) for timing exception (line number) (M767)

Severity

Warning

Description

There is no logical connection between the `from point` and the `through point`.

What Next

Modify a `through point` or `from point` to specify a path with a logical connection. Or no action is required.

M768

Message Text

This path has no logical connection between start and end-points. (M768)

Severity

Warning

Description

There is no logical connection between the `start point` and `end point`.

What Next

Specify the path which has a logical connection, else no action is required.

Error and Warning Messages M770 - M779

This section contains the following error messages and warnings:

- [M770](#)
- [M771](#)
- [M772](#)
- [M773](#)
- [M777](#)

M770

Message Text

Number scan cells overwrote load values during chain test (ID=id_number, pat_number=number). (M770)

Severity

Warning

Description

This message is reported if an unconstrained set or reset signal or an improperly specified clock corrupts a chain test. This check is performed by the `run_atpg`, `run_simulation`, `run_fault_sim`, and `run_diagnosis` commands. For diagnosis, this check is only performed on failing scan chains.

What Next

Constrain the clock, set or reset signal that is corrupting the scan cells, then re-run ATPG. If you are not using these patterns for chain diagnosis, you can avoid this message by disabling the chain test pattern using the `set_atpg -chain_test off` command.

M771

Message Text

```
Option <option_name> is invalid with the current diagnostic settings.  
(M771)
```

Severity

Warning

Description

This message appears because `<option_name>` could not be used with current diagnostics settings.

What Next

Starting with the 2008.09 release, the diagnostics engine default has changed. This is one possible reason for the message.

M772

Message Text

```
Chain diagnosis is not supported in this mode. (M772)
```

Severity

Error

Description

This message indicates that the chain diagnosis is not supported for the current diagnosis setting.

What Next

The diagnosis is waiting for logic failures. The failures in the failure log file are launching the chain diagnosis process, but the current settings are disabling the run. The likely source of this problem is that the pattern-based failures are using chain names and the `set_diagnosis -dftmax_chain_format` command. This configuration is not supported.

M773

Message Text

`N scan cell pairs have same triplets. (M773)`

Severity

Warning

Description

This indicates that the `N` scan chain pairs will be an alias for a subset of their scan cells. It gives an early indication of potential deficiency in diagnosis.

What Next

Alter the connectivity of the scan chain outputs for each pair of such scan chains. Increase the number of serializer flops or number of serializer segments to resolve this violation.

M777

Message Text

`Warning: 11 incorrect PO measures were removed. (M777)`

Severity

Warning

Description

This message occurs during the `run_simulation -override_difference` step when more than one PO measures is found in one pattern.

This situation is not supported for fault simulation. The M777 message indicates this situation and removes all PO measures other than the last one.

What Next

No action is required.

Error and Warning Messages M790 - M799

This section contains the following error messages and warnings:

- [M790](#)
- [M791](#)
- [M798](#)

M790

Message Text

Cannot change work dir while persistent faults are present. (M790)

Severity

Error

Description

This message is issued by the `set_faults -work_dir` command. It is issued when you attempt to change the fault cache work directory and the directory contains fault files. TestMAX ATPG will not move these files to a new directory.

What Next

You should set the work directory near the beginning of your session, before you add any faults.

Alternatively, you can use the following steps:

Save the faults for all fault models using the `write_faults` command.

Discard all cached faults by turning off persistent fault models (`set_faults -nopersistent_fault_models`).

Re-enable persistent fault models (`set_faults -persistent_fault_models`).

Restore your saved faults using the `read_faults` command.

For more information, see the Multi-Fault Grading beta document.

M791

Message Text

```
update_faults -direct_credit command is not available, because reason.  
(M791)
```

```
active fault model is a and the external fault model is b
```

Severity

Error

Description

These message variations are issued by the `update_faults -direct_credit` command for one of three reasons:

The fault model is not set to stuck-at.

The stuck-at fault list is empty.

Persistent fault models is turned off and no file name was specified.

What Next

Set the fault model to stuck-at and populate the fault list. The `update_faults` command does not add faults.

Specify the transition faults to update from by either activating persistent fault models and populating the transition fault list, or by specifying a transition fault list file (`update_faults -direct_credit -external file`).

M798

Message Text

```
Internal patterns were not simulated and may fail in simulation (M798)
```

```
Pattern set has been validated by run_simulation (M798)
```

```
Pattern set contains number failing patterns (M798)
```

Severity

Warning / Informational

Description

These message variations are reported to make sure you specify the `run_simulation` command before running the `write_patterns` command or if your generated pattern set contains failing patterns after the `run_simulation` process completes.

The first variation is a warning message that the `run_simulation` command was not run before the `write_patterns` command.

The second variation confirms that the `run_simulation` command successfully validated the patterns.

The third variation reports the number of failing patterns encountered by the `write_patterns` command after the `run_simulation` process.

What Next

When generating patterns, you should always run the `run_atpg`, `run_simulation`, and `write_patterns` commands in succession.

If the first variation of the M798 message is reported, the `write_patterns` command successfully wrote patterns, but there is a possibility the patterns may fail during a Verilog simulation.

If the third variation is reported, you should debug the `run_simulation` command failures before proceeding to Verilog simulation. For more information, see the FAQ - Troubleshooting Pattern Simulation Failures.

10

Messages M800 - M2832

This section contains the following error messages and warnings:

- [Messages M800 - M849](#)
- [Messages M850 - M899](#)
- [Messages M900 - M949](#)
- [Messages M950 - M999](#)
- [Messages M1000 - M1299](#)
- [Messages M1800 - M2832](#)

Messages M800 - M849

This section contains the following error messages and warnings:

- [M804](#)
- [M830](#)
- [M831](#)
- [M832](#)
- [M833](#)
- [M834](#)
- [M835](#)
- [M840](#)
- [M841](#)
- [M842](#)
- [M843](#)
- [M844](#)

- [M847](#)
- [M848](#)
- [M849](#)

M804

Message Text

Filtered *Number* candidates with match score outside the `score_distance` of 20. (M804)

Filtered *Number* candidates with match score less than the `min_score` of 50. (M804)

Filtered *Number* candidates that exceeded the `max_number` of 10. (M804)

Severity

Warning

Description

Variations of this message print when you filter candidate defects for class-based diagnosis reporting using the `-filter` option of the `set_diagnosis` command.

What Next

This is an informational message only. If you specified a parameter with the `-filter` option, note the number of candidates that are filtered after you specify the `run_diagnosis` command.

For more information, see the topic Class-Based Diagnosis Reporting in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M830

Message Text

A sequence of more than one latch was identified starting with latch `<inst>` (M830)

Severity

Warning

Description

This message is printed by the `report_lockup_latches` command. The specified instance is the start of a chain of two or more latches on the scan path that share the same clock and polarity. The additional latches might be redundant.

What Next

Check the chain of latches to determine whether they are really necessary in the design.

You can analyze this violation by launching the GSV and displaying the instance using the Shift pindata.

M831

Message Text

```
A lockup latch is missing between scan cells scan_cell1 clocked by clock1
(through pinpath_name1) le | te(time) and scan_cell2 clocked by clock2
(through pinpath_name2) le | te(time) (M831)
```

Severity

Warning

Description

This message is printed by the `report_lockup_latches` command. The two instances are adjacent scan cells belonging to two different clock domains that are both clocked on the leading (LE) or trailing edge (TE).

What Next

Check that the correct `report_lockup_latches` arguments were specified. If not, re-run the command with the correct arguments. Otherwise the violation should be analyzed.

You can analyze this violation by launching the GSV and displaying the two instances using the Shift pindata.

M832

Message Text

```
An extra lockup latch is added between scan cells scan_cell1 clocked by
clock1 (through pinpath_name1) LE (time) and scan_cell2 clocked by clock2
(through pinpath_name2) LE (time) (M832)
```

An extra lockup latch is added between scan cells `scan_cell1` clocked by `clock1` (through `pinpath_name1`) TE (`time`) and `scan_cell2` clocked by `clock2` (through `pinpath_name2`) TE (`time`) (M832)

Severity

Warning

Description

This message is printed by the `report_lockup_latches` command. The two instances are adjacent scan cells. An extra lockup latch is reported if any of the following conditions apply:

More than one lockup latch is identified in series.

A lockup latch is identified between two adjacent scan cells that are driven by same clock edges. Namely, LE to LE and TE to TE.

This extra lockup latch configuration is inefficient but does not create race conditions.

What Next

Extra lockup latch can be removed but not removing will not be timing unsafe.

You can analyze this violation by launching the GSV and displaying the two instances using the Shift pindata.

M833

Message Text

Incorrectly clocked lockup latch detected between scan cells `scan_cell1` clocked by `clock1` (through `pinpath_name1`) le | te(`time`) and `scan_cell2` clocked by `clock2` (through `pinpath_name2`) le | te(`time`) (M833)

Severity

Warning

Description

This message is printed by the `report_lockup_latches` command. The two instances are adjacent scan cells. An incorrectly clocked lockup latch is reported if any of the following conditions apply:

A lockup latch between two different scan cells belonging to two different clock domains belongs to a clock domain that is different than either the driving scan cell, or the observing scan cell.

A lockup latch between two different scan cells belonging to two different clock domains belongs to a clock domain that is the same as the driving scan cell, and whose clock is not inverted with respect to the driving scan cell.

A lockup latch between two different scan cells belonging to two different clock domains belongs to a clock domain that is the same as the observing scan cell, and whose clock is inverted with respect to the observing scan cell.

What Next

Check that the correct `report_lockup_latches` arguments were specified. If not, re-run the command with the correct arguments. Otherwise the violation should be analyzed.

You can analyze this violation by launching the GSV and displaying the two instances using the Shift pindata.

M834

Message Text

```
N patterns were rejected for exceeding Capture Power during simulation interval (M834).
```

```
N patterns were rejected for exceeding Shift Power during simulation interval (M834).
```

```
N patterns were rejected for not meeting Shift Power limit during simulation interval (M834).
```

Severity

Informational

Description

These message variations report the number of patterns rejected due to power considerations. They are not error messages or warnings.

The first variation reports the number of patterns (N) rejected for exceeding the capture power during the simulation interval.

The second variation reports the number of patterns rejected for exceeding the shift power during the simulation interval.

The third variation is specific to Shift Power Controller (SPC) and non-threaded ATPG. It reports the number of patterns rejected for not meeting the lower limit of the shift power during the simulation interval.

What Next

If there are excessive pattern rejections as reported by this message, the specified power budgets may be too low for the design. You may increase the power budget or consider alternatives such as various fill strategies or modifying the design by inserting SPC to better manage power.

M835

Message Text

```
Incorrect lockup latch implementation between scan cells scan_cell1  
clocked by clock1 (through pinpath_name1) le | te(time) and scan_cell2  
clocked by clock2 (through pinpath_name2) le | te(time) (M835)
```

Severity

Warning

Description

This message is printed by the `report_lockup_latches` command. The lockup latch between the specified scan cells is incorrect, but the problem could not be fully analyzed.

What Next

Check the logic between the two scan cells to determine the correct implementation.

You can analyze this violation by launching the GSV and displaying the instance using the Shift pindata.

M840

Message Text

```
Interval pattern is not currently defined. (M840)
```

Severity

Error

Description

This is an error message that is issued when using the `good_sim_results` option of the `set_pindata` command for JTAG ULTRA-LBIST circuits if the interval pattern has not been selected.

What Next

Set the desired interval pattern using the `set_simulation -interval_pattern` command.

M841

Message Text

Cannot simulate first interval which is used only for initialization.
(M841)

Severity

Error

Description

This is an error message that is issued when using the `good_sim_results` option of the `set_pindata` command for JTAG ULTRA-LBIST circuits if the selected interval pattern is intended to be used only for initialization.

What Next

Set an allowed non-initialization interval pattern using the `set_simulation -interval_pattern` command.

M842

Message Text

Interval pattern (A) exceeds number of patterns in interval (B). (M842)

Severity

Error

Description

This is an error message that is issued when using the `good_sim_results` option of the `set_pindata` command for JTAG ULTRA-LBIST circuits if the selected pattern exceeds the number of patterns in the selected interval pattern.

What Next

Change the selected interval pattern using `set_simulation -interval_pattern` command or the selected pattern using `good_sim_results` option of the `set_pindata` command.

M843

Message Text

Number of selected capture cycles (N1) exceeds number of allowed cycles (N2). (M843)

Severity

Error

Description

Note the following:

N1 is the selected number of capture cycles

N2 is the maximum allowed number of capture cycles

This is an error message that is issued when performing ATPG for JTAG ULTRA-LBIST circuits with internal clocking if the selected number of capture cycles exceeds the maximum allowed.

What Next

Change the selected number of captures cycles to be used by ATPG by using the `-jtag_lbist` option (third argument) of the `run_atpg` command. The command usage is:

```
run_atpg [-JTAG_Lbist <N1> <N2> <N3>]
```

where:

N1 is number of seeds (intervals)

N2 is number of patterns

N3 is the number of capture cycles

M844

Message Text

N primary faults were aborted due to excess care bits per shift. (M844)

Severity

Error

Description

Note the following:

N is the number of primary target faults that were aborted due to excess care bits per shift

This is an expert level warning message that is issued at the end of ATPG for DFTMAX Ultra circuits indicating that some primary target faults were aborted due to excess care bits per shift.

What Next

No action is required. The only effect is a potential loss of test coverage.

M847

Message Text

JTAG LBIST mode requires definition of min_ateclock_cycles. (M847)

Severity

Error

Description

This message is printed when the minimum number of system cycles for each pattern is not defined when running JTAG LBIST mode during DRC.

What Next

Set the minimum number of system cycles for each pattern either using the `-min_ateclock_cycles` option of the `set_atpg` command or in the STL procedure file.

M848

Message Text

Pattern *pat_number* failed nonhold pipe cell checking. (M848)

Severity

Error

Description

This message occurs during the `run_simulation` process when input pipelines in a design with DFTMAX compression fail to hold their values during capture cycles. A message is issued for every pattern with a failure. You can report pin-by-pin details by specifying the `set_atpg -verbose` command before `run_simulation`.

Pipelines that fail the R18 rule check (that is, non-hold pipelines) also fail this check if the value in the pipeline cells changes during capture cycles. You can confirm this by comparing the data generated with the `report_patterns` command for the last `num_input_pipeline_stages` load values with the activity on the pipeline cell displayed in the GSV using the command `set_pindata -good_sim_results pattern _number`.

What Next

This message indicates that the indicated patterns are incorrect and might fail serial VCS simulation. If the pattern source is an external file, the incorrect patterns might be generated from an older version of TestMAX ATPG, or erroneously modified outside TestMAX ATPG. If the pattern was generated by the `run_atpg` command, this message indicates a potential issue. If this is the case, contact Synopsys support to submit a test case.

M849

Message Text

Requested xchain N has no mask. (M849)

Severity

Error

Description

Note that N is the chain name

This message is in response to a request to treat a core scan chain as an X-chain using the command `add_cell_constraints xchain chain_name`. In this case, the chain requested does not have an associated mask register bit in the `mask_chain`.

What Next

The circuit needs to be enhanced to add a masking register bit to the compression architecture.

Messages M850 - M899

This section contains the following error messages and warnings:

- [M853](#)
- [M854](#)
- [M858](#)

- [M860](#)
- [M861](#)
- [M862](#)
- [M863](#)
- [M864](#)
- [M865](#)
- [M868](#)
- [M869](#)
- [M870](#)
- [M871](#)
- [M873](#)
- [M874](#)
- [M881](#)
- [M889](#)
- [M891](#)
- [M892](#)
- [M893](#)
- [M897](#)

M853

Message Text

Incorrect tester cycle calculated for pattern *name*. (M853)

Incorrect original id for pattern *name*. (M853)

Severity

Warning

Description

There is an inconsistency in the specified pattern set.

What Next

Make sure the patterns are in STIL format.

M854

Message Text

Pattern file name *name* is not unique. (M854)

Severity

Error

Description

Multiple pattern files can be read using the `-split` option of the `set_patterns` `-external` command. Each pattern file must have a unique name.

What Next

Rename pattern files so that each pattern file has unique name.

M858

Message Text

Unload_mode debug data stored with all chain outputs set to X. (M869)

Unload_mode debug data stored with chain outputs of compressor *name* set to binary values; others to X. (M869)

Unload_mode debug data stored with all chain outputs set to binary values. (M869)

Severity

Warning

Description

These messages are issued during DRC when R23 violations occur or when the `set_drc -store_unload_mode_data` command is used. These messages facilitate GSV debugging by defining the conditions used when storing the pin data displayed in the GSV. Depending on the type of violations encountered, the chain outputs are set to either random binary values or to X, as indicated by the message.

What Next

Check the description of the R23 violation message and the `set_drc -store_unload_mode_data` command for information on how to debug the DRC violations.

M860

Message Text

Strength Optimization is not supported for IDDQ Bridging Faults. (M860)

Severity

Warning

Description

This message indicates that ATPG or fault simulation will not take into account the strength optimization as requested by the following commands:

```
set_atpg -optimize_bridge_strength
```

```
run_fault_sim -strong_bridge
```

What Next

Strength optimization is not supported. Thus, do not use the commands or options `set_atpg-optimize_bridge_strength` or `run_fault_sim -strong_bridge` while generating or fault-simulating IDDQ bridging patterns.

M861

Message Text

Full-observe mode chain test patterns created for #chains=N/T. (M861)

Severity

Warning

Description

Note the following:

N indicates the number of chains covered by the chain test

T is the name of the failing sequential compressor

Every chain should load “011” and “100” somewhere during a load of the chain test patterns. Note that either “01100” or “10011” suffice as both triples appear, although they are overlapped. If a particular chain does not satisfy this target behavior, then a warning is issued.

What Next

No action is required. There is no reason that this behavior needs to be satisfied. Sometimes, chains are too short to allow these triples to propagate. Other times, the CarePRPG does not deliver the random data to support this effort. This is a rare case, however.

M862

Message Text

Multiple/no-observe mode chain test patterns created for #modes=N/T.
(M862)

Severity

Warning

Description

Note the following:

N indicates the number of modes used by the chain test

T indicates the total number of modes in the design

The different modes select different groups of scan chains to be unloaded into the MISR during the chain test pattern creation. Typically, all modes are used for this process, but this might not always be the case.

What Next

No action is required.

M863

Message Text

Single-observe mode chain test for chain=C of length=L might not use “01100”. (M863)

Severity

Warning

Description

Note the following:

C is name of the chain

L indicates the length of the chain

Every chain should load “011” and “100” somewhere during a load of the chain test patterns. Note that either “01100” or “10011” suffice as both triples appear, although they are overlapped. If a particular chain does not satisfy this target behavior, then a warning is issued. This warning indicates that this target behavior could not be satisfied because the chain is too short.

What Next

No action is required.

M864

Message Text

Chain test not supported in LBIST mode. (M864)

Chain test not supported with misr_measure once_per_test. (M864)

Severity

Warning

Description

The chain test is incompatible with the end-of-test MISR unload format of patterns. The `set_atpg -seq_compressor_chain_test` switch is ignored in this case.

The chain test is also incompatible with DFTMAX Ultra LBIST. The `set_atpg -seq_compressor_chain_test` switch is ignored in this case.

What Next

Using the `set_atpg -noseq_compressor_chain_test` will eliminate the M864 message in all cases. If chain test is desired, switch to a pattern style in which the MISR measures once per pattern using `set_atpg -misr_measure once_per_pattern`. Also, make sure you are using a non-LBIST mode of operation using `set_drc seq_comp_jtag_lbist_mode jtag` if JTAG access is being used.

M865

Message Text

Chain test in JTAG mode only supports single MISR unload per pattern.
(M865)

Severity

Warning

Description

DFTMAX Ultra accessed via JTAG only supports a single unload per pattern, and not multiple unloads per pattern. As such, the chain test using JTAG access will have slightly less resolution than non-JTAG accessed DFTMAX Ultra chain test patterns.

What Next

No action is required.

M868

Message Text

Unload compressor pipeline output traced back to input *N* (*name*) is missing *M* pipeline stages. (M868)

Severity

Warning

Description

Indicates that an unload compressor pipeline output was traced back to the input with the ID (*N*) and the name (*name*), but the output is missing *M* pipeline stages. All paths to the unload compressor must have the same number of pipeline stages, and must be consistent with the STL procedure file.

What Next

Edit the netlist so that all paths to the unload compressor have the same number of pipeline stages.

M869

Message Text

Unload_mode debug data stored with all chain outputs set to X. (M869)

Unload_mode debug data stored with chain outputs of compressor *name* set to binary values; others to X. (M869)

Unload_mode debug data stored with all chain outputs set to binary values. (M869)

Severity

Warning

Description

These messages are issued during DRC when R23 violations occur or when the `set_drc -store_unload_mode_data` command is used. These messages facilitate GSV debugging by defining the conditions used when storing the pin data displayed in the GSV. Depending on the type of violations encountered, the chain outputs are set to either random binary values or to X, as indicated by the message.

What Next

Check the description of the R23 violation message and the `set_drc -store_unload_mode_data` command for information on how to debug the DRC violations.

M870

Message Text

Retention Test is supported in only Scan Mode. (M870)

Severity

Error

Description

This is a run-time message that indicates ATPG was halted because an incorrect mode was specified for retention testing.

What Next

Re-run retention testing in Scan mode only.

M871

Message Text

No Retention cell has been defined in the netlist. (M871)

Severity

Warning

Description

This message occurs during retention testing when a retention cell is undefined in the netlist. This means that no cells were listed after the `'define retention` directive in the netlist. The patterns will not be useful because none of the scan cells are observed. TestMAX ATPG, in this case, does not stop the execution of the `run_atpg` command.

What Next

Make sure a retention cell is defined in the netlist by placing the ``define retention` directive before any retention cell definitions in the library and the ``undef retention` directive after the cell definitions.

M873

Message Text

Possibly incorrect load serializer parallel output specification:
`serializer index serializer_bit_name.` (M873)

Severity

Warning

Description

By default, TestMAX ATPG checks the validity of user-specified load and unload serializer registers in the STL procedure file when serializers are used during DRC. If an invalid bit is specified in the STL procedure file, a M873 message is issued and the specification for that bit is ignored.

What Next

Edit the STL procedure file and specify the correct serializer registers.

M874

Message Text

Possibly incorrect load serializer parallel output inversion specification: `serializer index serializer_bit_name invert.` (M874)

Severity

Warning

Description

By default, TestMAX ATPG checks the validity of user-specified load and unload serializer registers in the STL procedure file when serializers are used during DRC. If a valid bit is specified in the STL procedure file with incorrect inversion, a M874 message is issued and the specification for that bit is ignored.

What Next

Edit the STL procedure file and specify the correct serializer registers.

M881

Message Text

Using `-split` option with `set_patterns -netlist_independent` is disabled due to significant overhead. (M881)

Severity

Error

Description

This message occurs when pattern generation is disabled due to significant CPU time and pattern size overhead.

What Next

Split the patterns after the netlist pattern translation is complete, as shown in the following example script:

```
set_patterns -netlist_independent
write_patterns dftmax_patt.bin -compress gzip -replace

drc -force
run_drc scan_mode.spf
set_patterns -external dftmax_patt.bin
write_patterns scan_patt.bin -split 200 -compress gzip -replace
```

M889

Message Text

Invalid option `-mapping_table` -> no mapping table exists between logical and physical IDs. (M889)

Severity

Error

Description

You did not specify the `match_names` command, which creates a mapping table used in the static name matching flow.

What Next

Run the `match_names -verify all` command to build a mapping table and add the logical and physical IDs.

M891

Message Text

Mapping table database does not match current database. (M891)

Severity

Error

Description

You tried to create a mapping table without specifying the `-replace` option of the `match_names` command. As a result, the existing mapping table contains a different ID than the current PHDS database. This mismatch could lead to failed queries or queries that return incorrect information.

What Next

Make sure the database you used to create the mapping table is connected to the database used to query the current mapping table.

For more information, see the topic Name Matching in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M892

Message Text

Could not map any logical to physical IDs. (M892)

Severity

Warning

Description

There are two possible causes of this message:

When name matching rules are applied to the PHDS database, there are no matches between the logical and physical instances.

An older PHDS database does not have the functionality to provide the physical instance names.

What Next

Check the PHDS database used for name matching. You might need to use the `set_match_names` command to specify string substitutions to correct the mismatches.

For more information, see the topic Name Matching in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M893

Message Text

```
No physical data retrieved from PHDS. (M893)
```

Severity

Warning

Description

This message appears when TestMAX ATPG diagnostics is unable to retrieve any data from the PHDS database, including physical information for subnets or bridges.

What Next

If you are running chain diagnostics, or if you don't need subnets or bridges, you can safely ignore this message. If you aren't running chain diagnostics, you should first check the connections with the PHDS database. If there are no problems, you should check if any Y rule violations might affect the PHDS database.

M897

Message Text

```
Error: No mapping table available for querying PHDS with obscured names.  
(M897)
```

Severity

Error

Description

There is no mapping table for this design in the PHDS or inside of the image.

What Next

Create a mapping table and store it in the PHDS or the image.

Messages M900 - M949

This section contains the following error messages and warnings:

- [M900](#)
- [M902](#)
- [M903](#)
- [M904](#)
- [M905](#)
- [M906](#)
- [M907](#)
- [M912](#)
- [M913](#)
- [M914](#)
- [M915](#)
- [M916](#)
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- [M928](#)
- [M929](#)
- [M930](#)
- [M931](#)
- [M938](#)
- [M939](#)
- [M940](#)

- [M941](#)
- [M942](#)
- [M945](#)
- [M946](#)
- [M948](#)

M900

Message Text

`RHEL version <d>.<d> is not one of the QSC supported releases: 5.7, 5.9, 6.2, 6.4, 6.5, 6.6, 7.1. (M900)`

Severity

Error

Description

This message is issued when the OS version does not match the recommended version for PHDS generation.

What Next

Make sure your OS is one of the supported releases (5.7, 5.9, 6.2, 6.4, 6.5, 6.6, or 7.1).

M902

Message Text

`Contact layer information not available for via via_name (M902)`

Severity

Warning

Description

This message is issued during the access of the contact layer information for a given via when one of the following situations is encountered:

Parsing a net in the NET/SPECIALNET section of a DEF file

Accessing additional layers of a via for a single layer

Storing via information in a TestMAX ATPG image

What Next

Contact layer information must be defined before parsing and storing via information. You should always check the order in which the LEF files are read. The LEF file that contains all the layer information (often called a "technology" LEF file) must be read before reading other LEF and DEF files. You can use the `report_physical -technology` command to determine if the layers were read. Check the via definition in the LEF / DEF files and search for the associated contact layers in the technology LEF. All layers must be defined in the technology LEF file.

M903

Message Text

Via `via_name` is not defined (M903)

Severity

Error

Description

This message appears when an undefined via type is encountered while parsing the NET/SPECIALNET section of a DEF file. This error impacts the subsequent subnet extraction step because the net with a missing via is viewed as a disconnected net with no subnet information available.

What Next

Look for the missing via type in the output of the `report_physical -technology` command. If it is not reported as a via (either as an "Obj" or not at all), then search for the file that has the via definition and determine if this file was read into TestMAX ATPG.

M904

Message Text

Contact layer information incomplete for Via `via_name` (M904)

Severity

Warning

Description

This message appears if the contact layer information for a particular via is incomplete. A via should have two defined contact layers. If a defined layer is missing, the contact layer information for the via is incomplete.

What Next

Look for the missing contact layer in the output of the `report_physical -technology` command. If it is not reported, then search for the file containing the contact definition and determine if this file was read into TestMAX ATPG.

M905

Message Text

Via `via_name` is not a stacked via (M905)

Severity

Error

Description

This message appears when a via is erroneously recognized as a stacked via. It is a TestMAX ATPG internal error, not a LEF/DEF format-related issue.

What Next

No action is necessary.

M906

Message Text

Contact layer information is not available for stacked via `via_name` (M906)

Severity

Warning

Description

This message appears when contact layer information is missing for a recognized stacked via. The stacked via is created as a collection of basic via types either during DEF parsing or when analyzing the connectivity of a given net.

What Next

Look for the missing contact layer in the output of the `report_physical -technology` command. If it is not reported, then search for the file containing the contact definition and determine if this file was read into TestMAX ATPG.

M907

Message Text

Contact layer information incomplete for stacked via `via_name` (M907)

Severity

Warning

Description

This message appears when contact layer information is missing a recognized stacked via. Note that a stacked via is created as a collection of basic via types either during DEF parsing or when analyzing the connectivity of a given net.

What Next

Look for the missing contact layer in the output of the `report_physical -technology` command. If it is not reported, then search for the file containing the contact definition and determine if this file was read into TestMAX ATPG.

M912

Message Text

Pin information not available for net `net_name` (M912)

Severity

Warning

Description

This message appears when the driver or the load pin information is missing for a given net. This message appears during subnet extraction.

What Next

Look for the identified `net_name` in the pin information section of the corresponding DEF file.

M913

Message Text

No technology LEF data found (M913)

Severity

Error

Description

This message appears when you want to report technology data (i.e. layer and via information), but no such data is found.

What Next

Restart your session by reading in the LEF/DEF files, including the technology files.

M914

Message Text

Null string is not allowed as the first argument of `-sub_str` (M914)

Severity

Error

Description

This message appears when the first argument of the `-sub_str` option of the `set_match_names` command is a null string. This option replaces all occurrences of the first argument string with the second argument string. You cannot replace a null sub-string.

What Next

Add a valid string to the first argument of the `-sub_str` option of the `set_match_names` command.

M915

Message Text

Invalid logical module `module_name` (M915)

Severity

Warning

Description

This message appears when an invalid logical module name is the argument to the `-exclude_modules` option of the `set_match_names` command.

What Next

Use the `report_modules` command to find the correct logical module name.

M916

Message Text

Invalid prefix *prefix_string* for logical instance *instance_name* (M916)

Severity

Warning

Description

This message appears when an invalid prefix for a logical instance is specified using the `-sub_str` option of the `set_match_names` command. The `-sub_prefix` option replaces the prefix string from a logical instance name with another sub-string.

What Next

Use a valid prefix for the logical instance specified in the `-sub_prefix` option.

M927

Message Text

Cannot overwrite an existing file without using '`-replace`' option (M927)

Severity

Error

Description

This message appears when you use the `report_layout` command to overwrite a file without specifying the `-replace` option.

What Next

Make sure you use the `-replace` option with the `report_layout` command to overwrite a file.

M928

Message Text

No file name specified (M928)

Severity

Error

Description

This message appears when a file name is not specified as an argument to the `report_layout` command.

What Next

Specify a file name as an argument for the `report_layout` command.

M929

Message Text

The net *net_name* has no driver (M929)

Severity

Error

Description

This message appears when a driver pin is not found in the list of pins associated with a given net when using the verbose mode during subnet analysis.

What Next

Check the pin definition for the given net in the DEF file and see if a driver pin is in the pin list.

M930

Message Text

Driver pin *pin_name* of net *net_name* is not connected (M930)

Severity

Error

Description

This message appears in verbose mode during connectivity analysis as part of the subnet extraction step. In this case, the driver pin of a given net appears to be disconnected from the rest of the geometries. This message is similar to the M929 message, which appears when no driver pin is found for a given net. However, in the M930 message, the driver

pin is found but the connectivity analysis failed at the driver pin. TestMAX ATPG does not currently support the layout structure used for the connection of this pin.

What Next

No action is necessary.

M931

Message Text

Only *value1* among *value2* load pins of net *net_name* could be reached
(M931)

Severity

Error

Description

This message appears in verbose mode during connectivity analysis when performing subnet extraction. In this case, only *value1* out of *value2* load pins could be reached from the driver pin. A load pin appears disconnected from the rest of the net. TestMAX ATPG does not currently support the layout structure used for the connection of this pin.

What Next

No action is necessary.

M938

Message Text

YDF Candidates Schema retrieval from file *file_name* was not successful
(M938)

Severity

Error

Description

This message appears when appending new data to an existing Yield Explorer Data Format (YDF) DFTCandidate table, but the existing schema for the table is not retrieved properly.

What Next

Examine the YDF file used for appending the DFTCandidate table. Check the syntax or determine if the file is empty.

M939

Message Text

Could not create Via segment - physical model does not exist (M939)

Severity

Error

Description

This message appears when writing a physical data segment without a physical model in memory.

What Next

Provide a physical model by reading the LEF/DEF files.

M940

Message Text

The physical model is now deleted (M940)

Severity

Error

Description

This message appears when the existing TestMAX ATPG representation of a physical model is deleted when reading an image.

What Next

No action is necessary.

M941

Message Text

Could not find an existing YDF file *file_name.ydf*. (M941)

Severity

Warning

Description

This message appears when appending new data to a Yield Explorer data format (YDF) file, but the file does not exist. In this case, a new YDF file is created.

What Next

No action is required.

M942

Message Text

Unable to open YDF file *file_name* (M942)

Severity

Error

Description

This message appears when TestMAX ATPG writes to an existing file, but the file does not open.

What Next

No action is necessary.

M945

Message Text

Subnet annotation to logical view was not successful (M945)

Severity

Error

Description

This message appears when the subnet information is not read successfully into the TestMAX ATPG logical view. In this case, the logical view is incomplete or the logical and physical pin names do not match.

What Next

Check the accuracy of the logical and physical pin names of the design.

M946

Message Text

Could not find an existing candidate file *file_name* (M946)

Severity

Error

Description

This message appears when a particular diagnostic candidate file is not found.

What Next

Check the name of the file specified by the `write_ydf -candidates` command.

M948

Message Text

Connection to server lost, *n* queries failed (M948)

Severity

Warning

Description

This message warns that the connection to the PHDS database was lost and reports the number of failed queries. If 10 nets are queried and the query fails, the number of failed queries *n* is 10. The connection failure is usually caused by a network problem.

What Next

Repeatedly specify the `run_diagnosis` command until the connection returns.

Messages M950 - M999

This section contains the following error messages and warnings:

- [M950](#)
- [M951](#)
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M950

Message Text

Could not create Physical Database. (M950)

Could not create Physical Database. \$PHDS_ROOT variable not set. (M950)

```
Could not create Physical Database. $PHDS_ROOT does not exist. (M950)
```

```
Could not create Physical Database. $PHDS_ROOT/lefdef_reader does not  
exist. (M950)
```

```
Could not create Physical Database. $PHDS_ROOT/phds_loader does not  
exist. (M950)
```

Severity

Error

Description

The first version of this message indicates a problem occurred when creating the physical database using the `set_physical_db` or `write_physical_db` commands.

The second through fifth versions of this message indicate that the `$PHDS_ROOT` environment variable is not set correctly, and the physical database could not be created. The `$PHDS_ROOT` variable is set in the TestMAX ATPG wrapper in the installation tree in `$SYNOPSYS/bin/tmax`.

What Next

The first version of this message is likely caused by a problem with the input LEF or DEF files or some other erroneous input. You should verify that the LEF or DEF files exist or check the settings of the `set_physical_db` or `write_physical_db` commands.

If you get the second through fifth versions of this message, you need to verify and correctly set the `$PHDS_ROOT` environment variable.

M951

Message Text

```
Cannot report available layers and tolerance values. (M951)
```

Severity

Warning

Description

There is a problem with the technology information. Either the wrong technology files are used or the database is obsolete. In either case, all queries to the database will fail to return any information.

What Next

Verify that the correct technology files were read when creating the PHDS database.

M952

Message Text

Top level physical design name could not be retrieved successfully from PHDS. (M952)

Severity

Warning

Description

The PHDS database cannot be queried because of a problem accessing the top-level physical design name.

What Next

Verify the connection to the PHDS database, or check the DEF files to see if the top-level design name is defined correctly.

M953

Message Text

The PHDS used to create the mapping table and the PHDS on which the current TMAX session is connected to are different. (M953)

Severity

Error

Description

This message appears when you read an image or connect to a PHDS database, and an ID in the PHDS database does not match an ID in the mapping table.

What Next

Verify that the PHDS database connected to TestMAX ATPG corresponds to the PHDS database used to match the logical and physical names.

For more information, refer to the topic Setting Up a Connection to the PHDS Database in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M954

Message Text

DEF input directory *DEF_directory* does not exist. (M954)

Severity

Error

Description

This message appears when the input DEF directory specified for the `set_physical_db` command does not exist.

What Next

Correct the path you specified in the `-def_directory` option of the `set_physical_db` command.

For more information, refer to the topic Using TestMAX ATPG to Create a PHDS Database in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M955

Message Text

Could not connect to physical database. (M955)

Severity

Error

Description

This message appears when you are attempting to perform physical diagnostics and a connection to a PHDS database cannot be established or is disconnected.

What Next

For more information, refer to the topic Reading a PHDS Database in the TestMAX™ ATPG and TestMAX Diagnosis User Guide and make sure that you correctly specified the `set_physical_db` or `open_physical_db` commands. If the original connection to a PHDS database is disconnected, you can restart the DAP process using the `open_physical_db` command.

For more information, refer to the topic Using TestMAX ATPG to Create a PHDS Database in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M956

Message Text

Invalid device name and/or version specified: *device_name device_version.*
(M956).

Severity

Error

Description

This message appears when the parameters specified in the `-device` option of the `set_physical_db` command do not match a device or version in the PHDS database for which you want to connect.

What Next

Make sure you correctly specified the PHDS device name and device version parameters in the `-device` option of the `set_physical_db` command. For example:

```
set_physical_db -device {DES 4}
```

The device name or version should match the list printed by the `set_physical_db -hostname hostname -port_number port_number` command.

For more information, refer to the topic Reading a PHDS Database in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M957

Message Text

Cannot perform physical diagnosis. (M957).

Severity

Warning

Description

This message appears immediately after an M955 message when a disconnected PHDS database is preventing TestMAX ATPG from performing diagnosis.

What Next

For more information, refer to the topic Reading a PHDS Database in the TestMAX™ ATPG and TestMAX Diagnosis User Guide and make sure you correctly specified the `set_physical_db` or `open_physical_db` commands. If the original connection to a PHDS

database is disconnected, you also might need to respecify the required connection parameters. After fixing the M955 message, try running the diagnosis again.

For more information, refer to the topic Using TestMAX ATPG to Create a PHDS Database in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M958

Message Text

DEF file with top level design not specified. (M958)

Severity

Error

Description

This message appears during physical diagnostics when you specify the `write_physical_db` command without first specifying the correct top level DEF file name using the `-top_def_file` option of the `set_physical_db` command.

What Next

Make sure you specify the `-top_def_file` option of the `set_physical_db` command. For example:

```
set_physical_db -top_def_file top_def_filename
```

For more information, refer to the topic Using TestMAX ATPG to Create a PHDS Database in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

For more information, refer to the topic Using Physical Data for Diagnosis in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M959

Message Text

Could not stop Data Access Process. (M959)

Severity

Error

Description

This message appears when you attempt to use the `close_physical_db` command to close a PHDS DAP process on a host that is not connected to the current TestMAX ATPG session.

What Next

Make sure you specify the `close_physical_db` command on a host in which a TestMAX ATPG session is currently connected to a PHDS database.

For more information, refer to the following topics in the TestMAX™ ATPG and TestMAX Diagnosis User Guide:

Starting and Stopping the DAP Server Process

Using Physical Data for Diagnosis

M961

Message Text

Could not map driver pin '*driver_pin_name*' to gate. (M961)

Could not map logical driver pin '*driver_pin_name*' to physical net.
(M961)

Severity

Error

Description

Both versions of this message appear when an incorrect driver pin name is specified with the `report_physical` command and prevents the reporting of physical data.

What Next

Specify an existing driver pin name with the `report_physical` command. A common problem is to specify an instance name without the pin name, as shown in the following example:

```
report_physical -net I_RISC_CORE/I_CONTROL/U42
```

The following example successfully reports the physical data:

```
report_physical -net I_RISC_CORE/I_CONTROL/U42/ZN
```

For more information, refer to the topic Using Physical Data for Diagnosis in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M963

Message Text

Process *process_ID* returned unsuccessfully. (M963)

Maximum number of processes recreated, no new processes will be created.
(M963)

Severity

Error

Description

This message appears when a process stops unexpectedly when running physical diagnostics. The problem is most likely caused by a lost connection to the PHDS (physical design store) database.

What Next

Make sure that you can still connect to the PHDS database (for more information, see the topic Setting Up a Connection to the PHDS Database in the TestMAX™ ATPG and TestMAX Diagnosis User Guide). You can specify the `-max_retries` option of the `set_physical` command to enable TestMAX ATPG diagnostics to automatically attempt to reconnect to a PHDS database if it disconnects again. The following example specifies TestMAX ATPG diagnostics to attempt to connect to a PHDS database a maximum of eight times:

```
set_physical -max_retries 8
```

You can also specify the `-no_max_retries` option of the `set_physical` command to attempt to connect to the PHDS database an unlimited number of times. Use this option with caution because it can cause the extraction process to run indefinitely.

For more information, see the topic Setting Up and Running Physical Diagnosis in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M964

Message Text

Multi-core extraction completed unsuccessfully. (M964)

Severity

Error

Description

This message appears when the multicore process fails and only some of the physical data is extracted. The problem is most likely caused by a lost connection to the PHDS (physical design store) database.

What Next

Make sure that you can still connect to the PHDS database (for more information, see the topic [Setting Up a Connection to the PHDS Database in the TestMAX™ ATPG and TestMAX Diagnosis User Guide](#)). You can specify the `-max_retries` option of the `set_physical` command to automatically attempt to reconnect to a PHDS database if it disconnects again. The following example specifies TestMAX ATPG diagnostics to attempt to connect to a PHDS database a maximum of three times:

```
set_physical -max_retries 3
```

You can also specify the `-no_max_retries` option of the `set_physical` command to attempt to connect to the PHDS database an unlimited number of times. Use this option with caution because it can cause the extraction process to continue indefinitely.

For more information, see the topic [Setting Up and Running Physical Diagnosis in the TestMAX™ ATPG and TestMAX Diagnosis User Guide](#).

M967

Message Text

```
Could not map logical driver pin driver_pin_name to physical net. (M967)
```

Severity

Error

Description

This message appears when physical data for a net or pin has not been extracted prior to using the `report_physical` command.

What Next

The `report_physical` command can report only the physical data previously extracted by the `run_diagnosis`, `extract_nets`, or `extract_pins` commands. If the physical data is missing, you can use the `-names` option of the `extract_nets` command or the `-names` option of the `extract_pins` command to extract additional physical data for the `report_physical` command.

For more information, see the topic [Setting Up and Running Physical Diagnosis in the TestMAX™ ATPG and TestMAX Diagnosis User Guide](#).

M970

Message Text

Pattern counter of *number* bits cannot support requested *pattern_count* patterns (M970)

Severity

Error

Description

This message prints during the Ultra LBIST ATPG process when the requested number of patterns exceeds the highest number that can be stored in the pattern counter.

What Next

Use the `set_atpg` command to specify a lower pattern count and invoke the `run_atpg` command again.

M972

Message Text

Net *net_name* has excessive fanin (M972).

Severity

Error

Description

TestMAX ATPG attempted to create a model containing the identified net *net_name*. However this net exceeded the limit of 65,535 inputs, and the model was not created.

What Next

Reduce the inputs of the net by changing the netlist, net connections, or build options that modify the model. For more information, see the topic Building the ATPG Model in the TestMAX™ ATPG and TestMAX Diagnosis User Guide.

M975

Message Text

Scan enable could not be found (*Explanation*) (M975)

Severity

Warning

Description

This message indicates that TestMAX ATPG was unable to automatically identify the scan enable port used for the first scan cell in the first scan chain.

What Next

No action is required. However, if you are performing two-clock transition fault test generation using the `-nopi_changes` option of the `set_delay` command, make sure a pin constraint is placed on every scan enable port.

M976

Message Text

`Nopi_changes` option selected when scan enable (*port_name*) not constrained to non-shift state (*constrained_value*). (M976)

Severity

Warning

Description

When performing two-clock transition fault test generation using the `-nopi_changes` option of the `set_delay` command, TestMAX ATPG checks the user-defined or pre-calculated scan enable ports to make sure they are constrained. If TestMAX ATPG encounters an unconstrained scan enable port, it issues a M976 message. This check is not performed for JTAG circuits using sequential compression.

What Next

No action is required. However, you should make sure that a pin constraint is placed on the actual scan enable port(s) to optimize test generation.

M977

Message Text

`Nopi_changes` option selected when scan enable cannot be identified. (M977)

Severity

Warning

Description

When performing two-clock transition fault test generation using the `-nopi_changes` option of the `set_delay` command, TestMAX ATPG checks the user-defined or pre-calculated scan enable ports to make sure they are constrained. If TestMAX ATPG is unable to find user-defined or pre-calculated scan enable ports, it issues the M977 message. This check is not performed for JTAG circuits using sequential compression.

What Next

No action is required. However, you should make sure that a pin constraint is placed on the actual scan enable port(s) to optimize test generation.

M978

Message Text

Invalid operation in line number: truncation simulation will not be performed. (M978)

Severity

Error

Description

This message indicates that an invalid operation occurred while parsing a particular line in the file that defines the selected observe points used for a truncated simulation. This file is specified using the `set_simulation-truncate file_name` command.

The possible invalid operations are as follows:

Invalid file name (*file_name*) — The *file_name* argument of the `set_simulation-truncate` command is invalid.

Invalid primary output (PO) name (*po_name*) — A line with a single argument uses an invalid PO name.

Invalid chain name (*chain_name*) — The first entry of a line with two arguments is an invalid chain name.

Invalid chain position (*chain_position*) — The second entry of a line with two arguments is an invalid chain position.

Too many arguments (*number*) — The identified line has too many arguments.

No observe points were defined — The specified file contains no observe points.

What Next

Fix the identified problem and run the simulation again.

M979

Message Text

No arguments in line *number*: line is ignored. (M979)

Severity

Warning

Description

This message indicates that no arguments were found in the identified line of the file used to define the selected observe points for truncated simulation. This file is specified using the `set_simulation-truncate file_name` command.

What Next

No action is required if the line was intended to be blank.

M980

Message Text

Truncated simulation will not be performed on *number* shift cycles due to exceeding 2 cycles. (M980)

Severity

Warning

Description

This message indicates that the truncated simulation of the selected pattern failed because it has too many shift cycles.

What Next

No action is required, and the simulation is performed as normal on all gates. If you want to use truncated simulation, you must reduce the number of shift cycles to fewer than two cycles using the `-shift_cycles` option of the `set_simulation` command. The command usage is:

```
set_simulation [-shift_cycles number]
```

M981

Message Text

Truncated simulation will not be performed due to pattern cycles (*number*) exceeding 2. (M981)

Severity

Warning

Description

This message indicates that the truncated simulation of the selected pattern failed because it has too many capture cycles.

What Next

No action is required, and the simulation is performed as normal on all gates. If you want to use truncated simulation, select a different pattern that contains fewer than two capture cycles.

M988

Message Text

There are *N* generated patterns plus *P* pending observe needs (M988)

Severity

Warning

Description

When TetraMAX ATPG cannot meet a pattern's observe requirements with the next generated pattern, it places the pattern in a temporary buffer. ATPG tests each newly generated pattern to determine if it meets the observe needs of one of the *P* previously generated patterns in the temporary buffer. If there is a match, both patterns are added to the pattern buffer. At the end of the ATPG run (or if the ATPG process is interrupted), all patterns remaining in the temporary buffer require the addition of padding patterns.

At worst, *N* padding patterns are added to the previously generated patterns at the end of ATPG. If ATPG completes, the number of padding patterns is usually less than *N* patterns – although at least some padding patterns are usually added at the end of ATPG. The number printed for each interval shows the sum of generated patterns plus pending observe needs, which is the worst-case number of patterns required by ATPG up to that interval.

What Next

No action is required. This is an informational message only.

M990

Message Text

Deleting common clock exception. (M990)

SRC: *exception_type* TYPE: *clock_name1* (from) *clock_name2* (to)

Severity

Informational

Description

This message identifies a timing exception deleted because the `set_sdc -delete_common_clock_exceptions` command is specified. The timing exception is between two clocks that are defined on the same source object.

What Next

You should confirm that the timing exception is not needed. Only clock-to-clock timing exceptions in which both clocks are defined on the same source object are removed. These exceptions cannot be interpreted properly for purposes of ATPG, so it is appropriate to remove them. Here is an example of a timing exception that is safe to remove:

```
create_clock -name CLK1 -period 10 -waveform {2.0 4.0} {ck_inv2/OUT}
create_clock -name CLK2 -period 20 -waveform {8.0 10.0} {ck_inv2/OUT}
-add
set_clock_groups -async -name grp1 -group CLK1 -group CLK2
```

Some timing exceptions should not be removed, for example:

```
create_clock -name FE_CLK_master -period 200 -waveform { 45 55 145 155 }
[get_ports FE_CLK]
create_generated_clock -name FE_CLK_wft1 -edges { 1 2 5 } -source
[get_ports FE_CLK] [get_ports FE_CLK] -add -master_clock FE_CLK_master
create_generated_clock -name FE_CLK_wft2 -edges { 3 4 7 } -source
[get_ports FE_CLK] [get_ports FE_CLK] -add -master_clock FE_CLK_master
set_false_path -from [get_clock FE_CLK_wft1] -to [get_clock FE_CLK_wft2]
```

This timing exception might need to be rewritten to be interpreted appropriately by TestMAX ATPG.

M998

Message Text

Quiet chain test is not compatible with Shift Power Controller. Setting quiet_chain_test=no. (M998)

Severity

Informational

Description

This message is printed when the shift power controller (SPC) is in use, and the `set_atpg -quiet_chain_test` command is specified. It informs you that the `-quiet_chain_test` ATPG option was reset to its default state of no when using a SPC in a DFTMAX Ultra design.

What Next

No action is required. Chain test patterns are generated if enabled, and the SPC ensures that these patterns meet the budget specified by the `set_atpg -shift_controller_peak` command.

M999

Message Text

Specified shift controller budget of *D%* is less than the percentage of scan chains controlled by an SPC bit (*N*) and may not be achievable. (M999)

Severity

Warning

Description

This message appears when you set too low of a shift controller peak or budget using the `-shift_controller_peak` option of the `set_atpg` command. A low peak or budget specification may impact QoR or slow ATPG because of excessive pattern rejection.

The minimum shift controller peak or budget is calculated as a percentage using the following formula:

$$100 / \langle \text{total number of SPC control bits} \rangle$$

For example, if a design has 10 SPC control bits, the specified shift controller peak should not exceed 10 percent (100/10). The following example shows a budget specification that triggers an M999 warning message:

```
TEST-T> set_atpg -shift_controller_peak 4
```

```
Warning: Specified shift controller budget of 4% is less than the
percentage of scan chains controlled by an SPC bit (10%) and may not be
achievable. (M999)
```

What Next

Use the `-shift_controller_peak` option option to set a budget that exceeds the percentage of scan chains controlled by an SPC bit.

Messages M1000 - M1299

This section contains the following error messages and warnings:

- [M1001](#)
- [M1002](#)
- [M1003](#)
- [M1004](#)
- [M1005](#)
- [M1006](#)
- [M1007](#)
- [M1008](#)
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- [M1031](#)
- [M1032](#)
- [M1037](#)
- [M1038](#)
- [M1039](#)
- [M1250](#)
- [M1252](#)

M1001

Message Text

SPC Registers control *D* out of a possible *N* Scan Chains. ATPG may not succeed (M1001)

Severity

Warning

Description

This message is printed when shift power controller (SPC) registers are not in control of all possible internal chains in a design.

What Next

Recheck your DFT specifications to ensure that SPC is used across all internal chains present in the design.

M1002

Message Text

Clock constraints <constraints> and timing <timing> are both specified; timing <timing> is ignored. (M1002)

Severity

Warning

Description

This message is displayed when both the clock constraints and the synchronized timing block are specified when using TestMAX ATPG. The tool will continue to pattern generation using only the clock constraints.

What Next

This is a message to inform that two timing features are enabled together and synchronized timing is ignored.

M1003

Message Text

Constrained synchronized clocking is not supported in single/multicore ATPG. (M1003)

Severity

Error

Description

This message is displayed when the constrained synchronized clocking feature is used with TestMAX ATPG.

What Next

Try using the constrained synchronized clocking feature with TestMAX II.

M1004

Message Text

Clock constraints <constraints> and timing <timing> are both specified; use set_drc -constrained_clock_timing to run constrained synchronized clocking. (M1004)

Severity

Warning

Description

This message is displayed when both the clock constraints and the synchronized timing block are specified when using TestMAX II. The tool will continue by ignoring the timing block.

What Next

If the intention is to run constrained synchronized clocking feature, then enable it.

M1005

Message Text

Constrained synchronized clocking will run using constraints
<constraints> and clock timing <timing>. (M1005)

Severity

Warning

Description

This message is displayed when constrained synchronized clocking is specified for TestMAX II and both constraints and clock timing are specified.

What Next

Not action is required. This is an informational message only.

M1006

Message Text

Constrained synchronized clocking supports only single non-zero clocking
procedure; multiple usable clocking procedures exist. (M1006)

Severity

Warning

Description

This message is displayed when multiple non-zero clock constraints are specified in the SPF and the constrained synchronized clocking feature is enabled.

What Next

Remove the clock constraints that are not needed from the SPF.

M1007

Message Text

Clock timing is not specified for constrained synchronized clocking.
Using clock constraints <constraints>. (M1007)

Clock constraints are not specified for constrained synchronized clocking. Using clock timing <timing>. (M1007)

Severity

Warning

Description

This message is displayed when constrained synchronized clocking is specified and one of the clock constraints or timing block is missing when using TestMAX II. The tool will continue with the one specified.

What Next

Specify the timing block or the clock constraints block and then run the tool to generate patterns with the constrained synchronized clocking feature.

M1008

Message Text

Generated SDC constraints with <#clocks> clocks and <#groups> groups for Constrained Synchronized Clocking. (M1008)

Severity

Warning

Description

This message is displayed when constrained synchronized clocking is specified. It reports the number of clocks and groups used.

What Next

Not action is required. This is an informational message only.

M1009

Message Text

SynchronizedCycle value <value> is not valid; SynchronizedCycle value is set to 0. (M890)

Severity

Warning

Description

This message is displayed when an invalid value is specified for the SynchronizedCycle in the SPF. The value 0 is used instead.

What Next

No action is required. This is a message only.

M1010

Message Text

Upper limit of opened GSV should be set between the min number (1) and the max number allowed (4). (M1010).

The current GSV number (N1) has exceeded your set value (N2). (M1010)

Severity

Error

Description

An attempt was made to set the upper limit number of the opened GSV windows less than 1 or more than 4. Or, an attempt was made to set the upper limit number less than the current opened GSV number. N1 is the current value, N2 is the value users set.

What Next

Reset the value between the current opened GSV number and the max value 4. If you want to decrease the upper limit, you can close some GSVs to make sure the current GSV number is no more than the value you want to set.

M1017

Message Text

Parsing balanced clock group file at line 1: Read more than 47 pin path names in a line, split "balanced_clock_group" in multiple lines. (M1017)

Severity

Warning

Description

This warning message appears when the `report_lockup_latches` command with the `-balanced_gate_file` option tries reading more than 47 pin path names in a line. The balanced gate file is ignored in this case.

What Next

You can split the balanced clock groups in multiple lines.

It is possible to have one line for each pin, and multiple lines can refer to the same group name, for example:

```
balanced_clock_group my_clock_group1 clk_gate_0/main_gate/Z
balanced_clock_group my_clock_group1 clk_gate_1/main_gate/Z
balanced_clock_group my_clock_group1 clk_gate_2/main_gate/Z
balanced_clock_group my_clock_group2 clk_gate_3/main_gate/Z
balanced_clock_group my_clock_group2 clk_gate_4/main_gate/Z
```

M1018

Message Text

X capture from chain=C into MISR N of seq_compressor K in pattern P during shift S using mode=M. (M1018)

Severity

Warning

Description

Note the following:

C is name of the failing chain

N is name of the failing MISR (if available)

K is the name of the failing sequential compressor

P is the failing pattern id number for the current simulation interval

S is the failing shift position

M is the observe mode type

This warning message appears if the following conditions are met:

The `-seq_comp_allow_misrs_x` option of the `set_atpg` command is specified

The `run_simulation` command is specified for sequential-based compression patterns when an X value is captured into a measured MISR (multiple-input shift register).

The input value to a MISR depends on the values currently on the output of the chains and the current observe mode as determined by the XTOLPRPG (X-tolerant pseudo-random pattern generator). When this message is issued, the associated MISR is no longer simulated for the failing pattern and the potential simulation failures associated with it are not considered.

What Next

This message occurs only when running the `run_simulation` command. The problem is likely caused by constraints that affect pattern simulation (such as capture masks or cell constraints) that were not used during the ATPG process. If the patterns are intended to be used under these conditions, it is likely the patterns are bad and need to be recreated.

M1019

Message Text

Average shift power budget A%(B%) is equal to minimum shift power C%(D%) for compressor *name* (M1019).

Average shift power budget A%(B%) is lesser than minimum shift power C%(D%) for compressor *name* (M1019).

Severity

Warning

Description

Note the following:

A is the user-specified average shift power budget

B is the mapped average shift power budget value based on A

C is the calculated minimum average shift power budget for the IP

D is the mapped average shift power budget based on C

This message appears when you set too low a shift power budget to meet the average power using the `set_drc -seq_comp_power_mode` command for a sequential compressor (*name*) using JTAG-LBIST ATPG.

Based on the number of shifts (refer to sequential compressor rules checking in DRC) in the IP, you can calculate the minimum average shift power budget possible for the IP. The higher the number of shifts, the lower the average shift power budget for the IP.

You can set the number of power enable registers used to control shift power (refer to Power_Enable block in the SPF in the SeqCompressorStructures section).

Based on the possible values of all the registers, there are specific values of average shift power available. This is a fixed property of the IP.

A user-specified budget specified must be mapped to a specific value of the power enable registers, such that the average shift power value obtained by setting power enable registers is just less than the user-specified shift power.

Similarly, the minimum average shift power budget for the IP must be mapped to specific value of power enable registers.

The two versions of this message indicate the comparison between values B and D.

This second message indicates that JTAG-LBIST ATPG does not meet the average user-specified shift power budget. This message is appears during DRC when sequential compressor rules checking is performed.

What Next

Based on the message, identify the minimum average shift power possible for the IP (C%) and use the `set_drc -seq_comp_power_mode` command to specify a higher budget than specified before until you do not see the message.

M1020

Message Text

Cannot create assertion for pin <P>. Possibly removed during build optimizations. (M1020)

Severity

Warning

Description

This message appears whenever an assertion cannot be created because of a missing pin. Possible cause of this is that the pin was removed during build optimizations.

What Next

This is just a message to inform that the assertion was not created.

M1027

Message Text

The following setting is ignored for chain test patterns. (M1027)

`command_settings`

Severity

Warning

Description

This message warns that a certain command is not supported for chain test pattern generation.

What Next

You can safely ignore this message. However, you should be aware of the impact of the ignored command. For example, if the `set_atpg -shift_controller_peak -partition` command is ignored, then the generated chain test pattern might violate power budget for partitions.

M1029

Message Text

Instance name (id Gate Id) identified to be a potential cause of R7 violations. (M1029)

Severity

Warning

Description

If DRC encounters R7 violations, a quick backward trace is performed, and this message is printed if it identifies any gate that may be the source of X leading to R7. This message is intended to help the user debug and resolve R7 violations.

What Next

Use the gate identified in this message as part of R7 debug.

M1031

Message Text

Error: Cannot map to end-of-test MISR measures because some patterns have MISR at X. (M1031)

Severity

Error

Description

At the time of pattern mapping from MUPP->MUPT, it checks if some patterns (except the first two patterns) have X-ed out `mISR_unload`. Such patterns come from XLBIST to SEQ mapped pattern without the option `set_patterns -full_xlbist_mapping`.

Such patterns cannot be converted to MUPT patterns and M1031 error comes up.

What Next

To support XLBIST to SEQ MUPP to SEQ MUPT pattern mapping, you need to use the `set_patterns -full_xlbist_mapping` option while mapping XLBIST pattern to SEQ pattern.

M1032

Message Text

PI constraints (`#failing_gates/first_failing_gat_id`) are not satisfied during load_unload. (M1032)

PI constraints (`#failing_gates/first_failing_gat_id`) are not satisfied during reseed_load. (M1032)

External clocks (`#ext_clks_used_in_this_pattern/first_ext_clk`) are used in the pattern `pattern_number`. (M1032)

All PO's (`#unmasked_PO/first_unmasked_PO_gat_id`) are not masked. (M1032)

Severity

Information message

Description

For IDU2 designs, when `internal_load_delay` is greater than zero, the delay cycles (which are otherwise needed as there is a delay from Scan inputs to IDU inputs) in the final `load_unload` (before capture) call of a pattern are not added, when the design satisfies following criteria:

1. All PI constraints are satisfied during `load_unload` and `reseed_load`.
2. All PO's are masked.
3. No External clock is used for this pattern.

Criteria 1 is checked during DRC, and criteria 2 is checked before the starting of ATPG. Criteria 3 is checked per pattern.

If the pattern fails to satisfy one of the aforementioned conditions, M1032 is issued accordingly, and delay cycles are added to the final `load_unload` call.

What Next

The design, the protocol file or the TetraMAX script needs to be modified to satisfy the aforementioned criteria, if you want to optimize the `cycles_count` by not having the delay cycles in the `final_load_unload`.

The `add_pi_constraints` command or `remove_pi_constraints` command can be used to add or remove constraints on PIs.

The `add_PO_mask` or `set_delay -nopop_measure` commands can be used to mask POs.

M1037

Message Text

Reading back the Garbled Image written without the `-netlist_data` option is not supported in SEQ/XLBIST. (M1037)

Severity

Warning

Description

This warning message is displayed during `write_image` command if a garbled image is written without the “`-netlist_data`” option for SEQ/XLBIST designs. SEQ/XLBIST does not support reading garbled images that do not have netlist data. The image has to be written with the “`-netlist_data`” option to avoid an error during `read_image`.

What Next

Rewrite the image with the “-netlist_data” option to avoid the error when reading it backward.

M1038

Message Text

Reading Garbled image written without the -netlist_data option is not supported in SEQ/XLBIST. (M1038)

Severity

Error

Description

This error is displayed during `read_image` if a garbled image without netlist data is read for SEQ/XLBIST designs. SEQ/XLBIST does not support reading garbled images that are written without the “-netlist_data” option.

What Next

Rewrite the image with the -netlist_data option to proceed.

M1039

Message Text

Warning: Portfault boxed pin(s) dropped during build optimization.
(M1039)

Severity

Information message

Description

The `set_build -portfault_box` command does not guarantee that a pin is preserved. It preserves the inputs and outputs of a module but not necessarily the pins that are internal. The build may optimize these internal pins based on ATPG rules.

If any such pins are optimized during build, then the M1039 error message is displayed at two stages:

- `run_build`

A single Warning message if any number of portfault box pins are dropped during build optimizations:

```
Warning: Portfault boxed pin(s) dropped during build optimization.  
(M1039)
```

- `read_sdc`

With the `set_sdc -read_sdc_reporting` command, TMAX displays a Warning message followed by the list of pins dropped during build at the end of `read_sdc`.

```
Warning: Portfault boxed pin(s) were dropped during build  
optimization. (M1039)
```

```
Pin dropped during build: MY_module/top/core/des/Q[1] (M1039)
```

```
Pin dropped during build: MY_module/top/core/des/Q[2] (M1039)
```

```
... (list of pins dropped)
```

With the `set_sdc -noread_sdc_reporting` command (default), TMAX displays a Warning message and the way to display all pins dropped during build at the end of `read_sdc`.

```
Warning: Portfault boxed pin(s) were dropped during build  
optimization. (M1039)
```

Use "`set_sdc -read_sdc_reporting`" before `read_sdc` to view list of pins optimized in build.

What Next

The `get_pins` command does not return any pin if these pins are provided as arguments. This can cause the issues during `read_sdc`. SDC can be modified by replacing optimized pins with alternative pins.

M1250

Message Text

```
Clock <clock-name> will not be used in this interval due to excessive X-  
capture of number (> threshold-value). (M1250)
```

Severity

Warning

Description

This message prints when the `run_atpg` command is used with the `-jtag_lbist {s p c}` option and the `-enable_x_avoidance` option to generate XLBIST intervals. The clock `<clock-name>` is skipped in the present interval if the X-capture *number* is greater than the *threshold-value*.

What Next

No action is required. Pattern generation continues.

M1252

Message Text

Clock `<clock-name>` is not driven by OCC, it affects `<scan_cell_name>` and `<scan_cell_num>` more flops. (M1252)

Severity

Warning

Description

Utilizing the `set_drc -hierarchical_atpg` switch for pattern porting requires all scan cells to be driven by OCC.

This message specifies that the clock `<clock-name>` that affect `<scan_cell_name>` and `<scan_cell_num>` other flops is not driven by OCC.

What Next

Utilize the `set_drc -verbose` option to display all the flops and the clock name that are not driven by OCC, then proceed to modify them to be driven by the OCC clock.

Messages M1800 - M2832

This section contains the following error messages and warnings:

- [M1831](#)
- [M1832](#)
- [M1833](#)
- [M1835](#)
- [M1836](#)
- [M1837](#)

- [M1838](#)
- [M1839](#)
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- [M1844](#)
- [M1845](#)
- [M1846](#)
- [M1848](#)
- [M2831](#)
- [M2832](#)

M1831

Message Text

Pattern period value %d does not match with SPF timing value %d. (M1831)

Severity

Warning

Description

This message is printed during `write_pattern` command when the timing defined in SPF does not match with the STIL pattern (used with `set_pattern -external`).

What Next

Check whether the pattern file used with `set_pattern-external` has the same timing information as defined in SPF. If not, then fix the external pattern to generate the correct waveform timing information.

M1832

Message Text

An extra lockup latch is added between scan cells *scan_cell1* clocked by *clock1* (through *pinpath_name1*) LE (*time*) and *scan_cell2* clocked by *clock2* (through *pinpath_name2*) TE (*time*) (M1832)

An extra lockup latch is added between scan cells *scan_cell1* clocked by *clock1* (through *pinpath_name1*) TE (*time*) and *scan_cell2* clocked by *clock2* (through *pinpath_name2*) LE (*time*) (M1832)

Severity

Warning

Description

This message is printed by the `report_lockup_latches` command. The two instances are adjacent scan cells. An extra lockup latch is reported if the following condition applies:

A lockup latch is identified between two adjacent scan cells that are driven by different clock edges. Namely, LE to TE and TE to LE.

This extra lockup latch configuration creates a race condition, so the latch should be removed. When adjacent scan cells are driven by TE to LE or by LE to TE clocking, there is no race and no lockup latch should be used.

What Next

Check that the correct `report_lockup_latches` arguments were specified. If not, rerun the command with the correct arguments. Otherwise the violation should be analyzed.

You can analyze this violation by launching the GSV and displaying the two instances using the Shift pindata.

Extra lockup latches between LE to TE or TE to LE registers can create a race condition. Therefore, extra lockup latches should be removed from the design.

M1833

Message Text

Could not find requested procedure name (M1833).

Severity

Warning

Description

This message is issued when user calls `set_pindata` along with a procedure name (for example, `-reseed_shift`, `-occ_load`, and so on.) to identify the simulation values for a Gate but such procedure data has not been saved during DRC. The corresponding procedure *name* is printed in the message text.

What Next

The user may call `-report_proc_data` to list the procedure data that have been saved during DRC. The user should choose a “Pattern data” from output of the former command, to identify the list of available procedure names, that can be used alongside `-setpin_data`. Note that `-report_proc_data` is an undocumented command.

M1835

Message Text

Error: Payload_cycles_counter is loaded with value(V) less than the codec_drc_chain length(L). (M1835)

Severity

Error

Description

Note:

- V is the value of `Payload_cycles_counter` register
- L is the `codec_drc_chain` length

The error occurs when the `payload_cycles_counter` register is loaded with a value less than the `codec_drc_chain` length, by the end of `codec_drc` procedure decoding, when `codec_drc` procedure is controlled by IDU. To load the full `codec_drc_chain`, `payload_cycles_counter` should be loaded with value at least equal to the `codec_drc_chain` length.

What Next

This needs to be corrected in the netlist. Check from which register, the `payload_cycles_counter` is getting loaded for `codec_drc` procedure, and correct the value in the register.

M1836

Message Text

Warning: Scan enable gate appears to be constrained to SHIFT state (M1836).

Severity

Warning

Description

The warning occurs when scan enable gate is constrained to `SHIFT` state. This means that scan cells are in `SHIFT` state even during the capture cycle, thus causing the very low coverage.

What Next

By setting the pin-data to good pattern simulation, you can analyze why `scan_enable` gate is constrained to `SHIFT` state.

M1837

Message Text

Warning: N clock pairs are now considered asynchronous and affect test coverage and simulation values. (M1837)

Severity

Warning

Description

The warning occurs when there are clocks using asynchronous OCC structures that potentially impact fault coverage results. By default any two internal clocks are considered asynchronous, regardless of `PLLSource`, DRC violations, or any other SPF constructs. If ATPG pulses the clocks in parallel and they have a disturbed relationship, then additional masking is applied to ensure that the patterns pass simulation.

What Next

This is an information only message. You could inspect the fault universe with `report_summaries faults -per_clock_domain`, `report_clocks -intclocks` and `report_clocks -matrix -verbose` commands.

Consider implementing synchronous OCC to increase the coverage if possible.

M1838

Message Text

Implication learning was truncated due to memory limits. Results may differ in another run. (M1838)

Severity

Warning

Description

As implication learning is the single main consumer of memory during `run_build` multithreaded learning. This message indicates truncation of implication learning if memory limits are exceeded. Note that the stopping point may differ from run to run as parallel threads allocate memory in nondeterministic order.

Note:

Truncation is often followed by stopping.

What Next

Try to relaunch your job on another machine with more memory to avoid truncation or stopping of implication learning.

M1839

Message Text

Implication learning was stopped due to memory limits. Results may differ in another run. (M1839)

Severity

Warning

Description

As implication learning is the single main consumer of memory during `run_build` multithreaded learning. This message indicates stopping of implication learning if memory limits are exceeded.

Note:

The stopping point may differ from run to run as parallel threads allocate memory in nondeterministic order.

What Next

Try to relaunch your job on another machine with more memory to avoid truncation or stopping of implication learning.

M1840

Message Text

This message has following two flavors.

Error: `Payload_cycles_counter` is loaded with incorrect value (got=G/exp=E) for IDU N during `codec_drc` procedure. (M1840).

Error: `Local_cycles_counter` is loaded with incorrect value (got=G/exp=E) for IDU N during `codec_drc` procedure. (M1840)

Where G is the value on `Payload_cycles_counter` or `Local_cycles_counter`

E is the expected value of the `Payload_cycles_counter` or `Local_cycles_counter`

N is the name of the IDU

Severity

Error

Description

When pure copy IDUs (identified by “*PureCopy*” construct in SPF) are present, `Payload_cycles_counter` and `Local_cycles_counters` of all the IDUs should be loaded with same value (length of the longest `codec_chain`) at the end of the `codec_drc` procedure. This decoding is at the end of preamble simulation of the `codec_drc` procedure failing which this message is shows up.

What Next

To fix this error, the protocol file needs to be corrected such that

`Payload_cycles_counter` and `Local_cycles_counters` of all the IDUs are loaded with the same value.

M1841

Message Text

Without the “`-netlist_data`” option of `write_image`, `run_drc` may fail after `read_image`. (M1841)

Severity

Warning

Description

If the `-netlist_data` option is not given while writing the build image, then net names are not written into the image. The `run_drc` command might give v5 (unresolved reference) if such build image is used.

What Next

If DRC errors are similar to v5 (unresolved reference (No such net ***)), then they can be resolved by providing the `-netlist_data` option while writing the image.

M1842

Message Text

```
set_atpg -min_ate setting is not honoured. (M1842)
```

Severity

Warning

Description

When `capture_window` and `se_delay` are specified within the protocol file, `min_ate` is calculated as `capture_window + 2` and user-defined `min_ate` settings is not honored.

What Next

`Min_ate` can be set to the required value by changing the `capture_window` in the protocol file.

M1843

Message Text

```
CaptureWindow is not defined in the protocol file. (M1843)
```

Severity

Warning

Description

This message gets printed if 'CaptureWindow' is not defined in protocol file. For IDU2 designs, 'CaptureWindow' should be specified otherwise incorrect patterns may get generated.

What Next

Protocol file needs to be corrected.

M1844

Message Text

CARE_Shadow flop(G) of compressor S is not part of the occ_chain. (M1844)

Where, G indicates gate ID of the CareShadow cell

S indicates name of DFTMAX Seq CODEC associated with the CARE_shadow flop

Severity

Warning

Description

If a CODEC has Low_Power v3 hardware, all its CARE_Shadow flops should be part of the occ_chain.

What Next

Netlist needs to be corrected such that CARE_Shadow flops are part of the occ_chain. The "occ_chain_load_segments" option of the report_compressor command can be used to see the cells present on the occ_chain.

M1845

Message Text

LocalCyclesCounter of IDU X is not loaded with enough cycles during codec_drc procedure(G/E). (M1845)

Severity

Error

Description

Note the following:

G is the value of the LocalCyclesCounter loaded, during codec_drc.

E is the expected value that LocalCyclesCounter should be loaded with.

If codec_drc chains of different IDUs are sharing the scan-ins, LocalCyclesCounters of IDUs should be loaded as following:

$$\text{LocalCyclesCounter value} = \sum_{k=0}^{\text{index of the IDU in the spf}+1} \text{codec_drc_length}^k$$

Where index of the IDU is proportional to the positionId of the IDU.

What Next

Protocol file needs to be corrected such that LocalCyclesCounters are loaded with enough cycles.

M1846

Message Text

PHDS file: file_name missing. (M1846)

PHDS file: file_name size mismatch (expected=n got=m) (M1846) (M1846)

Severity

Error

Description

This message is displayed during `set_physical_db -device {name version}` after `set_physical_db -database dirpath`, when an essential file in PHDS is missing or file size is not as expected.

What Next

Check the PHDS generation log to ensure the generation is successful. For more information, see section “Using Physical Data for Diagnosis” in the “TestMAX™ ATPG and TestMAX Diagnosis User Guide.”

M1848

Message Text

User specified fault limit is N, retaining N collapsed faults for CHAIN_CAPTURE ATPG. (M1848)

Severity

Warning

Description

When the `set_drc -clock -chain_capture` or `set_drc -clock -retention_test` command is specified, this message is displayed during the `run_atpg` and `run_fault_sim` steps, if the following value is non-zero: `set_atpg -retention_test_fault_limit N`.

What Next

No action is necessary. Use the command to `set_atpg -retention_test_fault_limit` to 0 if you want to disable the fault count limit during `chain_capture` ATPG.

M2831

Message Text

Testability might degrade due to LE to TE transition between the scan cells `scan_cell1` clocked by `clock1` (through `pinpath_name1`) LE (time) and `scan_cell2` clocked by `clock2` (through `pinpath_name2`) TE (time). (M2831)

Severity

Warning

Description

This message is printed by the `report_lockup_latches` command. The two instances are adjacent scan cells belonging to two different or same clock domains. The `scan_cell1` cell is clocked on the leading edge (LE) and `scan_cell2` cell is clocked on the trailing edge (TE) and a lockup latch is not present between them.

What Next

You can analyze this violation by launching the GSV and displaying the two instances using the Shift pindata.

M2832

Message Text

Unpredictability introduced, extra lockup latch in the LE to TE transition of different clock domain between the scan cells `scan_cell1` clocked by `clock1` (through `pinpath_name1`) LE (time) and `scan_cell2` clocked by `clock2` (through `pinpath_name2`) TE(time). (M2832)

Severity

Warning

Description

This message is printed by the `report_lockup_latches` command . The two instances are adjacent scan cells belonging to two different clock domains. The cell `scan_cell11` is clocked on the leading edge (LE) and cell `scan_cell12` cell is clocked on the trailing edge (TE) and, a lockup latch is present between them.

What Next

You can analyze this violation by launching the GSV and displaying the two instances using the Shift pindata.